



Configuration Management Reference Guide for UnityCloud

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AMF CC Configuration

This module defines the configuration for AMF Call Control.

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Amf Cc

AMF Call Control

XPath: *cna-amf-cc:/amf-cc*

CLI Path: *amf-cc*

Service Area

Service area objects

XPath: *cna-amf-cc:/amf-cc/service-area*

CLI Path: *amf-cc service-area*

Interface

Interface objects

XPath: *cna-amf-cc:/amf-cc/interface*

CLI Path: *amf-cc interface*

N26

N26 interworking interface to EPC

Reference: 3GPP TS 29.274: 3rd Generation Partnership Project; Technical Specification Group Core Network and Terminals; 3GPP Evolved Packet System (EPS); Evolved General Packet Radio Service (GPRS) Tunneling Protocol for Control plane (GTPv2-C); Stage 3

XPath: *cna-amf-cc:/amf-cc/interface/n26*

CLI Path: *amf-cc interface n26*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
version	enumeration	15.9.0	16.11.0	3GPP spec applicable to this interface is TS 29.274, Default: 16.11.0
		16.5.0		
		16.11.0		
t3-response	uint16	100..15000	5000	Response timer for GTP messages on N26 interface. Default: 5000; Unit: milliseconds
n3-request	uint8	1..5	2	Number of retries for GTP messages on N26 interface(retry due to t3-response timer expiry). Default: 2

Attribute	Type	Range/Length/Pattern	Default	Description
src-amf-resource-release-timer	uint16	0..30000	10000	Defines the duration the source AMF waits before releasing the UE context; The timer is started after sending the Context Response to the target MME; Default: 10000; Unit: milliseconds
data-forwarding	enumeration	never indirect direct	never	Controls data forwarding related to the N26 interface. The target AMF uses this value and the Direct Forwarding Flag received from the source MME to indicate applicability of data forwarding to the SMF. The source AMF uses this value along with Direct Forwarding Flag and Direct Forwarding Path Availability to indicate applicability of data forwarding to the target MME. Procedures that include data forwarding are described in TS 23.502. Default: never
update-smf-direct-data-forwarding	enumeration	enable disable	enable	Indicates whether the SMF is notified with an Update SM Context Request message when direct data forwarding is applicable. Default: enable

Nas

5G Non Access Stratum

Reference: 3GPP TS 24.501: 3rd Generation Partnership Project; Technical Specification Group Core Network and Terminals; Non-Access-Stratum (NAS) protocol for 5G System (5GS); Stage 3

XPath: [cna-amf-cc:/amf-cc/interface/nas](#)

CLI Path: [amf-cc interface nas](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
version	enumeration	15.6.0 16.6.0 16.11.0	16.11.0	3GPP spec applicable to this interface is TS 24.501; Default: 16.11.0

Mm

5G-NAS Mobility Management Interface

XPath: [cna-amf-cc:/amf-cc/interface/nas/mm](#)

CLI Path: [amf-cc interface nas mm](#)

Default

Default Mobility Management Profile attributes

XPath: [cna-amf-cc:/amf-cc/interface/nas/mm/default](#)

CLI Path: [amf-cc interface nas mm default](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
ue-registration-retry-delay-t3502-value	uint8	1..60	12	Delay time before the UE retries a registration procedure after five consecutive failed attempts; Default: 12; Unit: minutes
ue-periodic-registration-t3512-value	uint32	1..19200	60	Periodic registration procedure timer. Values in range 32..310 must be a multiple of 10, values in range 311..1860 must be a multiple of 60, values in range 1861..18600 must be a multiple of 600, and 19200. Default 60; Unit: minutes Values in range 32..310 must be a multiple of 10, values in range 311..1860 must be a multiple of 60, values in range 1861..18600 must be a multiple of 600, and 19200.

Attribute	Type	Range/Length/Pattern	Default	Description
ue-periodic-registration-t3512-source	enumeration	local-config udm-subscription	local-config	Source for the periodic registration timer value sent to the UE as defined in 3GPP NAS standard section 8.2.7 of TS 24.501. If a subscribed value is not yet received from the UDM, then local-config will be used. Default: local-config
network-init-deregistration-t3522	uint16	100..15000	6000	Defines the duration the AMF waits for a response to a deregistration request sent to the UE; Default: 6000; Unit: milliseconds
network-init-deregistration-retry-n3522	uint8	0..5	2	Defines the number of retries the AMF performs for the network initiated deregistration procedure; Default: 2
registration-accept-t3550	uint16	100..15000	6000	Defines the duration the AMF waits for a response to a registration accept; Default: 6000; Unit: milliseconds
registration-accept-retry-n3550	uint8	0..5	3	Defines the number of retries the AMF performs for a registration accept; Default: 3
guti-lifetime	uint8	1..168	12	Defines the life time of a current 5G-GUTI; Default: 12; Unit: hours
identity-t3570	uint16	100..15000	6000	Defines the duration the AMF waits for a response to an identity request; Default: 6000; Unit: milliseconds
identity-retry-n3570	uint8	0..5	2	Defines the number of retries the AMF performs for an identity request; Default: 2
mobile-reachable-timer	union	['uint32,10..34560', 'Enumerations: t3512plus4 t3512times2plus1 ']	t3512plus4	Defines the duration the UE can stay IDLE without messaging the AMF; t3512plus4, t3512times2plus1, 30-34560; Default: t3512plus4; Unit: minutes The value must be greater than 'ue-periodic-registration-t3512-value'
tai-list-max	uint8	1..16	1	Defines the number of TAIs which can be stored for a particular UE; Default: 1
tai-list-allocation	enumeration	unrestricted within-plmn	unrestricted	Defines the PLMN based restriction for TAI list; Default: unrestricted
configuration-update-t3555	uint16	100..15000	6000	Defines the duration the AMF waits for a response to a configuration update command; Default: 6000; Unit: milliseconds
configuration-update-retry-n3555	uint8	0..4	4	Defines the number of retries the AMF performs for a configuration update command; Default: 4
full-network-name	string	1..128	3gppnetwork	Defines the full network name provided to UE in a configuration update command; Default: 3gppnetwork
short-network-name	string	1..128	3gppnetwork	Defines the short network name provided to UE in a configuration update command; Default: 3gppnetwork

Optional GUTI Reallocation

Attributes to control GUTI reallocation

At least one of the attributes 'ue-init-service-request', 'mobility-reg-request', 'periodic-reg-request' must be set to enable

XPath: [cna-amf-cc:/amf-cc/interface/nas/mm/default/optional-guti-reallocation](#)

CLI Path: [amf-cc interface nas mm default optional-guti-reallocation](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
ue-init-service-request	enumeration	enable disable	disable	Controls if the AMF allocates a new GUTI during UE initiated Service Request procedure. Default: disable
mobility-reg-request	enumeration	enable disable	enable	Controls if the AMF allocates a new GUTI during Mobility Registration procedure. Default: enable
periodic-reg-request	enumeration	enable disable	enable	Controls if the AMF allocates a new GUTI during Periodic Registration procedure. Default: enable

Non 3Gpp Values

Attributes to control non-3GPP procedures

XPath: *cna-amf-cc:/amf-cc/interface/nas/mm/default/non-3gpp-values*

CLI Path: *amf-cc interface nas mm default non-3gpp-values*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
deregistration-timer	uint32	10..34560	54	Defines the non-3GPP deregistration timer value. This is sent in registration accept towards UE. Default: 54

Equivalent Plmn

List of PLMNs considered equivalent

XPath: *cna-amf-cc:/amf-cc/interface/nas/mm/default/equivalent-plmn*

CLI Path: *amf-cc interface nas mm default equivalent-plmn*

List Properties:

Keys: plmn-id

Maximum number of elements: 15

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
plmn-id	string	(([0-9]{5,6}))		Equivalent PLMN identity; Range: numbers with 5-6 digits

Profile

List of Mobility Management Profiles

XPath: *cna-amf-cc:/amf-cc/interface/nas/mm/profile*

CLI Path: *amf-cc interface nas mm profile*

List Properties:

Keys: name

Maximum number of elements: 100

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	1..64		Name of the profile
ue-registration-retry-delay-t3502-value	uint8	1..60	12	Delay time before the UE retries a registration procedure after five consecutive failed attempts; Default: 12; Unit: minutes
ue-periodic-registration-t3512-value	uint32	1..19200	60	Periodic registration procedure timer. Values in range 32..310 must be a multiple of 10, values in range 311..1860 must be a multiple of 60, values in range 1861..18600 must be a multiple of 600, and 19200. Default 60; Unit: minutes Values in range 32..310 must be a multiple of 10, values in range 311..1860 must be a multiple of 60, values in range 1861..18600 must be a multiple of 600, and 19200.
ue-periodic-registration-t3512-source	enumeration	local-config udm-subscription	local-config	Source for the periodic registration timer value sent to the UE as defined in 3GPP NAS standard section 8.2.7 of TS 24.501.If a subscribed value is not yet received from the UDM, then local-config will be used. Default: local-config
network-init-deregistration-t3522	uint16	100..15000	6000	Defines the duration the AMF waits for a response to a deregistration request sent to the UE; Default: 6000; Unit: milliseconds
network-init-deregistration-retry-n3522	uint8	0..5	2	Defines the number of retries the AMF performs for the network initiated deregistration procedure; Default: 2

Attribute	Type	Range/Length/Pattern	Default	Description
registration-accept-t3550	uint16	100..15000	6000	Defines the duration the AMF waits for a response to a registration accept; Default: 6000; Unit: milliseconds
registration-accept-retry-n3550	uint8	0..5	3	Defines the number of retries the AMF performs for a registration accept; Default: 3
guti-lifetime	uint8	1..168	12	Defines the life time of a current 5G-GUTI; Default: 12; Unit: hours
identity-t3570	uint16	100..15000	6000	Defines the duration the AMF waits for a response to an identity request; Default: 6000; Unit: milliseconds
identity-retry-n3570	uint8	0..5	2	Defines the number of retries the AMF performs for an identity request; Default: 2
mobile-reachable-timer	union	['uint32,10..34560', 'Enumerations: t3512plus4 t3512times2plus1 ']	t3512plus4	Defines the duration the UE can stay IDLE without messaging the AMF; t3512plus4, t3512times2plus1, 30-34560; Default: t3512plus4; Unit: minutes The value must be greater than 'ue-periodic-registration-t3512-value'
tai-list-max	uint8	1..16	1	Defines the number of TAIs which can be stored for a particular UE; Default: 1
tai-list-allocation	enumeration	unrestricted within-plmn	unrestricted	Defines the PLMN based restriction for TAI list; Default: unrestricted
configuration-update-t3555	uint16	100..15000	6000	Defines the duration the AMF waits for a response to a configuration update command; Default: 6000; Unit: milliseconds
configuration-update-retry-n3555	uint8	0..4	4	Defines the number of retries the AMF performs for a configuration update command; Default: 4
full-network-name	string	1..128	3gppnetwork	Defines the full network name provided to UE in a configuration update command; Default: 3gppnetwork
short-network-name	string	1..128	3gppnetwork	Defines the short network name provided to UE in a configuration update command; Default: 3gppnetwork

Optional GUTI Reallocation

Attributes to control GUTI reallocation

At least one of the attributes 'ue-init-service-request', 'mobility-reg-request', 'periodic-reg-request' must be set to enable

XPath: [cna-amf-cc:/amf-cc/interface/nas/mm/profile/optional-guti-reallocation](#)

CLI Path: [amf-cc interface nas mm profile optional-guti-reallocation](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
ue-init-service-request	enumeration	enable disable	disable	Controls if the AMF allocates a new GUTI during UE initiated Service Request procedure. Default: disable
mobility-reg-request	enumeration	enable disable	enable	Controls if the AMF allocates a new GUTI during Mobility Registration procedure. Default: enable
periodic-reg-request	enumeration	enable disable	enable	Controls if the AMF allocates a new GUTI during Periodic Registration procedure. Default: enable

Non 3GPP Values

Attributes to control non-3GPP procedures

XPath: [cna-amf-cc:/amf-cc/interface/nas/mm/profile/non-3gpp-values](#)

CLI Path: [amf-cc interface nas mm profile non-3gpp-values](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
deregistration-timer	uint32	10..34560	54	Defines the non-3GPP deregistration timer value. This is sent in registration accept towards UE. Default: 54

Equivalent Plmn

List of PLMNs considered equivalent

XPath: *cna-amf-cc:/amf-cc/interface/nas/mm/profile/equivalent-plmn*

CLI Path: *amf-cc interface nas mm profile equivalent-plmn*

List Properties:

Keys: plmn-id

Maximum number of elements: 15

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
plmn-id	string	(([0-9]{5,6}))		Equivalent PLMN identity; Range: numbers with 5-6 digits

Global

Mobility Management global attributes

XPath: *cna-amf-cc:/amf-cc/interface/nas/mm/global*

CLI Path: *amf-cc interface nas mm global*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
nitz-update-triggers	leaf-list of enumeration	initial-registration inter-nodal-registration timezone-change	initial-registration timezone-change	Defines the list of triggers that will cause AMF to send an optional Configuration Update message to provide the network identity and timezone information to the UE. Default: [initial-registration timezone-change]
ims-vops-3gpp-setting-for-voice-centric-ue	enumeration	standard force-on	standard	Controls whether AMF tries to move the UE to S1 mode (or 2G/3G) for a voice centric device. A value of 'standard' uses the service PLMN and mobility-class to decide whether to move. A value of 'force-on' keeps the UE on the AMF. Default: standard

Signaling Rate Control

Signaling rate parameters for mobility management congestion control as defined by 3GPP TS 24.501 section 5.3.9.

XPath: *cna-amf-cc:/amf-cc/interface/nas/mm/global/signaling-rate-control*

CLI Path: *amf-cc interface nas mm global signaling-rate-control*

Initial Registration

Signaling rate control attributes for initial registration procedure

XPath: *cna-amf-cc:/amf-cc/interface/nas/mm/global/signaling-rate-control/initial-registration*

CLI Path: *amf-cc interface nas mm global signaling-rate-control initial-registration*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
rate-limit	union	['uint16,1..1000', 'Enumerations: disable ']	disable	Defines the maximum rate per second at which the applicable messages may be processed by an AMF Call Control microservice. The value disable indicates that rate control functionality is disabled. Default: disable Unit: messages per second
action	enumeration	ignore reject	reject	Defines the action to be taken when the signaling rate control limit has been exceeded. Default: reject
t3346-low	union	['uint16,2..11159', 'Enumerations: disable ']	4	Defines the lower end of the range for the T3346 value sent to the UE. A random number between t3346-low and t3346-high is included in the initial registration reject message as defined by 3GPP TS 24.501 section 8.2.9, and may be used by the UE to re-send the request after the specified time. The value disable indicates that the T3346 value is not included in the initial registration reject message. Default: 4; Unit: seconds

Attribute	Type	Range/Length/Pattern	Default	Description
t3346-high	uint16	3..11160	16	Defines the higher end of the range for the T3346 value to be sent to the UE. A random number between t3346-low and t3346-high is included in the initial registration reject message as defined by 3GPP TS 24.501 section 8.2.9, and may be used by the UE to re-send the request after the specified time. Default: 16; Unit: seconds The value must be greater than 't3346-low'

Security

5G-NAS Security Interface

XPath: *cna-amf-cc:/amf-cc/interface/nas/security*

CLI Path: *amf-cc interface nas security*

Default

Default Security Profile attributes

XPath: *cna-amf-cc:/amf-cc/interface/nas/security/default*

CLI Path: *amf-cc interface nas security default*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
security-t3560	uint16	100..15000	6000	Defines the duration the AMF waits for a response to authentication and security mode; Default: 6000; Unit: milliseconds
security-retry-n3560	uint8	0..5	2	Defines the number of retries the AMF performs for authentication request and security mode command; Default: 2
ciphering-algorithms	leaf-list of enumeration	nea0 nea1 nea2	nea2 nea0	Defines the list of 5G ciphering algorithms that can be used for NAS ciphering. Valid values are nea0, nea1 and nea2 in any order or combination, with order indicating preference. SNOW3G (nea1) algorithm for NAS ciphering is considered compromised, and is not recommended. Default: nea2,nea0 Ordered by: user
integrity-algorithms	leaf-list of enumeration	null nia1 nia2	nia1 nia2 null	Defines the list of 5G integrity algorithms that can be used for NAS integrity protection. Valid values are null, nia1 and nia2 in any order or combination, with order indicating preference. Default: nia1,nia2,null Ordered by: user
eps-ciphering-algorithms	leaf-list of enumeration	eea0 eea1 eea2	eea2 eea0	Defines the list of EPS ciphering algorithms that can be used for NAS ciphering. Valid values are eea0, eea1 and eea2 in any order or combination, with order indicating preference. SNOW3G (eea1) algorithm for NAS ciphering is considered compromised, and is not recommended. Default: eea2,eea0 Ordered by: user
eps-integrity-algorithms	leaf-list of enumeration	null eia1 eia2	eia1 eia2 null	Defines the list of EPS integrity algorithms that can be used for NAS integrity protection. Valid values are null, eia1 and eia2 in any order or combination, with order indicating preference. Default: eia1,eia2,null Ordered by: user
additional-auth-events	leaf-list of enumeration	intra-amf-mobility inter-amf-mobility periodic-registration service-request deregistration eps-mobility		Additional events for which AMF performs authentication
additional-security-for-eps-algorithms	enumeration	enable disable	disable	Controls support of an additional security procedure in order to provide the selected EPS security algorithms to the UE when the AMF supports N26 interface and the UE indicates support of S1-mode. This attribute is applicable only when 'amf-cc interface nas version' is set to '15.6.0'. A value of 'disable' is compliant to TS 24.501 v15.6.0. A value of 'enable' is compliant to v16.6.0. Default: disable

Profile

List of Security Profiles

XPath: *cna-amf-cc:/amf-cc/interface/nas/security/profile*

CLI Path: *amf-cc interface nas security profile*

List Properties:

Keys: name

Maximum number of elements: 100

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	1..64		Name of the profile
security-t3560	uint16	100..15000	6000	Defines the duration the AMF waits for a response to authentication and security mode; Default: 6000; Unit: milliseconds
security-retry-n3560	uint8	0..5	2	Defines the number of retries the AMF performs for authentication request and security mode command; Default: 2
ciphering-algorithms	leaf-list of enumeration	nea0 nea1 nea2	nea2 nea0	Defines the list of 5G ciphering algorithms that can be used for NAS ciphering. Valid values are nea0, nea1 and nea2 in any order or combination, with order indicating preference. SNOW3G (nea1) algorithm for NAS ciphering is considered compromised, and is not recommended. Default: nea2,nea0 Ordered by: user
integrity-algorithms	leaf-list of enumeration	null nia1 nia2	nia1 nia2 null	Defines the list of 5G integrity algorithms that can be used for NAS integrity protection. Valid values are null, nia1 and nia2 in any order or combination, with order indicating preference. Default: nia1,nia2,null Ordered by: user
eps-ciphering-algorithms	leaf-list of enumeration	eea0 eea1 eea2	eea2 eea0	Defines the list of EPS ciphering algorithms that can be used for NAS ciphering. Valid values are eea0, eea1 and eea2 in any order or combination, with order indicating preference. SNOW3G (eea1) algorithm for NAS ciphering is considered compromised, and is not recommended. Default: eea2,eea0 Ordered by: user
eps-integrity-algorithms	leaf-list of enumeration	null eia1 eia2	eia1 eia2 null	Defines the list of EPS integrity algorithms that can be used for NAS integrity protection. Valid values are null, eia1 and eia2 in any order or combination, with order indicating preference. Default: eia1,eia2,null Ordered by: user
additional-auth-events	leaf-list of enumeration	intra-amf-mobility inter-amf-mobility periodic-registration service-request deregistration eps-mobility		Additional events for which AMF performs authentication
additional-security-for-eps-algorithms	enumeration	enable disable	disable	Controls support of an additional security procedure in order to provide the selected EPS security algorithms to the UE when the AMF supports N26 interface and the UE indicates support of S1-mode. This attribute is applicable only when 'amf-cc interface nas version' is set to '15.6.0'. A value of 'disable' is compliant to TS 24.501 v15.6.0. A value of 'enable' is compliant to v16.6.0. Default: disable

Paging

Controls for the AMF paging behavior

XPath: *cna-amf-cc:/amf-cc/interface/nas/paging*

CLI Path: *amf-cc interface nas paging*

Default

Controls for the AMF's default paging behavior

XPath: *cna-amf-cc:/amf-cc/interface/nas/paging/default*

CLI Path: *amf-cc interface nas paging default*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
t3513-initial	uint16	1000..30000	5000	Defines the wait time for the initial page attempt. Default: 5000; Unit: milliseconds
t3513-increment	uint16	0..10000	2000	Defines the additional delay added to retried page attempts. Default: 2000; Unit: milliseconds
n3513-retry	uint8	0..4	2	Defines the number of paging retries the AMF performs before declaring paging failure. Default: 2
initial-paging-target	enumeration	last-known-gnb last-visited-gnbs last-known-ta ta-list	ta-list	Defines initial paging target. Default: ta-list
paging-priority	enumeration	PrioLevel1 PrioLevel2 PrioLevel3 PrioLevel4 PrioLevel5 PrioLevel6 PrioLevel7 PrioLevel8 none	none	Defines the paging priority. Lower value indicates higher priority as described in TS 38.413 section 9.3.1.78. Default: none

Profile Set

Set of paging profiles

XPath: *cna-amf-cc:/amf-cc/interface/nas/paging/profile-set*

CLI Path: *amf-cc interface nas paging profile-set*

List Properties:

Keys: name

Maximum number of elements: 4

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	1..64		Name of the paging profile set

Profile

Controls for the AMF's paging behavior

At least one 'paging profile-set profile trigger' must be configured

XPath: *cna-amf-cc:/amf-cc/interface/nas/paging/profile-set/profile*

CLI Path: *amf-cc interface nas paging profile-set profile*

List Properties:

Keys: name

Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{1,64})		Name of the paging profile
precedence	uint8			The precedence in the paging profile set of this paging profile. Lower value indicates higher precedence
t3513-initial	uint16	1000..30000	5000	Defines the wait time for the initial page attempt. Default: 5000; Unit: milliseconds
t3513-increment	uint16	0..10000	2000	Defines the additional delay added to retried page attempts. Default: 2000; Unit: milliseconds
n3513-retry	uint8	0..4	2	Defines the number of paging retries the AMF performs before declaring paging failure. Default: 2

Attribute	Type	Range/Length/Pattern	Default	Description
initial-paging-target	enumeration	last-known-gnb last-visited-gnbs last-known-ta ta-list	ta-list	Defines initial paging target. Default: ta-list
paging-priority	enumeration	PrioLevel1 PrioLevel2 PrioLevel3 PrioLevel4 PrioLevel5 PrioLevel6 PrioLevel7 PrioLevel8 none	none	Defines the paging priority. Lower value indicates higher priority as described in TS 38.413 section 9.3.1.78. Default: none

Triggers

XPath: [cna-amf-cc:/amf-cc/interface/nas/paging/profile-set/profile/triggers](#)

CLI Path: [amf-cc interface nas paging profile-set profile triggers](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
qi	leaf-list of uint8	0..255		List of 5G QoS identifiers used to match this paging profile
service	leaf-list of enumeration	sms ue-policy lcs		List of services used to match this paging profile
arp	leaf-list of uint8	1..15		List of Allocation & Retention Priorities (ARP) used to match this paging profile
ppi	leaf-list of uint8	0..7		List of Paging Policy Indicators (PPI) used to match this paging profile

Global

XPath: [cna-amf-cc:/amf-cc/interface/nas/paging/global](#)

CLI Path: [amf-cc interface nas paging global](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
max-visited-gnbs-tracked	uint8	2..16	5	Maximum number of visited gnbs to be tracked per UE for paging
visited-gnbs-allocation	enumeration	unrestricted tai-list	tai-list	Visited gnbs allocation method for paging

Ngap

NG application interface to 5G-AN

Reference: 3GPP TS 38.413: 3rd Generation Partnership Project; Technical Specification Group Radio Access Network; NG-RAN; NG Application Protocol (NGAP)

XPath: [cna-amf-cc:/amf-cc/interface/ngap](#)

CLI Path: [amf-cc interface ngap](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
version	enumeration	15.7.0 16.3.0 16.9.0	16.9.0	3GPP spec applicable to this interface is TS 38.413; Default: 16.9.0
time-to-wait	uint8	1,2,5,10,20,60	60	Time to wait; the minimum allowed waiting time after a setup or configuration update failure; Default: 60; Unit: seconds.
reset-ack-delay-timer	uint8	8..14	14	5G-AN reset ACK delay timer; Default: 14; Unit: seconds

Attribute	Type	Range/Length/Pattern	Default	Description
n2-release-delay-timer	union	[Enumerations: disabled, 'uint16,1000..5000']	5000	Defines the time the AMF waits before releasing the N2 connection after a NAS procedure; Default: 5000; Units: milliseconds
reloc-prep-timer	uint16	1000..30000	3000	Time that the AMF waits for the target gNodeB to respond to the handover request in an N2 based handover. Default: 3000; Unit: milliseconds
reloc-overall-timer	uint16	1000..30000	5000	Time that the AMF waits for the handover notify from the target gNodeB. Default: 5000; Unit: milliseconds
handover-resource-release-timer	uint16	1000..30000	2000	Time that the AMF waits before releasing the N2 connection on the source gNodeB during an N26 based handover procedure. Default: 2000; Unit: milliseconds
wait-for-mobility-registration-timer	uint16	1000..10000	5000	Time that the AMF waits for a mobility registration after an intra-AMF relocation with TAI list change or after an inter-AMF relocation. Default: 5000; Unit: milliseconds

Amf Profile

AMF profile

XPath: *cna-amf-cc:/amf-cc/amf-profile*

CLI Path: *amf-cc amf-profile*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
amf-name	string	[A-Za-z0-9 '() +, -./: = ?]*		AMF name communicated to the 5GAN
nf-instance-id	string	[0-9a-fA-F]{8}-[0-9a-fA-F]{4}-[0-9a-fA-F]{4}-[0-9a-fA-F]{12}		Unique identity of the AMF in the form of a uuid
amf-region-id	uint8	0..255		AMF region identifier - part of GUAMI
amf-set-id	uint16	0..1023		AMF set identifier - part of GUAMI
amf-pointer	leaf-list of uint8	0..63		AMF pointer information Minimum number of elements: 1 Maximum number of elements: 16
relative-capacity	uint8	0..255		The relative processing capacity of an AMF with respect to the other AMFs in the AMF Set
inter-plmn-fqdn	inet:domain-name			An FQDN that is included in the NF profile that can be used for inter PLMN routing.
locality	string	1..63		Location of the AMF
include-tai-in-nrf-registration	enumeration	enable disable	enable	Determines if the TAI list needs to be included in the NF Registration request (as defined by TS 29.510 section 5.2.2.2.2) sent to NRF. Default: enable

Serving Plmn

A list of serving Public Land Mobile Network (PLMN) identity having list of supported Single Network Slice Selection AssistanceInformation (SNSSAI) and Tracking Areas by the AMF

XPath: *cna-amf-cc:/amf-cc/amf-profile/serving-plmn*

CLI Path: *amf-cc amf-profile serving-plmn*

List Properties:

Keys: plmn-id

Minimum number of elements: 1

Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
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Attribute	Type	Range/Length/Pattern	Default	Description
plmn-id	string	(([0-9]{5,6}))		Supported PLMN identity; Range: numbers with 5-6 digits
non-3gpp-tai	string	(0x[A-Fa-f0-9]{4}) or (0x[A-Fa-f0-9]{6})		Non-3GPP Tracking Area Identifier

Tracking Area Default

Tracking area default configuration. Default configuration to be used by the application when the matching tracking-area configuration is missing.

XPath: *cna-amf-cc:/amf-cc/amf-profile/serving-plmn/tracking-area-default*

CLI Path: *amf-cc amf-profile serving-plmn tracking-area-default*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
ims-vops-support	enumeration	enable disable	enable	IMS voice over PS support enable or disable. Default: enable
timezone	enumeration	Africa/Abidjan Africa/Accra Africa/Addis_Ababa Africa/Algiers Africa/Asmara Africa/Asmera Africa/Bamako Africa/Bangui Africa/Banjul Africa/Bissau Africa/Blantyre Africa/Brazzaville Africa/Bujumbura Africa/Cairo Africa/Casablanca Africa/Ceuta Africa/Conakry Africa/Dakar Africa/Dar_es_Salaam Africa/Djibouti Africa/Douala Africa/El_Aaiun Africa/Freetown Africa/Gaborone Africa/Harare Africa/Johannesburg Africa/Juba Africa/Kampala Africa/Khartoum Africa/Kigali Africa/Kinshasa Africa/Lagos Africa/Libreville Africa/Lome Africa/Luanda Africa/Lubumbashi Africa/Lusaka Africa/Malabo Africa/Maputo Africa/Maseru Africa/Mbabane Africa/Mogadishu Africa/Monrovia Africa/Nairobi Africa/Ndjamena Africa/Niamey Africa/Nouakchott Africa/Ouagadougou Africa/Porto-Novo		TA timezone information

Attribute	Type	Range/Length/Pattern	Default	Description
		Africa/Sao_Tome		
		Africa/Timbuktu		
		Africa/Tripoli		
		Africa/Tunis		
		Africa/Windhoek		
		America/Adak		
		America/Anchorage		
		America/Anguilla		
		America/Antigua		
		America/Araguaina		
		America/Argentina/Buenos_Aires		
		America/Argentina/Catamarca		
		America/Argentina/ComodRivadavia		
		America/Argentina/Cordoba		
		America/Argentina/Jujuy		
		America/Argentina/La_Rioja		
		America/Argentina/Mendoza		
		America/Argentina/Rio_Gallegos		
		America/Argentina/Salta		
		America/Argentina/San_Juan		
		America/Argentina/San_Luis		
		America/Argentina/Tucuman		
		America/Argentina/Ushuaia		
		America/Aruba		
		America/Asuncion		
		America/Atikokan		
		America/Atka		
		America/Bahia		
		America/Bahia_Banderas		
		America/Barbados		
		America/Belem		
		America/Belize		
		America/Blanc-Sablon		
		America/Boa_Vista		
		America/Bogota		
		America/Boise		
		America/Buenos_Aires		
		America/Cambridge_Bay		
		America/Campo_Grande		
		America/Cancun		
		America/Caracas		
		America/Catamarca		
		America/Cayenne		
		America/Cayman		
		America/Chicago		
		America/Chihuahua		
		America/Coral_Harbour		
		America/Cordoba		
		America/Costa_Rica		
		America/Creston		
		America/Cuiaba		
		America/Curacao		
		America/Danmarkshavn		
		America/Dawson		
		America/Dawson_Creek		
		America/Denver		
		America/Detroit		
		America/Dominica		
		America/Edmonton		
		America/Eirunepe		
		America/El_Salvador		
		America/Ensenada		
		America/Fortaleza		
		America/Fort_Nelson		
		America/Fort_Wayne		

Attribute	Type	Range/Length/Pattern	Default	Description
		America/Glace_Bay		
		America/Godthab		
		America/Goose_Bay		
		America/Grand_Turk		
		America/Grenada		
		America/Guadeloupe		
		America/Guatemala		
		America/Guayaquil		
		America/Guyana		
		America/Halifax		
		America/Havana		
		America/Hermosillo		
		America/Indiana/Indianapolis		
		America/Indiana/Knox		
		America/Indiana/Marengo		
		America/Indiana/Petersburg		
		America/Indianapolis		
		America/Indiana/Tell_City		
		America/Indiana/Vevay		
		America/Indiana/Vincennes		
		America/Indiana/Winamac		
		America/Inuvik		
		America/Iqaluit		
		America/Jamaica		
		America/Jujuy		
		America/Juneau		
		America/Kentucky/Louisville		
		America/Kentucky/Monticello		
		America/Knox_IN		
		America/Kralendijk		
		America/La_Paz		
		America/Lima		
		America/Los_Angeles		
		America/Louisville		
		America/Lower_Princes		
		America/Maceio		
		America/Managua		
		America/Manaus		
		America/Marigot		
		America/Martinique		
		America/Matamoros		
		America/Mazatlan		
		America/Mendoza		
		America/Menominee		
		America/Merida		
		America/Metlakatla		
		America/Mexico_City		
		America/Miquelon		
		America/Moncton		
		America/Monterrey		
		America/Montevideo		
		America/Montreal		
		America/Montserrat		
		America/Nassau		
		America/New_York		
		America/Nipigon		
		America/Nome		
		America/Noronha		
		America/North_Dakota/Beulah		
		America/North_Dakota/Center		
		America/North_Dakota/New_Salem		
		America/Nuuk		
		America/Ojinaga		
		America/Panama		
		America/Pangnirtung		

Attribute	Type	Range/Length/Pattern	Default	Description
		America/Paramaribo		
		America/Phoenix		
		America/Port-au-Prince		
		America/Porto_Acre		
		America/Port_of_Spain		
		America/Porto_Velho		
		America/Puerto_Rico		
		America/Punta_Arenas		
		America/Rainy_River		
		America/Rankin_Inlet		
		America/Recife		
		America/Regina		
		America/Resolute		
		America/Rio_Branco		
		America/Rosario		
		America/Santa_Isabel		
		America/Santarem		
		America/Santiago		
		America/Santo_Domingo		
		America/Sao_Paulo		
		America/Scoresbysund		
		America/Shiprock		
		America/Sitka		
		America/St_Barthelemy		
		America/St_Johns		
		America/St_Kitts		
		America/St_Lucia		
		America/St_Thomas		
		America/St_Vincent		
		America/Swift_Current		
		America/Tegucigalpa		
		America/Thule		
		America/Thunder_Bay		
		America/Tijuana		
		America/Toronto		
		America/Tortola		
		America/Vancouver		
		America/Virgin		
		America/Whitehorse		
		America/Winnipeg		
		America/Yakutat		
		America/Yellowknife		
		Antarctica/Casey		
		Antarctica/Davis		
		Antarctica/DumontDUrville		
		Antarctica/Macquarie		
		Antarctica/Mawson		
		Antarctica/McMurdo		
		Antarctica/Palmer		
		Antarctica/Rothera		
		Antarctica/South_Pole		
		Antarctica/Syowa		
		Antarctica/Troll		
		Antarctica/Vostok		
		Arctic/Longyearbyen		
		Asia/Aden		
		Asia/Almaty		
		Asia/Amman		
		Asia/Anadyr		
		Asia/Aqtau		
		Asia/Aqtobe		
		Asia/Ashgabat		
		Asia/Ashkhabad		
		Asia/Atyrau		
		Asia/Baghdad		

Attribute	Type	Range/Length/Pattern	Default	Description
		Asia/Bahrain		
		Asia/Baku		
		Asia/Bangkok		
		Asia/Barnaul		
		Asia/Beirut		
		Asia/Bishkek		
		Asia/Brunei		
		Asia/Chita		
		Asia/Choibalsan		
		Asia/Chongqing		
		Asia/Colombo		
		Asia/Dacca		
		Asia/Damascus		
		Asia/Dhaka		
		Asia/Dili		
		Asia/Dubai		
		Asia/Dushanbe		
		Asia/Famagusta		
		Asia/Gaza		
		Asia/Harbin		
		Asia/Hebron		
		Asia/Ho_Chi_Minh		
		Asia/Hong_Kong		
		Asia/Hovd		
		Asia/Irkutsk		
		Asia/Istanbul		
		Asia/Jakarta		
		Asia/Jayapura		
		Asia/Jerusalem		
		Asia/Kabul		
		Asia/Kamchatka		
		Asia/Karachi		
		Asia/Kashgar		
		Asia/Kathmandu		
		Asia/Katmandu		
		Asia/Khandyga		
		Asia/Kolkata		
		Asia/Krasnoyarsk		
		Asia/Kuala_Lumpur		
		Asia/Kuching		
		Asia/Kuwait		
		Asia/Macao		
		Asia/Macau		
		Asia/Magadan		
		Asia/Makassar		
		Asia/Manila		
		Asia/Muscat		
		Asia/Nicosia		
		Asia/Novokuznetsk		
		Asia/Novosibirsk		
		Asia/Omsk		
		Asia/Oral		
		Asia/Phnom_Penh		
		Asia/Pontianak		
		Asia/Pyongyang		
		Asia/Qatar		
		Asia/Qostanay		
		Asia/Qyzylorda		
		Asia/Riyadh		
		Asia/Sakhalin		
		Asia/Samarkand		
		Asia/Seoul		
		Asia/Shanghai		
		Asia/Singapore		
		Asia/Srednekolymsk		

Attribute	Type	Range/Length/Pattern	Default	Description
		Asia/Taipei		
		Asia/Tashkent		
		Asia/Tbilisi		
		Asia/Tehran		
		Asia/Tel_Aviv		
		Asia/Thimbu		
		Asia/Thimphu		
		Asia/Tokyo		
		Asia/Tomsk		
		Asia/Ujung_Pandang		
		Asia/Ulaanbaatar		
		Asia/Ulan_Bator		
		Asia/Urumqi		
		Asia/Ust-Nera		
		Asia/Vientiane		
		Asia/Vladivostok		
		Asia/Yakutsk		
		Asia/Yangon		
		Asia/Yekaterinburg		
		Asia/Yerevan		
		Atlantic/Azores		
		Atlantic/Bermuda		
		Atlantic/Canary		
		Atlantic/Cape_Verde		
		Atlantic/Faeroe		
		Atlantic/Faroe		
		Atlantic/Jan_Mayen		
		Atlantic/Madeira		
		Atlantic/Reykjavik		
		Atlantic/South_Georgia		
		Atlantic/Stanley		
		Atlantic/St_Helena		
		Australia/ACT		
		Australia/Adelaide		
		Australia/Brisbane		
		Australia/Broken_Hill		
		Australia/Canberra		
		Australia/Currie		
		Australia/Darwin		
		Australia/Eucla		
		Australia/Hobart		
		Australia/LHI		
		Australia/Lindeman		
		Australia/Lord_Howe		
		Australia/Melbourne		
		Australia/North		
		Australia/NSW		
		Australia/Perth		
		Australia/Queensland		
		Australia/South		
		Australia/Sydney		
		Australia/Tasmania		
		Australia/Victoria		
		Australia/West		
		Australia/Yancowinna		
		Brazil/Acre		
		Brazil/DeNoronha		
		Brazil/East		
		Brazil/West		
		Canada/Atlantic		
		Canada/Central		
		Canada/Eastern		
		Canada/Mountain		
		Canada/Newfoundland		
		Canada/Pacific		

Attribute	Type	Range/Length/Pattern	Default	Description
		Canada/Saskatchewan		
		Canada/Yukon		
		CET		
		Chile/Continental		
		Chile/EasterIsland		
		CST6CDT		
		Cuba		
		EET		
		Egypt		
		Eire		
		EST		
		EST5EDT		
		Etc/GMT		
		Etc/GMT+0		
		Etc/GMT-0		
		Etc/GMT0		
		Etc/GMT+1		
		Etc/GMT-1		
		Etc/GMT+10		
		Etc/GMT-10		
		Etc/GMT+11		
		Etc/GMT-11		
		Etc/GMT+12		
		Etc/GMT-12		
		Etc/GMT-13		
		Etc/GMT-14		
		Etc/GMT+2		
		Etc/GMT-2		
		Etc/GMT+3		
		Etc/GMT-3		
		Etc/GMT+4		
		Etc/GMT-4		
		Etc/GMT+5		
		Etc/GMT-5		
		Etc/GMT+6		
		Etc/GMT-6		
		Etc/GMT+7		
		Etc/GMT-7		
		Etc/GMT+8		
		Etc/GMT-8		
		Etc/GMT+9		
		Etc/GMT-9		
		Etc/Greenwich		
		Etc/UCT		
		Etc/Universal		
		Etc/UTC		
		Etc/Zulu		
		Europe/Amsterdam		
		Europe/Andorra		
		Europe/Astrakhan		
		Europe/Athens		
		Europe/Belfast		
		Europe/Belgrade		
		Europe/Berlin		
		Europe/Bratislava		
		Europe/Brussels		
		Europe/Bucharest		
		Europe/Budapest		
		Europe/Busingen		
		Europe/Chisinau		
		Europe/Copenhagen		
		Europe/Dublin		
		Europe/Gibraltar		
		Europe/Guernsey		
		Europe/Helsinki		

Attribute	Type	Range/Length/Pattern	Default	Description
		Europe/Isle_of_Man		
		Europe/Istanbul		
		Europe/Jersey		
		Europe/Kaliningrad		
		Europe/Kiev		
		Europe/Kirov		
		Europe/Lisbon		
		Europe/Ljubljana		
		Europe/London		
		Europe/Luxembourg		
		Europe/Madrid		
		Europe/Malta		
		Europe/Mariehamn		
		Europe/Minsk		
		Europe/Monaco		
		Europe/Moscow		
		Europe/Nicosia		
		Europe/Oslo		
		Europe/Paris		
		Europe/Podgorica		
		Europe/Prague		
		Europe/Riga		
		Europe/Rome		
		Europe/Samara		
		Europe/San_Marino		
		Europe/Sarajevo		
		Europe/Saratov		
		Europe/Simferopol		
		Europe/Skopje		
		Europe/Sofia		
		Europe/Stockholm		
		Europe/Tallinn		
		Europe/Tirane		
		Europe/Tiraspol		
		Europe/Ulyanovsk		
		Europe/Uzhgorod		
		Europe/Vaduz		
		Europe/Vatican		
		Europe/Vienna		
		Europe/Vilnius		
		Europe/Volgograd		
		Europe/Warsaw		
		Europe/Zagreb		
		Europe/Zaporozhye		
		Europe/Zurich		
		GB		
		GB-Eire		
		GMT		
		GMT+0		
		GMT-0		
		GMT0		
		Greenwich		
		Hongkong		
		HST		
		Iceland		
		Indian/Antananarivo		
		Indian/Chagos		
		Indian/Christmas		
		Indian/Cocos		
		Indian/Comoro		
		Indian/Kerguelen		
		Indian/Mahe		
		Indian/Maldives		
		Indian/Mauritius		
		Indian/Mayotte		

Attribute	Type	Range/Length/Pattern	Default	Description
		Indian/Reunion		
		Iran		
		Israel		
		Jamaica		
		Japan		
		Kwajalein		
		Libya		
		MET		
		Mexico/BajaNorte		
		Mexico/BajaSur		
		Mexico/General		
		MST		
		MST7MDT		
		Navajo		
		NZ		
		NZ-CHAT		
		Pacific/Apia		
		Pacific/Auckland		
		Pacific/Bougainville		
		Pacific/Chatham		
		Pacific/Chuuk		
		Pacific/Easter		
		Pacific/Efate		
		Pacific/Enderbury		
		Pacific/Fakaofu		
		Pacific/Fiji		
		Pacific/Funafuti		
		Pacific/Galapagos		
		Pacific/Gambier		
		Pacific/Guadalcanal		
		Pacific/Guam		
		Pacific/Honolulu		
		Pacific/Johnston		
		Pacific/Kanton		
		Pacific/Kiritimati		
		Pacific/Kosrae		
		Pacific/Kwajalein		
		Pacific/Majuro		
		Pacific/Marquesas		
		Pacific/Midway		
		Pacific/Nauru		
		Pacific/Niue		
		Pacific/Norfolk		
		Pacific/Noumea		
		Pacific/Pago_Pago		
		Pacific/Palau		
		Pacific/Pitcairn		
		Pacific/Pohnpei		
		Pacific/Ponape		
		Pacific/Port_Moresby		
		Pacific/Rarotonga		
		Pacific/Saipan		
		Pacific/Samoa		
		Pacific/Tahiti		
		Pacific/Tarawa		
		Pacific/Tongatapu		
		Pacific/Truk		
		Pacific/Wake		
		Pacific/Wallis		
		Pacific/Yap		
		Poland		
		Portugal		
		PRC		
		PST8PDT		
		ROK		

Attribute	Type	Range/Length/Pattern	Default	Description
		Singapore		
		Turkey		
		UCT		
		Universal		
		US/Alaska		
		US/Aleutian		
		US/Arizona		
		US/Central		
		US/Eastern		
		US/East-Indiana		
		US/Hawaii		
		US/Indiana-Starke		
		US/Michigan		
		US/Mountain		
		US/Pacific		
		US/Pacific-New		
		US/Samoa		
		UTC		
		WET		
		W-SU		
		Zulu		
preferred-smf-locality	string			Preferred Locality for SMF Discovery

Emergency Support

Support of various emergency services

XPath: *cna-amf-cc:/amf-cc/amf-profile/serving-plmn/tracking-area-default/emergency-support*

CLI Path: *amf-cc amf-profile serving-plmn tracking-area-default emergency-support*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
emergency-services	enumeration	enable disable	enable	Controls support of emergency services. Default: enable
emergency-services-fallback	enumeration	enable disable	enable	Controls support for fallback for emergency services. Default: enable

Tracking Area

A list of Tracking Area range under the configured serving-plmn. tac-low and tac-high range should not be configured with overlapping range values, else the lookup will fetch the closest tac-low entry. Tac values with different number of leading zeros will be considered same, Example: 0x000123 and 0x0123 are same.

XPath: *cna-amf-cc:/amf-cc/amf-profile/serving-plmn/tracking-area*

CLI Path: *amf-cc amf-profile serving-plmn tracking-area*

List Properties:

Keys: tac-low

Maximum number of elements: 10000

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
tac-low	string	(0x[A-Fa-f0-9]{4}) or (0x[A-Fa-f0-9]{6})		The tac-low uniquely identifies a TA (Tracking Area) within a PLMN If 'tac-high' is configured, this attribute is considered as the lower end of the TAC range else it represents a single tac value. Values with different leading zeros are considered as single value.
tac-high	string	(0x[A-Fa-f0-9]{4}) or (0x[A-Fa-f0-9]{6})		TAC upper range. It is an optional field representing the upper boundary for a range of tracking areas from 'tac-low' to 'tac-high'. Values with different leading zeros are considered as single value.
ims-vops-support	enumeration	enable disable	enable	IMS voice over PS support enable or disable. Default: enable
timezone	enumeration	Africa/Abidjan		TA timezone information

Attribute	Type	Range/Length/Pattern	Default	Description
		Africa/Accra		
		Africa/Addis_Ababa		
		Africa/Algiers		
		Africa/Asmara		
		Africa/Asmera		
		Africa/Bamako		
		Africa/Bangui		
		Africa/Banjul		
		Africa/Bissau		
		Africa/Blantyre		
		Africa/Brazzaville		
		Africa/Bujumbura		
		Africa/Cairo		
		Africa/Casablanca		
		Africa/Ceuta		
		Africa/Conakry		
		Africa/Dakar		
		Africa/Dar_es_Salaam		
		Africa/Djibouti		
		Africa/Douala		
		Africa/El_Aaiun		
		Africa/Freetown		
		Africa/Gaborone		
		Africa/Harare		
		Africa/Johannesburg		
		Africa/Juba		
		Africa/Kampala		
		Africa/Khartoum		
		Africa/Kigali		
		Africa/Kinshasa		
		Africa/Lagos		
		Africa/Libreville		
		Africa/Lome		
		Africa/Luanda		
		Africa/Lubumbashi		
		Africa/Lusaka		
		Africa/Malabo		
		Africa/Maputo		
		Africa/Maseru		
		Africa/Mbabane		
		Africa/Mogadishu		
		Africa/Monrovia		
		Africa/Nairobi		
		Africa/Ndjamena		
		Africa/Niamey		
		Africa/Nouakchott		
		Africa/Ouagadougou		
		Africa/Porto-Novo		
		Africa/Sao_Tome		
		Africa/Timbuktu		
		Africa/Tripoli		
		Africa/Tunis		
		Africa/Windhoek		
		America/Adak		
		America/Anchorage		
		America/Anguilla		
		America/Antigua		
		America/Araguaina		
		America/Argentina/Buenos_Aires		
		America/Argentina/Catamarca		
		America/Argentina/ComodRivadavia		
		America/Argentina/Cordoba		
		America/Argentina/Jujuy		
		America/Argentina/La_Rioja		
		America/Argentina/Mendoza		

Attribute	Type	Range/Length/Pattern	Default	Description
		America/Argentina/Rio_Gallegos		
		America/Argentina/Salta		
		America/Argentina/San_Juan		
		America/Argentina/San_Luis		
		America/Argentina/Tucuman		
		America/Argentina/Ushuaia		
		America/Aruba		
		America/Asuncion		
		America/Atikokan		
		America/Atka		
		America/Bahia		
		America/Bahia_Banderas		
		America/Barbados		
		America/Belem		
		America/Belize		
		America/Blanc-Sablon		
		America/Boa_Vista		
		America/Bogota		
		America/Boise		
		America/Buenos_Aires		
		America/Cambridge_Bay		
		America/Campo_Grande		
		America/Cancun		
		America/Caracas		
		America/Catamarca		
		America/Cayenne		
		America/Cayman		
		America/Chicago		
		America/Chihuahua		
		America/Coral_Harbour		
		America/Cordoba		
		America/Costa_Rica		
		America/Creston		
		America/Cuiaba		
		America/Curacao		
		America/Danmarkshavn		
		America/Dawson		
		America/Dawson_Creek		
		America/Denver		
		America/Detroit		
		America/Dominica		
		America/Edmonton		
		America/Eirunepe		
		America/El_Salvador		
		America/Ensenada		
		America/Fortaleza		
		America/Fort_Nelson		
		America/Fort_Wayne		
		America/Glace_Bay		
		America/Godthab		
		America/Goose_Bay		
		America/Grand_Turk		
		America/Grenada		
		America/Guadeloupe		
		America/Guatemala		
		America/Guayaquil		
		America/Guyana		
		America/Halifax		
		America/Havana		
		America/Hermosillo		
		America/Indiana/Indianapolis		
		America/Indiana/Knox		
		America/Indiana/Marengo		
		America/Indiana/Petersburg		
		America/Indianapolis		

Attribute	Type	Range/Length/Pattern	Default	Description
		America/Indiana/Tell_City		
		America/Indiana/Vevay		
		America/Indiana/Vincennes		
		America/Indiana/Winamac		
		America/Inuvik		
		America/Iqaluit		
		America/Jamaica		
		America/Jujuy		
		America/Juneau		
		America/Kentucky/Louisville		
		America/Kentucky/Monticello		
		America/Knox_IN		
		America/Kralendijk		
		America/La_Paz		
		America/Lima		
		America/Los_Angeles		
		America/Louisville		
		America/Lower_Princes		
		America/Maceio		
		America/Managua		
		America/Manaus		
		America/Marigot		
		America/Martinique		
		America/Matamoros		
		America/Mazatlan		
		America/Mendoza		
		America/Menominee		
		America/Merida		
		America/Metlakatla		
		America/Mexico_City		
		America/Miquelon		
		America/Moncton		
		America/Monterrey		
		America/Montevideo		
		America/Montreal		
		America/Montserrat		
		America/Nassau		
		America/New_York		
		America/Nipigon		
		America/Nome		
		America/Noronha		
		America/North_Dakota/Beulah		
		America/North_Dakota/Center		
		America/North_Dakota/New_Salem		
		America/Nuuk		
		America/Ojinaga		
		America/Panama		
		America/Pangnirtung		
		America/Paramaribo		
		America/Phoenix		
		America/Port-au-Prince		
		America/Porto_Acre		
		America/Port_of_Spain		
		America/Porto_Velho		
		America/Puerto_Rico		
		America/Punta_Arenas		
		America/Rainy_River		
		America/Rankin_Inlet		
		America/Recife		
		America/Regina		
		America/Resolute		
		America/Rio_Branco		
		America/Rosario		
		America/Santa_Isabel		
		America/Santarem		

Attribute	Type	Range/Length/Pattern	Default	Description
		America/Santiago		
		America/Santo_Domingo		
		America/Sao_Paulo		
		America/Scoresbysund		
		America/Shiprock		
		America/Sitka		
		America/St_Barthelemy		
		America/St_Johns		
		America/St_Kitts		
		America/St_Lucia		
		America/St_Thomas		
		America/St_Vincent		
		America/Swift_Current		
		America/Tegucigalpa		
		America/Thule		
		America/Thunder_Bay		
		America/Tijuana		
		America/Toronto		
		America/Tortola		
		America/Vancouver		
		America/Virgin		
		America/Whitehorse		
		America/Winnipeg		
		America/Yakutat		
		America/Yellowknife		
		Antarctica/Casey		
		Antarctica/Davis		
		Antarctica/DumontDUrville		
		Antarctica/Macquarie		
		Antarctica/Mawson		
		Antarctica/McMurdo		
		Antarctica/Palmer		
		Antarctica/Rothera		
		Antarctica/South_Pole		
		Antarctica/Syowa		
		Antarctica/Troll		
		Antarctica/Vostok		
		Arctic/Longyearbyen		
		Asia/Aden		
		Asia/Almaty		
		Asia/Amman		
		Asia/Anadyr		
		Asia/Aqtau		
		Asia/Aqtobe		
		Asia/Ashgabat		
		Asia/Ashkhabad		
		Asia/Atyrau		
		Asia/Baghdad		
		Asia/Bahrain		
		Asia/Baku		
		Asia/Bangkok		
		Asia/Barnaul		
		Asia/Beirut		
		Asia/Bishkek		
		Asia/Brunei		
		Asia/Chita		
		Asia/Choibalsan		
		Asia/Chongqing		
		Asia/Colombo		
		Asia/Dacca		
		Asia/Damascus		
		Asia/Dhaka		
		Asia/Dili		
		Asia/Dubai		
		Asia/Dushanbe		

Attribute	Type	Range/Length/Pattern	Default	Description
		Asia/Famagusta		
		Asia/Gaza		
		Asia/Harbin		
		Asia/Hebron		
		Asia/Ho_Chi_Minh		
		Asia/Hong_Kong		
		Asia/Hovd		
		Asia/Irkutsk		
		Asia/Istanbul		
		Asia/Jakarta		
		Asia/Jayapura		
		Asia/Jerusalem		
		Asia/Kabul		
		Asia/Kamchatka		
		Asia/Karachi		
		Asia/Kashgar		
		Asia/Kathmandu		
		Asia/Katmandu		
		Asia/Khandyga		
		Asia/Kolkata		
		Asia/Krasnoyarsk		
		Asia/Kuala_Lumpur		
		Asia/Kuching		
		Asia/Kuwait		
		Asia/Macao		
		Asia/Macau		
		Asia/Magadan		
		Asia/Makassar		
		Asia/Manila		
		Asia/Muscat		
		Asia/Nicosia		
		Asia/Novokuznetsk		
		Asia/Novosibirsk		
		Asia/Omsk		
		Asia/Oral		
		Asia/Phnom_Penh		
		Asia/Pontianak		
		Asia/Pyongyang		
		Asia/Qatar		
		Asia/Qostanay		
		Asia/Qyzylorda		
		Asia/Riyadh		
		Asia/Sakhalin		
		Asia/Samarkand		
		Asia/Seoul		
		Asia/Shanghai		
		Asia/Singapore		
		Asia/Srednekolymsk		
		Asia/Taipei		
		Asia/Tashkent		
		Asia/Tbilisi		
		Asia/Tehran		
		Asia/Tel_Aviv		
		Asia/Thimbu		
		Asia/Thimphu		
		Asia/Tokyo		
		Asia/Tomsk		
		Asia/Ujung_Pandang		
		Asia/Ulaanbaatar		
		Asia/Ulan_Bator		
		Asia/Urumqi		
		Asia/Ust-Nera		
		Asia/Vientiane		
		Asia/Vladivostok		
		Asia/Yakutsk		

Attribute	Type	Range/Length/Pattern	Default	Description
		Asia/Yangon		
		Asia/Yekaterinburg		
		Asia/Yerevan		
		Atlantic/Azores		
		Atlantic/Bermuda		
		Atlantic/Canary		
		Atlantic/Cape_Verde		
		Atlantic/Faeroe		
		Atlantic/Faroe		
		Atlantic/Jan_Mayen		
		Atlantic/Madeira		
		Atlantic/Reykjavik		
		Atlantic/South_Georgia		
		Atlantic/Stanley		
		Atlantic/St_Helena		
		Australia/ACT		
		Australia/Adelaide		
		Australia/Brisbane		
		Australia/Broken_Hill		
		Australia/Canberra		
		Australia/Currie		
		Australia/Darwin		
		Australia/Eucla		
		Australia/Hobart		
		Australia/LHI		
		Australia/Lindeman		
		Australia/Lord_Howe		
		Australia/Melbourne		
		Australia/North		
		Australia/NSW		
		Australia/Perth		
		Australia/Queensland		
		Australia/South		
		Australia/Sydney		
		Australia/Tasmania		
		Australia/Victoria		
		Australia/West		
		Australia/Yancowinna		
		Brazil/Acre		
		Brazil/DeNoronha		
		Brazil/East		
		Brazil/West		
		Canada/Atlantic		
		Canada/Central		
		Canada/Eastern		
		Canada/Mountain		
		Canada/Newfoundland		
		Canada/Pacific		
		Canada/Saskatchewan		
		Canada/Yukon		
		CET		
		Chile/Continental		
		Chile/EasterIsland		
		CST6CDT		
		Cuba		
		EET		
		Egypt		
		Eire		
		EST		
		EST5EDT		
		Etc/GMT		
		Etc/GMT+0		
		Etc/GMT-0		
		Etc/GMT0		
		Etc/GMT+1		

Attribute	Type	Range/Length/Pattern	Default	Description
		Etc/GMT-1		
		Etc/GMT+10		
		Etc/GMT-10		
		Etc/GMT+11		
		Etc/GMT-11		
		Etc/GMT+12		
		Etc/GMT-12		
		Etc/GMT-13		
		Etc/GMT-14		
		Etc/GMT+2		
		Etc/GMT-2		
		Etc/GMT+3		
		Etc/GMT-3		
		Etc/GMT+4		
		Etc/GMT-4		
		Etc/GMT+5		
		Etc/GMT-5		
		Etc/GMT+6		
		Etc/GMT-6		
		Etc/GMT+7		
		Etc/GMT-7		
		Etc/GMT+8		
		Etc/GMT-8		
		Etc/GMT+9		
		Etc/GMT-9		
		Etc/Greenwich		
		Etc/UCT		
		Etc/Universal		
		Etc/UTC		
		Etc/Zulu		
		Europe/Amsterdam		
		Europe/Andorra		
		Europe/Astrakhan		
		Europe/Athens		
		Europe/Belfast		
		Europe/Belgrade		
		Europe/Berlin		
		Europe/Bratislava		
		Europe/Brussels		
		Europe/Bucharest		
		Europe/Budapest		
		Europe/Busingen		
		Europe/Chisinau		
		Europe/Copenhagen		
		Europe/Dublin		
		Europe/Gibraltar		
		Europe/Guernsey		
		Europe/Helsinki		
		Europe/Isle_of_Man		
		Europe/Istanbul		
		Europe/Jersey		
		Europe/Kaliningrad		
		Europe/Kiev		
		Europe/Kirov		
		Europe/Lisbon		
		Europe/Ljubljana		
		Europe/London		
		Europe/Luxembourg		
		Europe/Madrid		
		Europe/Malta		
		Europe/Mariehamn		
		Europe/Minsk		
		Europe/Monaco		
		Europe/Moscow		
		Europe/Nicosia		

Attribute	Type	Range/Length/Pattern	Default	Description
		Europe/Oslo		
		Europe/Paris		
		Europe/Podgorica		
		Europe/Prague		
		Europe/Riga		
		Europe/Rome		
		Europe/Samara		
		Europe/San_Marino		
		Europe/Sarajevo		
		Europe/Saratov		
		Europe/Simferopol		
		Europe/Skopje		
		Europe/Sofia		
		Europe/Stockholm		
		Europe/Tallinn		
		Europe/Tirane		
		Europe/Tiraspol		
		Europe/Ulyanovsk		
		Europe/Uzhgorod		
		Europe/Vaduz		
		Europe/Vatican		
		Europe/Vienna		
		Europe/Vilnius		
		Europe/Volgograd		
		Europe/Warsaw		
		Europe/Zagreb		
		Europe/Zaporozhye		
		Europe/Zurich		
		GB		
		GB-Eire		
		GMT		
		GMT+0		
		GMT-0		
		GMT0		
		Greenwich		
		Hongkong		
		HST		
		Iceland		
		Indian/Antananarivo		
		Indian/Chagos		
		Indian/Christmas		
		Indian/Cocos		
		Indian/Comoro		
		Indian/Kerguelen		
		Indian/Mahe		
		Indian/Maldives		
		Indian/Mauritius		
		Indian/Mayotte		
		Indian/Reunion		
		Iran		
		Israel		
		Jamaica		
		Japan		
		Kwajalein		
		Libya		
		MET		
		Mexico/BajaNorte		
		Mexico/BajaSur		
		Mexico/General		
		MST		
		MST7MDT		
		Navajo		
		NZ		
		NZ-CHAT		
		Pacific/Apia		

Attribute	Type	Range/Length/Pattern	Default	Description
		Pacific/Auckland		
		Pacific/Bougainville		
		Pacific/Chatham		
		Pacific/Chuuk		
		Pacific/Easter		
		Pacific/Efate		
		Pacific/Enderbury		
		Pacific/Fakaofo		
		Pacific/Fiji		
		Pacific/Funafuti		
		Pacific/Galapagos		
		Pacific/Gambier		
		Pacific/Guadalcanal		
		Pacific/Guam		
		Pacific/Honolulu		
		Pacific/Johnston		
		Pacific/Kanton		
		Pacific/Kiritimati		
		Pacific/Kosrae		
		Pacific/Kwajalein		
		Pacific/Majuro		
		Pacific/Marquesas		
		Pacific/Midway		
		Pacific/Nauru		
		Pacific/Niue		
		Pacific/Norfolk		
		Pacific/Noumea		
		Pacific/Pago_Pago		
		Pacific/Palau		
		Pacific/Pitcairn		
		Pacific/Pohnpei		
		Pacific/Ponape		
		Pacific/Port_Moresby		
		Pacific/Rarotonga		
		Pacific/Saipan		
		Pacific/Samoa		
		Pacific/Tahiti		
		Pacific/Tarawa		
		Pacific/Tongatapu		
		Pacific/Truk		
		Pacific/Wake		
		Pacific/Wallis		
		Pacific/Yap		
		Poland		
		Portugal		
		PRC		
		PST8PDT		
		ROK		
		Singapore		
		Turkey		
		UCT		
		Universal		
		US/Alaska		
		US/Aleutian		
		US/Arizona		
		US/Central		
		US/Eastern		
		US/East-Indiana		
		US/Hawaii		
		US/Indiana-Starke		
		US/Michigan		
		US/Mountain		
		US/Pacific		
		US/Pacific-New		
		US/Samoa		

Attribute	Type	Range/Length/Pattern	Default	Description
		UTC		
		WET		
		W-SU		
		Zulu		
preferred-smf-locality	string			Preferred Locality for SMF Discovery

Emergency Support

Support of various emergency services

XPath: *cna-amf-cc:/amf-cc/amf-profile/serving-plmn/tracking-area/emergency-support*

CLI Path: *amf-cc amf-profile serving-plmn tracking-area emergency-support*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
emergency-services	enumeration	enable disable	enable	Controls support of emergency services. Default: enable
emergency-services-fallback	enumeration	enable disable	enable	Controls support for fallback for emergency services. Default: enable

Snssai List

A list of supported SNSSAI, identified by two keys 'slice-service-type' and 'slice-differentiator'
Value 0xFFFFFFFF is reserved and cannot be used to configure slice-differentiator

XPath: *cna-amf-cc:/amf-cc/amf-profile/serving-plmn/snssai-list*

CLI Path: *amf-cc amf-profile serving-plmn snssai-list*

List Properties:

Keys: slice-service-type, slice-differentiator

Minimum number of elements: 1

Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
slice-service-type	union	['uint8,0 or 4 .. 255', 'Enumerations: eMBB URLLC MIoT ']		Key 1: 'slice-service-type' identifies the expected Network Slice behavior in terms of features and services
slice-differentiator	union	['string,(0x[0-9a-fA-F]{6})', 'Enumerations: not-specified ']		

Eps Interworking

EPS interworking parameters

XPath: *cna-amf-cc:/amf-cc/amf-profile/serving-plmn/eps-interworking*

CLI Path: *amf-cc amf-profile serving-plmn eps-interworking*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
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Attribute	Type	Range/Length/Pattern	Default	Description
supported-type	enumeration	with-n26 without-n26 none	with-n26	Defines support for EPS interworking. Default: with-n26 The interface 'amf-cc interface n26' must be configured if 'eps-interworking supported-type' is set to with-n26
return-preferred-ind-n26	enumeration	enable disable	disable	Controls the preferred return of the UE to the last known PLMN during N26 procedures. Default: disable

Emergency Support

AMF support for emergency services

XPath: *cna-amf-cc:/amf-cc/amf-profile/serving-plmn/emergency-support*

CLI Path: *amf-cc amf-profile serving-plmn emergency-support*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
emergency-user-option	enumeration	valid-ues-only authenticated-ues-only supi-required-ues all-ues	all-ues	User category to allow an emergency call. Default: all-ues
skip-authentication	enumeration	enable disable	disable	Skip authentication for UEs requesting emergency services. Default: disable Value of 'enable' is valid only when 'emergency-user-option' is either 'supi-only-ues' or 'all-ues'
ue-ambr-uplink	uint32	0..4000000000	64	Emergency uplink Aggregated Maximum Bit Rate (AMBR). Default: 64; Unit: kbps
ue-ambr-downlink	uint32	0..4000000000	64	Emergency downlink Aggregated Maximum Bit Rate (AMBR). Default: 64; Unit: kbps
mobile-reachable-timer	union	['uint32,10..34560', 'Enumerations: t3512plus1 ']	t3512plus1	Mobile reachable timer for emergency attached UEs. Default: t3512plus1; Unit: minutes
dnn-ni	string	0..100	emergency	Network Identifier (NI) portion of the Data Network Name (DNN) used for emergency services. This is combined with the serving PLMN to make the full DNN. Default: emergency

Emergency S Nssai

The S-NSSAI used for emergency services

XPath: *cna-amf-cc:/amf-cc/amf-profile/serving-plmn/emergency-support/emergency-s-nssai*

CLI Path: *amf-cc amf-profile serving-plmn emergency-support emergency-s-nssai*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
slice-service-type	union	['uint8,0 or 4 .. 255', 'Enumerations: eMBB URLLC MIoT ']	eMBB	Identifies the expected Network Slice behavior in terms of features and services; Default: eMBB
slice-differentiator	union	['string,(0x[0-9a-fA-F]{6})', 'Enumerations: not-specified ']	not-specified	Differentiates amongst multiple Network Slices of the same Slice Service type, and is a 3-octet string in hexadecimal format with a prefix of '0x', or 'not-specified' if the value is not specified. Value 0xFFFFF is reserved and cannot be used to configure slice-differentiator. Default: 'not-specified'

Nsi List

NSI Identities of the AMF

XPath: *cna-amf-cc:/amf-cc/amf-profile/nsi-list*

CLI Path: *amf-cc amf-profile nsi-list*

List Properties:

Keys: nsi-id
Minimum number of elements: 0
Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
nsi-id	string	1..63		NSI Identity

Amf Service

Services provided by the AMF. See 3GPP specification TS 29.518 for more information.
Reference: 3GPP TS 29.518: 3rd Generation Partnership Project; Technical Specification Group Core Network and Terminals; 5G System; Access and Mobility Management Services; Stage 3

XPath: *cna-amf-cc:/amf-cc/amf-service*

CLI Path: *amf-cc amf-service*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
oauth-token-validation	enumeration	never if-available mandatory	never	Defines the Oauth token validation supported for AMF services. Default: never

Namf Communications

NAMF Communication (Namf_Communication) Service

XPath: *cna-amf-cc:/amf-cc/amf-service/namf-comm*

CLI Path: *amf-cc amf-service namf-communications*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
version-supported	enumeration	16.11.0 16.5.0	16.11.0	3GPP specification versions supported for namf-comm
src-amf-resource-release-timer	uint16	0..30000	10000	Defines the duration the source AMF waits before releasing the UE context; The timer is started after UE context transfer request is received from target AMF; Default: 10000; Unit: milliseconds
include-seaf-data	enumeration	always exclude-when-ue-validated	exclude-when-ue-validated	On the Source AMF, this controls whether the Source AMF includes security context information in the form of SEAF (SEcurity Anchor Function) data in the UE Context Transfer response. On the Target AMF, this controls whether the SEAF data is expected in the response. The UE Context Transfer service operation is described in TS 29.518. A value of 'always' is compliant to Release 15 and prior. A value of 'exclude-when-ue-validated' is compliant to Release 16. Default: exclude-when-ue-validated
n1n2-transfer-area-validation	enumeration	enable disable	disable	Defines support for TAI validation based on areaOfValidity TAI list from SMF. If enabled, AMF sends update SMF context if validation fails or areaOfValidity TAI list IE is not sent from SMF otherwise if disabled then no action as areaOfValidity based validation is ignored. Default: disable

Namf Comm Version (choice)

XPath: *cna-amf-cc:/amf-cc/amf-service/namf-comm*

CLI Path: *amf-cc amf-service namf-communications*

Case : all-available

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
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Attribute	Type	Range/Length/Pattern	Default	Description
all-available	empty			Enable all available versions for namf-comm service. Choice: This all-available attribute is mutually exclusive with attributes [['version']]

Case : supported-versions

Version

This attribute is obsolete, not supported anymore

Choice: This version attribute is mutually exclusive with attributes [['all-available']]

XPath: [cna-amf-cc:/amf-cc/amf-service/namf-comm/version](#)

CLI Path: [amf-cc amf-service namf-communications version](#)

List Properties:

Keys: version

Minimum number of elements: 1

Maximum number of elements: 2

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
version	enumeration	v1		

Namf Eventexposure

NAMF Event Exposure (Namf_EventExposure) Service

XPath: [cna-amf-cc:/amf-cc/amf-service/namf-evts](#)

CLI Path: [amf-cc amf-service namf-eventexposure](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
version-supported	enumeration	16.11.0 16.5.0	16.11.0	3GPP specification versions supported for namf-evts
default-events-policy	leafref			Event exposure policy with default values Leaf Reference Path: cna-amf-cc:/amf-cc/amf-service/namf-evts/event-exposure-policy/name

Name Evts Version (choice)

XPath: [cna-amf-cc:/amf-cc/amf-service/namf-evts](#)

CLI Path: [amf-cc amf-service namf-eventexposure](#)

Case : all-available

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
all-available	empty			Enable all available versions for the namf-evts service. Choice: This all-available attribute is mutually exclusive with attributes [['version']]

Case : supported-versions

Version

This attribute is obsolete, not supported anymore

Choice: This version attribute is mutually exclusive with attributes [['all-available']]

XPath: [cna-amf-cc:/amf-cc/amf-service/namf-evts/version](#)

CLI Path: *amf-cc amf-service namf-eventexposure version*

List Properties:

Keys: version
Minimum number of elements: 1
Maximum number of elements: 2

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
version	enumeration	v1		

Events Policy

List of event exposure policies
The default policy 'amf-cc amf-service namf-evts default-events-policy' must be configured before configuring 'events-policy'

XPath: *cna-amf-cc:/amf-cc/amf-service/namf-evts/events-policy*

CLI Path: *amf-cc amf-service namf-eventexposure events-policy*

List Properties:

Keys: nf-id
Maximum number of elements: 20

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
nf-id	string	1..64		NF ID of the peer node
exposure-policy-name	leafref			Event exposure policy name Leaf Reference Path: <i>cna-amf-cc:/amf-cc/amf-service/namf-evts/event-exposure-policy/name</i>

Event Exposure Policy

Defines the event exposure policies

XPath: *cna-amf-cc:/amf-cc/amf-service/namf-evts/event-exposure-policy*

CLI Path: *amf-cc amf-service namf-eventexposure event-exposure-policy*

List Properties:

Keys: name
Maximum number of elements: 20

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	1..64		Name of the event exposure policy

Allowed Event Type

Allowed event type per policy

XPath: *cna-amf-cc:/amf-cc/amf-service/namf-evts/event-exposure-policy/allowed-event-type*

CLI Path: *amf-cc amf-service namf-eventexposure event-exposure-policy allowed-event-type*

List Properties:

Keys: event-type
Minimum number of elements: 1
Maximum number of elements: 12

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
event-type	enumeration	reachability-report		AMF event type
allowed-ue-type	enumeration	one-ue		UE type

Attribute	Type	Range/Length/Pattern	Default	Description
allowed-report-type	enumeration	one-time	one-time	Report type

Namf Mt

Namf_MT Service

XPath: *cna-amf-cc:/amf-cc/amf-service/namf-mt*

CLI Path: *amf-cc amf-service namf-mt*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
version-supported	enumeration	16.11.0 16.5.0	16.11.0	3GPP specification versions supported for namf-mt

Namf Mt Version (choice)

XPath: *cna-amf-cc:/amf-cc/amf-service/namf-mt*

CLI Path: *amf-cc amf-service namf-mt*

Case : all-available

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
all-available	empty			Enable all available versions for the namf-mt service. Choice: This all-available attribute is mutually exclusive with attributes [['version']]

Case : supported-versions

Version

This attribute is obsolete, not supported anymore.

Choice: This version attribute is mutually exclusive with attributes [['all-available']]

XPath: *cna-amf-cc:/amf-cc/amf-service/namf-mt/version*

CLI Path: *amf-cc amf-service namf-mt version*

List Properties:

Keys: version

Minimum number of elements: 1

Maximum number of elements: 2

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
version	enumeration	v1		

Namf Loc

AMF Location Service(Namf_Location) Service

XPath: *cna-amf-cc:/amf-cc/amf-service/namf-loc*

CLI Path: *amf-cc amf-service namf-loc*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
version-supported	enumeration	16.11.0 16.5.0	16.11.0	3GPP specification versions supported for namf-loc

Namf Loc Version (choice)

XPath: *cna-amf-cc:/amf-cc/amf-service/namf-loc*

CLI Path: *amf-cc amf-service namf-loc*

Case : all-available

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
all-available	empty			Enable all available versions for the namf-loc service. Choice: This all-available attribute is mutually exclusive with attributes [['version']]

Case : supported-versions

Version

This attribute is obsolete, not supported anymore.

Choice: This version attribute is mutually exclusive with attributes [['all-available']]

XPath: *cna-amf-cc:/amf-cc/amf-service/namf-loc/version*

CLI Path: *amf-cc amf-service namf-loc version*

List Properties:

Keys: version
Minimum number of elements: 1
Maximum number of elements: 2

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
version	enumeration	v1		

Canary

Canary objects

XPath: *cna-amf-cc:/amf-cc/canary*

CLI Path: *amf-cc canary*

Target Supi

List of target supi for canary deployment.

XPath: *cna-amf-cc:/amf-cc/canary/target-supi*

CLI Path: *amf-cc canary target-supi*

List Properties:

Keys: prefix
Maximum number of elements: 100

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
prefix	string	(([0-9]){15})		15 digit of target supi number

Consumed Service

Services consumed by the AMF

XPath: *cna-amf-cc:/amf-cc/consumed-service*

CLI Path: *amf-cc consumed-service*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
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Attribute	Type	Range/Length/Pattern	Default	Description
nf-instance-per-service-allowed	leaf-list of enumeration	udm		Identifies the NFs for which the selection of NF instances per service is allowed; Default: none;

Nausf Auth

Nausf UE Authentication Service (reference TS 29.509)

XPath: *cna-amf-cc:/amf-cc/consumed-service/nausf-auth*

CLI Path: *amf-cc consumed-service nausf-auth*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
version-supported	enumeration	16.9.0 16.5.0	16.9.0	3GPP specification versions supported for nausf-auth
request-timer	uint16	0..15000	6000	Defines the duration the AMF waits for a response from AUSF; Default: 6000; Unit: milliseconds

Nausf Auth Version (choice)

XPath: *cna-amf-cc:/amf-cc/consumed-service/nausf-auth*

CLI Path: *amf-cc consumed-service nausf-auth*

Case : all-available

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
all-available	empty			All available versions of nausf-auth are supported, with the highest version being used. Choice: This all-available attribute is mutually exclusive with attributes [['version']]

Case : supported-versions

Version

This attribute is obsolete, not supported anymore

Choice: This version attribute is mutually exclusive with attributes [['all-available']]

XPath: *cna-amf-cc:/amf-cc/consumed-service/nausf-auth/version*

CLI Path: *amf-cc consumed-service nausf-auth version*

List Properties:

Keys: version

Minimum number of elements: 1

Maximum number of elements: 2

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
version	enumeration	v1		

Supported Features

AUSF feature support

XPath: *cna-amf-cc:/amf-cc/consumed-service/nausf-auth/supported-features*

CLI Path: *amf-cc consumed-service nausf-auth supported-features*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
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Attribute	Type	Range/Length/Pattern	Default	Description
authentication-result-removal	enumeration	enable disable	disable	Defines if the Authentication result removal procedure should be performed during purge or SMC failure; Default: disable
ES3XX	enumeration	enable disable	enable	Defines Extended Support of HTTP 307/308 redirection; Default: enable

Nudm Sdm

Nudm Subscriber Data Management Service (reference TS 29.503)

XPath: [cna-amf-cc:/amf-cc/consumed-service/nudm-sdm](#)

CLI Path: [amf-cc consumed-service nudm-sdm](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
version-supported	enumeration	16.11.0 16.5.0	16.11.0	3GPP specification versions supported for nudm-sdm
request-timer	uint16	0..15000	6000	Defines the duration the AMF waits for an nudm-sdm response from UDM; Default: 6000; Unit: milliseconds

Nudm Sdm Version (choice)

XPath: [cna-amf-cc:/amf-cc/consumed-service/nudm-sdm](#)

CLI Path: [amf-cc consumed-service nudm-sdm](#)

Case : all-available

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
all-available	empty			All available versions of nudm-sdm are supported, with the highest version being used. Choice: This all-available attribute is mutually exclusive with attributes [['version']]

Case : supported-versions

Version

This attribute is obsolete, not supported anymore

Choice: This version attribute is mutually exclusive with attributes [['all-available']]

XPath: [cna-amf-cc:/amf-cc/consumed-service/nudm-sdm/version](#)

CLI Path: [amf-cc consumed-service nudm-sdm version](#)

List Properties:

Keys: version

Minimum number of elements: 1

Maximum number of elements: 2

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
version	enumeration	v1 v2		

Nudm Uecm

Nudm UE Context Management Service (reference TS 29.503)

XPath: [cna-amf-cc:/amf-cc/consumed-service/nudm-uecm](#)

CLI Path: [amf-cc consumed-service nudm-uecm](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
version-supported	enumeration	16.11.0 16.5.0	16.11.0	3GPP specification versions supported for nudm-uecm
request-timer	uint16	0..15000	6000	Defines the duration the AMF waits for an nudm-uecm response from UDM; Default: 6000; Unit: milliseconds

Nudm Uecm Version (choice)

XPath: [cna-amf-cc:/amf-cc/consumed-service/nudm-uecm](#)

CLI Path: [amf-cc consumed-service nudm-uecm](#)

Case : all-available

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
all-available	empty			All available versions of nudm-uecm are supported, with the highest version being used. Choice: This all-available attribute is mutually exclusive with attributes [['version']]

Case : supported-versions

Version

This attribute is obsolete, not supported anymore

Choice: This version attribute is mutually exclusive with attributes [['all-available']]

XPath: [cna-amf-cc:/amf-cc/consumed-service/nudm-uecm/version](#)

CLI Path: [amf-cc consumed-service nudm-uecm version](#)

List Properties:

Keys: version

Minimum number of elements: 1

Maximum number of elements: 2

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
version	enumeration	v1		

Nudm Pp

Nudm Paramter Provision Service (reference TS 29.503)

XPath: [cna-amf-cc:/amf-cc/consumed-service/nudm-pp](#)

CLI Path: [amf-cc consumed-service nudm-pp](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
version-supported	enumeration	16.11.0	16.11.0	3GPP specification versions supported for nudm-pp
request-timer	uint16	0..15000	6000	Defines the duration the AMF waits for an nudm-pp response from UDM; Default: 6000; Unit: milliseconds

Nudm Pp Version (choice)

XPath: [cna-amf-cc:/amf-cc/consumed-service/nudm-pp](#)

CLI Path: [amf-cc consumed-service nudm-pp](#)

Case : all-available

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
all-available	empty			All available versions of nudm-pp are supported, with the highest version being used. Choice: This all-available attribute is mutually exclusive with attributes [['version']]

Case : supported-versions

Version

This attribute is obsolete, not supported anymore

Choice: This version attribute is mutually exclusive with attributes [['all-available']]

XPath: *cna-amf-cc:/amf-cc/consumed-service/nudm-pp/version*

CLI Path: *amf-cc consumed-service nudm-pp version*

List Properties:

Keys: version

Minimum number of elements: 1

Maximum number of elements: 2

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
version	enumeration	v1		

Nsmf Pdusession

Nsmf PDU Session Service (reference TS 29.502)

XPath: *cna-amf-cc:/amf-cc/consumed-service/nsmf-pdusession*

CLI Path: *amf-cc consumed-service nsmf-pdusession*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
version-supported	enumeration	16.11.0 16.5.0	16.11.0	3GPP specification versions supported for nsmf-pdusession
request-timer	uint16	0..15000	6000	Defines the duration the AMF waits for an nsmf-pdusession response from SMF; Default: 6000; Unit: milliseconds

Nsmf Pdusession Version (choice)

XPath: *cna-amf-cc:/amf-cc/consumed-service/nsmf-pdusession*

***CLI Path: *amf-cc consumed-service nsmf-pdusession* ***

Case : all-available

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
all-available	empty			All available versions of nsmf-pdusession are supported, with the highest version being used. Choice: This all-available attribute is mutually exclusive with attributes [['version']]

Case : supported-versions

Version

This attribute is obsolete, not supported anymore

Choice: This version attribute is mutually exclusive with attributes [['all-available']]

XPath: *cna-amf-cc:/amf-cc/consumed-service/nsmf-pdusession/version*

CLI Path: *amf-cc consumed-service nsmf-pdusession version*

List Properties:

Keys: version
Minimum number of elements: 1
Maximum number of elements: 2

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
version	enumeration	v1		

Supported Features

SMF feature support

XPath: [cna-amf-cc:/amf-cc/consumed-service/nsmf-pdusession/supported-features](#)

CLI Path: [amf-cc consumed-service nsmf-pdusession supported-features](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
ES3XX	enumeration	enable disable	enable	Defines Extended Support of HTTP 307/308 redirection; Default: enable

Nnssf Nsselection

Nnssf NS Selection Service (reference TS 29.531)

XPath: [cna-amf-cc:/amf-cc/consumed-service/nnssf-nselection](#)

CLI Path: [amf-cc consumed-service nnssf-nselection](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
request-timer	uint16	0..15000	6000	Defines the duration the AMF waits for a nnssf-nselection response from NSSF; Default: 6000; Unit: milliseconds
version-supported	enumeration	16.8.0 16.4.0	16.8.0	3GPP specification versions supported for nnssf-nselection

Nnssf Nsselection Version (choice)

XPath: [cna-amf-cc:/amf-cc/consumed-service/nnssf-nselection](#)

CLI Path: [amf-cc consumed-service nnssf-nselection](#)

Case : all-available

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
all-available	empty			All available versions of nnssf-nselection are supported, with the highest version being used. Choice: This all-available attribute is mutually exclusive with attributes [['version']]

Case : supported-versions

Version

This attribute is obsolete, not supported anymore

Choice: This version attribute is mutually exclusive with attributes [['all-available']]

XPath: [cna-amf-cc:/amf-cc/consumed-service/nnssf-nselection/version](#)

CLI Path: [amf-cc consumed-service nnssf-nselection version](#)

List Properties:

Keys: version

Minimum number of elements: 1
Maximum number of elements: 2

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
version	enumeration	v2		

Namf Comm

Namf Communication Service (reference TS 29.518)

XPath: [cna-amf-cc:/amf-cc/consumed-service/namf-comm](#)

CLI Path: [amf-cc consumed-service namf-comm](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
version-supported	enumeration	16.11.0 16.5.0	16.11.0	3GPP specification versions supported for namf-comm
request-timer	uint16	0..15000	6000	Defines the duration the AMF waits for a response from a peer AMF; Default: 6000; Unit: milliseconds

Namf Comm Version (choice)

XPath: [cna-amf-cc:/amf-cc/consumed-service/namf-comm](#)

CLI Path: [amf-cc consumed-service namf-comm](#)

Case : all-available

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
all-available	empty			All available versions of namf-comm are supported, with the highest version being used. Choice: This all-available attribute is mutually exclusive with attributes [['version']]

Case : supported-versions

Version

This attribute is obsolete, not supported anymore

Choice: This version attribute is mutually exclusive with attributes [['all-available']]

XPath: [cna-amf-cc:/amf-cc/consumed-service/namf-comm/version](#)

CLI Path: [amf-cc consumed-service namf-comm version](#)

List Properties:

Keys: version
Minimum number of elements: 1
Maximum number of elements: 2

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
version	enumeration	v1		

Supported Features

Namf Comm feature support

XPath: [cna-amf-cc:/amf-cc/consumed-service/namf-comm/supported-features](#)

CLI Path: [amf-cc consumed-service namf-comm supported-features](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
ES3XX	enumeration	enable disable	enable	Defines Extended Support of HTTP 307/308 redirection; Default: enable

Nsmsf Sms

Nsmsf SMS Service (reference TS 29.540)

XPath: [cna-amf-cc:/amf-cc/consumed-service/nsmsf-sms](#)

CLI Path: [amf-cc consumed-service nsmsf-sms](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
version-supported	enumeration	16.10.0 16.5.0	16.10.0	3GPP specification versions supported for nsmsf-sms
request-timer	uint16	0..15000	6000	Defines the duration the AMF waits for a response from a peer SMSF; Default: 6000; Unit: milliseconds

Nsmsf Sms Version (choice)

XPath: [cna-amf-cc:/amf-cc/consumed-service/nsmsf-sms](#)

CLI Path: [amf-cc consumed-service nsmsf-sms](#)

Case : all-available

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
all-available	empty			All available versions of nsmsf-sms are supported, with the highest version being used. Choice: This all-available attribute is mutually exclusive with attributes [['version']]

Case : supported-versions

Version

This attribute is obsolete, not supported anymore

Choice: This version attribute is mutually exclusive with attributes [['all-available']]

XPath: [cna-amf-cc:/amf-cc/consumed-service/nsmsf-sms/version](#)

CLI Path: [amf-cc consumed-service nsmsf-sms version](#)

List Properties:

Keys: version

Minimum number of elements: 1

Maximum number of elements: 2

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
version	enumeration	v1		

Supported Features

SMSF feature support

XPath: [cna-amf-cc:/amf-cc/consumed-service/nsmsf-sms/supported-features](#)

CLI Path: [amf-cc consumed-service nsmsf-sms supported-features](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
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Attribute	Type	Range/Length/Pattern	Default	Description
ES3XX	enumeration	enable disable	enable	Defines Extended Support of HTTP 307/308 redirection; Default: enable

Npcf Uepolicycontrol

Npcf UE Policy Control Service (reference TS 29.525)

XPath: [cna-amf-cc:/amf-cc/consumed-service/npcf-uepolicycontrol](#)

CLI Path: [amf-cc consumed-service npcf-uepolicycontrol](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
version-supported	enumeration	16.10.0 16.5.0	16.10.0	3GPP specification versions supported for npcf-uepolicycontrol
request-timer	uint16	0..15000	6000	Defines the duration the AMF waits for a response from PCF; Default: 6000; Unit: milliseconds

Npcf Ue Policy Control Version (choice)

XPath: [cna-amf-cc:/amf-cc/consumed-service/npcf-uepolicycontrol](#)

CLI Path: [amf-cc consumed-service npcf-uepolicycontrol](#)

Case : all-available

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
all-available	empty			All available versions of npcf-uepolicycontrol are supported, with the highest version being used. Choice: This all-available attribute is mutually exclusive with attributes [['version']]

Case : supported-versions

Version

This attribute is obsolete, not supported anymore

Choice: This version attribute is mutually exclusive with attributes [['all-available']]

XPath: [cna-amf-cc:/amf-cc/consumed-service/npcf-uepolicycontrol/version](#)

CLI Path: [amf-cc consumed-service npcf-uepolicycontrol version](#)

List Properties:

Keys: version

Minimum number of elements: 1

Maximum number of elements: 2

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
version	enumeration	v1		

Npcf Ampolicycontrol

Npcf AM Policy Control Service (reference TS 29.507)

XPath: [cna-amf-cc:/amf-cc/consumed-service/npcf-ampolicycontrol](#)

CLI Path: [amf-cc consumed-service npcf-ampolicycontrol](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
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Attribute	Type	Range/Length/Pattern	Default	Description
version-supported	enumeration	16.11.0 16.5.0	16.11.0	3GPP specification versions supported for npcf-ampolicycontrol
request-timer	uint16	0..15000	6000	Defines the duration the AMF waits for a response from PCF; Default: 6000; Unit: milliseconds

Npcf Am Policy Control Version (choice)

XPath: [cna-amf-cc:/amf-cc/consumed-service/npcf-ampolicycontrol](#)

CLI Path: [amf-cc consumed-service npcf-ampolicycontrol](#)

Case : all-available

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
all-available	empty			All available versions of npcf-ampolicycontrol are supported, with the highest version being used. Choice: This all-available attribute is mutually exclusive with attributes [['version']]

Case : supported-versions

Version

This attribute is obsolete, not supported anymore

Choice: This version attribute is mutually exclusive with attributes [['all-available']]

XPath: [cna-amf-cc:/amf-cc/consumed-service/npcf-ampolicycontrol/version](#)

CLI Path: [amf-cc consumed-service npcf-ampolicycontrol version](#)

List Properties:

Keys: version

Minimum number of elements: 1

Maximum number of elements: 2

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
version	enumeration	v1		

Nlmf Loc

Nlmf Location Service (reference TS 29.273)

XPath: [cna-amf-cc:/amf-cc/consumed-service/nlmf-loc](#)

CLI Path: [amf-cc consumed-service nlmf-loc](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
version-supported	enumeration	16.9.0 16.4.0	16.9.0	3GPP specification versions supported for nlmf-loc
request-timer	uint16	0..15000	6000	Defines the duration the AMF waits for a response from LMF; Default: 6000; Unit: milliseconds

Nlmf Loc Version (choice)

XPath: [cna-amf-cc:/amf-cc/consumed-service/nlmf-loc](#)

CLI Path: [amf-cc consumed-service nlmf-loc](#)

Case : all-available

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
all-available	empty			All available versions of nlmf-loc are supported, with the highest version being used. Choice: This all-available attribute is mutually exclusive with attributes [['version']]

Case : supported-versions

Version

This attribute is obsolete, not supported anymore

Choice: This version attribute is mutually exclusive with attributes [['all-available']]

XPath: cna-amf-cc:/amf-cc/consumed-service/nlmf-loc/version

CLI Path: amf-cc consumed-service nlmf-loc version

List Properties:

Keys: version
Minimum number of elements: 1
Maximum number of elements: 2

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
version	enumeration	v1		

Ngmlc Loc

Ngmlc Location Service (reference TS 29.215)

XPath: cna-amf-cc:/amf-cc/consumed-service/ngmlc-loc

CLI Path: amf-cc consumed-service ngmlc-loc

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
version-supported	enumeration	16.7.0 16.2.0	16.7.0	3GPP specification versions supported for ngmlc-loc
request-timer	uint16	0..15000	6000	Defines the duration the AMF waits for a response from GMLC; Default: 6000; Unit: milliseconds

Ngmlc Loc Version (choice)

XPath: cna-amf-cc:/amf-cc/consumed-service/ngmlc-loc

***CLI Path: amf-cc consumed-service ngmlc-loc ***

Case : all-available

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
all-available	empty			All available versions of ngmlc-loc are supported, with the highest version being used. Choice: This all-available attribute is mutually exclusive with attributes [['version']]

Case : supported-versions

Version

This attribute is obsolete, not supported anymore

Choice: This version attribute is mutually exclusive with attributes [['all-available']]

XPath: cna-amf-cc:/amf-cc/consumed-service/ngmlc-loc/version

CLI Path: *amf-cc consumed-service ngmlc-loc version*

List Properties:

Keys: version
Minimum number of elements: 1
Maximum number of elements: 2

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
version	enumeration	v1		

Edr

XPath: *cna-amf-cc:/amf-cc/edr*

CLI Path: *amf-cc edr*

Default

default edr profiles values

XPath: *cna-amf-cc:/amf-cc/edr/default*

CLI Path: *amf-cc edr default*

Event Layer

XPath: *cna-amf-cc:/amf-cc/edr/default/event-layer*

CLI Path: *amf-cc edr default event-layer*

List Properties:

Keys: event-layer-name
Maximum number of elements: 20

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
event-layer-name	enumeration	mm		Event layer name - predefined set of supported event layers
		sm		
		ngap		
		lcs		
		sms		
		policy		

Event Type

XPath: *cna-amf-cc:/amf-cc/edr/default/event-layer/event-type*

CLI Path: *amf-cc edr default event-layer event-type*

List Properties:

Keys: event-type-name
Minimum number of elements: 1
Maximum number of elements: 20

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
event-type-name	enumeration	establishment		Event type name - predefined set of supported event types
		normal-release		
		abnormal-release		
		modification		
outcome-type-name	leaf-list of enumeration	success		outcome type name - predefined set of supported outcomes
		failure		
		reject		

Profile

Set of EDR profiles

XPath: *cna-amf-cc:/amf-cc/edr/profile*

CLI Path: *amf-cc edr profile*

List Properties:

Keys: name

Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	1..64		Name of the EDR profile

Event Layer

XPath: *cna-amf-cc:/amf-cc/edr/profile/event-layer*

CLI Path: *amf-cc edr profile event-layer*

List Properties:

Keys: event-layer-name

Maximum number of elements: 20

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
event-layer-name	enumeration	mm		Event layer name - predefined set of supported event layers
		sm		
		ngap		
		lcs		
		sms		
		policy		

Event Type

XPath: *cna-amf-cc:/amf-cc/edr/profile/event-layer/event-type*

CLI Path: *amf-cc edr profile event-layer event-type*

List Properties:

Keys: event-type-name

Minimum number of elements: 1

Maximum number of elements: 20

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
event-type-name	enumeration	establishment		Event type name - predefined set of supported event types
		normal-release		
		abnormal-release		
		modification		
outcome-type-name	leaf-list of enumeration	success		outcome type name - predefined set of supported outcomes
		failure		
		reject		

Mobility Policy

Mobility Policy

XPath: *cna-amf-cc:/amf-cc/mobility-policy*

CLI Path: *amf-cc mobility-policy*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
analyzer-name	string	1..64		Name of the subscriber analyzer

Default

Mobility class with default values

XPath: *cna-amf-cc:/amf-cc/mobility-policy/default*

CLI Path: *amf-cc mobility-policy default*

Access Control

Access Control Policy

XPath: *cna-amf-cc:/amf-cc/mobility-policy/default/access-control*

CLI Path: *amf-cc mobility-policy default access-control*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
access-reject	enumeration	undefined-mm-cause-0 pei-not-accepted-5 illegal-me-6 services-not-allowed-7 plmn-not-allowed-11 ta-not-allowed-12 roaming-not-allowed-in-this-ta-13 congestion-22 n1-mode-not-allowed-27 restricted-service-area-28 insufficient-resources-for-specific-slice-69	undefined-mm-cause-0	Access reject cause for the subscriber
nssai-inclusion-mode	enumeration	disable mode-a mode-b mode-c mode-d	disable	NSSAI inclusion mode in Registration Accept
rat-restrictions	leaf-list of enumeration	nr eutran		3GPP Radio Access Technology restrictions applicable for an UE
core-network-restrictions	leaf-list of enumeration	epc 5gc		Core network type restrictions applicable for a UE

Slice Selection

Slice Selection parameters

XPath: *cna-amf-cc:/amf-cc/mobility-policy/default/access-control/slice-selection*

CLI Path: *amf-cc mobility-policy default access-control slice-selection*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
selection-policy	enumeration	standard use-nssf-policy	standard	Indicates the policy for slice selection
reject-cause	union	['string,([0-9]{1,2} or 1[0-9]{1,2} or 2[0-4][0-9] or 25[0-5])(-([a-zA-Z]+(-[a-zA-Z]+)))', 'Enumerations: 7-5gs-service-not-allowed 11-plmn-not-allowed 12-roaming-not-allowed-in-ta 15-no-suitable-cell-in-ta 27-N1-mode-not-allowed ']	15-no-suitable-cell-in-ta	Describes cause code for registration-reject. Default: 15-no-suitable-cell-in-ta.
ran-reroute-preferred	enumeration	enable disable	disable	Controls RAN Reroute. Default: disable

Attribute	Type	Range/Length/Pattern	Default	Description
slice-negotiation-during-n26-handover	enumeration	enable disable	enable	Controls slice selection during N26 Handover. Default: enable

Fallback Snssai List

A list of S-NSSAI used if slice selection yields no results
Value 0xFFFFFFFF is reserved and cannot be used to configure slice-differentiator

XPath: *cna-amf-cc:/amf-cc/mobility-policy/default/access-control/slice-selection/fallback-snssai-list*

CLI Path: *amf-cc mobility-policy default access-control slice-selection fallback-snssai-list*

List Properties:
Keys: slice-service-type, slice-differentiator
Minimum number of elements: 0
Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
slice-service-type	union	['uint8,0 or 4 .. 255', 'Enumerations: eMBB URLLC MIoT ']		Key 1: 'slice-service-type' identifies the expected Network Slice behavior in terms of features and services
slice-differentiator	union	['string,(0x[0-9a-fA-F]{6})', 'Enumerations: not-specified ']		

Reroute

Reroute applicable for a UE

XPath: *cna-amf-cc:/amf-cc/mobility-policy/default/access-control/slice-selection/reroute*

CLI Path: *amf-cc mobility-policy default access-control slice-selection reroute*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
target-amf-set	string	[0-9]{3}-[0-9]{2,3}-[A-Fa-f0-9]{2}-[0-3][A-Fa-f0-9]{2}		Target AMF Set
candidate-amf-id-list	leaf-list of string	[0-9a-fA-F]{8}-[0-9a-fA-F]{4}-[0-9a-fA-F]{4}-[0-9a-fA-F]{4}-[0-9a-fA-F]{12}		List of candidate AMF IDs

Service Area Restrictions

Service Area restrictions applicable for a UE

XPath: *cna-amf-cc:/amf-cc/mobility-policy/default/access-control/service-area-restrictions*

CLI Path: *amf-cc mobility-policy default access-control service-area-restrictions*

Plmn

List of Service Area Restriction PLMN's

XPath: *cna-amf-cc:/amf-cc/mobility-policy/default/access-control/service-area-restrictions/plmn*

CLI Path: *amf-cc mobility-policy default access-control service-area-restrictions plmn*

List Properties:
Keys: plmn-id
Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
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Attribute	Type	Range/Length/Pattern	Default	Description
plmn-id	string	(([0-9]{5,6}))		Supported PLMN identity; Range: numbers with 5-6 digits

Restriction Type (choice)

XPath: *cna-amf-cc:/amf-cc/mobility-policy/default/access-control/service-area-restrictions/plmn*

***CLI Path: *amf-cc mobility-policy default access-control service-area-restrictions plmn* ***

Case : allowed-areas

Allowed Areas

Choice: This allowed-areas attribute is mutually exclusive with attributes [['not-allowed-areas']]

XPath: *cna-amf-cc:/amf-cc/mobility-policy/default/access-control/service-area-restrictions/plmn/allowed-areas*

CLI Path: *amf-cc mobility-policy default access-control service-area-restrictions plmn allowed-areas*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
tacs	leaf-list of string	(0x[A-Fa-f0-9]{4}) or (0x[A-Fa-f0-9]{6})		List of allowed Tracking Area Codes Choice: This tacs attribute is mutually exclusive with attributes [['not-allowed-areas']] Maximum number of elements: 16

Case : not-allowed-areas

Not Allowed Areas

Choice: This not-allowed-areas attribute is mutually exclusive with attributes [['allowed-areas']]

XPath: *cna-amf-cc:/amf-cc/mobility-policy/default/access-control/service-area-restrictions/plmn/not-allowed-areas*

CLI Path: *amf-cc mobility-policy default access-control service-area-restrictions plmn not-allowed-areas*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
tacs	leaf-list of string	(0x[A-Fa-f0-9]{4}) or (0x[A-Fa-f0-9]{6})		List of Tracking Area Codes not allowed Choice: This tacs attribute is mutually exclusive with attributes [['allowed-areas']] Maximum number of elements: 16

Service Control

Service Control Policy

XPath: *cna-amf-cc:/amf-cc/mobility-policy/default/service-control*

CLI Path: *amf-cc mobility-policy default service-control*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
sms-support	enumeration	enable disable	disable	Controls Short Message Service (SMS) support. Default: disable

Ims

IP Multimedia Subsystem parameters

XPath: *cna-amf-cc:/amf-cc/mobility-policy/default/service-control/ims*

CLI Path: *amf-cc mobility-policy default service-control ims*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
ims-vops-support	leaf-list of enumeration	3gpp-5gc fallback-eps non-3gpp-5gc		List of IMS Voice over PS support options
emergency-support	leaf-list of enumeration	fallback-eps 3gpp-5gc		Defines how UE obtains emergency service

Purge Trigger

List of purge triggers

XPath: *cna-amf-cc:/amf-cc/mobility-policy/default/service-control/purge-trigger*

CLI Path: *amf-cc mobility-policy default service-control purge-trigger*

List Properties:

Keys: deregistration-type

Maximum number of elements: 3

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
deregistration-type	enumeration	explicit-deregistration implicit-deregistration all		Deregistration type The entry with 'deregistration-type' set to all is already present, no other entry is allowed to be configured The 'deregistration-type' with value all is not allowed when entries with other deregistration types are already configured The 'deregistration-type' with value all is not allowed when entries with other deregistration types are already configured
inactive-age	union	[Enumerations: immediate ']	immediate	Time to purge. Default: immediate; Unit: hours

Lcs

LoCation Services(LCS) parameters

XPath: *cna-amf-cc:/amf-cc/mobility-policy/default/service-control/lcs*

CLI Path: *amf-cc mobility-policy default service-control lcs*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
lcs-support	enumeration	enable disable	disable	Controls location service support. Default: disable
nas-notification-timer-tlcsn	uint16	10..20	10	Defines the duration the AMF waits for a response to a Nas Notification invoke request sent to the UE; Default: 10; Unit: seconds

Network Induced Location Request

Network Induced Location Request (NI-LR) parameters

XPath: *cna-amf-cc:/amf-cc/mobility-policy/default/service-control/lcs/network-induced-location-request*

CLI Path: *amf-cc mobility-policy default service-control lcs network-induced-location-request*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
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Attribute	Type	Range/Length/Pattern	Default	Description
ni-lr-support	enumeration	disable enable-with-initial-positioning enable-without-initial-positioning	enable-with-initial-positioning	Controls Network Induced - Location Request procedure initiation. Default: enable-with-initial-positioning
horizontal-accuracy	string		50	Describes NI-LR measurement for horizontal accuracy. Default: 50; Unit: meters
vertical-accuracy	string		0	Describes NI-LR measurement for vertical accuracy. Default: 0; Unit: meters
emergency-target-sender	enumeration	source-node target-node	source-node	Specifies which node sends target node id to the GMLC. Default: source-node
response-time	enumeration	no-delay low-delay delay-tolerant	delay-tolerant	Controls initial NI-LR QoS response time. Default: delay-tolerant
qos-class	enumeration	best-effort assured	best-effort	Describes NI-LR QoS class type. Default: best-effort

Nas Policy

Non Access Stratum Policy

XPath: [cna-amf-cc:/amf-cc/mobility-policy/default/nas-policy](#)

CLI Path: [amf-cc mobility-policy default nas-policy](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
mm-profile-name	leafref			Mobility Management profile name per mobility class Leaf Reference Path: cna-amf-cc:/amf-cc/interface/nas/mm/profile/name
security-profile-name	leafref			Security profile name per mobility class Leaf Reference Path: cna-amf-cc:/amf-cc/interface/nas/security/profile/name
paging-profile-set-name	leafref			Name of the paging profile set for this mobility class Leaf Reference Path: cna-amf-cc:/amf-cc/interface/nas/paging/profile-set/name

Observability

XPath: [cna-amf-cc:/amf-cc/mobility-policy/default/observability](#)

CLI Path: [amf-cc mobility-policy default observability](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
edr-profile-name	leafref			edr profiles per mobility class Leaf Reference Path: cna-amf-cc:/amf-cc/edr/profile/name

Pcf Control

PCF Control attributes

XPath: [cna-amf-cc:/amf-cc/mobility-policy/default/pcf-control](#)

CLI Path: [amf-cc mobility-policy default pcf-control](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
reselect-pcf-on-amf-relocation	enumeration	enable disable	disable	Defines support for PCF re-selection on target AMF. Default: disable
forward-pcf-id-to-smf	enumeration	enable disable	disable	Defines PCF-Id to be forwarded to SMF or not. Default: disable

Ue Policy Control

UE Policy Control

XPath: *cna-amf-cc:/amf-cc/mobility-policy/default/pcf-control/ue-policy-control*

CLI Path: *amf-cc mobility-policy default pcf-control ue-policy-control*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
policy-association	enumeration	never always ue-requested	ue-requested	Defines support for UE Policy Association Establishment with PCF. Default: ue-requested

Am Policy Control

AM Policy Control

XPath: *cna-amf-cc:/amf-cc/mobility-policy/default/pcf-control/am-policy-control*

CLI Path: *amf-cc mobility-policy default pcf-control am-policy-control*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
policy-association	enumeration	never always	never	Defines support for AM Policy Association Establishment with PCF. Default: never

Class

Mobility Class

XPath: *cna-amf-cc:/amf-cc/mobility-policy/class*

CLI Path: *amf-cc mobility-policy class*

List Properties:

Keys: name

Maximum number of elements: 1000

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	1..64		Name of the mobility class

Access Control

Access Control Policy

XPath: *cna-amf-cc:/amf-cc/mobility-policy/class/access-control*

CLI Path: *amf-cc mobility-policy class access-control*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
access-reject	enumeration	undefined-mm-cause-0	undefined-mm-cause-0	Access reject cause for the subscriber
		pei-not-accepted-5		
		illegal-me-6		
		services-not-allowed-7		
		plmn-not-allowed-11		
		ta-not-allowed-12		
		roaming-not-allowed-in-this-ta-13		
		congestion-22		
		n1-mode-not-allowed-27		
		restricted-service-area-28		
		insufficient-resources-for-specific-slice-69		

Attribute	Type	Range/Length/Pattern	Default	Description
nssai-inclusion-mode	enumeration	disable mode-a mode-b mode-c mode-d	disable	NSSAI inclusion mode in Registration Accept
rat-restrictions	leaf-list of enumeration	nr eutran		3GPP Radio Access Technology restrictions applicable for an UE
core-network-restrictions	leaf-list of enumeration	epc 5gc		Core network type restrictions applicable for a UE

Slice Selection

Slice Selection parameters

XPath: *cna-amf-cc:/amf-cc/mobility-policy/class/access-control/slice-selection*

CLI Path: *amf-cc mobility-policy class access-control slice-selection*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
selection-policy	enumeration	standard use-nssf-policy	standard	Indicates the policy for slice selection
reject-cause	union	['string,([0-9]{1,2} or 1[0-9]{1,2} or 2[0-4][0-9] or 25[0-5])(-([a-zA-Z]+(-[a-zA-Z]+)))', 'Enumerations: 7-5gs-service-not-allowed 11-plmn-not-allowed 12-roaming-not-allowed-in-ta 15-no-suitable-cell-in-ta 27-N1-mode-not-allowed ']	15-no-suitable-cell-in-ta	Describes cause code for registration-reject. Default: 15-no-suitable-cell-in-ta.
ran-reroute-preferred	enumeration	enable disable	disable	Controls RAN Reroute. Default: disable
slice-negotiation-during-n26-handover	enumeration	enable disable	enable	Controls slice selection during N26 Handover. Default: enable

Fallback Snssai List

A list of S-NSSAI used if slice selection yields no results

Value 0xFFFFFFFF is reserved and cannot be used to configure slice-differentiator

XPath: *cna-amf-cc:/amf-cc/mobility-policy/class/access-control/slice-selection/fallback-snssai-list*

CLI Path: *amf-cc mobility-policy class access-control slice-selection fallback-snssai-list*

List Properties:

Keys: slice-service-type, slice-differentiator

Minimum number of elements: 0

Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
slice-service-type	union	['uint8,0 or 4 .. 255', 'Enumerations: eMBB URLLC MIoT ']		Key 1: 'slice-service-type' identifies the expected Network Slice behavior in terms of features and services
slice-differentiator	union	['string,(0x[0-9a-fA-F]{6})', 'Enumerations: not-specified ']		

Reroute

Reroute applicable for a UE

XPath: *cna-amf-cc:/amf-cc/mobility-policy/class/access-control/slice-selection/reroute*

CLI Path: *amf-cc mobility-policy class access-control slice-selection reroute*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
target-amf-set	string	[0-9]{3}-[0-9]{2,3}-[A-Fa-f0-9]{2}-[0-3][A-Fa-f0-9]{2}		Target AMF Set
candidate-amf-id-list	leaf-list of string	[0-9a-fA-F]{8}-[0-9a-fA-F]{4}-[0-9a-fA-F]{4}-[0-9a-fA-F]{4}-[0-9a-fA-F]{12}		List of candidate AMF IDs

Service Area Restrictions

Service Area restrictions applicable for a UE

XPath: *cna-amf-cc:/amf-cc/mobility-policy/class/access-control/service-area-restrictions*

CLI Path: *amf-cc mobility-policy class access-control service-area-restrictions*

Plmn

List of Service Area Restriction PLMN's

XPath: *cna-amf-cc:/amf-cc/mobility-policy/class/access-control/service-area-restrictions/plmn*

CLI Path: *amf-cc mobility-policy class access-control service-area-restrictions plmn*

List Properties:

Keys: plmn-id

Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
plmn-id	string	(([0-9]{5,6}))		Supported PLMN identity; Range: numbers with 5-6 digits

Restriction Type (choice)

XPath: *cna-amf-cc:/amf-cc/mobility-policy/class/access-control/service-area-restrictions/plmn*

CLI Path: *amf-cc mobility-policy class access-control service-area-restrictions plmn*

Case : allowed-areas

Allowed Areas

Choice: This allowed-areas attribute is mutually exclusive with attributes [['not-allowed-areas']]

XPath: *cna-amf-cc:/amf-cc/mobility-policy/class/access-control/service-area-restrictions/plmn/allowed-areas*

CLI Path: *amf-cc mobility-policy class access-control service-area-restrictions plmn allowed-areas*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
				List of allowed Tracking Area Codes
tacs	leaf-list of string	(0x[A-Fa-f0-9]{4}) or (0x[A-Fa-f0-9]{6})		Choice: This tacs attribute is mutually exclusive with attributes [['not-allowed-areas']] Maximum number of elements: 16

Case : not-allowed-areas

Not Allowed Areas

Choice: This not-allowed-areas attribute is mutually exclusive with attributes [['allowed-areas']]

XPath: *cna-amf-cc:/amf-cc/mobility-policy/class/access-control/service-area-restrictions/plmn/not-allowed-areas*

CLI Path: *amf-cc mobility-policy class access-control service-area-restrictions plmn not-allowed-areas*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
tacs	leaf-list of string	(0x[A-Fa-f0-9]{4}) or (0x[A-Fa-f0-9]{6})		List of Tracking Area Codes not allowed Choice: This tacs attribute is mutually exclusive with attributes [['allowed-areas']] Maximum number of elements: 16

Service Control

Service Control Policy

XPath: *cna-amf-cc:/amf-cc/mobility-policy/class/service-control*

CLI Path: *amf-cc mobility-policy class service-control*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
sms-support	enumeration	enable disable	disable	Controls Short Message Service (SMS) support. Default: disable

Ims

IP Multimedia Subsystem parameters

XPath: *cna-amf-cc:/amf-cc/mobility-policy/class/service-control/ims*

CLI Path: *amf-cc mobility-policy class service-control ims*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
ims-vops-support	leaf-list of enumeration	3gpp-5gc fallback-eps non-3gpp-5gc		List of IMS Voice over PS support options
emergency-support	leaf-list of enumeration	fallback-eps 3gpp-5gc		Defines how UE obtains emergency service

Purge Trigger

List of purge triggers

XPath: *cna-amf-cc:/amf-cc/mobility-policy/class/service-control/purge-trigger*

CLI Path: *amf-cc mobility-policy class service-control purge-trigger*

List Properties:

Keys: deregistration-type

Maximum number of elements: 3

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
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Attribute	Type	Range/Length/Pattern	Default	Description
deregistration-type	enumeration	explicit-deregistration implicit-deregistration all		Deregistration type The entry with 'deregistration-type' set to all is already present, no other entry is allowed to be configured The 'deregistration-type' with value all is not allowed when entries with other deregistration types are already configured The 'deregistration-type' with value all is not allowed when entries with other deregistration types are already configured

inactive-age	union	[Enumerations: immediate ']	immediate	Time to purge. Default: immediate; Unit: hours
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Lcs

LoCation Services(LCS) parameters

XPath: *cna-amf-cc:/amf-cc/mobility-policy/class/service-control/lcs*

CLI Path: *amf-cc mobility-policy class service-control lcs*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
lcs-support	enumeration	enable disable	disable	Controls location service support. Default: disable
nas-notification-timer-tlcsn	uint16	10..20	10	Defines the duration the AMF waits for a response to a Nas Notification invoke request sent to the UE; Default: 10; Unit: seconds

Network Induced Location Request

Network Induced Location Request (NI-LR) parameters

XPath: *cna-amf-cc:/amf-cc/mobility-policy/class/service-control/lcs/network-induced-location-request*

CLI Path: *amf-cc mobility-policy class service-control lcs network-induced-location-request*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
ni-lr-support	enumeration	disable enable-with-initial-positioning enable-without-initial-positioning	enable-with-initial-positioning	Controls Network Induced - Location Request procedure initiation. Default: enable-with-initial-positioning
horizontal-accuracy	string		50	Describes NI-LR measurement for horizontal accuracy. Default: 50; Unit: meters
vertical-accuracy	string		0	Describes NI-LR measurement for vertical accuracy. Default: 0; Unit: meters
emergency-target-sender	enumeration	source-node target-node	source-node	Specifies which node sends target node id to the GMLC. Default: source-node
response-time	enumeration	no-delay low-delay delay-tolerant	delay-tolerant	Controls initial NI-LR QoS response time. Default: delay-tolerant
qos-class	enumeration	best-effort assured	best-effort	Describes NI-LR QoS class type. Default: best-effort

Nas Policy

Non Access Stratum Policy

XPath: *cna-amf-cc:/amf-cc/mobility-policy/class/nas-policy*

CLI Path: *amf-cc mobility-policy class nas-policy*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
mm-profile-name	leafref			Mobility Management profile name per mobility class Leaf Reference Path: cna-amf-cc:/amf-cc/interface/nas/mm/profile/name
security-profile-name	leafref			Security profile name per mobility class Leaf Reference Path: cna-amf-cc:/amf-cc/interface/nas/security/profile/name
paging-profile-set-name	leafref			Name of the paging profile set for this mobility class Leaf Reference Path: cna-amf-cc:/amf-cc/interface/nas/paging/profile-set/name

Observability

XPath: [cna-amf-cc:/amf-cc/mobility-policy/class/observability](#)

CLI Path: [amf-cc mobility-policy class observability](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
edr-profile-name	leafref			edr profiles per mobility class Leaf Reference Path: cna-amf-cc:/amf-cc/edr/profile/name

Pcf Control

PCF Control attributes

XPath: [cna-amf-cc:/amf-cc/mobility-policy/class/pcf-control](#)

CLI Path: [amf-cc mobility-policy class pcf-control](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
reselect-pcf-on-amf-relocation	enumeration	enable disable	disable	Defines support for PCF re-selection on target AMF. Default: disable
forward-pcf-id-to-smf	enumeration	enable disable	disable	Defines PCF-Id to be forwarded to SMF or not. Default: disable

Ue Policy Control

UE Policy Control

XPath: [cna-amf-cc:/amf-cc/mobility-policy/class/pcf-control/ue-policy-control](#)

CLI Path: [amf-cc mobility-policy class pcf-control ue-policy-control](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
policy-association	enumeration	never always ue-requested	ue-requested	Defines support for UE Policy Association Establishment with PCF. Default: ue-requested

Am Policy Control

AM Policy Control

XPath: [cna-amf-cc:/amf-cc/mobility-policy/class/pcf-control/am-policy-control](#)

CLI Path: [amf-cc mobility-policy class pcf-control am-policy-control](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
policy-association	enumeration	never always	never	Defines support for AM Policy Association Establishment with PCF. Default: never

Session Policy

Session Policy

XPath: [cna-amf-cc:/amf-cc/session-policy](#)**CLI Path:** [amf-cc session-policy](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
analyzer-name	string	1..64		Name of the analyzer used for session policy classification.

Default

Session class with default values

XPath: [cna-amf-cc:/amf-cc/session-policy/default](#)**CLI Path:** [amf-cc session-policy default](#)

Access Control

Access Control Policy for sessions

XPath: [cna-amf-cc:/amf-cc/session-policy/default/access-control](#)**CLI Path:** [amf-cc session-policy default access-control](#)

S Nssai For Eps Interworking

This S-NSSAI is used in setting up the sessions on the SMF and provided to the UE during EPS to 5GS idle mode mobility or handover using the N26 interface.

SNSSAI is identified by two fields, 'slice-service-type' and 'slice-differentiator'.

Value FFFFFFFF is reserved and cannot be used to configure slice-differentiator

XPath: [cna-amf-cc:/amf-cc/session-policy/default/access-control/s-nssai-for-eps-interworking](#)**CLI Path:** [amf-cc session-policy default access-control s-nssai-for-eps-interworking](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
slice-service-type	union	['uint8,0 or 4 .. 255', 'Enumerations: eMBB URLLC MIoT ']	eMBB	Identifies the expected Network Slice behavior in terms of features and services; Default: eMBB
slice-differentiator	union	['string,(0x[0-9a-fA-F] {6})', 'Enumerations: not-specified ']	not-specified	Differentiates amongst multiple Network Slices of the same Slice Service type, and is a 3-octet string in hexadecimal format with a prefix of '0x', or 'not-specified' if the value is not specified. Value 0xFFFFFFFF is reserved and cannot be used to configure slice-differentiator. Default: 'not-specified'

Dnn Management Policy

Data Network Name (DNN) management policy

XPath: [cna-amf-cc:/amf-cc/session-policy/default/dnn-management-policy](#)**CLI Path:** [amf-cc session-policy default dnn-management-policy](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
dnn-usage-to-smf	enumeration	requested selected	selected	Controls whether requested or selected DNN will be sent in Create SM Context Request to SMF. If the UE did not request for a specific DNN, the selected DNN will be sent irrespective of the value of this attribute. Default: selected

Initial Dnn Ni Selection

Initial Data Network Name Network Identifier (DNN-NI) selection

XPath: [cna-amf-cc:/amf-cc/session-policy/default/dnn-management-policy/initial-dnn-ni-selection](#)

CLI Path: *amf-cc session-policy default dnn-management-policy initial-dnn-ni-selection*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
selection-method	enumeration	standard use-default-dnn use-subscribed-dnn use-configured-dnn	standard	Defines the Data Network Name Network Identifier (DNN-NI) selection method. Default: standard
subscribed-dnn-ni	string	(([a-zA-Z0-9])+((- or .)[A-Za-z0-9]+)*)		DNN Network Identifier to match the subscription
replacement-for-subscription-dnn-ni	string	(([a-zA-Z0-9])+((- or .)[A-Za-z0-9]+)*)		Replacement DNN-NI for the subscription DNN-NI
configured-dnn-ni	string	(([a-zA-Z0-9])+((- or .)[A-Za-z0-9]+)*)		Configured DNN Network Identifier

Dnn Info

DNN information to be used when 'configured-dnn-ni' is selected.

XPath: *cna-amf-cc:/amf-cc/session-policy/default/dnn-management-policy/initial-dnn-ni-selection/dnn-info*

CLI Path: *amf-cc session-policy default dnn-management-policy initial-dnn-ni-selection dnn-info*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
iwk-eps-ind	enumeration	enable disable	enable	EPS interworking indication for the selected DNN.

Fallback Dnn Ni Selection

Fallback Data Network Name Network Identifier (DNN-NI) selection

XPath: *cna-amf-cc:/amf-cc/session-policy/default/dnn-management-policy/fallback-dnn-ni-selection*

CLI Path: *amf-cc session-policy default dnn-management-policy fallback-dnn-ni-selection*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
selection-method	enumeration	standard use-default-dnn use-subscribed-dnn use-configured-dnn		Defines the Data Network Name Network Identifier (DNN-NI) selection method. Value cannot be same as initial-dnn-ni-selection The selection-method is same as intial-dnn-ni-selection
subscribed-dnn-ni	string	(([a-zA-Z0-9])+((- or .)[A-Za-z0-9]+)*)		DNN Network Identifier to match the subscription
replacement-for-subscription-dnn-ni	string	(([a-zA-Z0-9])+((- or .)[A-Za-z0-9]+)*)		Replacement DNN-NI for the subscription DNN-NI
configured-dnn-ni	string	(([a-zA-Z0-9])+((- or .)[A-Za-z0-9]+)*)		Configured DNN Network Identifier

Dnn Info

DNN information to be used when 'configured-dnn-ni' is selected.

XPath: *cna-amf-cc:/amf-cc/session-policy/default/dnn-management-policy/fallback-dnn-ni-selection/dnn-info*

CLI Path: *amf-cc session-policy default dnn-management-policy fallback-dnn-ni-selection dnn-info*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
iwk-eps-ind	enumeration	enable disable	enable	EPS interworking indication for the selected DNN.

Class

Session Class

XPath: *cna-amf-cc:/amf-cc/session-policy/class*

CLI Path: *amf-cc session-policy class*

List Properties:

Keys: name

Maximum number of elements: 1000

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	1..64		Name of the session class

Dnn Management Policy

Data Network Name (DNN) management policy

XPath: *cna-amf-cc:/amf-cc/session-policy/class/dnn-management-policy*

CLI Path: *amf-cc session-policy class dnn-management-policy*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
dnn-usage-to-smf	enumeration	requested selected	selected	Controls whether requested or selected DNN will be sent in Create SM Context Request to SMF. If the UE did not request for a specific DNN, the selected DNN will be sent irrespective of the value of this attribute. Default: selected

Initial Dnn Ni Selection

Initial Data Network Name Network Identifier (DNN-NI) selection

XPath: *cna-amf-cc:/amf-cc/session-policy/class/dnn-management-policy/initial-dnn-ni-selection*

CLI Path: *amf-cc session-policy class dnn-management-policy initial-dnn-ni-selection*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
selection-method	enumeration	standard use-default-dnn use-subscribed-dnn use-configured-dnn	standard	Defines the Data Network Name Network Identifier (DNN-NI) selection method. Default: standard
subscribed-dnn-ni	string	(([a-zA-Z0-9])+((- or .)[A-Za-z0-9]+)*)		DNN Network Identifier to match the subscription
replacement-for-subscription-dnn-ni	string	(([a-zA-Z0-9])+((- or .)[A-Za-z0-9]+)*)		Replacement DNN-NI for the subscription DNN-NI
configured-dnn-ni	string	(([a-zA-Z0-9])+((- or .)[A-Za-z0-9]+)*)		Configured DNN Network Identifier

Dnn Info

DNN information to be used when 'configured-dnn-ni' is selected.

XPath: *cna-amf-cc:/amf-cc/session-policy/class/dnn-management-policy/initial-dnn-ni-selection/dnn-info*

CLI Path: *amf-cc session-policy class dnn-management-policy initial-dnn-ni-selection dnn-info*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
iwk-eps-ind	enumeration	enable disable	enable	EPS interworking indication for the selected DNN.

Fallback Dnn Ni Selection

Fallback Data Network Name Network Identifier (DNN-NI) selection

XPath: *cna-amf-cc:/amf-cc/session-policy/class/dnn-management-policy/fallback-dnn-ni-selection*

CLI Path: *amf-cc session-policy class dnn-management-policy fallback-dnn-ni-selection*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
selection-method	enumeration	standard use-default-dnn use-subscribed-dnn use-configured-dnn		Defines the Data Network Name Network Identifier (DNN-NI) selection method. Value cannot be same as initial-dnn-ni-selection The selection-method is same as intial-dnn-ni-selection
subscribed-dnn-ni	string	(([a-zA-Z0-9])+((- or .) [A-Za-z0-9]+)*)		DNN Network Identifier to match the subscription
replacement-for-subscription-dnn-ni	string	(([a-zA-Z0-9])+((- or .) [A-Za-z0-9]+)*)		Replacement DNN-NI for the subscription DNN-NI
configured-dnn-ni	string	(([a-zA-Z0-9])+((- or .) [A-Za-z0-9]+)*)		Configured DNN Network Identifier

Dnn Info

DNN information to be used when 'configured-dnn-ni' is selected.

XPath: *cna-amf-cc:/amf-cc/session-policy/class/dnn-management-policy/fallback-dnn-ni-selection/dnn-info*

CLI Path: *amf-cc session-policy class dnn-management-policy fallback-dnn-ni-selection dnn-info*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
iwk-eps-ind	enumeration	enable disable	enable	EPS interworking indication for the selected DNN.

Plmn

Subscriber Public Land Mobile Network (PLMN) parameters

XPath: *cna-amf-cc:/amf-cc/plmn*

CLI Path: *amf-cc plmn*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
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Attribute	Type	Range/Length/Pattern	Default	Description
mcc-using-3-digit-mnc	leaf-list of uint16	1	100..999	302
				310
				311
				312
				313
				314
				315
				316
				330
				334
				338
				342
				344
				346
				348
				352
				354
				356
				360
				365
				366
				376
				708
				722
				732
				750

Default

Public Land Mobile Network (PLMN) profile with default values

XPath: *cna-amf-cc:/amf-cc/plmn/default*

CLI Path: *amf-cc plmn default*

Subscriber

Subscriber specific parameters

XPath: *cna-amf-cc:/amf-cc/plmn/default/subscriber*

CLI Path: *amf-cc plmn default subscriber*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
suppress-udm-update	enumeration	enable disable	disable	Controls if the AMF should suppress sending Nudm_UECM_Update service operation to UDM during EBI allocation or revocation procedure. Default: disable

Network

Network related parameters

XPath: *cna-amf-cc:/amf-cc/plmn/default/network*

CLI Path: *amf-cc plmn default network*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
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Attribute	Type	Range/Length/Pattern	Default	Description
peer-oauth2-required	leaf-list of enumeration	all ausf smf udm amf nssf pcf smsf lmf gmlc		List of Network Function (NF) Type of NF sevice producer peer that requires OAuth 2.0 based authorization of NF service access. Valid values are: 'all' or any combination of 'ausf', 'smf', 'udm', 'amf', 'nssf', 'pcf', 'smsf', 'lmf', 'gmlc'. When no value is configured, it indicates that all NF service producer peers do not require the OAuth 2.0 based authorization of NF service access. peer-oauth2-required can be either all or combination valid values but not both. Ordered by: user
include-dnn-operator-id	enumeration	enable disable	disable	Determines if the DNN operator-id needs to be included in the SMF discovery. Default is disable

Plmn Profile

Public Land Mobile Network (PLMN) profile

XPath: *cna-amf-cc:/amf-cc/plmn/plmn-profile*

CLI Path: *amf-cc plmn plmn-profile*

List Properties:

Keys: plmn-id

Maximum number of elements: 1000

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
plmn-id	string	((([0-9]{5,6}))		Unique PLMN identifier. Range: numbers with 5,6 digits

Subscriber

Subscriber specific parameters

XPath: *cna-amf-cc:/amf-cc/plmn/plmn-profile/subscriber*

CLI Path: *amf-cc plmn plmn-profile subscriber*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
suppress-udm-update	enumeration	enable disable	disable	Controls if the AMF should suppress sending Nudm_UECM_Update service operation to UDM during EBI allocation or revocation procedure. Default: disable

Network

Network related parameters

XPath: *cna-amf-cc:/amf-cc/plmn/plmn-profile/network*

CLI Path: *amf-cc plmn plmn-profile network*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
-----------	------	----------------------	---------	-------------

Attribute	Type	Range/Length/Pattern	Default	Description
peer-oauth2-required	leaf-list of enumeration	all ausf smf udm amf nssf pcf smsf lmf gmlc		List of Network Function (NF) Type of NF service producer peer that requires OAuth 2.0 based authorization of NF service access. Valid values are: 'all' or any combination of 'ausf', 'smf', 'udm', 'amf', 'nssf', 'pcf', 'smsf', 'lmf', 'gmlc'. When no value is configured, it indicates that all NF service producer peers do not require the OAuth 2.0 based authorization of NF service access. peer-oauth2-required can be either all or combination valid values but not both. Ordered by: user
include-dnn-operator-id	enumeration	enable disable	disable	Determines if the DNN operator-id needs to be included in the SMF discovery. Default is disable

Trace

Trace objects

XPath: [cna-amf-cc:/amf-cc/trace](#)

CLI Path: [amf-cc trace](#)

Session

Per UE tracing

XPath: [cna-amf-cc:/amf-cc/trace/session](#)

CLI Path: [amf-cc trace session](#)

List Properties:

Keys: target-type, target-value

Maximum number of elements: 100

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
target-type	enumeration	supi		Type of target for tracing
target-value	string	(imsi-[0-9]{5,15})		Value of target; For target type 'supi', format is 'imsi-(imsi digits)'. For target type 'supi', format is 'imsi-(imsi digits)'. Example:imsi-123456789123456
session-id	string	[0-9]{5,6}[.][0-9a-fA-F]{6}		ID to uniquely identify the trace session. Format: PLMN.trace_id, where PLMN is input as 5 or 6 digit BCD number and trace-id is input as a 3 byte hexadecimal number. Example: 310070.70F13A

Gnb Trace

gNodeB tracing of UE

Either 'trace-collection-entity-ip' or 'trace-collection-entity-uri' must be configured.

XPath: [cna-amf-cc:/amf-cc/trace/session/gnb-trace](#)

CLI Path: [amf-cc trace session gnb-trace](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
trace-depth	enumeration	minimum medium maximum minimum-no-vs medium-no-vs maximum-no-vs	minimum	Trace information level to be collected by the gNodeB. Default: med

Attribute	Type	Range/Length/Pattern	Default	Description
interfaces	leaf-list of enumeration	ng-c		Trace target interfaces for the gNodeB. Valid values are: 'all' or any order or combination of 'uu', 'ng-c', 'xn-c', 'f1-c', 'e1' interfaces can be either all or combination valid values but not both. Minimum number of elements: 1
		xn-c		
		uu		
		f1-c		
		e1		
		all		
trace-collection-entity-ip	inet:ip-address			Destination IP address for trace data transmission. This is ignored if 'trace-collection-entity-uri' is configured.
trace-collection-entity-uri	string			Destination URI for trace data transmission by gNodeB. If this is configured, the 'trace-collection-entity-ip' is ignored.

AMF MONITOR COMMON Configuration

This module defines the common configuration for AMF and MME monitor.

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AMF MONITOR Configuration

This module defines the configuration for AMF Monitor.

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- [amf-monitor](#)
 - [deregistration-time](#)
 - [critical-service-readiness-probe](#)
 - [critical-service-frequent-restart](#)
 - [load-profile](#)

Amf Monitor

AMF Monitor interface object

XPath: [cna-amf-monitor:/amf-monitor](#)

CLI Path: [amf-monitor](#)

Deregistration Time

Duration(in Secs) for which Monitor waits for a Critical Service to become available

XPath: [cna-amf-monitor:/amf-monitor/deregistration-time](#)

CLI Path: [amf-monitor deregistration-time](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
value	uint16	1..3600	300	Duration(in Secs) for which Monitor waits for a Critical Service to become available

Critical Service Readiness Probe

Configure Critical Service Readiness monitoring feature

critical-failure-count should be always greater than threshold-failure-count

XPath: [cna-amf-monitor:/amf-monitor/critical-service-readiness-probe](#)

CLI Path: [amf-monitor critical-service-readiness-probe](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
monitoring	enumeration	disabled enabled	disabled	Feature enabled or disabled

Attribute	Type	Range/Length/Pattern	Default	Description
check-interval	uint16	60..36000	300	The duration in seconds after which the CriticalService status should be rechecked.
threshold-failure-count	uint16	1..20000	5	Number of times a CriticalService is found NotReady after which Alarm/Alert is generated
critical-failure-count	uint16	1..20000	10	Number of times a CriticalService is found NotReady after which VNFRestart is triggered

Critical Service Frequent Restart

Configure Critical Service frequent Restart monitoring feature
status-check-interval should be always greater than stable-interval
critical-flip-count should be always greater than threshold-flip-count

XPath: [cna-amf-monitor:/amf-monitor/critical-service-frequent-restart](#)

CLI Path: [amf-monitor critical-service-frequent-restart](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
monitoring	enumeration	disabled enabled	disabled	Feature enabled or disabled
stable-interval	uint16	1..3600	120	The minimum runtime in seconds for a critical service to be considered as stable
status-check-interval	uint16	1..3600	180	The duration in seconds after which the critical service status should be rechecked.
threshold-flip-count	uint16	1..1000	3	Number of times a critical service is found unstable after which alarm/alert is generated
critical-flip-count	uint16	1..1000	20	Number of times a critical service is found unstable after which vnf restart is triggered

Load Profile

Configure AMF Load Profile

XPath: [cna-amf-monitor:/amf-monitor/load-profile](#)

CLI Path: [amf-monitor load-profile](#)

List Properties:

Keys: load-profile

Maximum number of elements: 1

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
load-profile	string			AMF Load Profile Name
reporting-frequency-interval	uint64		120	Frequency interval(seconds) at which the AMF load should be reported to NRF. Default 120s
reporting-frequency-change-factor	uint64	1..100	20	Load change factor(+/- load difference) which should result in reporting the AMF load to the NRF. Range 1..100. Default 20

AMF N2 Configuration

The module for entities implementing cna-amf-n2.

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- [amf-n2](#)
 - [trace](#)
 - [targets](#)
 - [global-ran-node](#)
 - [global-node-id](#)

- [n2](#)
 - [endpoint](#)
 - [sctp-profile](#)

Amf N2

Top level container for AMF N2 Configuration.

XPath: [cna-amf-n2:/amf-n2](#)

CLI Path: [amf-n2](#)

Trace

Trace objects

XPath: [cna-amf-n2:/amf-n2/trace](#)

CLI Path: [amf-n2 trace](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
default-probability	string		0.01	Probability of sampling the trace; valid values are 0.0 to 1.0 The 'default-probability' value must be less than 1.0

Targets

Trace targets

XPath: [cna-amf-n2:/amf-n2/trace/targets](#)

CLI Path: [amf-n2 trace targets](#)

Global Ran Node

Global RAN Node

One of ipv4 addresses or ipv6 addresses or global node Id must be set

XPath: [cna-amf-n2:/amf-n2/trace/targets/global-ran-node](#)

CLI Path: [amf-n2 trace targets global-ran-node](#)

List Properties:

Keys: name

Maximum number of elements: 512

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	1..254		Unique Global RAN Node name ; length: 1..254
ipv4-address-primary	inet:ipv4-address			Primary IPv4 address of the node
ipv4-address-secondary	inet:ipv4-address			Secondary IPv4 address of the node
ipv6-address-primary	inet:ipv6-address			Primary IPv6 address of the node
ipv6-address-secondary	inet:ipv6-address			Secondary IPv6 address of the node
probability	string		0.1	Probability of sampling the trace for the target; valid values are 0.0 to 1.0 The 'probability' value must be less than 1.0

Global Node Id

Identifier of the node

The 'global-node-id type' must be configured

The 'gnb-id' must be configured

XPath: *cna-amf-n2:/amf-n2/trace/targets/global-ran-node/global-node-id*

CLI Path: *amf-n2 trace targets global-ran-node global-node-id*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
plmn	string	([0-9]{5,6})		A unique PLMN identifier; valid values are numbers with 5..6 digits
type	enumeration	gnb		Global RAN Node type
gnb-id	uint32			A unique gNB identifier

N2

N2 interface to 5G access (5G AN)

XPath: *cna-amf-n2:/amf-n2/n2*

CLI Path: *amf-n2 n2*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
primary-endpoint-name	leafref			Primary SCTP endpoint on AMF for 5GAN to use on initial connection, it is mandatory. Subsequent connections may optionally use non-primary endpoints as dynamically directed by AMF. Leaf Reference Path: <i>cna-amf-n2:/amf-n2/n2/endpoint/name</i>
port	uint32	1..65535	38412	SCTP Port number; Default: 38412

Endpoint

SCTP endpoint for AMF side of N2

Either 'ipv4-fqdn' or 'ipv6-fqdn' must be provisioned

XPath: *cna-amf-n2:/amf-n2/n2/endpoint*

CLI Path: *amf-n2 n2 endpoint*

List Properties:

Keys: name

Minimum number of elements: 1

Maximum number of elements: 4

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	1..254		unique Endpoint name ; length: 1..254
weight	uint16	0..255	100	Relative weight of the endpoint on a scale of 0 to 255. 0 would indicate RAN to not use the TNLA represented by the endpoint. Default: 100
ipv4-fqdn	inet:domain-name			The IPv4 FQDN of the associated N2 endpoint. If provided it is published in the N2 Profile.
ipv6-fqdn	inet:domain-name			The IPv6 FQDN of the associated N2 endpoint. If provided it is published in the N2 Profile.
multi-home-ipv4-fqdn	inet:domain-name			The Multi-Home IPv4 FQDN of the associated N2 endpoint. If provided it is published in the N2 Profile.
multi-home-ipv6-fqdn	inet:domain-name			The Multi-Home IPv6 FQDN of the associated N2 endpoint. If provided it is published in the N2 Profile.
dscp-marking	string	([0-9] or [1-5][0-9] or [6][0-3]) or de or af[1-4][1-3] or cs[1-7] or ef	de	DSCP marking for N2. Valid values are 0..63, de, af11, af12, af13, af21, af22, af23, af31, af32, af33, af41, af42, af43, cs1, cs2, cs3, cs4, cs5, cs6, cs7, ef; Default: de
usage	enumeration	both	both	Indicates UE usage of this endpoint; Default: both

Sctp Profile

SCTP parameters for N2

XPath: [cna-amf-n2:/amf-n2/n2/endpoint/sctp-profile](#)

CLI Path: [amf-n2 n2 endpoint sctp-profile](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
rto-initial	uint32	1..5000	500	Retransmission Time Out (RTO) initial value. Default: 500; Unit: milliseconds The 'rto-initial' value should be in the range from 'rto-min' to 'rto-max'
rto-min	uint32	1..5000	250	Retransmission Time Out (RTO) minimum value. Default: 250; Unit: milliseconds
rto-max	uint32	1..60000	2000	Retransmission Time Out (RTO) maximum value. Default: 2000; Unit: milliseconds
path-max-retrans	uint16	1..20	5	Maximum number of consecutive retransmissions over a destination transport address of a peer endpoint before it is marked as inactive. Default: 5
association-init-max-transmit	uint16	1..10	3	Maximum number of times an INIT chunk or a COOKIE ECHO chunk is retransmitted before an endpoint aborts the initialization process and closes the association. Default: 3
association-max-retrans	uint16	1..20	6	Maximum number of consecutive retransmissions to a peer before an endpoint considers that the peer is unreachable and closes the association. Default: 6
no-delay	boolean		true	Specifies no delay. Default: true
max-out-stream	uint16	1..64	11	Specifies the maximum out stream. Default: 11
max-in-stream	uint16	1..64	11	Specifies the maximum in stream. Default: 11
hb-interval	uint16	1..120	30	Specifies the heartbeat interval. Default: 30; Unit: seconds
rx-buffer-size	uint32	25000..2097152	149000	Specifies the receive buffer size. Default: 149000; Unit: bytes
tx-buffer-size	uint32	16000..5509000	149000	Specifies the transmit buffer size. Default: 149000; Unit: bytes
sack-delay	uint16	1	10..500	200
sack-frequency	uint16	1..5	2	Specifies the SCTP SACK frequency value. Disable: 1; Default: 2; Unit: packets

CDRAGENT Configuration

The module for entities implementing cna-cdragent.

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- [cdr-agent](#)
 - [cdr-profile](#)
 - [file-rotation-profile](#)
 - [disk-management-profile](#)

Cdr Agent

Cdr-Agent Interface Object

XPath: [cna-cdragent:/cdr-agent](#)

CLI Path: [cdr-agent](#)

Cdr Profile

Configure CDR Agent CDR Profile

XPath: [cna-cdragent:/cdr-agent/cdr-profile](#)

CLI Path: [cdr-agent cdr-profile](#)

List Properties:

Keys: cdr-profile

Maximum number of elements: 512

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
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Attribute	Type	Range/Length/Pattern	Default	Description
cdr-profile	string			Local CDR charging profile Name
asn1-file-rotation-profile-name	string			Associated Local CDR Agent File Rotation Profile Name

File Rotation Profile

Configure CDR Agent File Rotation Profile

XPath: *cna-cdragent:/cdr-agent/file-rotation-profile*

CLI Path: *cdr-agent file-rotation-profile*

List Properties:

Keys: file-rotation-profile
Maximum number of elements: 512

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
file-rotation-profile	string			CDR Agent Common File Rotation Profile Name
file-format	enumeration	asn1		What type of file is this? asn1, csv ...
file-rotate-empty	enumeration	disabled enabled	disabled	Should files be rotated when empty?
file-rotate-frequency	enumeration	1min 5mins 15mins 30mins 1hour 2hours 3hours 6hour 12hours 1day 7days 14days 21days 30days 1year	1hour	Files should be rotated once every X time
file-rotate-num-records	uint32			Files should be rotated once the file has these many records.
file-rotate-size-bytes	uint32	1..1073741824	10485760	Files should be rotated once they reach size [bytes], max size 1GB
private-information	string			Optional private information to be appended to file-name.
file-extension	string			Optional file extension to be appended to file-name.
sequence-field-width	uint32	0..10	0	Defines the number of leading zeros to prepend to the filename sequence number

Disk Management Profile

Configure CDR Agent Disk Management Profile

XPath: *cna-cdragent:/cdr-agent/disk-management-profile*

CLI Path: *cdr-agent disk-management-profile*

List Properties:

Keys: disk-management-profile
Minimum number of elements: 1
Maximum number of elements: 1

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
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Attribute	Type	Range/Length/Pattern	Default	Description
disk-management-profile	string			Name of the disk management profile
disk-quota-low-threshold	uint32	1..100	70	Low threshold for disk quota in percentage, the usage falling below which will clear the critical alarm
disk-quota-high-threshold	uint32	1..100	90	High threshold for disk quota in percentage, the usage exceeding which will raise the critical alarm

Table of Contents

- [config-schema](#)
 - [config-schema-list](#)
 - [schema](#)
 - [banner](#)

Cfgmgr

Cfgmgr

XPath: [cna-cfgmgr:/config-schema](#)

CLI Path: [cfgmgr](#)

Schema

List of config schemas

XPath: [cna-cfgmgr:/config-schema/config-schema-list](#)

CLI Path: [cfgmgr schema](#)

List Properties:

Keys: service-type

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
service-type	string			
version	string			
instances	uint32			
time-first-installed	string	\d{4}-\d{2}-\d{2}T\d{2}:\d{2}:\d{2}(\.\d+)?(Z or [-]\d{2}:\d{2})		
time-last-removed	string	\d{4}-\d{2}-\d{2}T\d{2}:\d{2}:\d{2}(\.\d+)?(Z or [-]\d{2}:\d{2})		
status	string			Status of config schema for this service. Possible states include pending, loaded, and failed.

Schema

XPath: [cna-cfgmgr:/config-schema/schema](#)

CLI Path: [cfgmgr schema](#)

Banner

XPath: [cna-cfgmgr:/config-schema/banner](#)

CLI Path: [cfgmgr banner](#)

UPF COMMIT CHECK Configuration

This module defines a config 'false' container for commit-check, used to verify configuration changes before committing.

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- [commit-check](#)

Commit Check

A container used for pre-commit validation of configurations.

XPath: [cna-upf-commit-check:/commit-check](#)

CLI Path: [commit-check](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
message	string			A message giving more details about the result of the commit-check.

CHFAGENT Configuration

The module for entities implementing cna-chfagent.

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- [chf-agent](#)
 - [local-cdr-profile](#)
 - [event-trigger-template](#)
 - [chf-server-profile](#)
 - [chf-replay-profile](#)
 - [chf-profile](#)
 - [charging-profile](#)
 - [profile-analyzer](#)
 - [profile-key](#)
 - [imsi-range](#)
 - [imei-range](#)
 - [msisdn-range](#)
 - [chf-server-reachability](#)
 - [nchf-interface](#)

Chf Agent

Chf-Agent interface object

XPath: [cna-chfagent:/chf-agent](#)

CLI Path: [chf-agent](#)

Local Cdr Profile

Configure CHF Agent Local CDR Charging Profile

XPath: [cna-chfagent:/chf-agent/local-cdr-profile](#)

CLI Path: [chf-agent local-cdr-profile](#)

List Properties:

Keys: local-cdr-profile

Maximum number of elements: 512

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
local-cdr-profile	string			Local CDR charging profile Name
service-context-id	string	1..64	32255@3gpp.org	Service-Context-Id content (max length 64)
cdr-agent-profile-name	string			Associated Local CDR Agent Microservice Profile Name

Event Trigger Template

Configure Interface Interim Event Trigger Template for CHF

XPath: [cna-chfagent:/chf-agent/event-trigger-template](#)

CLI Path: *chf-agent event-trigger-template***List Properties:**

Keys: event-trigger-template

Maximum number of elements: 512

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
event-trigger-template	string			event trigger template name string
time-limit-close-trigger	uint32	1..20	1	Time Limit Close Trigger. If interim-updates enabled on the interface - x=[1-20]. Applies to offline Interfaces only. Close CDR when event occurs n=x times; container close if n < x.
qos-change-trigger	uint32	0..20	20	Enable/Disable QoS Change Close Trigger. x=[0]:Disabled. x=[1-20]:Enabled. Offline Interfaces: Close CDR when event repeats n=x times. Container close if n < x.
user-location-change-trigger	uint32	0..20	20	Enable/Disable User Location Change Close Trigger. x=[0]:Disabled. x=[1-20]:Enabled. Offline Interfaces: Close CDR when event repeats n=x times. Container close if n < x.
amf-change-trigger	uint32	0..20	20	Enable/Disable AMF Change Close Trigger. x=[0]:Disabled. x=[1-20]:Enabled. Offline Interfaces: Close CDR when event repeats n=x times. Container close if n < x.
change-of-ue-presence-in-pra-trigger	uint32	0..20	20	Enable/Disable Change Of UE Presence in Area Close Trigger. x=[0]:Disabled. x=[1-20]:Enabled. Offline Interfaces: Close CDR when event repeats n=x times. Container close if n < x.
change-of-3gpp-ps-data-off-status-trigger	uint32	0..20	20	Enable/Disable Change Of 3GPP PS Data Off Status Close Trigger. x=[0]:Disabled. x=[1-20]:Enabled. Offline Interfaces: Close CDR when event repeats n=x times. Container close if n < x.
expiry-of-data-time-limit-per-rating-group-trigger	uint32	0..20	20	Enable/Disable Expiry Of Data Time Limit Per Rating Group Trigger. x=[0]:Disabled. x=[1-20]:Enabled. Offline Interfaces: Close CDR when event repeats n=x times. Container close if n < x.
expiry-of-data-volume-limit-per-rating-group-trigger	uint32	0..20	20	Enable/Disable Expiry Of Volume Time Limit Per Rating Group Trigger. x=[0]:Disabled. x=[1-20]:Enabled. Offline Interfaces: Close CDR when event repeats n=x times. Container close if n < x.
expiry-of-data-event-limit-per-rating-group-trigger	uint32	0..20	20	Enable/Disable Expiry Of Data Limit Per Rating Group Trigger. x=[0]:Disabled. x=[1-20]:Enabled. Offline Interfaces: Close CDR when event repeats n=x times. Container close if n < x.
time-threshold-reached-trigger	uint32	0..20	20	Enable/Disable Time Threshold Reached Trigger. x=[0]:Disabled. x=[1-20]:Enabled. Offline Interfaces: Close CDR when event repeats n=x times. Container close if n < x.
volume-threshold-reached-trigger	uint32	0..20	20	Enable/Disable Volume Threshold Reached Trigger. x=[0]:Disabled. x=[1-20]:Enabled. Offline Interfaces: Close CDR when event repeats n=x times. Container close if n < x.
unit-threshold-reached-trigger	uint32	0..20	20	Enable/Disable Unit Threshold Reached Trigger. x=[0]:Disabled. x=[1-20]:Enabled. Offline Interfaces: Close CDR when event repeats n=x times. Container close if n < x.
time-quota-exhausted-trigger	uint32	0..20	20	Enable/Disable Time Quota Exhausted Trigger. x=[0]:Disabled. x=[1-20]:Enabled. Offline Interfaces: Close CDR when event repeats n=x times. Container close if n < x.
volume-quota-exhausted-trigger	uint32	0..20	20	Enable/Disable Volume Quota Exhausted Trigger. x=[0]:Disabled. x=[1-20]:Enabled. Offline Interfaces: Close CDR when event repeats n=x times. Container close if n < x.
unit-quota-exhausted-trigger	uint32	0..20	20	Enable/Disable Unit Quota Exhausted Trigger. x=[0]:Disabled. x=[1-20]:Enabled. Offline Interfaces: Close CDR when event repeats n=x times. Container close if n < x.
expiry-of-quota-validity-time-trigger	uint32	0..20	20	Enable/Disable Expiry Of Quota Validity Time Trigger. x=[0]:Disabled. x=[1-20]:Enabled. Offline Interfaces: Close CDR when event repeats n=x times. Container close if n < x.

Attribute	Type	Range/Length/Pattern	Default	Description
ue-time-zone-change-trigger	uint32	1..20	1	UE Time Zone Change Close Trigger. If interim-updates enabled on the interface - x=[1-20]. Applies to offline Interfaces only. Close CDR when event occurs n=x times; container close if n < x.
plmn-change-trigger	uint32	1..20	1	PLMN Change Trigger. If interim-updates enabled on the interface - x=[1-20]. Applies to offline Interfaces only. Close CDR when event occurs n=x times; container close if n < x.
rat-type-change-trigger	uint32	1..20	1	RAT Type Change Trigger. If interim-updates enabled on the interface - x=[1-20]. Applies to offline Interfaces only. Close CDR when event occurs n=x times; container close if n < x.
dnn-ambr-change-trigger	uint32	1..20	1	DNN AMBR Change Trigger. If interim-updates enabled on the interface - x=[1-20]. Applies to offline Interfaces only. Close CDR when event occurs n=x times; container close if n < x.
removal-of-upf-trigger	uint32	1..20	1	Removal Of UPF Trigger. If interim-updates enabled on the interface - x=[1-20]. Applies to offline Interfaces only. Close CDR when event occurs n=x times; container close if n < x.
management-intervention-trigger	uint32	1..20	1	Management Intervention Trigger. If interim-updates enabled on the interface - x=[1-20]. Applies to offline Interfaces only. Close CDR when event occurs n=x times; container close if n < x.
expiry-of-data-time-limit-per-pdu-session-trigger	uint32	1..20	1	Expiry Of Data Time Limit Per Pdu Session Trigger. If interim-updates enabled on the interface - x=[1-20]. Applies to offline Interfaces only. Close CDR when event occurs n=x times; container close if n < x.
expiry-of-data-volume-limit-per-pdu-session-trigger	uint32	1..20	1	Expiry Of Data Volume Limit Per Pdu Session Trigger. If interim-updates enabled on the interface - x=[1-20]. Applies to offline Interfaces only. Close CDR when event occurs n=x times; container close if n < x.
expiry-of-data-event-limit-per-pdu-session-trigger	uint32	1..20	1	Expiry Of Data Event Limit Per Pdu Session Trigger. If interim-updates enabled on the interface - x=[1-20]. Applies to offline Interfaces only. Close CDR when event occurs n=x times; container close if n < x.

Chf Server Profile

Configure the CHF server profile

XPath: *cna-chfagent:/chf-agent/chf-server-profile*

CLI Path: *chf-agent chf-server-profile*

List Properties:

Keys: chf-server-profile

Maximum number of elements: 512

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
chf-server-profile	string			CHF server profile name
admin-state	enumeration	DOWN UP	UP	Admin state of CHF server
fqdn	string			FQDN of CHF server URI as per 3GPP specifications
ipv4-address	string			IPv4 address of CHF server
port	uint16		80	Port number
schema	enumeration	HTTP2 HTTPS	HTTP2	Port schema

Chf Replay Profile

Configure the CHF replay profile. This profile defines the parameters used for the replay functionality. When this replay profile is configured under a CHF profile, the replay functionality will be enabled for that CHF profile with the parameters configured here.

XPath: *cna-chfagent:/chf-agent/chf-replay-profile*

CLI Path: *chf-agent chf-replay-profile*

List Properties:

Keys: chf-replay-profile

Maximum number of elements: 512

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
chf-replay-profile	string			CHF replay profile name
partition-count	uint32			Number of partitions for the topic. Refer to official documentation of Kafka Message Broker for more information on this parameter.
replication-factor	uint32		1	Replication factor defines the number of copies of a given replay message in a Kafka cluster. Refer to official documentation of Kafka Message Broker for more information on this parameter.
retention-period	uint64		604800000	Defines the time period (milliseconds) for which a message is stored on a topic before it is discarded to free up space. Refer to official documentation of Kafka Message Broker for more information on this parameter.
retention-size	int64		-1	Maximum size in bytes a topic partition can grow before discarding old messages. Refer to official documentation of Kafka Message Broker for more information on this parameter.
replay-window-size	uint32			Maximum number of outstanding messages which will be replayed to the CHF server. Configuring this parameter to 0 will disable the replaying of the messages to the CHF server.

Chf Profile

Configure CHF profile

XPath: [cna-chfagent:/chf-agent/chf-profile](#)**CLI Path:** [chf-agent chf-profile](#)**List Properties:**

Keys: chf-profile

Maximum number of elements: 512

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
chf-profile	string			CHF Profile Name
primary-chf-server-profile	leafref			Associated primary CHF server profile Leaf Reference Path: cna-chfagent:/chf-agent/chf-server-profile/chf-server-profile
secondary-chf-server-profile	leafref			Associated secondary CHF server profile primary-chf-server-profile must be configured before secondary-chf-server-profile Leaf Reference Path: cna-chfagent:/chf-agent/chf-server-profile/chf-server-profile
http-request-timeout	uint16	1..20	5	HTTP request timeout interval in seconds
replay-profile	leafref			Associated CHF replay profile. This configuration enables the replay of Nchf interface usage to CHF servers configured under this CHF profile are not reachable. The replay functionality is enabled through Kafka Message Broker in which a topic with the CHF profile name will be created. Refer to the configured replay profile for more information about replay parameters. Leaf Reference Path: cna-chfagent:/chf-agent/chf-replay-profile/chf-replay-profile

Charging Profile

Configure CHF Agent charging common profile

XPath: [cna-chfagent:/chf-agent/charging-profile](#)**CLI Path:** [chf-agent charging-profile](#)**List Properties:**

Keys: charging-profile

Maximum number of elements: 512

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
charging-profile	string			CHF Agent Common Charging profile Name
partial-record-method	enumeration	individual default	default	Partial Charging Method
cdr-output-type	enumeration	chf-record pgw-record	chf- record	CDR Output Type
associated-chf-profile	leafref			Associated CHF Profile Leaf Reference Path: cna-chfagent:/chf-agent/chf-profile/chf-profile
associated-local-cdr-profile	leafref			Associated Local CDR Profile Leaf Reference Path: cna-chfagent:/chf-agent/local-cdr-profile/local-cdr-profile
event-trigger-template	leafref			Associated Event Trigger Template Leaf Reference Path: cna-chfagent:/chf-agent/event-trigger-template/event-trigger-template

Profile Analyzer

Configure charging profile analyzer

XPath: [cna-chfagent:/chf-agent/profile-analyzer](#)

CLI Path: [chf-agent profile-analyzer](#)

List Properties:

Keys: profile-analyzer

Maximum number of elements: 4096

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
profile-analyzer	string			Profile Analyzer Name
charging-profile	leafref			Charging Profile Name Leaf Reference Path: cna-chfagent:/chf-agent/charging-profile/charging-profile
priority	uint16	1..4096		Priority of the charging profile has to be unique across the Profile-Analyzers.

Profile Key

Charging Profile Key. Method of procedure for modifying the Profile-Key: Remove the whole profile-key 'no profile-key <>'; commit; configure the profile-key with the updated configuration 'profile-key <>'; commit

XPath: [cna-chfagent:/chf-agent/profile-analyzer/profile-key](#)

CLI Path: [chf-agent profile-analyzer profile-key](#)

List Properties:

Keys: profile-key

Minimum number of elements: 1

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
profile-key	string			Profile Key Name
dnn-id	string			DNN ID

Attribute	Type	Range/Length/Pattern	Default	Description
rat-type	enumeration	nr utran geran wlan gan hspaevolution eutra virtual nbiot-eutran		Rat-Type
roaming-user	boolean			Roaming User or not
charging-characteristics	uint32	0 .. 65535		Charging Characteristics
nf-plmn-id	uint64			NF PLMN ID
nf-functionality	string			NF Functionality
supi-plmn-id	uint64			SUPI PLMN ID
pdu-type	enumeration	ipv4v6 ipv4 ipv6 unstr ether		PDU Type

Imsi Range

IMSI Range

XPath: *cna-chfagent:/chf-agent/profile-analyzer/profile-key/imsi-range*

CLI Path: *chf-agent profile-analyzer profile-key imsi-range*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
low	uint64			IMSI lower value
high	uint64			IMSI higher value

Imei Range

IMEI Range

XPath: *cna-chfagent:/chf-agent/profile-analyzer/profile-key/imei-range*

CLI Path: *chf-agent profile-analyzer profile-key imei-range*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
low	uint64			IMEI lower value
high	uint64			IMEI higher value

Msisdn Range

MSISDN Range

XPath: *cna-chfagent:/chf-agent/profile-analyzer/profile-key/msisdn-range*

CLI Path: *chf-agent profile-analyzer profile-key msisdn-range*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
low	uint64			MSISDN lower value
high	uint64			MSISDN higher value

Chf Server Reachability

Configure information related to reachability of CHF servers

XPath: *cna-chfagent:/chf-agent/chf-server-reachability*

CLI Path: *chf-agent chf-server-reachability*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
failover-status-codes	leaf-list of uint32			The set of HTTP status codes received for which the Nchf request needs to be failovered to an alternate CHF server
blacklisted-server-expiry-interval	uint32	1..86400	60	Duration in seconds after which the CHF Server can be considered to be available for sending the Nchf requests to it. Status: deprecated
blocklisted-server-expiry-interval	uint32	1..86400	60	Duration in seconds after which the CHF Server can be considered to be available for sending the Nchf requests to it.
reachability-failures-percentage	uint32	1..100	100	The percentage of Nchf request failures within a duration defined by reachability-failures-duration configuration, after which the server is blocklisted.
reachability-failures-duration	uint32		5	The duration in seconds which is considered to monitor the percentage of Nchf request failures, defined by the reachability-failures-percentage, to check if given CHF server needs to be blocklisted or not.

Nchf Interface

NCHF Interface Configuration

XPath: *cna-chfagent:/chf-agent/nchf-interface*

CLI Path: *chf-agent nchf-interface*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
dscp-marking	string	([0-9] or [1-5][0-9] or [6][0-3]) or df or af[1-4][1-3] or cs[1-7] or ef	df	DSCP marking, Default: df; Valid values are: 0..63, df (DSCP-0), af11 (DSCP-10), af12 (DSCP-12), af13 (DSCP-14), af21 (DSCP-18), af22 (DSCP-20), af23 (DSCP-22), af31 (DSCP-26), af32 (DSCP-28), af33 (DSCP-30), af41 (DSCP-34), af42 (DSCP-36), af43 (DSCP-38), cs1 (DSCP-8), cs2 (DSCP-16), cs3 (DSCP-24), cs4 (DSCP-32), cs5 (DSCP-40), cs6 (DSCP-48), cs7 (DSCP-56), ef (DSCP-46)

PPE Configuration

The module for entities implementing cna-ppe

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 - ip-pool
 - nas-termination-profile
 - gre-profile
 - service
 - network-context

Ppe

Affirmed N3IWF Packet Processing Engine

XPath: *cna-ppe:/upf*

CLI Path: *ppe*

Service Construct

Affirmed UPF Service-Construct

XPath: *cna-ppe:/upf/service-construct*

CLI Path: *ppe service-construct*

Network Context

Network-Context instance.
Configure network context object

XPath: *cna-ppe:/upf/service-construct/network-context*

CLI Path: *ppe service-construct network-context*

List Properties:
Keys: name
Maximum number of elements: 4096

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	1..64		Name of this network-context. This name must be unique.
type	bits			Type of this network-context. Possible values are {N3}, {N6}, {N4},{N9},and {NWU}.
header-sanity-check	enumeration	false true	false	Checks for packet header consistency when this flag is set to true

Nat Pool

Configure nat pool, Warning!!! name of the nat-pool must be unique across network-context

XPath: *cna-ppe:/upf/service-construct/network-context/nat-pool*

CLI Path: *ppe service-construct network-context nat-pool*

List Properties:
Keys: name
Maximum number of elements: 4096

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			Name of this nat-pool, Warning!!! This name must be unique across network-context.
nat-mode	enumeration	Many_to_One One_to_One	Many_to_One	Setting that controls whether NAT or NAPT is applied for subscribers using this NAT pool
nat-allocation-policy	enumeration	statistical guaranteed	statistical	Policy that controls when and how NAT resources are assigned to UEs when using this NAT pool
nat-filter	enumeration	not_set address_port_dependent	address_port_dependent	Nat filter type for this nat pool
max-ports-per-session	uint32	0..64511	0	Number of ports allowed per UE. Default 0 implies unlimited ports per UE.
l3am-nat-ips-per-ppe	uint32		1	Number of ip's per ppe to be allocated

Ip Pool

IP Pools configuration, Warning!!! name of the ip-pool must be unique across network-context
Must specify NAT Pool when ip-pool-type is 'NAT_IP_Pool'

XPath: *cna-ppe:/upf/service-construct/network-context/ip-pool*

CLI Path: *ppe service-construct network-context ip-pool*

List Properties:
Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	1..64		Name of this ip-pool, Warning!!! This name must be unique across network-context.
type	enumeration	UE_IP_Pool	UE_IP_Pool	Type of the IP pool
range-start	inet:ip-address			Starting value for IP address of this IP pool
range-end	inet:ip-address			Ending value for IP address of this IP pool
route-summary-prefix-length	uint32		0	Route summary prefix length of this IP pool
advertise	boolean		false	Advertise reachability of this UE IP Pool over dynamic routing
l3am-chunk-size	uint32		8192	Size of the UE IP pool chunks that will be used in resource blocks for Layer 3 Affinity mode - for resource management with multiple PPE instances
nat-pool	leafref			NAT pool associated with an IP sub-pool of type NAT-IP-Pool Leaf Reference Path: cna-ppe:/upf/service-construct/network-context/nat-pool/name

Nas Termination Profile

Configure nas termination profile

XPath: [cna-ppe:/upf/service-construct/nas-termination-profile](#)

CLI Path: [ppe service-construct nas-termination-profile](#)

List Properties:

Keys: name

Minimum number of elements: 0

Maximum number of elements: 255

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	1..64		nas termination profile name
nas-port	uint32	0 .. 65535		Configure NAS TCP port
ipv4-loopback-template-name	string	1..64		Name of the ipv4 Loopback Template to be used for Nas TCP Termination.
ipv6-loopback-template-name	string	1..64		Name of the ipv6 Loopback Template to be used for Nas TCP Termination.
network-context	string	1..64		Network context to be used for NAS profile

Gre Profile

Configure gre profile

XPath: [cna-ppe:/upf/service-construct/gre-profile](#)

CLI Path: [ppe service-construct gre-profile](#)

List Properties:

Keys: name

Minimum number of elements: 0

Maximum number of elements: 255

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			gre profile name
type	enumeration	POINT_TO_POINT MULTI_POINT	MULTI_POINT	gre tunnel mode type
ip-version	enumeration	IPv4 IPv6		Type of IP adress [IPv4/IPv6]

Attribute	Type	Range/Length/Pattern	Default	Description
source	string	1..64		Name of the ipv4 Loopback Template
destination	string	1..64		
instance	uint32	0 .. 128		gre instance
network-context	string	1..64		Network context to be used for GRE profile

Service

N3IWF Service Configuration

XPath: [cna-ppe:/upf/service](#)

CLI Path: [ppe service](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
admin-state	enumeration	disabled enabled	enabled	Admin state of the N3IWF Service. default state is "Enabled".

Network Context

list of Network-Contexts - selected from "/upf/service-construct/network-context" list

XPath: [cna-ppe:/upf/service/network-context](#)

CLI Path: [ppe service network-context](#)

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: cna-ppe:/upf/service-construct/network-context/name

DIAMETER BASE Configuration

The module for entities implementing diameter_base.

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- diameter-base
 - diameter-group
 - peer
 - vendor-specific-application-id
 - routing-table
 - application-id
 - next-hop
 - vendor-specific-application-id
 - next-hop
 - sctp-profile
 - tcp-profile

Diameter Base

Diameter interface object

XPath: [cna-diameter-base:/diameter-base](#)

CLI Path: [diameter-base](#)

Diameter Group

Configure Diameter Group of servers/clients

Either 'auth-application-id' or 'acct-application-id' or 'vendor-specific-application-id' must be configured

XPath: *cna-diameter-base/diameter-base/diameter-group***CLI Path:** *diameter-base diameter-group***List Properties:**

Keys: group-name

Maximum number of elements: 512

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
group-name	string			Local Diameter Group Name
group-role	enumeration	client server	client	Diameter Group role, client/server
load-balancing-algorithm	enumeration	round-robin least-loaded	round-robin	Load balancing algorithm
origin-realm	string	1..128		Local Realm Name
origin-host	string	1..128		Local HostName in FQDN
dscp-marking	string	([0-9] or [1-5][0-9] or [6][0-3]) or df or af[1-4][1-3] or cs[1-7] or ef	0	DSCP marking. Default: 0; Valid values are: 0..63, df, af11, af12, af13, af21, af22, af23, af31, af32, af33, af41, af42, af43, cs1, cs2, cs3, cs4, cs5, cs6, cs7, ef
connection-attempt-timeout	uint16	1..30	3	Time interval between successive connection attempts. Unit: Seconds
cea-timeout	uint32	1..120	5	Wait time for receiving a Capability Exchange Answer. Unit: Seconds
dwr-timer	uint32	1..300	15	Time interval between successive Device Watchdog Requests that monitors peer's health periodically. Unit: Seconds
dwr-retry-timer	uint32	1..300	1	Time interval between Device Watchdog Requests retries, triggered at first expiry of dwr-timer with shorter interval to detect peer failure sooner. Unit: Seconds
dwr-retry-count	uint32	0..1000	3	Maximum number of Device Watchdog Requests retry attempts to the server before declaring the peer down
dpr-reconnect-timeout	uint16	1..65535	5	Wait time to reconnect again after Disconnect Peer Request received from the peer. Unit: Seconds
app-msg-retransmit-count	uint16	1..20	5	Maximum number of times diameter application message can be retransmitted. Default: 5
app-msg-retransmit-timeout	uint16	1..30	5	Time interval between successive application message retransmission attempts. Unit: Seconds
product-name	string		AN- CN- 2020	Diameter Product Name
vendor-id	uint32		10415	Diameter Vendor Id

Attribute	Type	Range/Length/Pattern	Default	Description
auth-application-id	leaf-list of enumeration	gx		Authentication application Id
		s6a		
		s6b		
		s6d		
		sta		
		swm		
		flowauth		
		sxab		
		s6t		
		t6a		
		s6m		
		t4		
		sd		
		slg		
		s13		
acct-application-id	leaf-list of enumeration	gz		Accounting application Id
		gy		

Peer

Configure diameter peer

XPath: [cna-diameter-base:/diameter-base/diameter-group/peer](#)

CLI Path: [diameter-base diameter-group peer](#)

List Properties:

Keys: peer-name
Maximum number of elements: 512

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
peer-name	string			Peer Name
dest-hostname	string	1..128		Configure peer hostname
dest-realm	string	1..128		Realm name
dest-ip-address	inet:ip-address			Address of Peer. Can be IPv4/IPv6
dest-port	uint32	1000..56000	3868	Signaling port number
transport	enumeration	tcp sctp	tcp	Transport type
sctp-profile	leafref			Associated SCTP Profile Leaf Reference Path: cna-diameter-base:/diameter-base/sctp-profile/profile-name
tcp-profile	leafref			Associated TCP Profile Leaf Reference Path: cna-diameter-base:/diameter-base/tcp-profile/profile-name

Vendor Specific Application Id

Supported vendor Id list
Either 'vsa-auth-application-id' or 'vsa-acct-application-id' must be configured

XPath: [cna-diameter-base:/diameter-base/diameter-group/vendor-specific-application-id](#)

CLI Path: [diameter-base diameter-group vendor-specific-application-id](#)

List Properties:

Keys: vendor-id
Maximum number of elements: 512

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
vendor-id	uint32			Supported vendor id

Attribute	Type	Range/Length/Pattern	Default	Description
vsa-acct-application-id	leaf-list of uint32			Specify accounting application number
vsa-auth-application-id	leaf-list of uint32			Specify authentication application number

Routing Table

Routing table configuration

XPath: [cna-diameter-base:/diameter-base/diameter-group/routing-table](#)

CLI Path: [diameter-base diameter-group routing-table](#)

Application Id

Contains list of Routing table Entries

XPath: [cna-diameter-base:/diameter-base/diameter-group/routing-table/application-id](#)

CLI Path: [diameter-base diameter-group routing-table application-id](#)

List Properties:

Keys: application-id, realm, host-name

Maximum number of elements: 512

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
application-id	uint32			Diameter Application ID(3/4/16777251 etc), Any of current Diameter Group supported authentication or accounting application id
realm	string			Realm Name, Configure as 'any' to provision any realm is reachable via this routing entry
host-name	string			Diameter End host name, Configure as 'any' to provision any host is reachable via this routing entry

Next Hop

Next Hop

XPath: [cna-diameter-base:/diameter-base/diameter-group/routing-table/application-id/next-hop](#)

CLI Path: [diameter-base diameter-group routing-table application-id next-hop](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
primary-peer	leaf-list of leafref			Leaf Reference Path: cna-diameter-base:/diameter-base/diameter-group/peer/peer-name Minimum number of elements: 1 Maximum number of elements: 256
secondary-peer	leaf-list of leafref			Configure primary-peer before configuring secondary-peer Leaf Reference Path: cna-diameter-base:/diameter-base/diameter-group/peer/peer-name Minimum number of elements: 0 Maximum number of elements: 256

Vendor Specific Application Id

Contains list of Routing table Entries

XPath: [cna-diameter-base:/diameter-base/diameter-group/routing-table/vendor-specific-application-id](#)

CLI Path: [diameter-base diameter-group routing-table vendor-specific-application-id](#)

List Properties:

Keys: vsa-application-id, realm, host-name

Maximum number of elements: 512

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
vsa-application-id	uint32			Vendor Supported Diameter Application id (Integer value)
realm	string			Realm Name, Configure as 'any' to provision any realm is reachable via this routing entry
host-name	string			Diameter End host name, Configure as 'any' to provision any host is reachable via this routing entry

Next Hop

Next Hop

XPath: [cna-diameter-base:/diameter-base/diameter-group/routing-table/vendor-specific-application-id/next-hop](#)

CLI Path: [diameter-base diameter-group routing-table vendor-specific-application-id next-hop](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
primary-peer	leaf-list of leafref			Leaf Reference Path: cna-diameter-base:/diameter-base/diameter-group/peer/peer-name Minimum number of elements: 1 Maximum number of elements: 256
secondary-peer	leaf-list of leafref			Configure primary-peer before configuring secondary-peer Leaf Reference Path: cna-diameter-base:/diameter-base/diameter-group/peer/peer-name Minimum number of elements: 0 Maximum number of elements: 256

Sctp Profile

Configure SCTP profile

XPath: [cna-diameter-base:/diameter-base/sctp-profile](#)

CLI Path: [diameter-base sctp-profile](#)

List Properties:

Keys: profile-name

Maximum number of elements: 512

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
profile-name	string			SCTP Profile Name
rto-initial	uint32	1..5000	500	Retransmission Time Out (RTO) initial value. Default: 500; Unit: milliseconds The 'rto-initial' value should be in the range from 'rto-min' to 'rto-max'
rto-min	uint32	1..5000	250	Retransmission Time Out (RTO) minimum value. Default: 250; Unit: milliseconds
rto-max	uint32	1..60000	2000	Retransmission Time Out (RTO) maximum value. Default: 2000; Unit: milliseconds
path-max-retrans	uint16	1..20	5	Maximum number of consecutive retransmissions over a destination transport address of a peer endpoint before it is marked as inactive. Default: 5
association-init-max-transmit	uint16	1..10	3	Maximum number of times an INIT chunk or a COOKIE ECHO chunk is retransmitted before an endpoint aborts the initialization process and closes the association. Default: 3
association-max-retrans	uint16	1..20	6	Maximum number of consecutive retransmissions to a peer before an endpoint considers that the peer is unreachable and closes the association. Default: 6
no-delay	boolean		true	Specifies no delay. Default: true
max-out-stream	uint16	1..64	11	Specifies the maximum out stream. Default: 11
max-in-stream	uint16	1..64	11	Specifies the maximum in stream. Default: 11

Attribute	Type	Range/Length/Pattern	Default	Description
hb-interval	uint16	1..120	30	Specifies the heartbeat interval. Default: 30; Unit: seconds
rx-buffer-size	uint32	25000..2097152	149000	Specifies the receive buffer size. Default: 149000; Unit: bytes
tx-buffer-size	uint32	16000..5509000	149000	Specifies the transmit buffer size. Default: 149000; Unit: bytes
sack-delay	uint16	1	10..500	200
sack-frequency	uint16	1..5	2	Specifies the SCTP SACK frequency value. Disable: 1; Default: 2; Unit: packets

Tcp Profile

Configure TCP profile

XPath: [cna-diameter-base:/diameter-base/tcp-profile](#)

CLI Path: [diameter-base tcp-profile](#)

List Properties:

Keys: profile-name
Maximum number of elements: 512

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
profile-name	string			TCP Profile Name
keep-alive-interval	uint16	1..7200	10	Specifies the keep alive interval. Default: 10; Unit: seconds
rx-buffer-size	uint32	25000..2097152	149000	Specifies the receive buffer size. Default: 149000; Unit: bytes
tx-buffer-size	uint32	25000..2097152	149000	Specifies the transmit buffer size. Default: 149000; Unit: bytes

DNS AGENT Configuration

This module defines the configuration for DNS Agent.

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Dns Agent

DNS Agent.

XPath: [cna-dns-agent:/dns-agent](#)

CLI Path: [dns-agent](#)

Default View

XPath: [cna-dns-agent:/dns-agent/default-view](#)

CLI Path: [dns-agent default-view](#)

General

DNS application parameters.

XPath: *cna-dns-agent/dns-agent/default-view/dns-general*

CLI Path: *dns-agent default-view general*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
min-cache-ttl	uint32		0	The minimum time to cache ordinary (positive) answers. Default: 0; Unit: seconds.
max-cache-ttl	uint32		604800	The maximum time to cache ordinary (positive) answers. A value of zero could result in failures when intermediate records are needed for responses. Default: 604800 (one week); Unit: seconds. The value must be greater than or equal to min-cache-ttl.
min-ncache-ttl	uint32	0..90	0	Minimum retention time to store negative answers to reduce network traffic and increase performance. Default: 0; Unit: seconds.
max-ncache-ttl	uint32	0..604800	10800	Maximum retention time to store negative answers to reduce network traffic and increase performance. Default: 10800 (3 hours); Unit: seconds. The value must be greater than or equal to min-cache-ttl.
max-cache-size	uint16	2..20	5	The maximum amount of memory to use for dynamic cache (in megabytes). When the amount of data in the cache reaches this limit, records will be expired prematurely based on an LRU strategy. Default: 5; Unit: megabytes.
resolver-retry-interval	uint16	500..2000	800	The base retry interval (in milliseconds). The base value is used for request timeouts for each request until the number of request timeouts reaches resolver-nonbackoff-tries at which point the value is doubled for each subsequent retry. The actual timeout value used is limited to a minimum of 1200 ms and a maximum of 9000 ms. The maximum time for a query including retries is limited to resolver-query-timeout. Default: 800; Unit: milliseconds.
resolver-nonbackoff-tries	uint16	1..25	3	The number of request timeouts for a particular query to allow before applying exponential backoff to resolver-retry-interval. Default: 3.
resolver-query-timeout	uint32	10000..30000	10000	The amount of time in milliseconds that the resolver will spend attempting to resolve a query (including retries and/or recursion) before failing. Default: 10000; Unit: milliseconds.
dscp-marking	string	([0-9] or [1-5][0-9] or [6][0-3]) or df or af[1-4][1-3] or cs[1-7] or ef	df	DSCP Marking, which is obsolete and has no effect now. Default: df; Valid values are: 0..63, df, af11, af12, af13, af21, af22, af23, af31, af32, af33, af41, af42, af43, cs1, cs2, cs3, cs4, cs5, cs6, cs7, ef
tcp-only	boolean		false	If true only attempt queries with TCP; otherwise attempt UDP with fallback to TCP. Default: false.
edns-udp-size	uint32	512..4096	4096	The maximum advertised EDNS UDP buffer size in bytes, to control the size of packets received from authoritative servers in response to recursive queries. The initial size used to first query a server is 512. If successful, subsequent sizes used are 1232, 1432, and 4096, but not exceeding edns-udp-size. Default: 4096.
stale-answer-enable	boolean		true	Enable the returning of stale cached answers when servers are not answering. Default: true.
max-stale-ttl	uint32	1..max	604800	The maximum time to retain records past their normal expiry to return them as stale records when the servers that resolve the associated query are not reachable. Default: 604800 (one week); Unit: seconds.
stale-answer-ttl	uint32	1..max	1	The TTL returned on stale answers. Default 1; Unit: seconds. The value must be less than max-stale-ttl.

Record

DNS records locally defined which are checked before dynamic cache or external server.

XPath: *cna-dns-agent/dns-agent/default-view/dns-record*

CLI Path: *dns-agent default-view record*

A

Address record for IPv4, up to 4000 supported.

XPath: [cna-dns-agent:/dns-agent/default-view/dns-record/a-record](#)

CLI Path: [dns-agent default-view record a](#)

List Properties:

Keys: node-name, node-ip-address
Maximum number of elements: 4000

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
node-name	inet:domain-name			DNS name for this record.
node-ip-address	inet:ipv4-address			IPv4 address to which this 'node-name' will resolve.

Aaaa

Address record for IPv6, up to 4000 supported.

XPath: [cna-dns-agent:/dns-agent/default-view/dns-record/aaaa-record](#)

CLI Path: [dns-agent default-view record aaaa](#)

List Properties:

Keys: node-name, node-ip-address
Maximum number of elements: 4000

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
node-name	inet:domain-name			DNS name for this record.
node-ip-address	inet:ipv6-address			IPv6 address to which this 'node-name' will resolve.

Naptr

Name Authority Pointer (NAPTR) record, up to 8000 supported.

XPath: [cna-dns-agent:/dns-agent/default-view/dns-record/naptr-record](#)

CLI Path: [dns-agent default-view record naptr](#)

List Properties:

Keys: domain-name, services, replacement
Maximum number of elements: 8000

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
domain-name	inet:domain-name			DNS domain for this record.
services	string	() or ([a-zA-Z][a-zA-Z0-9+-.]{0,31})(:[a-zA-Z][a-zA-Z0-9+-.]{0,31})*)		Services offered by this record with format as defined in TS 29.303 and RFC 3958. For example, the services parameter for PGW can be represented like this: x-3gpp-pgw:x-s5-gtp:x-s5-pmip. Length: 0..255
replacement	inet:domain-name			Replacement expression for this record.
order	uint16		1	Order that this record should be processed. Default: 1; Range: 0..65535
preference	uint16		1	Preference that this record has within the group with the same order. Default: 1; Range: 0..65535
flags	enumeration	none A		Flags applied to this record.
status	enumeration	enable disable	enable	Administrative status for this record. Disable means this record will not be selected. Default: enable

Server

DNS external servers, up to 10 are supported.

XPath: *cna-dns-agent:/dns-agent/default-view/dns-server*

CLI Path: *dns-agent default-view server*

List Properties:

Keys: server-name

Maximum number of elements: 10

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
server-name	string	(a-zA-Z*)		Name for this DNS server; Length: 1..32.
server-ip-address	inet:ip-address			IP address for this DNS server.
domain	inet:domain-name			Optional domain for this DNS server. If a domain is configured here, then DNS server selection is first based on longest domain match in terms of complete labels (between dots) searching from right to left. Secondly, if there is no matching domain criteria, then DNS servers with no domain configured are selected. Thirdly, if there are no servers configured with a domain, then all servers are considered based on weight. Finally, if matching servers are not responding, then non-matching servers may be selected.

View

DNS views.

XPath: *cna-dns-agent:/dns-agent/dns-view*

CLI Path: *dns-agent view*

List Properties:

Keys: view-name

Maximum number of elements: 2

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
view-name	string	(a-zA-Z*)		Name for this DNS view; Length: 1..32.

General

DNS application parameters.

XPath: *cna-dns-agent:/dns-agent/dns-view/dns-general*

CLI Path: *dns-agent view general*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
min-cache-ttl	uint32		0	The minimum time to cache ordinary (positive) answers. Default: 0; Unit: seconds.
max-cache-ttl	uint32		604800	The maximum time to cache ordinary (positive) answers. A value of zero could result in failures when intermediate records are needed for responses. Default: 604800 (one week); Unit: seconds. The value must be greater than or equal to min-cache-ttl.
min-ncache-ttl	uint32	0..90	0	Minimum retention time to store negative answers to reduce network traffic and increase performance. Default: 0; Unit: seconds.
max-ncache-ttl	uint32	0..604800	10800	Maximum retention time to store negative answers to reduce network traffic and increase performance. Default: 10800 (3 hours); Unit: seconds. The value must be greater than or equal to min-cache-ttl.

Attribute	Type	Range/Length/Pattern	Default	Description
max-cache-size	uint16	2..20	5	The maximum amount of memory to use for dynamic cache (in megabytes). When the amount of data in the cache reaches this limit, records will be expired prematurely based on an LRU strategy. Default: 5; Unit: megabytes.
resolver-retry-interval	uint16	500..2000	800	The base retry interval (in milliseconds). The base value is used for request timeouts for each request until the number of request timeouts reaches resolver-nonbackoff-tries at which point the value is doubled for each subsequent retry. The actual timeout value used is limited to a minimum of 1200 ms and a maximum of 9000 ms. The maximum time for a query including retries is limited to resolver-query-timeout. Default: 800; Unit: milliseconds.
resolver-nonbackoff-tries	uint16	1..25	3	The number of request timeouts for a particular query to allow before applying exponential backoff to resolver-retry-interval. Default: 3.
resolver-query-timeout	uint32	10000..30000	10000	The amount of time in milliseconds that the resolver will spend attempting to resolve a query (including retries and/or recursion) before failing. Default: 10000; Unit: milliseconds.
dscp-marking	string	([0-9] or [1-5][0-9] or [6][0-3]) or df or af[1-4] [1-3] or cs[1-7] or ef	df	DSCP Marking, which is obsolete and has no effect now. Default: df; Valid values are: 0..63, df, af11, af12, af13, af21, af22, af23, af31, af32, af33, af41, af42, af43, cs1, cs2, cs3, cs4, cs5, cs6, cs7, ef
tcp-only	boolean		false	If true only attempt queries with TCP; otherwise attempt UDP with fallback to TCP. Default: false.
edns-udp-size	uint32	512..4096	4096	The maximum advertised EDNS UDP buffer size in bytes, to control the size of packets received from authoritative servers in response to recursive queries. The initial size used to first query a server is 512. If successful, subsequent sizes used are 1232, 1432, and 4096, but not exceeding edns-udp-size. Default: 4096.
stale-answer-enable	boolean		true	Enable the returning of stale cached answers when servers are not answering. Default: true.
max-stale-ttl	uint32	1..max	604800	The maximum time to retain records past their normal expiry to return them as stale records when the servers that resolve the associated query are not reachable. Default: 604800 (one week); Unit: seconds.
stale-answer-ttl	uint32	1..max	1	The TTL returned on stale answers. Default 1; Unit: seconds. The value must be less than max-stale-ttl.

Record

DNS records locally defined which are checked before dynamic cache or external server.

XPath: [cna-dns-agent/dns-agent/dns-view/dns-record](#)

CLI Path: [dns-agent view record](#)

A

Address record for IPv4, up to 4000 supported.

XPath: [cna-dns-agent/dns-agent/dns-view/dns-record/a-record](#)

CLI Path: [dns-agent view record a](#)

List Properties:

Keys: node-name, node-ip-address

Maximum number of elements: 4000

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
node-name	inet:domain-name			DNS name for this record.
node-ip-address	inet:ipv4-address			IPv4 address to which this 'node-name' will resolve.

Aaaa

Address record for IPv6, up to 4000 supported.

XPath: [cna-dns-agent/dns-agent/dns-view/dns-record/aaaa-record](#)

CLI Path: *dns-agent view record aaaa*

List Properties:

Keys: node-name, node-ip-address
Maximum number of elements: 4000

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
node-name	inet:domain-name			DNS name for this record.
node-ip-address	inet:ipv6-address			IPv6 address to which this 'node-name' will resolve.

Naptr

Name Authority Pointer (NAPTR) record, up to 8000 supported.

XPath: *cna-dns-agent/dns-agent/dns-view/dns-record/naptr-record*

CLI Path: *dns-agent view record naptr*

List Properties:

Keys: domain-name, services, replacement
Maximum number of elements: 8000

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
domain-name	inet:domain-name			DNS domain for this record.
services	string	() or ([a-zA-Z][a-zA-Z0-9+-.]{0,31}(:[a-zA-Z][a-zA-Z0-9+-.]{0,31})*)		Services offered by this record with format as defined in TS 29.303 and RFC 3958. For example, the services parameter for PGW can be represented like this: x-3gpp-pgw:x-s5-gtp:x-s5-pmip. Length: 0..255
replacement	inet:domain-name			Replacement expression for this record.
order	uint16		1	Order that this record should be processed. Default: 1; Range: 0..65535
preference	uint16		1	Preference that this record has within the group with the same order. Default: 1; Range: 0..65535
flags	enumeration	none A		Flags applied to this record.
status	enumeration	enable disable	enable	Administrative status for this record. Disable means this record will not be selected. Default: enable

Server

DNS external servers, up to 10 are supported.

XPath: *cna-dns-agent/dns-agent/dns-view/dns-server*

CLI Path: *dns-agent view server*

List Properties:

Keys: server-name
Maximum number of elements: 10

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
server-name	string	(a-zA-Z*)		Name for this DNS server; Length: 1..32.
server-ip-address	inet:ip-address			IP address for this DNS server.

Attribute	Type	Range/Length/Pattern	Default	Description
domain	inet:domain-name			Optional domain for this DNS server. If a domain is configured here, then DNS server selection is first based on longest domain match in terms of complete labels (between dots) searching from right to left. Secondly, if there is no matching domain criteria, then DNS servers with no domain configured are selected. Thirdly, if there are no servers configured with a domain, then all servers are considered based on weight. Finally, if matching servers are not responding, then non-matching servers may be selected.

Server Monitoring

Parameters to control active monitoring of DNS server status by sending periodic null queries.

XPath: [cna-dns-agent:/dns-agent/dns-server-monitoring](#)

CLI Path: [dns-agent server-monitoring](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
interval	uint16	0	30..600	60
timeout	uint8	1..5	2	Time to wait for a reply from a DNS server. Unit: seconds; Default 2.
retry	uint8	1..4	2	Number of times to retry the status query. After retries are exhausted, the DNS server status is marked as down and an alarm is raised. Default 2.

EVENT EXPOSURE Configuration

The module for entities implementing event-exposure.

Table of Contents

- [event-exposure](#)
 - [service](#)
 - [profile](#)

Event Exposure

event-exposure

XPath: [cna-event-exposure:/event-exposure](#)

CLI Path: [event-exposure](#)

Service

Event Exposure Service config

XPath: [cna-event-exposure:/event-exposure/service](#)

CLI Path: [event-exposure service](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
producer-profile	leafref			EVENT EXPOSURE profile used when SMF sends notifications Leaf Reference Path: cna-event-exposure:/event-exposure/profile/name
nnef-pfdmanagement-oauth-required	boolean		false	OAuth Required for NEF PFD Management Interface
max-pfds-pfdm-msg	uint32	1 .. 10000	1000	Max number of PFDs to be sent in one PFD Message to PFCP
max-size-pfdm-msg	uint32	100 .. max	10000	Max Size of the PFD Message to PFCP in bytes
nnef-pfdmanagement-subscribe-to-nef	enumeration	disabled enabled	enabled	Enable NEF Subscribe for PFD changes(Enable/Disable)

Attribute	Type	Range/Length/Pattern	Default	Description
nnef-pfdmanagement-caching-time-support	enumeration	disabled enabled	disabled	Enable PFD Caching Time Support(Enable/Disable)
nnef-pfdmanagement-default-caching-time-duration	uint32	0 .. 2592000	86400	default caching duration to be used when NEF does not provide in seconds
nnef-pfdmanagement-nrf-subscription-required	boolean		false	NRF subscription required for NEF PFD Management Interface
nnef-pfdmanagement-max-load-threshold	uint32	0 .. 100	100	Maximum load threshold in percentage for NEF PFD Management Interface
include-3gpp-peer-info	boolean		false	Support to send 3gpp peer info header on NEF PFD Management Interface

Profile

Event Exposure Profile - contains timers/counts used by Event Exposure service

XPath: [cna-event-exposure:/event-exposure/profile](#)

CLI Path: [event-exposure profile](#)

List Properties:

Keys: name

Maximum number of elements: 1

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			Profile name
max-notification-re-transmits	uint32	0 .. 5	3	

GTPCAGENT Configuration

The module for entities implementing cna-gtpcagent

Table of Contents

- [gtpcagent](#)
 - [gtpc-interface](#)
 - [retry-config-profile](#)
 - [trace](#)
 - [trace-keys](#)

Gtpcagent

gtpcagent configuration

XPath: [cna-gtpcagent:/gtpcagent](#)

CLI Path: [gtpcagent](#)

Gtpc Interface

GTPC interface configuration

XPath: [cna-gtpcagent:/gtpcagent/gtpc-interface](#)

CLI Path: [gtpcagent gtpc-interface](#)

List Properties:

Keys: interface-name

Maximum number of elements: 23

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
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Attribute	Type	Range/Length/Pattern	Default	Description
interface-name	string			Interface name
interface-type	enumeration	N26-MME S3-MME SV-MME S11-MME S10 GN-GP-MME S5-S8-PGW S2A-PGW S2B-PGW S5-S8-S2A-S2B-PGW N26-AMF S11-SGW S5-S8-SGW S4-SGW GN-GP-GGSN S2A-TWAG S2B-EPDG S3-SGSN S4-SGSN GN-GP-SGSN SV-MSC S101-EUTRAN S101 S3-S10-MME S4-S11-SGW		Type of the GTPC interface
n3-request	uint8	0..20	2	n3-request count
t3-response	uint32	100..3600000	5000	t3-response time in milliseconds
version	enumeration	v1 v2	v2	GTP-C version supported on this interface
echo-interval	uint16	0..300	5	echo-interval in minutes. Use value 0 for disabling this behaviour
t3-echo-response	uint32	100..360000	15000	t3-echo-response in milliseconds.
n3-echo-request	uint8	1..20	8	n3-echo-request : number of attempts to send echo request message
inactivity-timer	uint16	1..36000	8	time in minutes to wait after GTP path failure for a peer
t3-path-recovery-echo-response	uint16	0	100..60000	5000
dscp-marking	string	([0-9] or [1-5][0-9] or [6][0-3]) or df or af[1-4] [1-3] or cs[1-7] or ef	df	DSCP marking, Default: df; Valid values are: 0..63, df (DSCP-0), af11 (DSCP-10), af12 (DSCP-12), af13 (DSCP-14), af21 (DSCP-18), af22 (DSCP-20), af23 (DSCP-22), af31 (DSCP-26), af32 (DSCP-28), af33 (DSCP-30), af41 (DSCP-34), af42 (DSCP-36), af43 (DSCP-38), cs1 (DSCP-8), cs2 (DSCP-16), cs3 (DSCP-24), cs4 (DSCP-32), cs5 (DSCP-40), cs6 (DSCP-48), cs7 (DSCP-56), ef (DSCP-46)
consumer-retry-count	uint8	0..20	3	number of times to retry the http msg towards consumer for peer initiated call flows
associated-retry-config	leafref			associated retry configuration for peer initiated call flows Leaf Reference Path: cna-gtpcagent:/gtpcagent/retry-config-profile/retry-profile

Retry Config Profile

configure retry profile for managing http retries towards consumer in peer initiated call flows

retry profile has to be configured with either one of the lists: msg-types supported or unsupported

XPath: *cna-gtpcagent:/gtpcagent/retry-config-profile*

CLI Path: *gtpcagent retry-config-profile*

List Properties:

Keys: retry-profile
Maximum number of elements: 128

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
retry-profile	string			retry configuration profile name
supported-gtpc-message-types	leaf-list of uint32	0 .. 255		GTPC message types in Peer Initiated Call flows supported for http retry towards consumer, Value 255 can be used to indicate all message types cannot configure this list when unsupported-gtpc-message-types is already present
unsupported-gtpc-message-types	leaf-list of uint32	0 .. 255		GTPC message types in peer initiated call flows not supported for http retry towards consumer, Value 255 can be used to indicate all message types cannot configure this list when supported-gtpc-message-types is already present

Trace

GTPC AGENT Tracing Configuration

XPath: *cna-gtpcagent:/gtpcagent/trace*

CLI Path: *gtpcagent trace*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
default-sampling-rate	uint32	0 .. 100	1	default Rate for GTPC Agent Traces in percentage

Trace Keys

Name
Tracing Key List

XPath: *cna-gtpcagent:/gtpcagent/trace/trace-keys*

CLI Path: *gtpcagent trace trace-keys*

List Properties:

Keys: name
Maximum number of elements: 128

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		
priority	uint32	1 .. 255	255	Sampling priority from 1 to 255. Lower value indicates higher priority.
sampling-rate	uint32	0 .. 100	100	Sampling Rate for GTPC Agent Traces in percentage
interface-name	leaf-list of string			Interface name as configured in the gtpc-interface configuration
local-endpoint-ip	leaf-list of inet:ip-address			local endpoint IP address
gtpc-message-type	leaf-list of uint32	0 .. 255		GTPC message type
gtpc-version	enumeration	v1 v2		GTPC version v1 or v2

HELLO WORLD Configuration

The module for entities implementing cna-hello-world.

Table of Contents

- [hello-world](#)
 - [test-list](#)
 - [test-container-1](#)
 - [test-list-1](#)
 - [test-container-2](#)
 - [test-container](#)

Hello World

a object

XPath: [cna-hello-world:/hello-world](#)

CLI Path: [hello-world](#)

Test List

test object

XPath: [cna-hello-world:/hello-world/test-list](#)

CLI Path: [hello-world test-list](#)

List Properties:

Keys: name
Maximum number of elements: 8

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	1..254		Name ; Range: 1..254
ip-address	inet:ip-address			IP address for this DNS server
weight	uint16	1..10	10	Weight for this object. Default: 10; Unit: relative
test-bool	boolean		true	Test bool info
version	enumeration	1.0.0	1.0.0	Version
		2.0.0		
		3.0.0		
new-attr	uint16	1..10	10	new one: Default: 10
test1-pwd-encrypted	string		testpassword	Test encrypted password

Test Container 1

Test container info 1.

XPath: [cna-hello-world:/hello-world/test-list/test-container-1](#)

CLI Path: [hello-world test-list test-container-1](#)

Test List 1

test object

XPath: [cna-hello-world:/hello-world/test-list/test-container-1/test-list-1](#)

CLI Path: [hello-world test-list test-container-1 test-list-1](#)

List Properties:

Keys: name1
Maximum number of elements: 8

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name1	string	1..254		Name ; Range: 1..254

Test Container 2

Test container info 2.

XPath: [cna-hello-world:/hello-world/test-list/test-container-1/test-list-1/test-container-2](#)

CLI Path: [hello-world test-list test-container-1 test-list-1 test-container-2](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
test3-pwd-encrypted	string		testpassword	Test encrypted password

Test Container

Test container info.

XPath: [cna-hello-world:/hello-world/test-container](#)

CLI Path: [hello-world test-container](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
ip-address-attr	inet:ip-address			IP address
admin-state	enumeration	enabled disabled maintenance	enabled	
int-attr	uint16	30..5000	300	Int-attr infor. Default: 300; Unit: minutes
test-leaf-list	leaf-list of enumeration	test-elem-1 test-elem-2 test-elem-3		test leaf-list Minimum number of elements: 0 Maximum number of elements: 3
test-hex-string	string	([0-9a-fA-F]{2}(:[0-9a-fA-F]{2})*)?		test string
test2-pwd-encrypted	string		testpassword	Test encrypted password
test-ordered-by-user	leaf-list of string			test ordered-by user Ordered by: user

TEST WARNING Configuration

The module for entities implementing cna-test-warning.

Table of Contents

- [test-warning](#)
 - [test-container](#)

Test Warning

test warning messages object

XPath: [cna-test-warning:/test-warning](#)

CLI Path: [test-warning](#)

Test Container

Test container info.

XPath: [cna-test-warning:/test-warning/test-container](#)

CLI Path: [test-warning test-container](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
bool-attr	boolean		true	Bool attribute info
ip-address-attr	inet:ip-address		0.0.0.0	IP address

Attribute	Type	Range/Length/Pattern	Default	Description
uint-attr	uint16	30..5000	300	Int-attr infor. Default: 300; Unit: minutes
int-attr	int8	1..16	1	Int-attr infor. Default: 300; Unit: minutes
enum-attr	enumeration	sgw pgw ggsn		Enum-attr
string-attr	string			String-attr

IKEMGR Configuration

This module defines the configuration related to IKE service and its constructs

Table of Contents

- ike
 - authenticationProfile
 - dpdProfile
 - ike-interface
 - localAuthentication
 - remoteAuthentication
 - ikeTransform
 - ipsecTransform
 - ike-global
 - cookie-protection-thresholds

Ike

IKE Service and Constructs configuration

XPath: *ikemgr:/ike*

CLI Path: *ike*

Auth Profile

IKE Authentication Profile Configuration

XPath: *ikemgr:/ike/authenticationProfile*

CLI Path: *ike auth-profile*

List Properties:

Keys: name
Maximum number of elements: 128

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			IKE authentication profile name
authentication-type	enumeration	eap5g		ike authentication type
identity	union	['inet:ip-address', 'inet:domain-name', 'string,(.@\.\.?)', 'string, (\)']		specify the IKE identity(IP Address, FQDN, EMAIL or WildCard("*")) to be used for the authentication profile. NOTE: Wildcard is only relevant in remote-authetication where we want to select this profile as fallback for any identity sent by remote peer

Dpd Profile

IKE DPD Profile Configuration

XPath: *ikemgr:/ike/dpdProfile*

CLI Path: *ike dpd-profile*

List Properties:

Keys: name
Maximum number of elements: 128

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			IKE DPD profile name
dpd-delay	uint32	60..86400	90	specify inactivity timeout(seconds) after which the DPD should be performed
dpd-resp-timeout	uint16	3..30	15	specify maximum time(seconds) to wait for the DPD response from peer (NOTE: DPD requests are retried exponetially with initial wait time of 1 second)
inactivity-type	enumeration	ike-only	ike-only	specify inactivity type to initiate a IKE DPD Combined (IKE and IPSEC) type inactivity for DPD is not supported yet

Ike Interface

IKE Interface Configuration
please specify atleast one of ipv4-loopback-interface-name or ipv6-loopback-interface-name
n3iwf can only be deployed as responder

XPath: [ikemgr:/ike/ike-interface](#)

CLI Path: [ike ike-interface](#)

List Properties:

Keys: name
Maximum number of elements: 128

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			IKE interface name
network-context	string			network context name for the ike termination. This must be one of '/interface-manager/network-context' names
ipv4-loopback-interface-name	string			name of loopback IPV4 address interface for IKE termination. This must be one of '/interface-manager/network-context{networkContextName}/loopback-ipv4-interface' and of type 'dynamic' loopback-scope
ipv6-loopback-interface-name	string			name of loopback IPV6 address interface for IKE termination. This must be one of '/interface-manager/network-context{networkContextName}/loopback-ipv6-interface' and of type 'dynamic' loopback-scope
ike-version	enumeration	v2	v2	IKE protocol Version only IKE V2 is supported for now
ike-mode	enumeration	responder	responder	IKE operation mode only responder mode is supported for now
deployment-mode	enumeration	n3iwf peer-to-site	n3iwf	IKE deployment mode
ike-life-time	uint32	60..2592000	86400	lifetime of the IKE SA (seconds)
ipsec-life-time	uint32	60..2592000	3600	lifetime of the IPSEC SA (seconds)
max-retransmission-timeout	uint16	3..30	15	specify retransmission-timeout(seconds). Retransmissions are retried in exponential manner starting with 1 second upto the specified timeout
ike-rekey	enumeration	disabled enabled	enabled	enable or disable IKE rekey
ipsec-rekey	enumeration	disabled enabled	enabled	enable or disable IPSEC rekey
dpd-profile	leafref			specify the dpd-profile to use with this interface Leaf Reference Path: ikemgr:/ike/dpdProfile/name
mobike-support	enumeration	disabled enabled	disabled	enable or disable MOBIKE support
mobike-return-route-check	enumeration	disabled enabled	disabled	enable or disable MOBIKE return routability check

Attribute	Type	Range/Length/Pattern	Default	Description
dynamic-peer-addr-update	enumeration	disabled ike-only	disabled	specify the preference to update dyanmic peer address change (IP, Port or Both) for non MOBIKE negotiated tunnels
ike-fragmentation	enumeration	disabled enabled	enabled	enable or disable IKE fragmentation
ike-fragment-size	uint16	0	160..9216	1280
backoff-timer	int32	0..3600		Specify the back-off timer to be sent to UE in case of congestion. If no value is specified, it is assumed backoff-timer is deactivated
tunnel-context-cache-timer	uint32	0	120..86400	600

Local Authentication

local authentication configuration

eap5g and pki-plus-eap5g are only allowed for n3iwf deployment mode

XPath: [ikemgr:/ike/ike-interface/localAuthentication](#)

CLI Path: [ike ike-interface local-authentication](#)

List Properties:

Keys: name

Minimum number of elements: 1

Maximum number of elements: 1

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			specify name
authentication-type	enumeration	eap5g		local authentication type
eap-only-auth	enumeration	supported disabled	supported	eap-only authentication supported/disabled
auth-profile	leafref			authentication profile to use for local authentication select authentication-profile with matching authentication-type Leaf Reference Path: ikemgr:/ike/authenticationProfile/name

Remote Authentication

remote authentication configuration

remote-authentication types eap5g and pki-plus-eap5g are only allowed for n3iwf deployment mode

please select atleast one authentication-profile for psk authentication-type

please select atleast one authentication-profile for pki authentication-type

XPath: [ikemgr:/ike/ike-interface/remoteAuthentication](#)

CLI Path: [ike ike-interface remote-authentication](#)

List Properties:

Keys: name

Minimum number of elements: 1

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			specify name
authentication-type	enumeration	eap5g		remote authentication type pki-plus-eap5g is not valid for remote-authentication
auth-profile-selector	enumeration	local-identity remote-identity	local-identity	option to select an authentication profile to validate AUTH payload. For 'local-identity' auth-profile specified in local authentication is used

Attribute	Type	Range/Length/Pattern	Default	Description
auth-profile-list	leaf-list of leafref			authentication profile to use for local authentication please select authentication-profile with matching authentication-type please select authentication-profile with matching authentication-type Leaf Reference Path: ikemgr:/ike/authenticationProfile/name

Ike Transform

IKE transform selections

XPath: [ikemgr:/ike/ike-interface/ikeTransform](#)

CLI Path: [ike ike-interface ike-transform](#)

List Properties:

Keys: name
Minimum number of elements: 1
Maximum number of elements: 1

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			
encr-algo	leaf-list of enumeration	aes128 aes192 aes256 aesctr128 aesctr192 aesctr256 aes128gcm16 aes192gcm16 aes256gcm16		Encryption algorithms Minimum number of elements: 1 Maximum number of elements: 10 Ordered by: user
integ-algo	leaf-list of enumeration	sha1 sha256 sha384 sha512		Integrity algorithms Minimum number of elements: 1 Maximum number of elements: 10 Ordered by: user
dh-group	leaf-list of enumeration	none group-5 group-14 group-15 group-16 group-17 group-18 group-19 group-20 group-21 group-22 group-23 group-24 group-ike-sa		Diffie-Helman groups Please select dh-groups other than 'none' and 'group-ike-sa' for IKE transform Minimum number of elements: 1 Maximum number of elements: 10 Ordered by: user

Ipsec Transform

IPsec transform selections

XPath: [ikemgr:/ike/ike-interface/ipsecTransform](#)

CLI Path: [ike ike-interface ipsec-transform](#)

List Properties:

Keys: name
Minimum number of elements: 1
Maximum number of elements: 1

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
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Attribute	Type	Range/Length/Pattern	Default	Description
name	string			
encr-algo	leaf-list of enumeration	aes128 aes192 aes256 aesctr128 aesctr192 aesctr256 aes128gcm16 aes192gcm16 aes256gcm16		Encryption algorithms Minimum number of elements: 1 Maximum number of elements: 20 Ordered by: user
integ-algo	leaf-list of enumeration	sha1 sha256 sha384 sha512		Integrity algorithms Minimum number of elements: 1 Maximum number of elements: 10 Ordered by: user
pfs-dh-group	leaf-list of enumeration	none group-5 group-14 group-15 group-16 group-17 group-18 group-19 group-20 group-21 group-22 group-23 group-24 group-ike-sa		IPSEC Diffie-Helman groups Minimum number of elements: 1 Maximum number of elements: 10 Ordered by: user
esn-support	enumeration	supported-prefer-esn supported-prefer-non-esn enforced disabled	supported-prefer-esn	ESN (enhanced-sequence-number) support preference

Ike Global

IKE Service globals configurations

XPath: [ikemgr:/ike/ike-global](#)

CLI Path: [ike ike-global](#)

Cookie Protection Thresholds

IKEV2 cookie protection thresholds configuration

XPath: [ikemgr:/ike/ike-global/cookie-protection-thresholds](#)

CLI Path: [ike ike-global cookie-protection-thresholds](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
half-open-sa-count	uint32		4294967295	For IKEv2, specifies the number of half-open SAs, above which cookie-challenges are enabled. Value 0 indicates cookie challenge is sent for all SAs

INTERFACE MGR Configuration

The module for cna-interface_mgr.

Table of Contents

- [interface-manager](#)
 - [network-context](#)
 - [ip-interface](#)

- [ipv4-address-template](#)
- [ipv6-address-template](#)
- [loopback-ipv4-interface](#)
- [loopback-ipv6-interface](#)
- [port-profile](#)

Interface Manager

Interface Manager configuration

Interface Manager handles the deployment of network resources on network entities in a NF. In this context, "network resources" include Network Contexts, vlans and IP addresses. "network entities" are such things as Ethernet Interfaces and Forwarding Tables that may dynamically appear and disappear. To accommodate the dynamic nature of network entities, network resources are not explicitly bound to network entities. The Interface Manager will assign resources to network entities on demand, using configuration information that describes sets of addresses available for deployment on vlans, loopbacks, etc. When a network entity disappears, its network resources will be recovered and reused. When Interface Manager deploys a VLAN in a Network Context, it creates a VLAN sub-interface on every "physical" ethernet interface in the federation. Each sub-interface will have the same VLAN ID, but each will be assigned an IP address belonging to a unique IP subnet. A Network Context is deployed by instantiating IPv4 and IPv6 Forwarding Table "network entities", (i.e. FIBs). Every VLAN sub-interface is bound to an IPv4/IPv6 pair of these tables based on its Network Context. Loopback interfaces, representing one or more loopback addresses, are also deployed on demand. Configuration allows the creation of multiple loopback interfaces. Each loopback interface is deployed by creating a loopback interface in the IPv4 or IPv6 FIB associated with the loopback's Network Context.

XPath: [cna-interface-mgr:/interface-manager](#)

CLI Path: [interface-manager](#)

Network Context

Configure the Network Contexts supported by this NF

Configure the Network Contexts supported by this NF. Each Network Context represents an entire IPv4 address space and an entire IPv6 address space. The address spaces of all Network Contexts are disjoint

XPath: [cna-interface-mgr:/interface-manager/network-context](#)

CLI Path: [interface-manager network-context](#)

List Properties:

Keys: nc-name
Maximum number of elements: 4096

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
nc-name	string	[-A-Za-z0-9#_()<>{}]*		Network Context name - type is
pod-type	enumeration	ulb		Specify Pod-type if this network-context needs to be used for a specific type of pod - No need to configure for IPPE or PPE

Ip Interface

Configure the ip-interface that will be deployed in this Network Context

Configure the ip-interface that will be created on every ethernet interface in this Network Context if the interface-label is not specified. Otherwise, the ip-interface will be created only on the interface-label matching interface.

ip interface name must be unique

ip interface using this interface-label and vlan is already configured

must not use a mix of tagged and untagged ip-interfaces with the same interface-label.

must not use a mix of tagged and untagged ip-interfaces with the same interface-label.

XPath: [cna-interface-mgr:/interface-manager/network-context/ip-interface](#)

CLI Path: [interface-manager network-context ip-interface](#)

List Properties:

Keys: interface-name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
interface-name	string	[-A-Za-z0-9#_()<>{}]*		Name of the interface - type is

Attribute	Type	Range/Length/Pattern	Default	Description
vlan-id	uint32	0..4094	0	802.1Q VLAN Identifier - range [0, 4094] - 0 implies an "untagged" VLAN interface. The VLAN Identifier for this interface. The acceptable range is [0, 4094] and the value must be unique across all Network Contexts. The default VLAN-ID is 0, which is used to specifies an untagged interface.
interface-label	string			Optionally specify a label to identify the h/w interface where this subinterface will be created.
ip-mtu	uint32	1280..9216	1500	Optionally set the Layer 3 MTU of the sub-interface - range [1280, 9216] - default is 1500
shared-subnet	boolean		true	shared subnet enabled Enables or disables shared subnet.
provision-admin-state	enumeration	disabled enabled	enabled	admin-state for provisioning this interface.

Ipv4 Address Template

Configure a set of IPv4 addresses available for use when deploying this ip-interface

Configure a set of IPv4 addresses available for use when deploying this ip-interface

XPath: [cna-interface-mgr:/interface-manager/network-context/ip-interface/ipv4-address-template](#)

CLI Path: [interface-manager network-context ip-interface ipv4-address-template](#)

List Properties:

Keys: ipv4-template-name

Maximum number of elements: 8

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
ipv4-template-name	string	[-A-Za-z0-9#_()<>{}]*		Name of IPv4 address template - type is
base-address	string	((([0-9] or [1-9][0-9] or 1[0-9][0-9] or 2[0-4][0-9] or 25[0-5]).){3}([0-9] or [1-9][0-9] or 1[0-9][0-9] or 2[0-4][0-9] or 25[0-5]))		IPv4 end-point address, (subnet and host), of first address created by template - dotted-quad format - mandatory
prefix-length	uint16	1 .. 31		IPv4 subnet length of all addresses created by template - range [1, 31] - mandatory
count	uint16	0 .. 128		Total number of addresses that can be generated by this template - range is [0, 128] - 0 disables the template - mandatory

Ipv6 Address Template

Configure a set of IPv6 addresses available for use when deploying this ip-interface

Configure a set of IPv6 addresses available for use when deploying this ip-interface.

XPath: [cna-interface-mgr:/interface-manager/network-context/ip-interface/ipv6-address-template](#)

CLI Path: [interface-manager network-context ip-interface ipv6-address-template](#)

List Properties:

Keys: ipv6-template-name

Maximum number of elements: 8

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
ipv6-template-name	string	[-A-Za-z0-9#_()<>{}]*		Name of IPv6 address template - type is
base-address	string	(((: or [0-9a-fA-F]{0,4}):)(([0-9a-fA-F]{0,4}):){0,5}(((([0-9a-fA-F]{0,4}):)?(: or [0-9a-fA-F]{0,4})) or (((25[0-5] or 2[0-4][0-9] or [01]?[0-9]?[0-9])).){3}(25[0-5] or 2[0-4][0-9] or [01]?[0-9]?[0-9])))		IPv6 end-point address, (subnet and host), of first address created by template - colon separated hexet format - mandatory

Attribute	Type	Range/Length/Pattern	Default	Description
prefix-length	uint16	1 .. 127		IPv6 subnet length of all addresses created by template - range [1, 127] - mandatory
count	uint16	0 .. 128		Total number of addresses that can be generated by this template - range is [0, 128] - 0 disables the template - mandatory

Loopback Ipv4 Interface

Configure IPv4 loopback interfaces that will be deployed in this Network Context

Configure IPv4 loopback interfaces that will be deployed in this Network Context. Each loopback entry will be deployed as an interface in the Network Context FIB. A loopback interface is defined by a "loopback-address" and a "prefix-length". The "loopback-address", masked by the "prefix-length", defines a loopback subnet. All host addresses in the subnet will be loopback addresses, including the "all 0's" and "all 1's" host addresses. "prefix-length" is optional, defaulting to 32 if not specified. Multicast addresses are invalid and will not be deployed. The "loopback-address", masked by "prefix-length", represents the routing prefix used to advertise reachability to the loopback addresses supported by the interface. "admin-state" controls the administrative state of the loopback interface. If set to "disabled", all traffic received on the loopback interface will be dropped, and no traffic will be originated from the interface.

XPath: [cna-interface-mgr:/interface-manager/network-context/loopback-ipv4-interface](#)

CLI Path: [interface-manager network-context loopback-ipv4-interface](#)

List Properties:

Keys: ipv4-loopback-interface-name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
ipv4-loopback-interface-name	string	[-A-Za-z0-9#_()<>{}]*		Name of IPv4 loopback interface, () - must be unique across all IPv4 and IPv6 loopback interfaces in the Network Context ipv4-loopback-interface-name is already being used in the loopback-ipv6-interface table. It the must be unique in this network-context
loopback-start-address	string	(([0-9] or [1-9][0-9] or 1[0-9][0-9] or 2[0-4][0-9] or 25[0-5]).){3}([0-9] or [1-9][0-9] or 1[0-9][0-9] or 2[0-4][0-9] or 25[0-5])		IPv4 address of loopback interface - start address for the range
loopback-end-address	string	(([0-9] or [1-9][0-9] or 1[0-9][0-9] or 2[0-4][0-9] or 25[0-5]).){3}([0-9] or [1-9][0-9] or 1[0-9][0-9] or 2[0-4][0-9] or 25[0-5])		IPv4 address of loopback interface - end address for the range
admin-state	enumeration	disabled enabled	enabled	Administrative state of loopback interface - optional - default is enabled
loopback-scope	enumeration	global pod dynamic	global	Scope of loopback interface - default is global
loopbacks-per-pod	uint16	1 .. 128	1	Number of loopback interfaces assigned to a ppe when the scope is type pod
tdef	boolean		false	Indicates if this loopback template is used for TDEF services - default is false
route-tag	uint32	0 .. max	0	Configure route-tag that is applied to all individual loopback instances
route-tag-aggregate	uint32	0 .. max	0	Configure an aggregate route-tag that is applied to the entire loopback pool
route-summary-prefix-length	uint32	0 .. 32	0	Configure a route summary prefix length to be used for the loopback ip pool
advertise-aggregate	boolean		false	Indicates if advertising aggregate route for loopback pool is allowed - default is false

Loopback Ipv6 Interface

Configure IPv6 loopback interfaces that will be deployed in this Network Context

Configure IPv6 loopback interfaces that will be deployed in this Network Context. Each loopback entry will be deployed as an interface in the Network Context

FIB. A loopback interface is defined by a "loopback-address" and a "prefix-length". The "loopback-address", masked by the "prefix-length", defines a loopback subnet. All host addresses in the subnet will be loopback addresses, including the "all 0's" and "all 1's" host addresses. "prefix-length" is optional, defaulting to 128 if not specified. Link-Local and Multicast addresses are invalid and will not be deployed. The "loopback-address", masked by "prefix-length", represents the routing prefix used to advertise reachability to the loopback addresses supported by the interface. "admin-state" controls the administrative state of the loopback interface. If set to "disabled", all traffic received on the loopback interface will be dropped, and no traffic will be originated from the interface.

XPath: [cna-interface-mgr:/interface-manager/network-context/loopback-ipv6-interface](#)

CLI Path: [interface-manager network-context loopback-ipv6-interface](#)

List Properties:

Keys: ipv6-loopback-interface-name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
ipv6-loopback-interface-name	string	[-A-Za-z0-9#_()<>{}]*		Name of IPv6 loopback interface, () - must be unique across all IPv4 and IPv6 loopback interfaces in the Network Context ipv6-loopback-interface-name is already being used in the loopback-ipv4-interface table. It the must be unique in this network-context
loopback-start-address	string	(((: or [0-9a-fA-F]{0,4})):([0-9a-fA-F]{0,4}):{0,5}(((([0-9a-fA-F]{0,4}):(: or [0-9a-fA-F]{0,4}))) or (((25[0-5] or 2[0-4][0-9] or [01]?[0-9]?[0-9])).{3}(25[0-5] or 2[0-4][0-9] or [01]?[0-9]?[0-9])))		IPv4 address of loopback interface - start address for the range
loopback-end-address	string	(((: or [0-9a-fA-F]{0,4})):([0-9a-fA-F]{0,4}):{0,5}(((([0-9a-fA-F]{0,4}):(: or [0-9a-fA-F]{0,4}))) or (((25[0-5] or 2[0-4][0-9] or [01]?[0-9]?[0-9])).{3}(25[0-5] or 2[0-4][0-9] or [01]?[0-9]?[0-9])))		IPv6 address of loopback interface - start address for the range
admin-state	enumeration	disabled enabled	enabled	Administrative state of loopback interface - optional - default is enabled
loopback-scope	enumeration	global pod dynamic	global	Scope of loopback interface - default is global
loopbacks-per-pod	uint16	1 .. 128	1	Number of loopback interfaces assigned to a ppe when the scope is type pod
tdef	boolean		false	Indicates if this loopback template is used for TDEF services - default is false
route-tag	uint32	0 .. max	0	Configure route-tag that is applied to all individual loopback instances
route-tag-aggregate	uint32	0 .. max	0	Configure an aggregate route-tag that is applied to the entire loopback pool
route-summary-prefix-length	uint32	0 .. 128	0	Configure a route summary prefix length to be used for the loopback ip pool
advertise-aggregate	boolean		false	Indicates if advertising aggregate route for loopback pool is allowed - default is false

Port Profile

Configure the Port Range Profile

XPath: [cna-interface-mgr:/interface-manager/port-profile](#)

CLI Path: [interface-manager port-profile](#)

List Properties:

Keys: profile-name

Maximum number of elements: 4096

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
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Attribute	Type	Range/Length/Pattern	Default	Description
profile-name	string	[-A-Za-z0-9#_()<>{}]*		Port Profile name - type is
start-port-range	uint32	0 .. 65535	1	Start value for the port range
end-port-range	uint32	0 .. 65535	1	End value for the port range
ports-per-pod	uint32	1 .. 500	1	Number of ports allocated to a ppe

L3RM Configuration

The module for L3RM.

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L3Rm

L3RM Service configuration

XPath: [l3rm:/l3rm](#)

CLI Path: [l3rm](#)

LB PPE Configuration

The module for LB PPE.

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Lb Ppe

LB PPE configuration

XPath: [lb-ppe:/lb-ppe](#)

CLI Path: [lb-ppe](#)

MME CC Configuration

This module defines the configuration for MME Call Control.

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Mme Cc

MME Call Control

XPath: [cna-mme-cc:/mme-cc](#)

CLI Path: [mme-cc](#)

Interface

Interface objects

XPath: [cna-mme-cc:/mme-cc/interface](#)

CLI Path: [mme-cc interface](#)

Nas

Non Access Stratum

Reference: 3GPP TS 24.301: Non-Access-Stratum (NAS) protocol for Evolved Packet System (EPS); Stage 3

XPath: [cna-mme-cc:/mme-cc/interface/nas](#)

CLI Path: [mme-cc interface nas](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
version	enumeration	13.10.0	15.7.0	3GPP spec applicable to this interface is TS 24.301. Default: 15.7.0
		15.2.0		
		15.7.0		

Mm

NAS EPS Mobility Management (EMM) interface

XPath: [cna-mme-cc:/mme-cc/interface/nas/mm](#)

CLI Path: [mme-cc interface nas mm](#)

Global

Global NAS EPS Mobility Management (EMM) attributes

XPath: [cna-mme-cc:/mme-cc/interface/nas/mm/global](#)

CLI Path: *mme-cc interface nas mm global*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
return-to-last-used-plmn-after-csfb	enumeration	enable disable	disable	Controls if the 'return to the last used PLMN after CSFB' function is supported as described in 3GPP spec TS23.272. Default: disable
csfb-window	uint8	1..5	2	Defines the amount of time allowed for CSFB HO/RAU after Extended Service Request. Default: 2; Unit: seconds
tai-list-max	uint8	1..16	1	Defines the number of TAIs which can be stored for a particular UE. Default: 1
guti-allocation-intra-tau	enumeration	enable disable	enable	Controls if the MME allocates a new GUTI during intra-MME TAU. Default: enable
location-update-intra-mme-tau	enumeration	standard always	standard	Specifies whether MME sends Location Update Request (LUR) to SGs VLR on every combined intra-MME TAU (always), or only when needed (standard). Setting the location update on all combined intra-MME TAU can increase traffic to the VLR, but does help with cases in which the VLR may be out of sync with respect to the UE location. As specified in TS 29.118 section 5.2.2.2, the LUR does not need to be sent to the VLR in all cases (typically needed if location changes), but the configuration option of "always" will override the VLR-Reliable condition checks and force the LUR to be sent. This attribute does not affect the LUR sending behavior on inter-MME TAU or periodic TAU. Default: standard
attach-without-imei	enumeration	allow reject	allow	Controls if MME allows attach when UE does not provide IMEISV. For MME the IMEISV will be requested in the Security Mode Command and the Identity Request messages, but the IMEISV is optional in the responses as defined by TS 24.301 sections 5.4.3 and 5.4.4. Attach without IMEI means that legal intercept based on IMEI may not always be possible. Default: allow
mm-info-triggers	leaf-list of enumeration	attach inter-mme-tau dst-change config-change timezone-change		Triggers that will cause the MME to send the EMM Information message. Valid values are: a list containing 'attach','inter-mme-tau','dst-change','config-change','timezone-change'
id-req-to-old-mme-for-5g-mm-registered-ue	enumeration	enable disable	enable	Controls if the new MME should send Identification Request message to old MME, when a UE sends Attach Request with unknown GUTI with type as native and UE status with N1-mode as 5G-MM-Registered. If enabled, new MME sends Identification Request message to old MME, otherwise new MME sends Identity Request to UE. Default: enable
full-network-name	string	[A-Za-z0-9 '() +, - . / : = ?]*	3gppnetwork	Defines the full length name of the network. Default: 3gppnetwork; max 127 characters from upper and lower case letters, digits, space, apostrophe, and any of '() +, - . / : = ?'
short-network-name	string	[A-Za-z0-9 '() +, - . / : = ?]*	3gppnetwork	Defines the abbreviated name of the network. Default: 3gppnetwork; max 32 characters from upper and lower case letters, digits, space, apostrophe, and any of '() +, - . / : = ?'

Signaling Rate Control

Signaling rate parameters for mobility management congestion control as defined by 3GPP TS 24.301 section 5.3.9.

XPath: *cna-mme-cc:/mme-cc/interface/nas/mm/global/signaling-rate-control*

CLI Path: *mme-cc interface nas mm global signaling-rate-control*

Attach

Signaling rate control parameters for Attach procedure

XPath: *cna-mme-cc:/mme-cc/interface/nas/mm/global/signaling-rate-control/attach*

CLI Path: *mme-cc interface nas mm global signaling-rate-control attach*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
rate-limit	union	['uint16,1..1000', 'Enumerations: disable ']	disable	Defines the maximum rate per second at which the applicable messages may be processed by MME Call Control microservice. A value of disable indicates that the rate control functionality is disabled. Default: disable; Unit: messages per second
action	enumeration	ignore reject	reject	Defines the action to be taken when the signaling rate control limit is exceeded. Default: reject
t3346-low	union	['uint16,2..11159', 'Enumerations: disable ']	4	Defines the lower end of the range for the T3346 value sent to the UE when the Attach is rejected due to signaling rate control. A random value between t3346-low and t3346-high is included in the Attach reject message as defined by 3GPP TS 24.301 section 8.2.3. The value disable indicates that the T3346 value is not included in the Attach reject message. Default: 4; Unit: seconds
t3346-high	uint16	3..11160	16	Defines the higher end of the range for the T3346 value sent to the UE when the attach is rejected due to signaling rate control. A random value between t3346-low and t3346-high is included in the Attach reject message as defined by 3GPP TS 24.301 section 8.2.3. Default: 16; Unit: seconds The value must be greater than 't3346-low'

Emm Cause Mapping

NAS EMM Cause Code Mapping

XPath: [cna-mme-cc;/mme-cc/interface/nas/mm/global/emm-cause-mapping](#)**CLI Path:** [mme-cc interface nas mm global emm-cause-mapping](#)**List Properties:**

Keys: error

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
error	enumeration	hss-user-unknown hss-unknown-subscription-gprs-data hss-unknown-subscription-no-gprs-data hss-rat-not-allowed hss-roaming-not-allowed-odb-vplmn hss-roaming-not-allowed-odb-hplmn hss-roaming-not-allowed-odb-all-apn hss-roaming-not-allowed eir-equipment-unknown hss-authentication-data-unavailable hss-access-restricted hss-operator-determined-barring hss-unknown dns-mapping-failure hss-authorization-rejected hss-application-unsupported hss-command-unsupported hss-out-of-space hss-too-busy hss-unable-to-comply hss-unknown-serving-node hss-roaming-restricted-in-mme-due-to-unsupported-feature		EMM error description

Attribute	Type	Range/Length/Pattern	Default	Description
cause-code	union	['string,([a-zA-Z]+(-[a-zA-Z]+)*-)?([1-9]0 or [1-9]{1,2} or 1[0-9]{1,2} or 2[0-4][0-9] or 25[0-5])', 'Enumerations: congestion-22 cs-domain-not-available-18 cs-domain-temp-not-available-39 csfb-not-allowed-38 eps-and-non-eps-service-not-allowed-8 eps-service-not-allowed-7 eps-service-not-allowed-in-this-plmn-14 esm-failure-19 illegal-me-6 illegal-ue-3 implicitly-detach-10 imsi-unknown-in-hss-2 mac-failure-20 msc-temp-not-reachable-16 network-failure-17 no-eps-bearer-context-activated-40 no-suitable-cells-in-ta-15 not-authorized-for-this-csg-25 plmn-not-allowed-11 roaming-not-allowed-in-this-ta-13 security-mode-rejected-24 sync-failure-21 tracking-area-not-allowed-12 ue-id-cannot-be-derived-9 ue-security-cap-mismatch-23 ']		EMM cause code default varies per error

Default

Default NAS EPS Mobility Management (EMM) profile attributes

Both 'full-network-name' and 'short-network-name' must be configured together

XPath: *cna-mme-cc:/mme-cc/interface/nas/mm/default*

CLI Path: *mme-cc interface nas mm default*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
full-network-name	string	1..127		Full length name of the network. Defined by TS 24.301 section 8.2.13. Reverts to value configured under 'mme-cc interface nas mm global' if not configured here.
short-network-name	string	1..32		Abbreviated name of the network. Defined by TS 24.301 section 8.2.13. Reverts to value configured under 'mme-cc interface nas mm global' if not configured here.
ue-retry-delay-t3402-value	uint8	1..60	12	Delay time before the UE retries an attach or tracking area update after five consecutive failed attempts. Default: 12; Unit: minutes
ue-periodic-tau-t3412-value	uint8	1..186	54	Periodic tracking area update timer. Default: 54; Unit: minutes; Note: Very low values of this timer will result in frequent Periodic TAU procedures initiated by the UE
ue-periodic-tau-t3412-source	enumeration	local-config hss-subscription	local-config	Source for the periodic tau timer value sent to the UE as defined in 3GPP NAS standard section 5.5.1 of TS 24.301. If a subscribed value is not yet received from the HSS, then the local-config will be used. Default: local-config
nwk-init-detach-t3422	uint8	1..15	6	Defines the duration the MME waits for a response to a detach request sent to the UE. Default: 6; Unit: seconds

Attribute	Type	Range/Length/Pattern	Default	Description
nwk-init-detach-retry-n3422	uint8	0..5	2	Defines the number of retries the MME performs for the network initiated detach procedure. Default: 2
guti-allocation-t3450	uint8	1..15	6	Defines the duration the MME waits for a response to an attach accept, tau accept with new GUTI or a GUTI reallocation command sent to the UE. Default: 6; Unit: seconds
guti-allocation-retry-n3450	uint8	0..5	3	Defines the number of retries the MME performs for an attach accept, tau accept with new GUTI or a GUTI reallocation command. Default: 3
guti-lifetime	uint8	1..168	12	Defines the life time of a current GUTI. Default: 12; Unit: hours
identity-t3470	uint8	1..15	6	Defines the duration the MME waits for a response to the identity request sent to the UE. Default: 6; Unit: seconds
identity-retry-n3470	uint8	0..5	2	Defines the number of retries the MME performs for the identity procedure. Default: 2
mobile-reachable-timer	union	['uint16,30..300', 'Enumerations: t3412plus4 t3412times2plus1 ']	t3412plus4	Defines the duration the UE can stay ECM-IDLE without messaging the MME. Default: t3412plus4; Unit: minutes The value must be greater than 'ue-periodic-tau-t3412-value'
mobile-reachable-timer-extended	union	['uint32,30..18790', 'Enumerations: t3412extplus4 ']	t3412extplus4	Defines the duration the UE can stay ECM-IDLE without messaging the MME. Default: t3412extplus4; Unit: minutes
src-mme-resource-release-timer	uint8	0..30	15	Defines the amount of time the MME waits before deactivating the bearers on the old MME during a SGW relocation procedure. Default: 15; Unit: seconds
cancel-location-wait-timer	uint8	10..30	10	Defines the amount of time the MME waits for the cancel location request from the HSS during an inter-MME TAU on the old MME. Default: 10; Unit: seconds
ue-service-reject-t3442-value	uint16	2..300	60	Defines the timer for UE delay after a service reject. Default: 60; Unit: seconds
ta-list-allocation	enumeration	unrestricted within-plmn within-la	unrestricted	Defines the Tracking Area (TA) list allocation method. Default: unrestricted
cancel-location-detach-cause	enumeration	none imsi-unknown-in-hss-2 illegal-ue-3 illegal-me-6 eps-service-not-allowed-7 eps-and-non-eps-service-not-allowed-8 plmn-not-allowed-11 tracking-area-not-allowed-12 roaming-not-allowed-in-this-ta-13 eps-service-not-allowed-in-this-plmn-14 no-suitable-cells-in-ta-15	eps-service-not-allowed-7	Defines the detach cause used when a cancel location request is received with cancellation type set as subscription withdrawn. Default: eps-service-not-allowed-7

Attribute	Type	Range/Length/Pattern	Default	Description
csfb-not-preferred-trigger	enumeration	none ims-enable ims-and-emg-enable	none	Defines when to include the CS Fallback Not Preferred indication in the Additional Update Result IE in an Attach or TAU (Tracking Area Update) Accept message during a combined attach procedure or combined TAU procedure. Refer to 3GPP spec TS24.301 for further details on the usage of the CS Fallback Not Preferred indication. Default: none
emm-timer-increment-nb	uint16	0..300	0	NAS EMM timer increment for UE in NB-S1 mode. Defines the amount to increment EMM timers for a UE using NB-S1 (narrowband) mode, as described in TS 24.301 section 4.7. This affects the following EMM timers: T3422, T3450, T3460, T3470. Default: 0; Unit: seconds
equivalent-plmn	leaf-list of string	(([0-9]{5,6}))		List of PLMNs considered equivalent Maximum number of elements: 15

Power Saving Control

Power saving control parameters

XPath: [cna-mme-cc:/mme-cc/interface/nas/mm/default/power-saving-control](#)

CLI Path: [mme-cc interface nas mm default power-saving-control](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
concurrent-psm-edrx	enumeration	edrx-only psm-only	edrx-only	PSM with eDRX. Default: edrx-only

Psm Active Time T3324

PSM timer configuration

XPath: [cna-mme-cc:/mme-cc/interface/nas/mm/default/power-saving-control/psm-active-time-t3324](#)

CLI Path: [mme-cc interface nas mm default power-saving-control psm-active-time-t3324](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
status	enumeration	enable disable	disable	Controls support of PSM. Default: disable
min-value	uint16	0..11160	60	Defines T3324 timer minimum value. Default: 60; Unit: seconds
max-value	uint16	0..11160	11160	Defines T3324 timer maximum value. Default: 11160; Unit: seconds The value of 'max-value' must be greater than the value of 'min-value'

T3412 Extended

Periodic Tracking Area Update (TAU) timer extended configuration

XPath: [cna-mme-cc:/mme-cc/interface/nas/mm/default/power-saving-control/t3412-extended](#)

CLI Path: [mme-cc interface nas mm default power-saving-control t3412-extended](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
min-value	union	['uint16,1..34556', 'Enumerations: t3412plus10 ']	t3412plus10	Periodic TAU timer extended minimum value. Default: t3412plus10; Unit: minutes; Note: Very low values of this timer will result in frequent Periodic TAU procedures initiated by the UE The value of 'min-value' must be greater than or equal to the value of 'mme-cc interface nas mm ue-periodic-tau-t3412-value'

Attribute	Type	Range/Length/Pattern	Default	Description
max-value	uint16	1..34556	34556	Periodic TAU timer extended maximum value. Default: 34556; Unit: minutes; Note: Very low values of this timer will result in frequent Periodic TAU procedures initiated by the UE The value of 'max-value' must be greater than the value of 'min-value' The value of 'max-value' must be greater than the value of 'mme-cc interface nas mm ue-periodic-tau-t3412-value'
default-value	uint16	1..34556	190	Periodic TAU timer extended. Default: 190; Unit: minutes; Note: Very low values of this timer will result in frequent Periodic TAU procedures initiated by the UE The value of 'default-value' must be greater than or equal to the value of 'min-value' The value of 'default-value' must be greater than or equal to the value of 'mme-cc interface nas mm ue-periodic-tau-t3412-value' The value of 'default-value' must be less than or equal to the value of 'max-value'
status	enumeration	enable disable	enable	Controls whether the optional t3412 extended timer is negotiated by MME. During ATTACH and TAU procedures, MME includes the T3412 extended timer in the ATTACH or TAU accept, if this control is enabled. The extended T3412 timer value is chosen based on the following criteria in this order: HSS subscription, the UE requested extended T3412 when PSM is negotiated, or the MME default configuration default-value. Default: enable

Extended Drx

Extended Discontinuous Reception (eDRX) parameters

XPath: [cna-mme-cc:/mme-cc/interface/nas/mm/default/power-saving-control/extended-drx](#)

CLI Path: [mme-cc interface nas mm default power-saving-control extended-drx](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
status	enumeration	enable disable	disable	Controls support of eDRX. Default: disable
max-paging-occasions	uint16	1..5	1	Defines the maximum number of eDRX paging occasions that can be attempted before failing the paging procedure. If the configured emm-paging-profile retries are exhausted before the paging-occasions are exhausted, then the procedure fails at that point. The paging retries restarts with the configured paging retry targets for each paging occasion, with one occasion per paging hyperframe. Default: 1

Edrx Nb

eDRX cycle length duration in seconds for devices in NB-S1 mode

XPath: [cna-mme-cc:/mme-cc/interface/nas/mm/default/power-saving-control/extended-drx/edrx-nb](#)

CLI Path: [mme-cc interface nas mm default power-saving-control extended-drx edrx-nb](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
min-value	enumeration	20.48 40.96 81.92 163.84 327.68 655.36 1310.72 2621.44 5242.88 10485.76	20.48	Minimum allowed eDRX cycle length duration for NB-S1 mode. Default: 20.48; Unit: seconds The value of 'min-value' must be less than or equal to the value of 'max-value'

Attribute	Type	Range/Length/Pattern	Default	Description
max-value	enumeration	20.48	10485.76	Maximum allowed eDRX cycle length duration for NB-S1 mode. Default: 10485.76; Unit: seconds The value of 'max-value' must be greater than or equal to the value of 'min-value'
		40.96		
		81.92		
		163.84		
		327.68		
		655.36		
		1310.72		
		2621.44		
		5242.88		
		10485.76		
default-value	enumeration	20.48	20.48	The default eDRX cycle length duration for NB-S1 mode. Default: 20.48; Unit: seconds The default value must be greater than or equal to the value of 'min-value' and less than or equal to the value of 'max-value'
		40.96		
		81.92		
		163.84		
		327.68		
		655.36		
		1310.72		
		2621.44		
		5242.88		
		10485.76		

Ptw Nb

Paging time window in seconds for devices in NB-S1 mode

XPath: *cna-mme-cc:/mme-cc/interface/nas/mm/default/power-saving-control/extended-drx/ptw-nb*

CLI Path: *mme-cc interface nas mm default power-saving-control extended-drx ptw-nb*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
min-value	enumeration	2.56	5.12	Minimum allowed paging time window length for NB-S1 mode. Default: 5.12; Unit: seconds The value of 'min-value' must be less than or equal to the value of 'max-value'
		5.12		
		7.68		
		10.24		
		12.8		
		15.36		
		17.92		
		20.48		
		23.04		
		25.6		
		28.16		
		30.72		
		33.28		
		35.84		
		38.4		
max-value	enumeration	40.96	40.96	Maximum allowed paging time window length for NB-S1 mode. Default: 40.96; Unit: seconds The value of 'max-value' must be greater than or equal to the value of 'min-value'
		2.56		
		5.12		
		7.68		
		10.24		
		12.8		
		15.36		
		17.92		
		20.48		
		23.04		
		25.6		
		28.16		
		30.72		
		33.28		
		35.84		
		38.4		
		40.96		

Attribute	Type	Range/Length/Pattern	Default	Description
default-value	enumeration	2.56	5.12	Default paging time window length for NB-S1 mode. Default: 5.12; Unit: seconds The default value must be greater than or equal to the value of 'min-value' and less than or equal to the value of 'max-value'
		5.12		
		7.68		
		10.24		
		12.8		
		15.36		
		17.92		
		20.48		
		23.04		
		25.6		
		28.16		
		30.72		
		33.28		
		35.84		
		38.4		
		40.96		

Profile

List of NAS EPS Mobility Management (EMM) profiles

Both 'full-network-name' and 'short-network-name' must be configured together

XPath: *cna-mme-cc:/mme-cc/interface/nas/mm/profile*

CLI Path: *mme-cc interface nas mm profile*

List Properties:

Keys: name

Maximum number of elements: 100

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{1,64})		Name of the MM Profile
full-network-name	string	1..127		Full length name of the network. Defined by TS 24.301 section 8.2.13. Reverts to value configured under 'mme-cc interface nas mm global' if not configured here.
short-network-name	string	1..32		Abbreviated name of the network. Defined by TS 24.301 section 8.2.13. Reverts to value configured under 'mme-cc interface nas mm global' if not configured here.
ue-retry-delay-t3402-value	uint8	1..60	12	Delay time before the UE retries an attach or tracking area update after five consecutive failed attempts. Default: 12; Unit: minutes
ue-periodic-tau-t3412-value	uint8	1..186	54	Periodic tracking area update timer. Default: 54; Unit: minutes; Note: Very low values of this timer will result in frequent Periodic TAU procedures initiated by the UE
ue-periodic-tau-t3412-source	enumeration	local-config hss-subscription	local-config	Source for the periodic tau timer value sent to the UE as defined in 3GPP NAS standard section 5.5.1 of TS 24.301. If a subscribed value is not yet received from the HSS, then the local-config will be used. Default: local-config
nwk-init-detach-t3422	uint8	1..15	6	Defines the duration the MME waits for a response to a detach request sent to the UE. Default: 6; Unit: seconds
nwk-init-detach-retry-n3422	uint8	0..5	2	Defines the number of retries the MME performs for the network initiated detach procedure. Default: 2

Attribute	Type	Range/Length/Pattern	Default	Description
guti-allocation-t3450	uint8	1..15	6	Defines the duration the MME waits for a response to an attach accept, tau accept with new GUTI or a GUTI reallocation command sent to the UE. Default: 6; Unit: seconds
guti-allocation-retry-n3450	uint8	0..5	3	Defines the number of retries the MME performs for an attach accept, tau accept with new GUTI or a GUTI reallocation command. Default: 3
guti-lifetime	uint8	1..168	12	Defines the life time of a current GUTI. Default: 12; Unit: hours
identity-t3470	uint8	1..15	6	Defines the duration the MME waits for a response to the identity request sent to the UE. Default: 6; Unit: seconds
identity-retry-n3470	uint8	0..5	2	Defines the number of retries the MME performs for the identity procedure. Default: 2
mobile-reachable-timer	union	['uint16,30..300', 'Enumerations: t3412plus4 t3412times2plus1 ']	t3412plus4	Defines the duration the UE can stay ECM-IDLE without messaging the MME. Default: t3412plus4; Unit: minutes The value must be greater than 'ue-periodic-tau-t3412-value'
mobile-reachable-timer-extended	union	['uint32,30..18790', 'Enumerations: t3412extplus4 ']	t3412extplus4	Defines the duration the UE can stay ECM-IDLE without messaging the MME. Default: t3412extplus4; Unit: minutes
src-mme-resource-release-timer	uint8	0..30	15	Defines the amount of time the MME waits before deactivating the bearers on the old MME during a SGW relocation procedure. Default: 15; Unit: seconds
cancel-location-wait-timer	uint8	10..30	10	Defines the amount of time the MME waits for the cancel location request from the HSS during an inter-MME TAU on the old MME. Default: 10; Unit: seconds
ue-service-reject-t3442-value	uint16	2..300	60	Defines the timer for UE delay after a service reject. Default: 60; Unit: seconds
ta-list-allocation	enumeration	unrestricted within-plmn within-la	unrestricted	Defines the Tracking Area (TA) list allocation method. Default: unrestricted
cancel-location-detach-cause	enumeration	none imsi-unknown-in-hss-2 illegal-ue-3 illegal-me-6 eps-service-not-allowed-7 eps-and-non-eps-service-not-allowed-8 plmn-not-allowed-11 tracking-area-not-allowed-12 roaming-not-allowed-in-this-ta-13 eps-service-not-allowed-in-this-plmn-14 no-suitable-cells-in-ta-15	eps-service-not-allowed-7	Defines the detach cause used when a cancel location request is received with cancellation type set as subscription withdrawn. Default: eps-service-not-allowed-7

Attribute	Type	Range/Length/Pattern	Default	Description
csfb-not-preferred-trigger	enumeration	none ims-enable ims-and-emg-enable	none	Defines when to include the CS Fallback Not Preferred indication in the Additional Update Result IE in an Attach or TAU (Tracking Area Update) Accept message during a combined attach procedure or combined TAU procedure. Refer to 3GPP spec TS24.301 for further details on the usage of the CS Fallback Not Preferred indication. Default: none
emm-timer-increment-nb	uint16	0..300	0	NAS EMM timer increment for UE in NB-S1 mode. Defines the amount to increment EMM timers for a UE using NB-S1 (narrowband) mode, as described in TS 24.301 section 4.7. This affects the following EMM timers: T3422, T3450, T3460, T3470. Default: 0; Unit: seconds
equivalent-plmn	leaf-list of string	(([0-9]{5,6}))		List of PLMNs considered equivalent Maximum number of elements: 15

Power Saving Control

Power saving control parameters

XPath: [cna-mme-cc:/mme-cc/interface/nas/mm/profile/power-saving-control](#)

CLI Path: [mme-cc interface nas mm profile power-saving-control](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
concurrent-psm-edrx	enumeration	edrx-only psm-only	edrx-only	PSM with eDRX. Default: edrx-only

Psm Active Time T3324

PSM timer configuration

XPath: [cna-mme-cc:/mme-cc/interface/nas/mm/profile/power-saving-control/psm-active-time-t3324](#)

CLI Path: [mme-cc interface nas mm profile power-saving-control psm-active-time-t3324](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
status	enumeration	enable disable	disable	Controls support of PSM. Default: disable
min-value	uint16	0..11160	60	Defines T3324 timer minimum value. Default: 60; Unit: seconds
max-value	uint16	0..11160	11160	Defines T3324 timer maximum value. Default: 11160; Unit: seconds The value of 'max-value' must be greater than the value of 'min-value'

T3412 Extended

Periodic Tracking Area Update (TAU) timer extended configuration

XPath: [cna-mme-cc:/mme-cc/interface/nas/mm/profile/power-saving-control/t3412-extended](#)

CLI Path: [mme-cc interface nas mm profile power-saving-control t3412-extended](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
min-value	union	['uint16,1..34556', 'Enumerations: t3412plus10 ']	t3412plus10	Periodic TAU timer extended minimum value. Default: t3412plus10; Unit: minutes; Note: Very low values of this timer will result in frequent Periodic TAU procedures initiated by the UE The value of 'min-value' must be greater than or equal to the value of 'mme-cc interface nas mm ue-periodic-tau-t3412-value'

Attribute	Type	Range/Length/Pattern	Default	Description
max-value	uint16	1..34556	34556	Periodic TAU timer extended maximum value. Default: 34556; Unit: minutes; Note: Very low values of this timer will result in frequent Periodic TAU procedures initiated by the UE The value of 'max-value' must be greater than the value of 'min-value' The value of 'max-value' must be greater than the value of 'mme-cc interface nas mm ue-periodic-tau-t3412-value'
default-value	uint16	1..34556	190	Periodic TAU timer extended. Default: 190; Unit: minutes; Note: Very low values of this timer will result in frequent Periodic TAU procedures initiated by the UE The value of 'default-value' must be greater than or equal to the value of 'min-value' The value of 'default-value' must be greater than or equal to the value of 'mme-cc interface nas mm ue-periodic-tau-t3412-value' The value of 'default-value' must be less than or equal to the value of 'max-value'
status	enumeration	enable disable	enable	Controls whether the optional t3412 extended timer is negotiated by MME. During ATTACH and TAU procedures, MME includes the T3412 extended timer in the ATTACH or TAU accept, if this control is enabled. The extended T3412 timer value is chosen based on the following criteria in this order: HSS subscription, the UE requested extended T3412 when PSM is negotiated, or the MME default configuration default-value. Default: enable

Extended Drx

Extended Discontinuous Reception (eDRX) parameters

XPath: [cna-mme-cc:/mme-cc/interface/nas/mm/profile/power-saving-control/extended-drx](#)

CLI Path: [mme-cc interface nas mm profile power-saving-control extended-drx](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
status	enumeration	enable disable	disable	Controls support of eDRX. Default: disable
max-paging-occasions	uint16	1..5	1	Defines the maximum number of eDRX paging occasions that can be attempted before failing the paging procedure. If the configured emm-paging-profile retries are exhausted before the paging-occasions are exhausted, then the procedure fails at that point. The paging retries restarts with the configured paging retry targets for each paging occasion, with one occasion per paging hyperframe. Default: 1

Edrx Nb

eDRX cycle length duration in seconds for devices in NB-S1 mode

XPath: [cna-mme-cc:/mme-cc/interface/nas/mm/profile/power-saving-control/extended-drx/edrx-nb](#)

CLI Path: [mme-cc interface nas mm profile power-saving-control extended-drx edrx-nb](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
min-value	enumeration	20.48	20.48	Minimum allowed eDRX cycle length duration for NB-S1 mode. Default: 20.48; Unit: seconds The value of 'min-value' must be less than or equal to the value of 'max-value'
		40.96		
		81.92		
		163.84		
		327.68		
		655.36		
		1310.72		
		2621.44		
		5242.88		
		10485.76		

Attribute	Type	Range/Length/Pattern	Default	Description
max-value	enumeration	20.48	10485.76	Maximum allowed eDRX cycle length duration for NB-S1 mode. Default: 10485.76; Unit: seconds The value of 'max-value' must be greater than or equal to the value of 'min-value'
		40.96		
		81.92		
		163.84		
		327.68		
		655.36		
		1310.72		
		2621.44		
		5242.88		
		10485.76		
default-value	enumeration	20.48	20.48	The default eDRX cycle length duration for NB-S1 mode. Default: 20.48; Unit: seconds The default value must be greater than or equal to the value of 'min-value' and less than or equal to the value of 'max-value'
		40.96		
		81.92		
		163.84		
		327.68		
		655.36		
		1310.72		
		2621.44		
		5242.88		
		10485.76		

Ptw Nb

Paging time window in seconds for devices in NB-S1 mode

XPath: *cna-mme-cc:/mme-cc/interface/nas/mm/profile/power-saving-control/extended-drx/ptw-nb*

CLI Path: *mme-cc interface nas mm profile power-saving-control extended-drx ptw-nb*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
min-value	enumeration	2.56	5.12	Minimum allowed paging time window length for NB-S1 mode. Default: 5.12; Unit: seconds The value of 'min-value' must be less than or equal to the value of 'max-value'
		5.12		
		7.68		
		10.24		
		12.8		
		15.36		
		17.92		
		20.48		
		23.04		
		25.6		
		28.16		
		30.72		
		33.28		
		35.84		
		38.4		
max-value	enumeration	40.96	40.96	Maximum allowed paging time window length for NB-S1 mode. Default: 40.96; Unit: seconds The value of 'max-value' must be greater than or equal to the value of 'min-value'
		2.56		
		5.12		
		7.68		
		10.24		
		12.8		
		15.36		
		17.92		
		20.48		
		23.04		
		25.6		
		28.16		
		30.72		
		33.28		
		35.84		
		38.4		
		40.96		

Attribute	Type	Range/Length/Pattern	Default	Description
default-value	enumeration	2.56	5.12	Default paging time window length for NB-S1 mode. Default: 5.12; Unit: seconds The default value must be greater than or equal to the value of 'min-value' and less than or equal to the value of 'max-value'
		5.12		
		7.68		
		10.24		
		12.8		
		15.36		
		17.92		
		20.48		
		23.04		
		25.6		
		28.16		
		30.72		
		33.28		
		35.84		
		38.4		
		40.96		

Security

NAS Security interface

XPath: [cna-mme-cc:/mme-cc/interface/nas/security](#)

CLI Path: [mme-cc interface nas security](#)

Global

Global NAS Security attributes

XPath: [cna-mme-cc:/mme-cc/interface/nas/security/global](#)

CLI Path: [mme-cc interface nas security global](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
vector-retrieval-failure	enumeration	ignore fail	ignore	Defines what the MME does when there is a failure to retrieve authentication vectors for authentication during a mobility management procedure. Default: ignore
imeisv-req-sec-mode-cmd	enumeration	on off	on	Controls if the MME is allowed to request IMEISV during the security mode command procedure. Default: on
nas-token-window	uint8	0..10	5	Defines the NAS token window. Default: 5

Default

Default NAS Security attributes

XPath: [cna-mme-cc:/mme-cc/interface/nas/security/default](#)

CLI Path: [mme-cc interface nas security default](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
security-context-lifetime	union	['uint8,1..168', 'Enumerations: disable ']	disable	Defines the lifetime of a security context. Default: disable; Unit: hours
security-t3460	uint8	1..15	6	Defines the duration the MME waits for a response to an authentication request or a security mode command. Default: 6; Unit: seconds
security-retry-n3460	uint8	0..5	2	Defines the number of retries the MME performs for the authentication and security mode procedures. Default: 2

Attribute	Type	Range/Length/Pattern	Default	Description
optional-authentication	leaf-list of enumeration	service-request detach tau mme-relocation		Defines when the MME performs optional authentication procedures. Valid values are one or more of the following: service-request,mme-relocation,detach,tau
ciphering-algorithms	leaf-list of enumeration	eea0 eea1 eea2	eea1 eea2 eea0	Defines the list of ciphering algorithms that can be used for NAS ciphering. Valid values in any order or combination, with order indicating preference are: eea0,eea1,eea2 Ordered by: user
integrity-algorithms	leaf-list of enumeration	null eia1 eia2	eia1 eia2 null	Defines the list of integrity algorithms that can be used for NAS integrity protection. Valid values in any order or combination, with order indicating preference are: null,eia1,eia2 Ordered by: user

Profile

List of NAS Security profiles

XPath: *cna-mme-cc:/mme-cc/interface/nas/security/profile*

CLI Path: *mme-cc interface nas security profile*

List Properties:

Keys: name

Maximum number of elements: 100

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{1,64})		Name of the Security Profile
security-context-lifetime	union	['uint8,1..168', 'Enumerations: disable ']	disable	Defines the lifetime of a security context. Default: disable; Unit: hours
security-t3460	uint8	1..15	6	Defines the duration the MME waits for a response to an authentication request or a security mode command. Default: 6; Unit: seconds
security-retry-n3460	uint8	0..5	2	Defines the number of retries the MME performs for the authentication and security mode procedures. Default: 2
optional-authentication	leaf-list of enumeration	service-request detach tau mme-relocation		Defines when the MME performs optional authentication procedures. Valid values are one or more of the following: service-request,mme-relocation,detach,tau
ciphering-algorithms	leaf-list of enumeration	eea0 eea1 eea2	eea1 eea2 eea0	Defines the list of ciphering algorithms that can be used for NAS ciphering. Valid values in any order or combination, with order indicating preference are: eea0,eea1,eea2 Ordered by: user
integrity-algorithms	leaf-list of enumeration	null eia1 eia2	eia1 eia2 null	Defines the list of integrity algorithms that can be used for NAS integrity protection. Valid values in any order or combination, with order indicating preference are: null,eia1,eia2 Ordered by: user

Sm

NAS EPS Session Management (ESM) interface

XPath: *cna-mme-cc:/mme-cc/interface/nas/sm*

CLI Path: *mme-cc interface nas sm*

Global

Global NAS EPS Session Management (ESM) attributes

XPath: *cna-mme-cc:/mme-cc/interface/nas/sm/global*

CLI Path: *mme-cc interface nas sm global*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
topological-search	enumeration	enable disable	enable	Controls if the MME will favor topologically proximate and/or colocated PGW/SGW. Default: enable
dual-addr-bearer	enumeration	enable disable	enable	Controls the dual address bearer functionality. Default: enable
use-pgwid-from-hss	enumeration	enable disable	enable	Controls if the MME uses the PGW ID from the HSS during PGW selection. Default: enable
gw-reselection-retries	uint8	0..10	2	Defines the maximum number of GW retries for reselection. Default: 2
max-bearers-per-ue	uint8	1..7	7	Defines the maximum number of dedicated bearers a UE can establish. Default: 7
max-pdn-conn-per-ue	uint8	1..8	8	Defines the maximum number of PDN connections a UE can establish. Default: 8

Nodal Qos

Nodal QoS attributes

XPath: *cna-mme-cc:/mme-cc/interface/nas/sm/global/nodal-qos***CLI Path:** *mme-cc interface nas sm global nodal-qos*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
single-qci1-bearer	enumeration	enable disable	disable	Controls the ability to restrict the number of QCI=1 bearers per UE (enable means only 1 QCI=1 bearer is allowed). Default: disable

Default

Default NAS EPS Session Management (ESM) attributes

XPath: *cna-mme-cc:/mme-cc/interface/nas/sm/default***CLI Path:** *mme-cc interface nas sm default*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
activate-bearer-t3485	uint8	1..30	5	Defines the time the MME waits for a response to the default or dedicated bearer activate request sent to the UE. Default: 5; Unit: seconds
activate-bearer-retry-n3485	uint8	0..5	2	Defines the number of times the MME retries the default or dedicated bearer activate request to the UE. Default: 2
modify-bearer-t3486	uint8	1..30	5	Defines the time the MME waits for the response to the bearer modification request sent to the UE. Default: 5; Unit: seconds
modify-bearer-retry-n3486	uint8	0..5	2	Defines the number of times the MME retries the bearer modification request to the UE. Default: 2
esm-info-t3489	uint8	1..15	4	Defines the time the MME waits for a response to the ESM information request sent to the UE. Default: 4; Unit: seconds
esm-info-retry-n3489	uint8	0..5	2	Defines the number of times the MME retries the ESM information request to the UE. Default: 2
deactivate-bearer-t3495	uint8	1..15	5	Defines the time the MME waits for a response to the bearer deactivate request sent to the UE. Default: 5; Unit: seconds
deactivate-bearer-retry-n3495	uint8	0..5	2	Defines the number of times the MME retries the bearer deactivate request to the UE. Default: 2

Attribute	Type	Range/Length/Pattern	Default	Description
deactivate-bearer-page-fail	enumeration	enable disable	disable	Controls whether the MME deactivates the GBR bearers upon page to the UE failure. Default: disable
deactivate-bearer-enodeb-fail	enumeration	enable disable	disable	Controls whether the MME deactivates the GBR bearers if the eNodeB fails or resets. Default: disable
page-ue-for-deactivation	enumeration	enable disable	enable	Page UE for deactivation. Default enable
delay-gbr-bearer-deactivation-timer	union	['uint8,1..30', 'Enumerations: disable ']	disable	Delay GBR bearer deactivation timer. Valid values are: disable, 1..30; Default: disable; Unit: seconds
esm-timer-increment-nb	uint16	0..300	0	NAS ESM timer increment for UE in NB-S1 mode. This attribute specifies the amount to increment ESM timers for a UE using NB-S1 (narrowband) mode, as described in TS 24.301 section 4.7. This affects the following ESM timers: T3485, T3486, T3489, T3495. Default: 0; Unit: seconds

Ext Qci Conversion

Mapping of extended QoS class identifier (QCI) values to standardized QCI values.

XPath: [cna-mme-cc:/mme-cc/interface/nas/sm/default/mme-nas-esm-ext-qci-conversion](#)

CLI Path: [mme-cc interface nas sm default ext-qci-conversion](#)

List Properties:

Keys: ext-qci

Maximum number of elements: 134

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
ext-qci	uint8	65	66	69
std-qci	uint8	1..9		Standardized QCI value mapped from the extended QCI value; Range: 1-9

Paging

Paging control

XPath: [cna-mme-cc:/mme-cc/interface/nas/paging](#)

CLI Path: [mme-cc interface nas paging](#)

Global

Global paging attributes

XPath: [cna-mme-cc:/mme-cc/interface/nas/paging/global](#)

CLI Path: [mme-cc interface nas paging global](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
ppf-reset-timer	union	['uint16,1..600', 'Enumerations: disable ']	600	Defines the paging proceed flag reset timer. Default: 600; Unit: seconds
max-ppf-failure-threshold	uint8	0..10	10	Defines the threshold that, if met or exceeded due to an accumulation of 'mme-cc interface nas paging profile-set profile ppf-failure-increment' for a failed Paging procedure, causes the Paging Proceed Flag to be set to 'false' for a UE. When the value is set to zero, MME ignores Page Proceed Flag. Default: 10

Attribute	Type	Range/Length/Pattern	Default	Description
max-visited-enbs-tracked	uint8	2..10	2	Defines the maximum number of last visited eNodeBs that are tracked on a per-UE basis, for paging purposes. Default: 2
visited-enbs-allocation	enumeration	unrestricted ta-list	ta-list	Visited eNodeBs allocation method. Default: ta-list
hyper-system-frame-start-time	string	[0-9]*	19700101000000	UTC time for start of hyper-system-frame zero, to provide loose synchronization with the eNodeB (e.g. for eDRX). If the eNB also uses UTC, the exact same timestamp shall be configured on the eNB and the MME. If the eNodeB side uses GPS time instead of UTC, the configured value on the MME shall account for the leap seconds. For example, if the configured start time on the eNB is on or before 19810701, the start time on the MME shall be 18 seconds before the start time on the eNB. If the configured start time on the eNB is after 19810701, the number of leap seconds needed to be applied to the hyper-system-frame-start-time on the MME would be less. When new leap seconds are introduced, this value will need to be adjusted accordingly. Valid values are: digits in the format YYYYMMDDhhmmss. Default: 19700101000000
nb-ptw-shift-time	uint16	0..8000	0	Defines the Paging Transmission Window (PTW) shift time for NB to improve the paging success rate in the case of CIoT. Default: 0; Unit: milliseconds
nb-ptw-shift-threshold	enumeration	20.48 40.96 81.92 163.84 327.68 655.36 1310.72 2621.44 5242.88 10485.76	20.48	Specifies the threshold of PTW which in turn determines the NB PTW shift time to be applied or not. Default: 20.48; Unit: seconds

Default

Default paging attributes

XPath: *cna-mme-cc:/mme-cc/interface/nas/paging/default*

CLI Path: *mme-cc interface nas paging default*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
ppf-failure-increment	uint8	0..10	10	Defines the increment that accumulates for a failed Paging procedure that counts towards the 'nas paging global max-ppf-failure-threshold' to determine when to set the Paging Proceed Flag to 'false'. Default: 10
ppf-override	enumeration	enable disable	disable	Defines the ability to ignore the Paging Proceed Flag for this profile. Default: disable
t3413-initial	uint16	1000..30000	5000	Defines the wait time for the initial page attempt. Default: 5000; Unit: milliseconds
t3413-increment	uint16	0..10000	2000	Defines the additional delay added to retried page attempts. Default: 2000; Unit: milliseconds
n3413-retry	uint8	0..4	2	Defines the number of paging retries the MME performs before declaring paging failure. Default: 2
initial-paging-target	enumeration	last-known-enb last-visited-enbs last-known-ta ta-list recommended-enbs	ta-list	Defines initial paging target. Default: ta-list The 'initial-paging-target' should not have value 'recommended-enbs'

Attribute	Type	Range/Length/Pattern	Default	Description
first-retry-target	enumeration	last-known-enb last-visited-enbs last-known-ta ta-list recommended-enbs		Defines paging target for first retry The 'first-retry-target' should not have value 'recommended-enbs'
second-retry-target	enumeration	last-known-enb last-visited-enbs last-known-ta ta-list recommended-enbs		Defines paging target for second retry The 'second-retry-target' should not have value 'recommended-enbs'
third-retry-target	enumeration	last-known-enb last-visited-enbs last-known-ta ta-list recommended-enbs		Defines paging target for third retry The 'third-retry-target' should not have value 'recommended-enbs'
fourth-retry-target	enumeration	last-known-enb last-visited-enbs last-known-ta ta-list recommended-enbs		Defines paging target for fourth retry The 'fourth-retry-target' should not have value 'recommended-enbs'
paging-priority	enumeration	PrioLevel1 PrioLevel2 PrioLevel3 PrioLevel4 PrioLevel5 PrioLevel6 PrioLevel7 PrioLevel8 none	none	Defines the paging priority. Lower value indicates higher priority as described in TS 36.413 section 9.2.1.78. Default: none
emm-paging-timer-increment-nb	uint16	0..300	0	NAS EMM paging timer increment for UE in NB-S1 mode. This attribute specifies the amount to increment EMM paging timer for a UE using NB-S1 (narrowband) mode, as described in TS 24.301 section 4.7. This affects the T3413 timer value. Default: 0; Unit: seconds
emm-paging-timer-increment-wb-ce	uint16	0..300	0	NAS EMM paging timer increment for UE which supports WB CE mode B. This attribute specifies the amount to increment EMM paging timer for a UE using WB-S1 (wideband) in CE (coverage enhancement) mode, as described in TS 24.301 section 4.8. This affects the T3413 timer value. Default: 0; Unit: seconds

Profile Set

List of paging profile sets

XPath: *cna-mme-cc:/mme-cc/interface/nas/paging/profile-set*

CLI Path: *mme-cc interface nas paging profile-set*

List Properties:

Keys: name

Maximum number of elements: 4

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{1,64})		Name of the paging profile set

Profile

Paging profile

XPath: *cna-mme-cc:/mme-cc/interface/nas/paging/profile-set/profile*

CLI Path: *mme-cc interface nas paging profile-set profile*

List Properties:

Keys: name

Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	[A-Za-z0-9.-]*		Name of this paging profile; Range: 1..32 characters
features	leaf-list of enumeration	edrx		Defines the list of features.
precedence	uint8			The precedence in the paging profile set of this paging profile. Lower value indicates higher precedence
ppf-failure-increment	uint8	0..10	10	Defines the increment that accumulates for a failed Paging procedure that counts towards the 'nas paging global max-ppf-failure-threshold' to determine when to set the Paging Proceed Flag to 'false'. Default: 10
ppf-override	enumeration	enable disable	disable	Defines the ability to ignore the Paging Proceed Flag for this profile. Default: disable
t3413-initial	uint16	1000..30000	5000	Defines the wait time for the initial page attempt. Default: 5000; Unit: milliseconds
t3413-increment	uint16	0..10000	2000	Defines the additional delay added to retried page attempts. Default: 2000; Unit: milliseconds
n3413-retry	uint8	0..4	2	Defines the number of paging retries the MME performs before declaring paging failure. Default: 2
initial-paging-target	enumeration	last-known-enb last-visited-enbs last-known-ta ta-list recommended-enbs	ta-list	Defines initial paging target. Default: ta-list The 'initial-paging-target' should not have value 'recommended-enbs'
first-retry-target	enumeration	last-known-enb last-visited-enbs last-known-ta ta-list recommended-enbs		Defines paging target for first retry The 'first-retry-target' should not have value 'recommended-enbs'
second-retry-target	enumeration	last-known-enb last-visited-enbs last-known-ta ta-list recommended-enbs		Defines paging target for second retry The 'second-retry-target' should not have value 'recommended-enbs'
third-retry-target	enumeration	last-known-enb last-visited-enbs last-known-ta ta-list recommended-enbs		Defines paging target for third retry The 'third-retry-target' should not have value 'recommended-enbs'
fourth-retry-target	enumeration	last-known-enb last-visited-enbs last-known-ta ta-list recommended-enbs		Defines paging target for fourth retry The 'fourth-retry-target' should not have value 'recommended-enbs'
paging-priority	enumeration	PrioLevel1 PrioLevel2 PrioLevel3 PrioLevel4 PrioLevel5 PrioLevel6 PrioLevel7 PrioLevel8 none	none	Defines the paging priority. Lower value indicates higher priority as described in TS 36.413 section 9.2.1.78. Default: none

Attribute	Type	Range/Length/Pattern	Default	Description
emm-paging-timer-increment-nb	uint16	0..300	0	NAS EMM paging timer increment for UE in NB-S1 mode. This attribute specifies the amount to increment EMM paging timer for a UE using NB-S1 (narrowband) mode, as described in TS 24.301 section 4.7. This affects the T3413 timer value. Default: 0; Unit: seconds

Triggers

Defines the triggers for which this profile applies

XPath: *cna-mme-cc:/mme-cc/interface/nas/paging/profile-set/profile/triggers*

CLI Path: *mme-cc interface nas paging profile-set profile triggers*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
qci	leaf-list of enumeration	qci1		Defines the list of QoS Class Identifier (QCI) triggers. QCI values 128 and above are operator defined. Each trigger value is unique across all paging profiles. Each 'qci' trigger value must be unique across the paging profiles.
		qci2		
		qci3		
		qci4		
		qci5		
		qci6		
		qci7		
		qci8		
		qci9		
		qci80		
		qci82		
		qci83		
		qci65		
		qci66		
		qci69		
		qci70		
		qci128		
		qci129		
		qci130		
		qci131		
		qci132		
		qci133		
		qci134		
		qci135		
		qci136		
		qci137		
		qci138		
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		qci142		
		qci143		
		qci144		
		qci145		
		qci146		
		qci147		
		qci148		
		qci149		
		qci150		
		qci151		
		qci152		
		qci153		
		qci154		
		qci155		
		qci156		
		qci157		
		qci158		
		qci159		
		qci160		

Attribute	Type	Range/Length/Pattern	Default	Description
		qci161		
		qci162		
		qci163		
		qci164		
		qci165		
		qci166		
		qci167		
		qci168		
		qci169		
		qci170		
		qci171		
		qci172		
		qci173		
		qci174		
		qci175		
		qci176		
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		qci214		
		qci215		
		qci216		
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		qci218		
		qci219		
		qci220		
		qci221		
		qci222		
		qci223		
		qci224		
		qci225		

Attribute	Type	Range/Length/Pattern	Default	Description
		qci226 qci227 qci228 qci229 qci230 qci231 qci232 qci233 qci234 qci235 qci236 qci237 qci238 qci239 qci240 qci241 qci242 qci243 qci244 qci245 qci246 qci247 qci248 qci249 qci250 qci251 qci252 qci253 qci254		
service	leaf-list of enumeration	sgs-cs sgs-sms gw-data gw-sig		Defines the list of service triggers. Each service trigger value is unique across the paging profiles.. Each 'service' trigger value must be unique across the paging profiles
arp	leaf-list of enumeration	arp-1 arp-2 arp-3 arp-4 arp-5 arp-6 arp-7 arp-8 arp-9		Defines the list of arp triggers. Each arp trigger value is unique across the paging profiles.. Each 'arp' trigger value must be unique across the paging profiles.

S1Ap

S1 Application Interface

Reference: 3GPP TS 36.413: S1 Application Protocol (S1AP)

XPath: *cna-mme-cc:/mme-cc/interface/s1ap*

CLI Path: *mme-cc interface s1ap*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
version	enumeration	13.6.0 15.3.0 15.9.0	15.9.0	3GPP spec applicable to this interface is TS 36.413. Default: 15.9.0
s1-release-delay-timer	union	['uint8,1..5', 'Enumerations: disable ']	disable	Controls the time the MME waits before releasing the S1-MME. Default: disable; Unit: seconds

Attribute	Type	Range/Length/Pattern	Default	Description
reloc-prep-timer	uint8	1..30	3	Time that the MME waits for the target eNodeB to respond to the handover request in an S1 based handover. Default: 3; Unit: seconds
reloc-overall-timer	uint8	1..30	5	Time that the MME waits for the handover notify from the target eNodeB. Default: 5; Unit: seconds
indirect-tunnel-hold-timer	uint8	1..30	5	Time that the target MME keeps the indirect forwarding tunnel during a relocation procedure that uses indirect forwarding. Default: 5; Unit: seconds
handover-resource-release-timer	uint16	100..30000	2000	Time that the MME waits before releasing the resource used or setup during the handover, such as the S1 connection on the source enb, sessions that are relocated and indirect forwarding tunnels if set up. Default: 2000; Unit: milliseconds
ue-context-release-timer	uint8	1..30	5	Time that the MME waits for the eNodeB to respond to a UE context release command. Default: 5; Unit: seconds
ue-context-modification-timer	uint8	1..30	5	Time that the MME waits for the eNodeB to respond to a UE context modification request. Default: 5; Unit: seconds
wait-for-tau-timer	uint8	1..10	5	Time that the MME waits for a TAU after an intra-mme relocation with TAI list change or after an inter-mme relocation. Default: 5; Unit: seconds The value of 'wait-for-tau-timer' should be less than or equal to the value of 'indirect-tunnel-hold-timer'.
inter-rat-indirect-forwarding	enumeration	never always inter-plmn-inter-rat	never	Controls the setup of an indirect forwarding tunnel for downlink data during an intra 3GPP inter RAT handover. Default: never
intra-rat-indirect-forwarding	enumeration	enable disable	enable	Controls the setup of an indirect forwarding tunnel for downlink data during an intra 3GPP intra RAT handover. Default: enable
indirect-data-fwd-ip-address	inet:ip-address		0.0.0.0	Indirect data forwarding IP address. Default: 0.0.0.0
csg-expiration-release-timer	uint8	1..30	5	Time that the MME waits for an S1 handover or S1 release when a UE context modification request is sent. Default: 5; Unit: seconds
paging-drx-default	enumeration	enb-provided none	none	Controls inclusion of the paging DRX (Discontinuous Reception) IE in the S1AP paging message sent from the MME as defined in TS 36.413 section 9.1.6. Normally the value sent is based on the value received from the UE in the attach request or TAU request messages. In the absence of any value from the UE, the MME may send the default value provided by the eNB in the S1 setup message, or send no DRX IE. Default: none
ciot-smart-s1-release-timer	uint16	1..600	10	Time that a CIoT UE, when not utilizing S1-U data transfer for active sessions, can go without signaling or data activity before its S1 connection is released by the MME. Default: 10; Unit: seconds
s1-release-after-reroute	enumeration	enable disable	disable	Controls the action to not release S1 connection after REROUTE NAS Request is sent by the MME. Depending on eNB implementation, it may be necessary to not release S1 connection after REROUTE NAS REQUEST is sent by MME. As defined by TS 36.413 section 8.6.2.5, the S1AP message REROUTE NAS REQUEST is sent by MME to eNB to request for a rerouting of the INITIAL UE MESSAGE to the MME in the indicated Dedicated Core Network (DCN). In this case the MME shall not release the S1 by default, but some eNB implementations may work better with the explicit release. Default: disable

Duplicate S1 Release Cause (choice)

XPath: [cna-mme-cc:/mme-cc/interface/s1ap](#)

CLI Path: [mme-cc interface s1ap](#)

Case : dup-s1-release-radio-network-cause

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
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Attribute	Type	Range/Length/Pattern	Default	Description
dup-s1-release-radio-network-cause	union	<pre>['uint8,0..38', 'Enumerations: unspecified-0 release-due-to-eutran-generated-reason-3 radio-connection-with-ue-lost-21 failure-in-radio-interface-procedure-26 interaction-with-other-procedure-29 ']</pre>		Specify a cause from the radio network group to use when releasing a duplicate s1 connection. The fields dup-s1-release-radio-network-cause and dup-s1-release-nas-cause are mutually exclusive. S1AP causes are defined by 3GPP TS 36.413 section 9.2.1.3. Choice: This dup-s1-release-radio-network-cause attribute is mutually exclusive with attributes [['dup-s1-release-nas-cause']]

Case : dup-s1-release-nas-cause

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
dup-s1-release-nas-cause	union	<pre>['uint8,0..4', 'Enumerations: normal-release-0 unspecified-3 ']</pre>		Specify a cause from the NAS group to use when releasing a duplicate s1 connection. The fields dup-s1-release-radio-network-cause and dup-s1-release-nas-cause are mutually exclusive. S1AP causes are defined by 3GPP TS 36.413 section 9.2.1.3. Choice: This dup-s1-release-nas-cause attribute is mutually exclusive with attributes [['dup-s1-release-radio-network-cause']]

S6

S6 Interface

XPath: [cna-mme-cc:/mme-cc/interface/s6](#)

CLI Path: [mme-cc interface s6](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
version	enumeration	13.11.0 15.9.0	15.9.0	3GPP spec applicable to this interface is TS 29.272. Default: 15.9.0
request-timer	uint8	1..10	5	Amount of time the MME waits for an S6 request sent towards the HSS. Default: 5; Unit: seconds
request-retry	uint8	0..5	2	Number of retries the MME performs for an S6 request. Default: 2
air-vector-limit	uint8	1..4	1	Number of vectors requested during each procedure. Default: 1
include-dest-host	enumeration	enable disable	disable	Controls whether the MME includes dest-host information in the S6a request message. Default: disable
t-ads-retrieval	enumeration	enable disable	disable	Controls support of T-ADS data retrieval. Default: disable
pgw-identity-type	enumeration	host-name ip-address	host-name	Defines the PGW identity type sent to HSS in ULR or NOR. Default: host-name
additional-msisdn	enumeration	enable disable	disable	Controls the usage of Additional-MSISDN flag bit in ULR message sent to HSS as per TS 29.272 section 5.2.1.1.2. If set to enable, the Additional-MSISDN flag bit in ULR is set to 1 which indicates to the HSS that the MME supports the feature. This feature may be needed for cases in which Multi-SIM service is used in conjunction with SRVCC. It affects the processing of the CS-PS request message on the Sv interface to the MSC (TS 29.280, 5.2.2), as well as the GTP Forward Relocation Request message (TS 29.274, 7.3.1), and the IDR and DSR messages exchanged with the HSS. Default: disable

Hss Routing

HSS Routing Parameters

XPath: [cna-mme-cc:/mme-cc/interface/s6/hss-routing](#)

CLI Path: [mme-cc interface s6 hss-routing](#)

List Properties:

Keys: imsi-range
Maximum number of elements: 1000

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
imsi-range	string	(([0-9]{3,15}) or ([0-9]{3,15}[-][0-9]{3,15}))		The value is specified in one of the 2 formats below: Single IMSI prefix - IMSI prefix 3 to 15 digits long; example: 999, 899000123674898 IMSI prefix range - Lower and upper end of the range separated by a '-'; example: 990-999, 899000123674898-899000123679999 In case multiple values match a given IMSI during lookup, the value with the longest prefix match is selected.
hss-realm	string	1..127		Realm of the HSS
group-name	leafref			Peer grouping for the HSS-realm Leaf Reference Path: cna-mme-cc:/mme-cc/interface/s6/diameter-group/group-name

Hss Routing Default

HSS routing configuration with default values

XPath: [cna-mme-cc:/mme-cc/interface/s6/hss-routing-default](#)

CLI Path: [mme-cc interface s6 hss-routing-default](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
group-name	leafref			Peer grouping for the HSS-realm Leaf Reference Path: cna-mme-cc:/mme-cc/interface/s6/diameter-group/group-name
hss-realm	string	1..127		Realm of HSS

Diameter Group

Diameter Group Parameters

XPath: [cna-mme-cc:/mme-cc/interface/s6/diameter-group](#)

CLI Path: [mme-cc interface s6 diameter-group](#)

List Properties:

Keys: group-name
Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
group-name	string	3..127		Peer grouping for the hss-realm
mme-hostname	string	3..127		FQDN of the peer
mme-realm	string	3..127		Realm of this MME

S11

S11 Interface

XPath: [cna-mme-cc:/mme-cc/interface/s11](#)

CLI Path: [mme-cc interface s11](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
gtpv2-version	enumeration	13.11.0 15.4.0 15.9.0	15.9.0	3GPP spec used for this interface is TS 29.274. Default: 15.9.0

Attribute	Type	Range/Length/Pattern	Default	Description
t3-response-session-creation	uint16	100..15000	5000	Response timer for session creation procedure messages. Default: 5000; Unit: milliseconds
n3-request-session-creation	uint8	0..4	1	Number of retries for the session creation procedure due to 't3-response-session-creation' timer expiration. It excludes the initial attempt. Default: 1
t3-response-bearer-modification	uint16	100..15000	5000	Response timer for bearer modification procedure messages. Default: 5000; Unit: milliseconds
n3-request-bearer-modification	uint8	0..4	1	Number of retries for the bearer modification procedure due to 't3-response-bearer-modification' timer expiration. It excludes the initial attempt. Default: 1
include-enodeb-id-in-uli	enumeration	enable disable	disable	Controls whether the MME includes the macro eNodeB ID in the User Location Information (ULI) when it is included in GTP messages. Default: disable
uli-inclusion	enumeration	standard force-in-mbr force-in-all force-in-all-and-handover	standard	Enables MME to include the User Location Information (ULI) in the GTP messages. uliForceInMbr_c: MME provides ULI in every Modify Bearer Request message. uliForceInAll_c: MME provides ULI in all the GTP messages. uliForceInAllAndHandover_c: MME provides ULI in all the GTP messages and Create Session Request message during Handover procedure. uliStandard_c: The default option, MME provides ULI in the GTP message if it is requested by the PGW.
ue-timezone-inclusion	enumeration	standard force-in-all	standard	Defines how the MME sends ue-timezone in GTPv2 messages. Default: standard
ran-nas-cause-inclusion	enumeration	enable disable	disable	Controls whether the MME provides RAN/NAS cause in S11 Delete Session Request and Delete Bearer Command messages. Default: disable

Sgs

SGs interface

XPath: [cna-mme-cc:/mme-cc/interface/sgs](#)

CLI Path: [mme-cc interface sgs](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
version	enumeration	11.11.0	11.11.0	3GPP spec applicable to this interface is TS 29.118. Default: 11.11.0
mme-name	string			Defines the MME name IE as per 3GPP TS 23.003 section 19.4.2.4 to be sent on the SGs interface (TS 29.118). Length: 54 char;Format: mmec(MMEC).mmegi(MMEGI).mme.epc.mnc(MNC).mcc(MCC).3gppnetwork.org;Example: mmec01.mmegi8001.mme.epc.mnc001.mcc001.3gppnetwork.org (zero padding should be used if needed).Note: this attribute is optional, if it is not configured the MME will derive the name IE automatically from the mme-codeand mme-groupid under the mme-profile and plmn configuration for simple systems in which there is only one possible value for each field.
loc-upd-guard-ts6-1	uint8	10..90	10	Specifies the amount of time the MME allows for a response from the VLR. Default: 10; Unit: seconds
loc-upd-retry-ns6	uint8	0..5	2	Number of location update retries. Default: 2
eps-detach-guard-ts8	uint8	1..30	4	Guards explicit IMSI detach from EPS services. Default: 4; Unit: seconds
eps-detach-retry-ns8	uint8	0..5	2	Number of EPS detach retries. Default: 2

Attribute	Type	Range/Length/Pattern	Default	Description
explicit-imsi-detach-guard-ts9	uint8	1..30	4	Guards explicit IMSI detach from non-EPS services. Default: 4; Unit: seconds
explicit-imsi-detach-retry-ts9	uint8	0..5	2	Number of explicit IMSI detach retries. Default: 2
implicit-imsi-detach-guard-ts10	uint8	1..30	4	Guards implicit IMSI detach from non-EPS services. Default: 4; Unit: seconds
implicit-imsi-detach-retry-ts10	uint8	0..5	2	Number of implicit IMSI detach retries. Default: 2
mme-reset-ind-guard-ts12-2	uint8	1..120	4	Guards the MME reset procedure. Default: 4; Unit: seconds
mme-reset-ind-retry-ts12	uint8	0..5	2	Number of MME reset indication retries. Default: 2
mme-reset-paging-type	enumeration	cs-paging ps-paging	cs-paging	Defines the MME reset paging type. Default: cs-paging
mme-reset-indication	boolean		true	Controls the MME reset indication. Default: true
page-response-during-mobility	enumeration	standard ignore-page	standard	Specifies MME response to a mobile terminated SGsAP-PAGING-REQUEST during a mobility procedure or handover. Sending the page reject can result in a race condition in which the MT call is missed. Setting this to ignore-page, allows the SGS VLR to retry the page so the call can complete. Standard behavior is to respond with a paging reject or unreachable message based on the SGs Association state and UE state. Default: standard
page-response-while-detached	enumeration	standard reject-1-imsi-detach-eps-services reject-3-imsi-unknown reject-6-ue-unreachable reject-13-ue-reject response-ue-unreachable ignore-page	standard	Specifies MME response to a mobile terminated SGsAP-PAGING-REQUEST when a UE is detached. Some responses may result in a VLR retry, and others may not, as described in TS 29.118. Depending on the scenario, this can lead to a missed MT call if a VLR retry is not triggered. TS 29.118 section 9.4.18 defines SGs causes. The paging reject SGs cause which is expected to trigger a VLR paging retry is 13 (MT CSFB call reject by UE). Other cases that may lead to a VLR paging retry include ignore, and a response of SGs-UE-Unreachable message. A paging reject with cause 1 (IMSI detached for EPS services) may cause the VLR to page on the circuit switched side. Standard behavior is to respond with a paging reject message with an SGs cause corresponding to the last UE detach type (1, 2, 4, or 5, depending on the combination of EPS and non-EPS services). Default: standard
remap-network-failure	enumeration	enable disable	disable	Controls location update reject cause remapping from network-failure-17 to cs-domain-not-available-18, for VLR that are not compliant to 3GPP TS 29.118. Default: disable
ue-unreachable-mode	enumeration	enable disable	disable	Controls whether the MME sends SGs-UE-Unreachable message when UE does not respond to Paging due to VLR initiated SGs-PAGING-Request. Default: disable
sms-only	boolean		false	Controls SMS only services (true means the MME supports SMS-only UE). Default: false
explicit-eps-detach-csfb-iratho	enumeration	enable disable	disable	Controls if MME sends SGsAP-EPS-DETACH-INDICATION to the VLR after an inter-RAT mobility event. As described in TS 29.118 section 5.4.2, the MME does not normally send the detach indication in the cases of CSFB or inter-RAT change away from LTE. However, for networks in which a large percentage of the UE don't do a fast return to LTE (by tracking area update), the UE may miss subsequent calls if the MSC still has the UE location on MME, and it may be desired to enable this option. For networks in which most UE do perform fast return to LTE, enabling this option will add latency to reception of their next downlink SGs message, and it is recommended to leave this disabled. Default: disable

S10

S10 Interface

XPath: *cna-mme-cc:/mme-cc/interface/s10***CLI Path:** *mme-cc interface s10*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
gtpv2-version	enumeration	13.11.0 15.9.0	15.9.0	3GPP spec used for this interface is TS 29.274. Default: 15.9.0
t3-response-context-request	uint16	100..15000	5000	Response timer for context request procedure messages. Default: 5000; Unit: milliseconds
n3-request-context-request	uint8	1..5	2	Number of requests for context request procedure messages. Default: 2
t3-response-forward-relocation	uint16	100..15000	5000	[future] Response timer for forward relocation procedure messages. Default: 5000; Unit: milliseconds
n3-request-forward-relocation	uint8	1..5	2	[future] Number of requests for forward relocation procedure messages. Default: 2

S3

S3 Interface

XPath: *cna-mme-cc:/mme-cc/interface/s3***CLI Path:** *mme-cc interface s3*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
gtpv2-version	enumeration	13.11.0 15.4.0	13.11.0	[future] 3GPP spec used for this interface is TS 29.274. Default: 13.11.0
t3-response-context-request	uint16	100..15000	5000	Response timer for context request procedure messages. Default: 5000; Unit: milliseconds
n3-request-context-request	uint8	1..5	2	Number of requests for context request procedure messages. Default: 2
t3-response-forward-relocation	uint16	100..15000	5000	[future] Response timer for forward relocation procedure messages. Default: 5000; Unit: milliseconds
n3-request-forward-relocation	uint8	1..5	2	[future] Number of requests for forward relocation procedure messages. Default: 2

Gn

Gn Interface

XPath: *cna-mme-cc:/mme-cc/interface/gn***CLI Path:** *mme-cc interface gn*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
gtpv1-version	enumeration	11.11.0 13.7.0	11.11.0	[future] 3GPP spec used for this interface is TS29.060. Default: 11.11.0
t3-response-context-request	uint16	100..15000	5000	Response timer for context request procedure messages. Default: 5000; Unit: milliseconds
n3-request-context-request	uint8	1..5	2	Number of requests for context request procedure messages. Default: 2
t3-response-forward-relocation	uint16	100..15000	5000	[future] Response timer for forward relocation procedure messages. Default: 5000; Unit: milliseconds
n3-request-forward-relocation	uint8	1..5	2	[future] Number of requests for forward relocation procedure messages. Default: 2

N26

N26 Interface

XPath: *cna-mme-cc:/mme-cc/interface/n26*

CLI Path: *mme-cc interface n26*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
gtpv2-version	enumeration	15.9.0	15.9.0	3GPP spec used for this interface is TS 29.274. Default: 15.9.0
return-preferred-indication	enumeration	enable disable	disable	Controls a preferred return of the UE to the last used EPS PLMN at a later access change to an EPS shared network. If enabled, the AMF stores the last used EPS PLMN ID in UE Context. Default: disable
t3-response-context-request	uint16	100..15000	5000	Response timer for context request procedure messages. Default: 5000; Unit: milliseconds
n3-request-context-request	uint8	1..5	2	Number of requests for context request procedure messages. Default: 2
t3-response-forward-relocation	uint16	100..15000	5000	[future] Response timer for forward relocation procedure messages. Default: 5000; Unit: milliseconds
n3-request-forward-relocation	uint8	1..5	2	[future] Number of requests for forward relocation procedure messages. Default: 2

Sv

Sv Interface

XPath: *cna-mme-cc:/mme-cc/interface/sv*

CLI Path: *mme-cc interface sv*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
gtpv2-version	enumeration	13.11.0 15.9.0	15.9.0	3GPP spec used for this interface is TS 29.274. Default: 15.9.0
vlr-discovery	enumeration	dns local-config last-known-tai	dns	Determines the MSC/VLR discovery mode for SRVCC as defined by TS 23.216 section 5.3.3.2.3. The default method is to use the target RAI to perform a DNS NAPTR query to locate the MSC as defined by TS 29.303 section 4.3.3.7. Alternatively, if local-config is chosen, a DNS lookup is avoided, and instead the target LAI is extracted from the target RAI and is used to look up a vlr-pool under 'mme-cc mme-profile plmn location-area' which is then used to find an sv-ip in a vlr under 'mme-cc vlr'. A third option is to use the last known TAI of the UE to look up the location-area under 'mme-cc mme-profile plmn tracking-area', which then is used to map to a vlr-pool which then maps to an sv-ip under a configured 'mme-cc vlr'. Default: dns

Plmn

Subscriber Public Land Mobile Network (PLMN) parameters

XPath: *cna-mme-cc:/mme-cc/plmn*

CLI Path: *mme-cc plmn*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
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Attribute	Type	Range/Length/Pattern	Default	Description
mcc-using-3-digit-mnc	leaf-list of uint16	1	100..999	302
				310
				311
				312
				313
				314
				315
				316
				330
				334
				338
				342
				344
				346
				348
				352
				354
				356
				360
				365
				366
				376
				708
				722
				732
				750

Default

Public Land Mobile Network (PLMN) profile with default values

XPath: *cna-mme-cc:/mme-cc/plmn/default*

CLI Path: *mme-cc plmn default*

Subscriber

Subscriber specific parameters

XPath: *cna-mme-cc:/mme-cc/plmn/default/subscriber*

CLI Path: *mme-cc plmn default subscriber*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
suppress-hss-notify	enumeration	enable disable enable-for-non-ims-apn	disable	Controls when the MME should suppress sending Notify Request (NOR) messages to the HSS. As defined by TS 29.272 section 5.2.5, there are many events for which NOR should be sent, but it is not specific to APN type. However, for some networks, it is desired to only send NOR for IMS APN (i.e. 'ims' in APN-NI), as selected by the enable-for-non-ims-apn option. Default: disable
ims-support	enumeration	enable disable enable-if-ims-subscribed	disable	Controls per PLMN IMS voice over PS support. Default: disable
wlan-offloadability	enumeration	enable disable	disable	Controls per PLMN Wireless Local Area Network (WLAN) offloadability support. This attribute indicates if the UE is allowed to offload the traffic of the APN to WLAN as indicated by the HSS WLAN-offloadability AVP in TS 29.272 section 7.3.181. Default: disable
ims-local-breakout	enumeration	allow disallow	allow	Controls if IMS local breakout is allowed. Default: allow
local-breakout	enumeration	allow disallow	allow	Controls if local breakout is allowed. Default: allow

Attribute	Type	Range/Length/Pattern	Default	Description
local-breakout-apn-list	leaf-list of string	[a-zA-Z0-9*][(-)] or [A-Za-z0-9*]]*)*		List of APNs allowed for local breakout while using wildcard subscription Maximum number of elements: 16
hss-t-ads	enumeration	enable disable	enable	Controls the HSS Terminating Access Domain Selection (T-ADS) retrieval on a per PLMN basis. When enabled and when HSS requests T-ADS data in the IDR message, the MME replies with T-ADS data (RAT-Type, Last-UE-Activity-Time and IMS-Voice-Over-PS-Sessions-Supported) in the IDA response. Note that t-ads-retrieval attribute under 'interface s6' must be enabled along with this configuration. Default: enable

Network

Network related parameters

XPath: *cna-mme-cc:/mme-cc/plmn/default/network*

CLI Path: *mme-cc plmn default network*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
native-mme-group-ids	leaf-list of uint16			Native MME group ID list having maximum of 100 mme group-ids for this PLMN. Maximum number of elements: 100
dns-preference	enumeration	unknown s-naptr address-only	unknown	Defines the DNS procedure preference for DNS queries. Default: unknown
pgw-host-format	enumeration	includes-realm does-not-include-realm	includes-realm	Controls the PGW host FQDN construction to be used in the DNS NAPTR query. Default: includes-realm
pgw-name-format	enumeration	includes-interface does-not-include-interface	includes-interface	Controls if the PGW hostname received from the HSS should be trimmed before performing the DNS S-NAPTR query. Default: includes-interface
inter-plmn-restriction	enumeration	disable eps-service-not-allowed-7 eps-and-non-eps-service-not-allowed-8 ue-id-cannot-be-derived-from-network-9 plmn-not-allowed-11 tracking-area-not-allowed-12 roaming-not-allowed-in-this-ta-13 eps-service-not-allowed-in-this-plmn-14 no-suitable-cells-in-ta-15 network-failure-17 esm-failure-19	disable	EMM cause to reject the inter PLMN TAU. If value is set to disable, no restriction is applicable. Default: disable

Plmn Profile

Public Land Mobile Network (PLMN) profile

XPath: *cna-mme-cc:/mme-cc/plmn/plmn-profile*

CLI Path: *mme-cc plmn plmn-profile*

List Properties:

Keys: plmn-id

Maximum number of elements: 1000

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
plmn-id	string	(([0-9]{5,6}))		Unique PLMN identifier. Range: numbers with 5.6 digits

Subscriber

Subscriber specific parameters

XPath: *cna-mme-cc/mme-cc/plmn/plmn-profile/subscriber*

CLI Path: *mme-cc plmn plmn-profile subscriber*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
suppress-hss-notify	enumeration	enable disable enable-for-non-ims-apn	disable	Controls when the MME should suppress sending Notify Request (NOR) messages to the HSS. As defined by TS 29.272 section 5.2.5, there are many events for which NOR should be sent, but it is not specific to APN type. However, for some networks, it is desired to only send NOR for IMS APN (i.e. 'ims' in APN-NI), as selected by the enable-for-non-ims-apn option. Default: disable
ims-support	enumeration	enable disable enable-if-ims-subscribed	disable	Controls per PLMN IMS voice over PS support. Default: disable
wlan-offloadability	enumeration	enable disable	disable	Controls per PLMN Wireless Local Area Network (WLAN) offloadability support. This attribute indicates if the UE is allowed to offload the traffic of the APN to WLAN as indicated by the HSS WLAN-offloadability AVP in TS 29.272 section 7.3.181. Default: disable
ims-local-breakout	enumeration	allow disallow	allow	Controls if IMS local breakout is allowed. Default: allow
local-breakout	enumeration	allow disallow	allow	Controls if local breakout is allowed. Default: allow
local-breakout-apn-list	leaf-list of string	[a-zA-Z0-9*][(-) or [A-Za-z0-9*]]*		List of APNs allowed for local breakout while using wildcard subscription Maximum number of elements: 16
hss-t-ads	enumeration	enable disable	enable	Controls the HSS Terminating Access Domain Selection (T-ADS) retrieval on a per PLMN basis. When enabled and when HSS requests T-ADS data in the IDR message, the MME replies with T-ADS data (RAT-Type, Last-UE-Activity-Time and IMS-Voice-Over-PS-Sessions-Supported) in the IDA response. Note that t-ads-retrieval attribute under 'interface s6' must be enabled along with this configuration. Default: enable

Network

Network related parameters

XPath: *cna-mme-cc/mme-cc/plmn/plmn-profile/network*

CLI Path: *mme-cc plmn plmn-profile network*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
native-mme-group-ids	leaf-list of uint16			Native MME group ID list having maximum of 100 mme group-ids for this PLMN. Maximum number of elements: 100
dns-preference	enumeration	unknown s-naptr address-only	unknown	Defines the DNS procedure preference for DNS queries. Default: unknown
pgw-host-format	enumeration	includes-realm does-not-include-realm	includes-realm	Controls the PGW host FQDN construction to be used in the DNS NAPTR query. Default: includes-realm
pgw-name-format	enumeration	includes-interface does-not-include-interface	includes-interface	Controls if the PGW hostname received from the HSS should be trimmed before performing the DNS S-NAPTR query. Default: includes-interface

Attribute	Type	Range/Length/Pattern	Default	Description
inter-plmn-restriction	enumeration	disable	disable	EMM cause to reject the inter PLMN TAU. If value is set to disable, no restriction is applicable. Default: disable
		eps-service-not-allowed-7		
		eps-and-non-eps-service-not-allowed-8		
		ue-id-cannot-be-derived-from-network-9		
		plmn-not-allowed-11		
		tracking-area-not-allowed-12		
		roaming-not-allowed-in-this-ta-13		
		eps-service-not-allowed-in-this-plmn-14		
		no-suitable-cells-in-ta-15		
		network-failure-17		
		esm-failure-19		

Vlr

Visitor Location Register (VLR) table
IP addresses in 'sv-ip-address-list' must be non-zero

XPath: *cna-mme-cc:/mme-cc/vlr*

CLI Path: *mme-cc vlr*

List Properties:
Keys: name
Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	1..255		Name of the VLR
vlr-type	leaf-list of enumeration	sgs		VLR type
		sv		Minimum number of elements: 1
vlr-number	string	[0-9]{1,15}		VLR number
sv-ip-address-list	leaf-list of inetip-address			List of up to four IP addresses for this VLR, used when accessing the VLR over the Sv interface Minimum number of elements: 1 Maximum number of elements: 4
nri	string	(0x[0-3][0-9a-f]{2} or (unassigned))	unassigned	TMSI based NRI provided by UE NRI value can be 'unassigned' or it must be unique across all 'mme-cc vlr' objects

Vlr Pool

Defines a pool of VLRs where a VLR is selected using the IMSI hash algorithm as described in 3GPP TS 23.236 section 5.5.4

XPath: *cna-mme-cc:/mme-cc/vlr-pool*

CLI Path: *mme-cc vlr-pool*

List Properties:
Keys: name
Maximum number of elements: 32

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	1..31		VLR pool name

Attribute	Type	Range/Length/Pattern	Default	Description
default-vlr-name	leafref			VLR identifier (reference to cn-mme path to list vlr vlr-name) Leaf Reference Path: cna-mme-cc:/mme-cc/vlr/name
non-broadcast-lai	string	([0-9]{5,6}). or normal-lai	normal-lai	Reserved LAI not broadcasted in network but used as the old LAI during mobility with VLR change for UEs with no TMSI based NRI. Default: normal-lai

Vlr Selection

Range of values computed from the IMSI hash algorithm $[(\text{IMSI} \div 10) \bmod 1000]$ which map uniquely on a per pool basis to a given VLR

XPath: [cna-mme-cc:/mme-cc/vlr-pool/vlr-selection](#)

CLI Path: [mme-cc vlr-pool vlr-selection](#)

List Properties:

Keys: digits-low, digits-high

Maximum number of elements: 32

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
digits-low	uint32			Lowest value computed from the IMSI hash algorithm
digits-high	uint32			Highest value computed from the IMSI hash algorithm
vlr-name	leafref			VLR identifier (reference to cn-mme path to vlr vlr-name) Leaf Reference Path: cna-mme-cc:/mme-cc/vlr/name

Mme Profile

MME profile

XPath: [cna-mme-cc:/mme-cc/mme-profile](#)

CLI Path: [mme-cc mme-profile](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
mme-name	string	[A-Za-z0-9 '() +, -./:=?]*		MME name communicated to the eNB
mme-groupid	uint16			MME group identifier. When 'native-mme-groups' is disabled, valid range is 32768 to 65535.
mme-code	leaf-list of uint8			List of MME codes. Minimum number of elements: 1 Maximum number of elements: 128
relative-capacity	uint8	0..255	100	This attribute provides a mechanism to set the MME relative capacity (as defined by TS 36.413 section 9.2.3.17). The relative capacity provided to an eNodeB is derived from this value when relative capacity values are not configured or configured with 'none' at the tracking area level (service-area tracking-area). A higher value indicates more capacity relative to other MMEs in the same pool, but it can also be used to influence the MME selection algorithm on the eNodeB based on geographic proximity for example when configured on a per tracking area basis. If a modification results in a changed derived value, an 'MME Configuration Update' procedure is triggered. Default: 100
locality	string	[A-Za-z0-9 '() +, -./:=?]*		Location of the MME.
native-mme-groups	enumeration	enable disable	disable	Controls the option to use native MME groups listed in 'mme-cc plmn plmn-profile network native-mme-group-ids'. Default: disable For MME group ID under 32768, native-mme-groups must be enabled

Plmn

Supported Public Land Mobile Network (PLMN); Range: numbers with 5-6 digits

XPath: [cna-mme-cc:/mme-cc/mme-profile/mme-plmn](#)

CLI Path: [mme-cc mme-profile plmn](#)

List Properties:

Keys: plmn-id

Minimum number of elements: 1

Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
plmn-id	string	((([0-9]){5,6}))		Unique PLMN identifier; Range: numbers with 5..6 digits When 'mme-cc mme-profile mme-groupid' value is less than 32768, the 'mme-cc plmn plmn-profile' must be defined for this plmn and mme-groupid value must exist in 'mme-cc plmn plmn-profile native-mme-group-ids' list When 'mme-cc mme-profile mme-groupid' value is less than 32768, the 'mme-cc plmn plmn-profile' must be defined for this plmn and mme-groupid value must exist in 'mme-cc plmn plmn-profile native-mme-group-ids' list

Tracking Area Default

Tracking area default configuration. Default configuration to be used by the application when the matching tracking-area configuration is missing.

XPath: *cna-mme-cc:/mme-cc/mme-profile/mme-plmn/tracking-area-default***CLI Path:** *mme-cc mme-profile plmn tracking-area-default*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
lac	string	0x[0-9a-f]{1,4}		Location Area Code (LAC); The values are expressed as a string of hexadecimal characters (with lower case alphabets) of length 1..4 with use of leading zeros if necessary. The value is prefixed with '0x' to convey the hexadecimal representation. For example: 0xfa10, 0x04cb, 0x2344.
lai-plmn-id	string	((([0-9]){5,6}))		PLMN of Location Area Identifier
relative-capacity	string	none or ([0-9] or [1-9][0-9] or [1][0-9][0-9] or [2][0-4][0-9] or [2][5][0-5])	none	This attribute provides a mechanism to set the MME relative capacity (as defined by TS 36.413 section 9.2.3.17) on a per tracking area basis. The relative capacity provided to an eNodeB is derived from this value. If a value of 'none' is configured for a tracking area, the relative capacity is derived from the configured value at the mme-profile level (mme-profile relative-capacity). For an eNodeB serving multiple tracking areas, the relative capacity is derived from the configured values of the tracking areas, using the lowest value. If the relative capacity is not configured or configured with 'none' for one or more of the serving tracking areas, the relative capacity is derived from the configured values of the tracking areas and the configured value at the mme-profile level, using the lowest value. If a modification results in a changed derived value, an 'MME Configuration Update' procedure is triggered. Default: none
ims-support	enumeration	enable disable	disable	Controls support for IP Multimedia Subsystem (IMS) voice over PS. Default: disable
emergency-support	enumeration	enable disable	enable	Controls support for emergency voice over PS. Default: enable
timezone	enumeration	Africa/Abidjan Africa/Accra Africa/Addis_Ababa Africa/Algiers Africa/Asmara Africa/Asmera Africa/Bamako Africa/Bangui Africa/Banjul Africa/Bissau Africa/Blantyre Africa/Brazzaville Africa/Bujumbura Africa/Cairo Africa/Casablanca Africa/Ceuta		LINUX timezone values

Attribute	Type	Range/Length/Pattern	Default	Description
		Africa/Conakry		
		Africa/Dakar		
		Africa/Dar_es_Salaam		
		Africa/Djibouti		
		Africa/Douala		
		Africa/El_Aaiun		
		Africa/Freetown		
		Africa/Gaborone		
		Africa/Harare		
		Africa/Johannesburg		
		Africa/Juba		
		Africa/Kampala		
		Africa/Khartoum		
		Africa/Kigali		
		Africa/Kinshasa		
		Africa/Lagos		
		Africa/Libreville		
		Africa/Lome		
		Africa/Luanda		
		Africa/Lubumbashi		
		Africa/Lusaka		
		Africa/Malabo		
		Africa/Maputo		
		Africa/Maseru		
		Africa/Mbabane		
		Africa/Mogadishu		
		Africa/Monrovia		
		Africa/Nairobi		
		Africa/Ndjamena		
		Africa/Niamey		
		Africa/Nouakchott		
		Africa/Ouagadougou		
		Africa/Porto-Novo		
		Africa/Sao_Tome		
		Africa/Timbuktu		
		Africa/Tripoli		
		Africa/Tunis		
		Africa/Windhoek		
		America/Adak		
		America/Anchorage		
		America/Anguilla		
		America/Antigua		
		America/Araguaina		
		America/Argentina/Buenos_Aires		
		America/Argentina/Catamarca		
		America/Argentina/ComodRivadavia		
		America/Argentina/Cordoba		
		America/Argentina/Jujuy		
		America/Argentina/La_Rioja		
		America/Argentina/Mendoza		
		America/Argentina/Rio_Gallegos		
		America/Argentina/Salta		
		America/Argentina/San_Juan		
		America/Argentina/San_Luis		
		America/Argentina/Tucuman		
		America/Argentina/Ushuaia		
		America/Aruba		
		America/Asuncion		
		America/Atikokan		
		America/Atka		
		America/Bahia		
		America/Bahia_Banderas		
		America/Barbados		
		America/Belem		
		America/Belize		

Attribute	Type	Range/Length/Pattern	Default	Description
		America/Blanc-Sablon		
		America/Boa_Vista		
		America/Bogota		
		America/Boise		
		America/Buenos_Aires		
		America/Cambridge_Bay		
		America/Campo_Grande		
		America/Cancun		
		America/Caracas		
		America/Catamarca		
		America/Cayenne		
		America/Cayman		
		America/Chicago		
		America/Chihuahua		
		America/Coral_Harbour		
		America/Cordoba		
		America/Costa_Rica		
		America/Creston		
		America/Cuiaba		
		America/Curacao		
		America/Danmarkshavn		
		America/Dawson		
		America/Dawson_Creek		
		America/Denver		
		America/Detroit		
		America/Dominica		
		America/Edmonton		
		America/Eirunepe		
		America/El_Salvador		
		America/Ensenada		
		America/Fortaleza		
		America/Fort_Nelson		
		America/Fort_Wayne		
		America/Glace_Bay		
		America/Godthab		
		America/Goose_Bay		
		America/Grand_Turk		
		America/Grenada		
		America/Guadeloupe		
		America/Guatemala		
		America/Guayaquil		
		America/Guyana		
		America/Halifax		
		America/Havana		
		America/Hermosillo		
		America/Indiana/Indianapolis		
		America/Indiana/Knox		
		America/Indiana/Marengo		
		America/Indiana/Petersburg		
		America/Indianapolis		
		America/Indiana/Tell_City		
		America/Indiana/Vevay		
		America/Indiana/Vincennes		
		America/Indiana/Winamac		
		America/Inuvik		
		America/Iqaluit		
		America/Jamaica		
		America/Jujuy		
		America/Juneau		
		America/Kentucky/Louisville		
		America/Kentucky/Monticello		
		America/Knox_IN		
		America/Kralendijk		
		America/La_Paz		
		America/Lima		

Attribute	Type	Range/Length/Pattern	Default	Description
		America/Los_Angeles		
		America/Louisville		
		America/Lower_Princes		
		America/Maceio		
		America/Managua		
		America/Manaus		
		America/Marigot		
		America/Martinique		
		America/Matamoros		
		America/Mazatlan		
		America/Mendoza		
		America/Menominee		
		America/Merida		
		America/Metlakatla		
		America/Mexico_City		
		America/Miquelon		
		America/Moncton		
		America/Monterrey		
		America/Montevideo		
		America/Montreal		
		America/Montserrat		
		America/Nassau		
		America/New_York		
		America/Nipigon		
		America/Nome		
		America/Noronha		
		America/North_Dakota/Beulah		
		America/North_Dakota/Center		
		America/North_Dakota/New_Salem		
		America/Nuuk		
		America/Ojinaga		
		America/Panama		
		America/Pangnirtung		
		America/Paramaribo		
		America/Phoenix		
		America/Port-au-Prince		
		America/Porto_Acre		
		America/Port_of_Spain		
		America/Porto_Velho		
		America/Puerto_Rico		
		America/Punta_Arenas		
		America/Rainy_River		
		America/Rankin_Inlet		
		America/Recife		
		America/Regina		
		America/Resolute		
		America/Rio_Branco		
		America/Rosario		
		America/Santa_Isabel		
		America/Santarem		
		America/Santiago		
		America/Santo_Domingo		
		America/Sao_Paulo		
		America/Scoresbysund		
		America/Shiprock		
		America/Sitka		
		America/St_Barthelemy		
		America/St_Johns		
		America/St_Kitts		
		America/St_Lucia		
		America/St_Thomas		
		America/St_Vincent		
		America/Swift_Current		
		America/Tegucigalpa		
		America/Thule		

Attribute	Type	Range/Length/Pattern	Default	Description
		America/Thunder_Bay		
		America/Tijuana		
		America/Toronto		
		America/Tortola		
		America/Vancouver		
		America/Virgin		
		America/Whitehorse		
		America/Winnipeg		
		America/Yakutat		
		America/Yellowknife		
		Antarctica/Casey		
		Antarctica/Davis		
		Antarctica/DumontDURville		
		Antarctica/Macquarie		
		Antarctica/Mawson		
		Antarctica/McMurdo		
		Antarctica/Palmer		
		Antarctica/Rothera		
		Antarctica/South_Pole		
		Antarctica/Syowa		
		Antarctica/Troll		
		Antarctica/Vostok		
		Arctic/Longyearbyen		
		Asia/Aden		
		Asia/Almaty		
		Asia/Amman		
		Asia/Anadyr		
		Asia/Aqtau		
		Asia/Aqtobe		
		Asia/Ashgabat		
		Asia/Ashkhabad		
		Asia/Atyrau		
		Asia/Baghdad		
		Asia/Bahrain		
		Asia/Baku		
		Asia/Bangkok		
		Asia/Barnaul		
		Asia/Beirut		
		Asia/Bishkek		
		Asia/Brunei		
		Asia/Chita		
		Asia/Choibalsan		
		Asia/Chongqing		
		Asia/Colombo		
		Asia/Dacca		
		Asia/Damascus		
		Asia/Dhaka		
		Asia/Dili		
		Asia/Dubai		
		Asia/Dushanbe		
		Asia/Famagusta		
		Asia/Gaza		
		Asia/Harbin		
		Asia/Hebron		
		Asia/Ho_Chi_Minh		
		Asia/Hong_Kong		
		Asia/Hovd		
		Asia/Irkutsk		
		Asia/Istanbul		
		Asia/Jakarta		
		Asia/Jayapura		
		Asia/Jerusalem		
		Asia/Kabul		
		Asia/Kamchatka		
		Asia/Karachi		

Attribute	Type	Range/Length/Pattern	Default	Description
		Asia/Kashgar		
		Asia/Kathmandu		
		Asia/Katmandu		
		Asia/Khandyga		
		Asia/Kolkata		
		Asia/Krasnoyarsk		
		Asia/Kuala_Lumpur		
		Asia/Kuching		
		Asia/Kuwait		
		Asia/Macao		
		Asia/Macau		
		Asia/Magadan		
		Asia/Makassar		
		Asia/Manila		
		Asia/Muscat		
		Asia/Nicosia		
		Asia/Novokuznetsk		
		Asia/Novosibirsk		
		Asia/Omsk		
		Asia/Oral		
		Asia/Phnom_Penh		
		Asia/Pontianak		
		Asia/Pyongyang		
		Asia/Qatar		
		Asia/Qostanay		
		Asia/Qyzylorda		
		Asia/Riyadh		
		Asia/Sakhalin		
		Asia/Samarkand		
		Asia/Seoul		
		Asia/Shanghai		
		Asia/Singapore		
		Asia/Srednekolymsk		
		Asia/Taipei		
		Asia/Tashkent		
		Asia/Tbilisi		
		Asia/Tehran		
		Asia/Tel_Aviv		
		Asia/Thimbu		
		Asia/Thimphu		
		Asia/Tokyo		
		Asia/Tomsk		
		Asia/Ujung_Pandang		
		Asia/Ulaanbaatar		
		Asia/Ulan_Bator		
		Asia/Urumqi		
		Asia/Ust-Nera		
		Asia/Vientiane		
		Asia/Vladivostok		
		Asia/Yakutsk		
		Asia/Yangon		
		Asia/Yekaterinburg		
		Asia/Yerevan		
		Atlantic/Azores		
		Atlantic/Bermuda		
		Atlantic/Canary		
		Atlantic/Cape_Verde		
		Atlantic/Faeroe		
		Atlantic/Faroe		
		Atlantic/Jan_Mayen		
		Atlantic/Madeira		
		Atlantic/Reykjavik		
		Atlantic/South_Georgia		
		Atlantic/Stanley		
		Atlantic/St_Helena		

Attribute	Type	Range/Length/Pattern	Default	Description
		Australia/ACT		
		Australia/Adelaide		
		Australia/Brisbane		
		Australia/Broken_Hill		
		Australia/Canberra		
		Australia/Currie		
		Australia/Darwin		
		Australia/Eucla		
		Australia/Hobart		
		Australia/LHI		
		Australia/Lindeman		
		Australia/Lord_Howe		
		Australia/Melbourne		
		Australia/North		
		Australia/NSW		
		Australia/Perth		
		Australia/Queensland		
		Australia/South		
		Australia/Sydney		
		Australia/Tasmania		
		Australia/Victoria		
		Australia/West		
		Australia/Yancowinna		
		Brazil/Acre		
		Brazil/DeNoronha		
		Brazil/East		
		Brazil/West		
		Canada/Atlantic		
		Canada/Central		
		Canada/Eastern		
		Canada/Mountain		
		Canada/Newfoundland		
		Canada/Pacific		
		Canada/Saskatchewan		
		Canada/Yukon		
		CET		
		Chile/Continental		
		Chile/EasterIsland		
		CST6CDT		
		Cuba		
		EET		
		Egypt		
		Eire		
		EST		
		EST5EDT		
		Etc/GMT		
		Etc/GMT+0		
		Etc/GMT-0		
		Etc/GMT0		
		Etc/GMT+1		
		Etc/GMT-1		
		Etc/GMT+10		
		Etc/GMT-10		
		Etc/GMT+11		
		Etc/GMT-11		
		Etc/GMT+12		
		Etc/GMT-12		
		Etc/GMT-13		
		Etc/GMT-14		
		Etc/GMT+2		
		Etc/GMT-2		
		Etc/GMT+3		
		Etc/GMT-3		
		Etc/GMT+4		
		Etc/GMT-4		

Attribute	Type	Range/Length/Pattern	Default	Description
		Etc/GMT+5		
		Etc/GMT-5		
		Etc/GMT+6		
		Etc/GMT-6		
		Etc/GMT+7		
		Etc/GMT-7		
		Etc/GMT+8		
		Etc/GMT-8		
		Etc/GMT+9		
		Etc/GMT-9		
		Etc/Greenwich		
		Etc/UCT		
		Etc/Universal		
		Etc/UTC		
		Etc/Zulu		
		Europe/Amsterdam		
		Europe/Andorra		
		Europe/Astrakhan		
		Europe/Athens		
		Europe/Belfast		
		Europe/Belgrade		
		Europe/Berlin		
		Europe/Bratislava		
		Europe/Brussels		
		Europe/Bucharest		
		Europe/Budapest		
		Europe/Busingen		
		Europe/Chisinau		
		Europe/Copenhagen		
		Europe/Dublin		
		Europe/Gibraltar		
		Europe/Guernsey		
		Europe/Helsinki		
		Europe/Isle_of_Man		
		Europe/Istanbul		
		Europe/Jersey		
		Europe/Kaliningrad		
		Europe/Kiev		
		Europe/Kirov		
		Europe/Lisbon		
		Europe/Ljubljana		
		Europe/London		
		Europe/Luxembourg		
		Europe/Madrid		
		Europe/Malta		
		Europe/Mariehamn		
		Europe/Minsk		
		Europe/Monaco		
		Europe/Moscow		
		Europe/Nicosia		
		Europe/Oslo		
		Europe/Paris		
		Europe/Podgorica		
		Europe/Prague		
		Europe/Riga		
		Europe/Rome		
		Europe/Samara		
		Europe/San_Marino		
		Europe/Sarajevo		
		Europe/Saratov		
		Europe/Simferopol		
		Europe/Skopje		
		Europe/Sofia		
		Europe/Stockholm		
		Europe/Tallinn		

Attribute	Type	Range/Length/Pattern	Default	Description
		Europe/Tirane		
		Europe/Tiraspol		
		Europe/Ulyanovsk		
		Europe/Uzhgorod		
		Europe/Vaduz		
		Europe/Vatican		
		Europe/Vienna		
		Europe/Vilnius		
		Europe/Volgograd		
		Europe/Warsaw		
		Europe/Zagreb		
		Europe/Zaporozhye		
		Europe/Zurich		
		GB		
		GB-Eire		
		GMT		
		GMT+0		
		GMT-0		
		GMT0		
		Greenwich		
		Hongkong		
		HST		
		Iceland		
		Indian/Antananarivo		
		Indian/Chagos		
		Indian/Christmas		
		Indian/Cocos		
		Indian/Comoro		
		Indian/Kerguelen		
		Indian/Mahe		
		Indian/Maldives		
		Indian/Mauritius		
		Indian/Mayotte		
		Indian/Reunion		
		Iran		
		Israel		
		Jamaica		
		Japan		
		Kwajalein		
		Libya		
		MET		
		Mexico/BajaNorte		
		Mexico/BajaSur		
		Mexico/General		
		MST		
		MST7MDT		
		Navajo		
		NZ		
		NZ-CHAT		
		Pacific/Apia		
		Pacific/Auckland		
		Pacific/Bougainville		
		Pacific/Chatham		
		Pacific/Chuuk		
		Pacific/Easter		
		Pacific/Efate		
		Pacific/Enderbury		
		Pacific/Fakaofu		
		Pacific/Fiji		
		Pacific/Funafuti		
		Pacific/Galapagos		
		Pacific/Gambier		
		Pacific/Guadalcanal		
		Pacific/Guam		
		Pacific/Honolulu		

Attribute	Type	Range/Length/Pattern	Default	Description
		Pacific/Johnston		
		Pacific/Kanton		
		Pacific/Kiritimati		
		Pacific/Kosrae		
		Pacific/Kwajalein		
		Pacific/Majuro		
		Pacific/Marquesas		
		Pacific/Midway		
		Pacific/Nauru		
		Pacific/Niue		
		Pacific/Norfolk		
		Pacific/Noumea		
		Pacific/Pago_Pago		
		Pacific/Palau		
		Pacific/Pitcairn		
		Pacific/Pohnpei		
		Pacific/Ponape		
		Pacific/Port_Moresby		
		Pacific/Rarotonga		
		Pacific/Saipan		
		Pacific/Samoa		
		Pacific/Tahiti		
		Pacific/Tarawa		
		Pacific/Tongatapu		
		Pacific/Truk		
		Pacific/Wake		
		Pacific/Wallis		
		Pacific/Yap		
		Poland		
		Portugal		
		PRC		
		PST8PDT		
		ROK		
		Singapore		
		Turkey		
		UCT		
		Universal		
		US/Alaska		
		US/Aleutian		
		US/Arizona		
		US/Central		
		US/Eastern		
		US/East-Indiana		
		US/Hawaii		
		US/Indiana-Starke		
		US/Michigan		
		US/Mountain		
		US/Pacific		
		US/Pacific-New		
		US/Samoa		
		UTC		
		WET		
		W-SU		
		Zulu		

Tracking Area

A list of Tracking Area ranges under the configured plmn, tac-low and tac-high range should not be configured with overlapping range values, else the lookup will fetch the closest tac-low entry.

XPath: *cna-mme-cc:/mme-cc/mme-profile/mme-plmn/tracking-area*

CLI Path: *mme-cc mme-profile plmn tracking-area*

List Properties:

Keys: tac-low

Maximum number of elements: 4000

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
tac-low	string	0x[0-9a-f]{4}		Tracking Area Code (TAC); The Tracking Area Code uniquely identifies a TA (Tracking Area) within a PLMN (Public Land Mobile Network); If 'tac-high' is configured, this attribute is considered as the lower end of the TAC range; The values are expressed as a string of hexadecimal characters (with lower case alphabets) of length 4 with use of leading zeros if necessary; The value is prefixed with '0x' to convey the hexadecimal representation; For example: 0xfa10, 0x04cb, 0x2344.
tac-high	string	0x[0-9a-f]{4}		TAC upper range; It is an optional field representing the upper boundary for a range of tracking areas from 'tac-low' to 'tac-high'; If not provided, it takes the value of 'tac-low'; The values are expressed as a string of hexadecimal characters (with lower case alphabets) of length 4 with use of leading zeros if necessary; The value is prefixed with '0x' to convey the hexadecimal representation; For example: 0xfa10, 0x04cb, 0x2344.
lac	string	0x[0-9a-f]{1,4}		Location Area Code (LAC); The values are expressed as a string of hexadecimal characters (with lower case alphabets) of length 1..4 with use of leading zeros if necessary. The value is prefixed with '0x' to convey the hexadecimal representation. For example: 0xfa10, 0x04cb, 0x2344.
lai-plmn-id	string	(([0-9]{5,6}))		PLMN of Location Area Identifier
relative-capacity	string	none or ([0-9] or [1-9][0-9] or [1][0-9][0-9] or [2][0-4][0-9] or [2][5][0-5])	none	This attribute provides a mechanism to set the MME relative capacity (as defined by TS 36.413 section 9.2.3.17) on a per tracking area basis. The relative capacity provided to an eNodeB is derived from this value. If a value of 'none' is configured for a tracking area, the relative capacity is derived from the configured value at the mme-profile level (mme-profile relative-capacity). For an eNodeB serving multiple tracking areas, the relative capacity is derived from the configured values of the tracking areas, using the lowest value. If the relative capacity is not configured or configured with 'none' for one or more of the serving tracking areas, the relative capacity is derived from the configured values of the tracking areas and the configured value at the mme-profile level, using the lowest value. If a modification results in a changed derived value, an 'MME Configuration Update' procedure is triggered. Default: none
ims-support	enumeration	enable disable	disable	Controls support for IP Multimedia Subsystem (IMS) voice over PS. Default: disable
emergency-support	enumeration	enable disable	enable	Controls support for emergency voice over PS. Default: enable
timezone	enumeration	Africa/Abidjan Africa/Accra Africa/Addis_Ababa Africa/Algiers Africa/Asmara Africa/Asmera Africa/Bamako Africa/Bangui Africa/Banjul Africa/Bissau Africa/Blantyre Africa/Brazzaville Africa/Bujumbura Africa/Cairo Africa/Casablanca Africa/Ceuta Africa/Conakry Africa/Dakar Africa/Dar_es_Salaam Africa/Djibouti Africa/Douala Africa/El_Aaiun Africa/Freetown Africa/Gaborone Africa/Harare		LINUX timezone values

Attribute	Type	Range/Length/Pattern	Default	Description
		Africa/Johannesburg		
		Africa/Juba		
		Africa/Kampala		
		Africa/Khartoum		
		Africa/Kigali		
		Africa/Kinshasa		
		Africa/Lagos		
		Africa/Libreville		
		Africa/Lome		
		Africa/Luanda		
		Africa/Lubumbashi		
		Africa/Lusaka		
		Africa/Malabo		
		Africa/Maputo		
		Africa/Maseru		
		Africa/Mbabane		
		Africa/Mogadishu		
		Africa/Monrovia		
		Africa/Nairobi		
		Africa/Ndjamena		
		Africa/Niamey		
		Africa/Nouakchott		
		Africa/Ouagadougou		
		Africa/Porto-Novo		
		Africa/Sao_Tome		
		Africa/Timbuktu		
		Africa/Tripoli		
		Africa/Tunis		
		Africa/Windhoek		
		America/Adak		
		America/Anchorage		
		America/Anguilla		
		America/Antigua		
		America/Araguaina		
		America/Argentina/Buenos_Aires		
		America/Argentina/Catamarca		
		America/Argentina/ComodRivadavia		
		America/Argentina/Cordoba		
		America/Argentina/Jujuy		
		America/Argentina/La_Rioja		
		America/Argentina/Mendoza		
		America/Argentina/Rio_Gallegos		
		America/Argentina/Salta		
		America/Argentina/San_Juan		
		America/Argentina/San_Luis		
		America/Argentina/Tucuman		
		America/Argentina/Ushuaia		
		America/Aruba		
		America/Asuncion		
		America/Atikokan		
		America/Atka		
		America/Bahia		
		America/Bahia_Banderas		
		America/Barbados		
		America/Belem		
		America/Belize		
		America/Blanc-Sablon		
		America/Boa_Vista		
		America/Bogota		
		America/Boise		
		America/Buenos_Aires		
		America/Cambridge_Bay		
		America/Campo_Grande		
		America/Cancun		
		America/Caracas		

Attribute	Type	Range/Length/Pattern	Default	Description
		America/Catamarca		
		America/Cayenne		
		America/Cayman		
		America/Chicago		
		America/Chihuahua		
		America/Coral_Harbour		
		America/Cordoba		
		America/Costa_Rica		
		America/Creston		
		America/Cuiaba		
		America/Curacao		
		America/Danmarkshavn		
		America/Dawson		
		America/Dawson_Creek		
		America/Denver		
		America/Detroit		
		America/Dominica		
		America/Edmonton		
		America/Eirunepe		
		America/El_Salvador		
		America/Ensenada		
		America/Fortaleza		
		America/Fort_Nelson		
		America/Fort_Wayne		
		America/Glace_Bay		
		America/Godthab		
		America/Goose_Bay		
		America/Grand_Turk		
		America/Grenada		
		America/Guadeloupe		
		America/Guatemala		
		America/Guayaquil		
		America/Guyana		
		America/Halifax		
		America/Havana		
		America/Hermosillo		
		America/Indiana/Indianapolis		
		America/Indiana/Knox		
		America/Indiana/Marengo		
		America/Indiana/Petersburg		
		America/Indianapolis		
		America/Indiana/Tell_City		
		America/Indiana/Vevay		
		America/Indiana/Vincennes		
		America/Indiana/Winamac		
		America/Inuvik		
		America/Iqaluit		
		America/Jamaica		
		America/Jujuy		
		America/Juneau		
		America/Kentucky/Louisville		
		America/Kentucky/Monticello		
		America/Knox_IN		
		America/Kralendijk		
		America/La_Paz		
		America/Lima		
		America/Los_Angeles		
		America/Louisville		
		America/Lower_Princes		
		America/Maceio		
		America/Managua		
		America/Manaus		
		America/Marigot		
		America/Martinique		
		America/Matamoros		

Attribute	Type	Range/Length/Pattern	Default	Description
		America/Mazatlan		
		America/Mendoza		
		America/Menominee		
		America/Merida		
		America/Metlakatla		
		America/Mexico_City		
		America/Miquelon		
		America/Moncton		
		America/Monterrey		
		America/Montevideo		
		America/Montreal		
		America/Montserrat		
		America/Nassau		
		America/New_York		
		America/Nipigon		
		America/Nome		
		America/Noronha		
		America/North_Dakota/Beulah		
		America/North_Dakota/Center		
		America/North_Dakota/New_Salem		
		America/Nuuk		
		America/Ojinaga		
		America/Panama		
		America/Pangnirtung		
		America/Paramaribo		
		America/Phoenix		
		America/Port-au-Prince		
		America/Porto_Acre		
		America/Port_of_Spain		
		America/Porto_Velho		
		America/Puerto_Rico		
		America/Punta_Arenas		
		America/Rainy_River		
		America/Rankin_Inlet		
		America/Recife		
		America/Regina		
		America/Resolute		
		America/Rio_Branco		
		America/Rosario		
		America/Santa_Isabel		
		America/Santarem		
		America/Santiago		
		America/Santo_Domingo		
		America/Sao_Paulo		
		America/Scoresbysund		
		America/Shiprock		
		America/Sitka		
		America/St_Barthelemy		
		America/St_Johns		
		America/St_Kitts		
		America/St_Lucia		
		America/St_Thomas		
		America/St_Vincent		
		America/Swift_Current		
		America/Tegucigalpa		
		America/Thule		
		America/Thunder_Bay		
		America/Tijuana		
		America/Toronto		
		America/Tortola		
		America/Vancouver		
		America/Virgin		
		America/Whitehorse		
		America/Winnipeg		
		America/Yakutat		

Attribute	Type	Range/Length/Pattern	Default	Description
		America/Yellowknife		
		Antarctica/Casey		
		Antarctica/Davis		
		Antarctica/DumontDURville		
		Antarctica/Macquarie		
		Antarctica/Mawson		
		Antarctica/McMurdo		
		Antarctica/Palmer		
		Antarctica/Rothera		
		Antarctica/South_Pole		
		Antarctica/Syowa		
		Antarctica/Troll		
		Antarctica/Vostok		
		Arctic/Longyearbyen		
		Asia/Aden		
		Asia/Almaty		
		Asia/Amman		
		Asia/Anadyr		
		Asia/Aqtau		
		Asia/Aqtobe		
		Asia/Ashgabat		
		Asia/Ashkhabad		
		Asia/Atyrau		
		Asia/Baghdad		
		Asia/Bahrain		
		Asia/Baku		
		Asia/Bangkok		
		Asia/Barnaul		
		Asia/Beirut		
		Asia/Bishkek		
		Asia/Brunei		
		Asia/Chita		
		Asia/Choibalsan		
		Asia/Chongqing		
		Asia/Colombo		
		Asia/Dacca		
		Asia/Damascus		
		Asia/Dhaka		
		Asia/Dili		
		Asia/Dubai		
		Asia/Dushanbe		
		Asia/Famagusta		
		Asia/Gaza		
		Asia/Harbin		
		Asia/Hebron		
		Asia/Ho_Chi_Minh		
		Asia/Hong_Kong		
		Asia/Hovd		
		Asia/Irkutsk		
		Asia/Istanbul		
		Asia/Jakarta		
		Asia/Jayapura		
		Asia/Jerusalem		
		Asia/Kabul		
		Asia/Kamchatka		
		Asia/Karachi		
		Asia/Kashgar		
		Asia/Kathmandu		
		Asia/Katmandu		
		Asia/Khandyga		
		Asia/Kolkata		
		Asia/Krasnoyarsk		
		Asia/Kuala_Lumpur		
		Asia/Kuching		
		Asia/Kuwait		

Attribute	Type	Range/Length/Pattern	Default	Description
		Asia/Macao		
		Asia/Macau		
		Asia/Magadan		
		Asia/Makassar		
		Asia/Manila		
		Asia/Muscat		
		Asia/Nicosia		
		Asia/Novokuznetsk		
		Asia/Novosibirsk		
		Asia/Omsk		
		Asia/Oral		
		Asia/Phnom_Penh		
		Asia/Pontianak		
		Asia/Pyongyang		
		Asia/Qatar		
		Asia/Qostanay		
		Asia/Qyzylorda		
		Asia/Riyadh		
		Asia/Sakhalin		
		Asia/Samarkand		
		Asia/Seoul		
		Asia/Shanghai		
		Asia/Singapore		
		Asia/Srednekolymsk		
		Asia/Taipei		
		Asia/Tashkent		
		Asia/Tbilisi		
		Asia/Tehran		
		Asia/Tel_Aviv		
		Asia/Thimbu		
		Asia/Thimphu		
		Asia/Tokyo		
		Asia/Tomsk		
		Asia/Ujung_Pandang		
		Asia/Ulaanbaatar		
		Asia/Ulan_Bator		
		Asia/Urumqi		
		Asia/Ust-Nera		
		Asia/Vientiane		
		Asia/Vladivostok		
		Asia/Yakutsk		
		Asia/Yangon		
		Asia/Yekaterinburg		
		Asia/Yerevan		
		Atlantic/Azores		
		Atlantic/Bermuda		
		Atlantic/Canary		
		Atlantic/Cape_Verde		
		Atlantic/Faeroe		
		Atlantic/Faroe		
		Atlantic/Jan_Mayen		
		Atlantic/Madeira		
		Atlantic/Reykjavik		
		Atlantic/South_Georgia		
		Atlantic/Stanley		
		Atlantic/St_Helena		
		Australia/ACT		
		Australia/Adelaide		
		Australia/Brisbane		
		Australia/Broken_Hill		
		Australia/Canberra		
		Australia/Currie		
		Australia/Darwin		
		Australia/Eucla		
		Australia/Hobart		

Attribute	Type	Range/Length/Pattern	Default	Description
		Australia/LHI		
		Australia/Lindeman		
		Australia/Lord_Howe		
		Australia/Melbourne		
		Australia/North		
		Australia/NSW		
		Australia/Perth		
		Australia/Queensland		
		Australia/South		
		Australia/Sydney		
		Australia/Tasmania		
		Australia/Victoria		
		Australia/West		
		Australia/Yancowinna		
		Brazil/Acre		
		Brazil/DeNoronha		
		Brazil/East		
		Brazil/West		
		Canada/Atlantic		
		Canada/Central		
		Canada/Eastern		
		Canada/Mountain		
		Canada/Newfoundland		
		Canada/Pacific		
		Canada/Saskatchewan		
		Canada/Yukon		
		CET		
		Chile/Continental		
		Chile/EasterIsland		
		CST6CDT		
		Cuba		
		EET		
		Egypt		
		Eire		
		EST		
		EST5EDT		
		Etc/GMT		
		Etc/GMT+0		
		Etc/GMT-0		
		Etc/GMT0		
		Etc/GMT+1		
		Etc/GMT-1		
		Etc/GMT+10		
		Etc/GMT-10		
		Etc/GMT+11		
		Etc/GMT-11		
		Etc/GMT+12		
		Etc/GMT-12		
		Etc/GMT-13		
		Etc/GMT-14		
		Etc/GMT+2		
		Etc/GMT-2		
		Etc/GMT+3		
		Etc/GMT-3		
		Etc/GMT+4		
		Etc/GMT-4		
		Etc/GMT+5		
		Etc/GMT-5		
		Etc/GMT+6		
		Etc/GMT-6		
		Etc/GMT+7		
		Etc/GMT-7		
		Etc/GMT+8		
		Etc/GMT-8		
		Etc/GMT+9		

Attribute	Type	Range/Length/Pattern	Default	Description
		Etc/GMT-9		
		Etc/Greenwich		
		Etc/UCT		
		Etc/Universal		
		Etc/UTC		
		Etc/Zulu		
		Europe/Amsterdam		
		Europe/Andorra		
		Europe/Astrakhan		
		Europe/Athens		
		Europe/Belfast		
		Europe/Belgrade		
		Europe/Berlin		
		Europe/Bratislava		
		Europe/Brussels		
		Europe/Bucharest		
		Europe/Budapest		
		Europe/Busingen		
		Europe/Chisinau		
		Europe/Copenhagen		
		Europe/Dublin		
		Europe/Gibraltar		
		Europe/Guernsey		
		Europe/Helsinki		
		Europe/Isle_of_Man		
		Europe/Istanbul		
		Europe/Jersey		
		Europe/Kaliningrad		
		Europe/Kiev		
		Europe/Kirov		
		Europe/Lisbon		
		Europe/Ljubljana		
		Europe/London		
		Europe/Luxembourg		
		Europe/Madrid		
		Europe/Malta		
		Europe/Mariehamn		
		Europe/Minsk		
		Europe/Monaco		
		Europe/Moscow		
		Europe/Nicosia		
		Europe/Oslo		
		Europe/Paris		
		Europe/Podgorica		
		Europe/Prague		
		Europe/Riga		
		Europe/Rome		
		Europe/Samara		
		Europe/San_Marino		
		Europe/Sarajevo		
		Europe/Saratov		
		Europe/Simferopol		
		Europe/Skopje		
		Europe/Sofia		
		Europe/Stockholm		
		Europe/Tallinn		
		Europe/Tirane		
		Europe/Tiraspol		
		Europe/Ulyanovsk		
		Europe/Uzhgorod		
		Europe/Vaduz		
		Europe/Vatican		
		Europe/Vienna		
		Europe/Vilnius		
		Europe/Volgograd		

Attribute	Type	Range/Length/Pattern	Default	Description
		Europe/Warsaw		
		Europe/Zagreb		
		Europe/Zaporozhye		
		Europe/Zurich		
		GB		
		GB-Eire		
		GMT		
		GMT+0		
		GMT-0		
		GMT0		
		Greenwich		
		Hongkong		
		HST		
		Iceland		
		Indian/Antananarivo		
		Indian/Chagos		
		Indian/Christmas		
		Indian/Cocos		
		Indian/Comoro		
		Indian/Kerguelen		
		Indian/Mahe		
		Indian/Maldives		
		Indian/Mauritius		
		Indian/Mayotte		
		Indian/Reunion		
		Iran		
		Israel		
		Jamaica		
		Japan		
		Kwajalein		
		Libya		
		MET		
		Mexico/BajaNorte		
		Mexico/BajaSur		
		Mexico/General		
		MST		
		MST7MDT		
		Navajo		
		NZ		
		NZ-CHAT		
		Pacific/Apia		
		Pacific/Auckland		
		Pacific/Bougainville		
		Pacific/Chatham		
		Pacific/Chuuk		
		Pacific/Easter		
		Pacific/Efate		
		Pacific/Enderbury		
		Pacific/Fakaofu		
		Pacific/Fiji		
		Pacific/Funafuti		
		Pacific/Galapagos		
		Pacific/Gambier		
		Pacific/Guadalcanal		
		Pacific/Guam		
		Pacific/Honolulu		
		Pacific/Johnston		
		Pacific/Kanton		
		Pacific/Kiritimati		
		Pacific/Kosrae		
		Pacific/Kwajalein		
		Pacific/Majuro		
		Pacific/Marquesas		
		Pacific/Midway		
		Pacific/Nauru		

Attribute	Type	Range/Length/Pattern	Default	Description
		Pacific/Niue		
		Pacific/Norfolk		
		Pacific/Noumea		
		Pacific/Pago_Pago		
		Pacific/Palau		
		Pacific/Pitcairn		
		Pacific/Pohnpei		
		Pacific/Ponape		
		Pacific/Port_Moresby		
		Pacific/Rarotonga		
		Pacific/Saipan		
		Pacific/Samoa		
		Pacific/Tahiti		
		Pacific/Tarawa		
		Pacific/Tongatapu		
		Pacific/Truk		
		Pacific/Wake		
		Pacific/Wallis		
		Pacific/Yap		
		Poland		
		Portugal		
		PRC		
		PST8PDT		
		ROK		
		Singapore		
		Turkey		
		UCT		
		Universal		
		US/Alaska		
		US/Aleutian		
		US/Arizona		
		US/Central		
		US/Eastern		
		US/East-Indiana		
		US/Hawaii		
		US/Indiana-Starke		
		US/Michigan		
		US/Mountain		
		US/Pacific		
		US/Pacific-New		
		US/Samoa		
		UTC		
		WET		
		W-SU		
		Zulu		

Location Area

Mapping between the location area code and the VLR pool table

XPath: [cna-mme-cc/mme-cc/mme-profile/mme-plmn/location-area](#)

CLI Path: [mme-cc mme-profile plmn location-area](#)

List Properties:

Keys: lac-low

Maximum number of elements: 1000

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
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Attribute	Type	Range/Length/Pattern	Default	Description
lac-low	string	0x[1-9a-f][0-9a-f]{0,3}		Location Area Code (LAC). The Location Area Code uniquely identifies a LA (Location Area) within a PLMN (Public Land Mobile Network). If 'lac-high' is configured, this attribute is considered as the lower end of the TAC range; The values are expressed as a string of hexadecimal characters (with lower case alphabets) of length 1..4 with use of leading zeros if necessary. The value is prefixed with '0x' to convey the hexadecimal representation. For example: 0xfa10, 0x04cb, 0x2344.
lac-high	string	0x[1-9a-f][0-9a-f]{0,3}		LAC upper range. It is an optional field representing the upper boundary for a range of location areas from 'lac-low' to 'lac-high'. If not provided, it takes the value of 'lac-low'; The values are expressed as a string of hexadecimal characters (with lower case alphabets) of length 1..4 with use of leading zeros if necessary. The value is prefixed with '0x' to convey the hexadecimal representation. For example: 0xfa10, 0x04cb, 0x2344.
vlr-pool-name	leafref			VLR pool name Leaf Reference Path: cna-mme-cc:/mme-cc/vlr-pool/name
include-as-mme-group-id	enumeration	enable disable		Controls inclusion of LAC as additional MME Group ID in the S1 Setup Response and MME Configuration Update messages sent to eNB. This can help for inter-RAT changes from 2G/3G to 4G in which the eNB may use LAC to help with MME selection. As defined by S1AP spec TS 36.413 the S1 Setup Response (section 9.1.8.5) and the MME Configuration Update (section 9.1.8.10) both include Served GroupID lists. The MME Group ID is 16 bits and so is the LAC. If this option is enabled for a particular location area served by this MME it will be included in a separate IE in the S1AP messages after the traditional Served GroupID list IE which includes the actual group ID. Default: disable

Nsa En Dc

Non-stand-alone (NSA) feature config

XPath: [cna-mme-cc:/mme-cc/mme-profile/mme-plmn/nsa-en-dc](#)

CLI Path: [mme-cc mme-profile plmn nsa-en-dc](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
gw-selection-failure-action	enumeration	reject retry-without-nc-nr	retry-without-nc-nr	Action to be taken if a GW with matching NC-NR is not found during the GW selection procedure for a DCNR UE. Default: retry-without-nc-nr
non-dcnr-ue-gw-selection-retry	enumeration	include-ncnr exclude-ncnr	include-ncnr	GW selection retry based on with or without the 'nc-nr' enhanced service parameters for a non-DCNR UE. Default: include-ncnr
gtp-fallback-cause-codes	leaf-list of union	['uint8,64..239', 'Enumerations: no-response-66 system-failure-72 no-resources-available-73 missing-or-unknown-apn-78 all-dynamic-addresses-are-occupied-84 protocol-type-not-supported-86 no-memory-available-91 request-rejected-94 pdn-gateway-not-responding-100 apn-congestion-113 timed-out-request-122 ']		List of GTPv2 cause codes received from nc-nr GW's in Create Session Response message that trigger fallback to a GW which does not have nc-nr support

Attribute	Type	Range/Length/Pattern	Default	Description
gtp-fallback-action	enumeration	fallback-without-nc-nr reject	reject	Describes action to be taken if all nc-nr capable GW are down. Default: reject

Emergency Support

MME support for emergency services

XPath: *cna-mme-cc:/mme-cc/mme-profile/mme-plmn/emergency-support*

CLI Path: *mme-cc mme-profile plmn emergency-support*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
emg-user-option	enumeration	valid-ues-only only-authenticated-ues imsi-only-ues all-ues	all-ues	User category to allow an IMS emergency call. Default: all-ues
apn-network-id	string	(([a-zA-Z0-9])+((- or .) [A-Za-z0-9]+)*)	emergency	Emergency Access Point Name (APN) network ID. Default: emergency
apn-operator-id	string	1..100	use-serving-plmn	Emergency Access point Name (APN) operator ID. Default: use-serving-plmn
qci-default-bearer	uint8	5..9	5	Emergency QoS Class Identifier (QCI) for default bearers. Default: 5
arp-priority-default-bearer	uint8	1..5	1	Emergency Allocation and Retention Priority (ARP) priority for default bearers. Default: 1
apn-ul-ambr	uint32	0..256000	32	Emergency APN uplink Aggregated Maximum Bit Rate (AMBR). Default: 32; Unit: kbps
apn-dl-ambr	uint32	0..256000	32	Emergency APN downlink Aggregated Maximum Bit Rate (AMBR). Default: 32; Unit: kbps
pgw-hostname	string	3..127	none	Fully Qualified Domain Name (FQDN) of the PGW. Default: none
pdn-type	enumeration	ipv4 ipv6 ipv4ipv6	ipv4	Packet Data Network (PDN) type chosen for the UE emergency session. Default: ipv4
pgw-ip-address1	inet:ip-address		0.0.0.0	IPv4/IPv6 address of the primary PGW. Default: 0.0.0.0
pgw-ip-address2	inet:ip-address		0.0.0.0	IPv4/IPv6 address of the alternative PGW. Default: 0.0.0.0
skip-authentication	enumeration	enable disable	enable	Controls whether authentication may be skipped during an emergency attach. Applicable when emg-user-option is imsi-only-ues or all-ues. Default: enable
mobile-reachable-timer	union	['uint16,30..300', 'Enumerations: t3412plus1 ']	t3412plus1	Mobile reachable timer for emergency attached UEs. Default: t3412plus1; Unit: minutes
deactivate-emg-pdn-after-srvcc	enumeration	enable disable	enable	Controls if the IMS emergency session should be deactivated after the transfer of emergency voice bearers to the MSC. Default: enable
apn-selection-mode	enumeration	nwk-provided-subscription-not-verified nwk-provided-subscription-verified	nwk-provided-subscription-not-verified	Emergency APN selection mode IE value. Controls the value of the APN Selection Mode IE in the Create Session Request message as defined by 3GPP TS 29.274 (8.58). This attribute does not change the actual selection algorithm, only the value presented in the IE. Default: nwk-provided-subscription-not-verified

Mobility Policy

Mobility Policy

XPath: *cna-mme-cc:/mme-cc/mobility-policy*

CLI Path: *mme-cc mobility-policy*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
analyzer-name	string	1..64		Name of the mobility analyzer

Default

Mobility class with default values

XPath: *cna-mme-cc:/mme-cc/mobility-policy/default*

CLI Path: *mme-cc mobility-policy default*

Access Control

Access Control Policy

XPath: *cna-mme-cc:/mme-cc/mobility-policy/default/access-control*

CLI Path: *mme-cc mobility-policy default access-control*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
access-reject	enumeration	undefined-cause-0 eps-service-not-allowed-7 eps-and-non-eps-service-not-allowed-8 ue-id-cannot-be-derived-from-network-9 plmn-not-allowed-11 tracking-area-not-allowed-12 roaming-not-allowed-in-this-ta-13 eps-service-not-allowed-in-this-plmn-14 no-suitable-cells-in-ta-15 network-failure-17 esm-failure-19	undefined-cause-0	Access reject cause for the subscriber
closed-subscriber-group	enumeration	enable disable	disable	Controls the closed subscriber group for this mobility class. Default: disable
inter-rat-ho-restriction-list	leaf-list of enumeration	eutran nrIn5gs all		Radio Access Type (RAT) types for which Inter RAT Hand Over (HO) is not allowed. Valid values are: 'all' or up to 1 of the following RAT types 'eutran' and 'nrIn5gs'. Maximum number of elements: 1
secondary-rat-en-dc	enumeration	standard prohibit	standard	This attribute provides the control to allow or prohibit the E-UTR-NR Dual Connectivity (EN-DC) functionality for this class of subscribers when 'nsa-en-dc' feature is enabled. MME support for EN-DC in general is configured under 'mme-cc feature nsa-en-dc'. Default: standard
user-type	enumeration	home-user visiting-user	home-user	Defines the type of the subscriber. Default: home-user

Ran Reroute

Reroute subscribers to other MME

XPath: *cna-mme-cc:/mme-cc/mobility-policy/default/access-control/ran-reroute*

CLI Path: *mme-cc mobility-policy default access-control ran-reroute*

Triggers

List of triggers to reroute subscribers to the configured MME.

XPath: *cna-mme-cc:/mme-cc/mobility-policy/default/access-control/ran-reroute/triggers*

CLI Path: *mme-cc mobility-policy default access-control ran-reroute triggers*

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	enumeration	always user-unknown dcnr-capable n1-capable		Reroute condition for subscribers. Valid values are always, user-unknown, dcnr-capable, n1-capable. The entry with trigger 'name' value 'always' is not allowed when the entries with other trigger names are already configured The entry with trigger 'name' value 'always' is not allowed when the entries with other trigger names are already configured The entry with trigger 'name' value 'always' is not allowed when the entries with other trigger names are already configured
dest-mme-group-id	uint16			MME group identifier to use as the destination of the reroute The 'dest-mme-group-id' should not be the same as configured under 'mme-cc mme-profile mme-groupid'

Service Control

Service Control Policy

XPath: *cna-mme-cc:/mme-cc/mobility-policy/default/service-control*

CLI Path: *mme-cc mobility-policy default service-control*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
sgs	enumeration	enable disable	enable	Control for MME to perform Combined Attach/TAU for a UE, over SGS Interface. Default: enable
force-combined-on-request	enumeration	enable disable	disable	Controls support for combined attach for the UE with data only SIMs even when the combined service is not available. Default: disable.
ims-vops-agreement	enumeration	enable disable enable-if-srvcc-subscribed	disable	Controls support for IMS voice over PS. Default: disable
emergency-support	enumeration	enable disable	enable	Controls support for emergency voice over PS. Default: enable
voice-domain-profile-name	leafref			Name of voice domain profile Leaf Reference Path: cna-mme-cc:/mme-cc/mobility-policy/voice-domain-profile/name
rfsp-index	union	['uint16,1..256', 'Enumerations: use-subscribed-value none ']	use-subscribed-value	RAT frequency selection priority index. Default: use-subscribed-value

Ciot

Cellular Internet of Things (CIoT) parameters

XPath: *cna-mme-cc:/mme-cc/mobility-policy/default/service-control/ciot*

CLI Path: *mme-cc mobility-policy default service-control ciot*

Ciot Eps Optimization

Feature list for CloT capable UEs.

XPath: *cna-mme-cc:/mme-cc/mobility-policy/default/service-control/ciot/ciot-eps-optimization*

CLI Path: *mme-cc mobility-policy default service-control ciot ciot-eps-optimization*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
control-plane-optimization	enumeration	enable disable	disable	Controls support of Control Plane Optimization. Default: disable
s1u-data-transfer	enumeration	enable disable	disable	Controls support of S1-U data transfer for UEs. If enabled and negotiated with a UE, it allows establishment of traditional S1-U bearers which are not subject to CloT optimizations. Default: disable
extended-pco	enumeration	enable disable	disable	Controls support of Extended Protocol Configuration Option (PCO). Default: disable

Sgw Extended DI Buffering

Extended downlink data buffering configuration.

XPath: *cna-mme-cc:/mme-cc/mobility-policy/default/service-control/ciot/sgw-extended-dl-buffering*

CLI Path: *mme-cc mobility-policy default service-control ciot sgw-extended-dl-buffering*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
status	enumeration	enable disable	enable	Controls support of extended downlink data buffering on the gateway as indicated by the MME in a DL Buffering Duration IE, during DDN ACK messaging, as described in 3GPP TS 29.272, section 7.2.11.2. Default:enable
policy-source	enumeration	subscription mme	mme	Extended downlink data buffering policy source. Valid values are subscription and mme. Default: mme
suggested-packet-count	union	['Enumerations: none , 'uint16,1..1024']	5	Extended downlink data buffering suggested packet count for data transferred over s1-u interface. Default: 5

Purge Trigger

List of purge triggers

XPath: *cna-mme-cc:/mme-cc/mobility-policy/default/service-control/purge-trigger*

CLI Path: *mme-cc mobility-policy default service-control purge-trigger*

List Properties:

Keys: detach-type

Maximum number of elements: 3

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
detach-type	enumeration	oam-forced explicit implicit all		Detach type The 'detach-type' with value all is not allowed when entries with other detach types are already configured The 'detach-type' with value all is not allowed when entries with other detach types are already configured The 'detach-type' with value all is not allowed when entries with other detach types are already configured
inactive-age	union	['Enumerations: immediate , ']	immediate	Time to purge. Default: immediate

Nas Policy

Non Access Stratum Policy

XPath: *cna-mme-cc:/mme-cc/mobility-policy/default/nas-policy*

CLI Path: *mme-cc mobility-policy default nas-policy*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
mm-profile-name	leafref			Name of the MM Profile Leaf Reference Path: cna-mme-cc:/mme-cc/interface/nas/mm/profile/name
paging-profile-set-name	leafref			Name of the paging profile set Leaf Reference Path: cna-mme-cc:/mme-cc/interface/nas/paging/profile-set/name
security-profile-name	leafref			Name of the Security Profile Leaf Reference Path: cna-mme-cc:/mme-cc/interface/nas/security/profile/name

Class

Mobility Class

XPath: *cna-mme-cc:/mme-cc/mobility-policy/class*

CLI Path: *mme-cc mobility-policy class*

List Properties:

Keys: name

Maximum number of elements: 1000

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	1..64		Name of the mobility class

Access Control

Access Control Policy

XPath: *cna-mme-cc:/mme-cc/mobility-policy/class/access-control*

CLI Path: *mme-cc mobility-policy class access-control*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
access-reject	enumeration	undefined-emmm-cause-0	undefined-emmm-cause-0	Access reject cause for the subscriber
		eps-service-not-allowed-7		
		eps-and-non-eps-service-not-allowed-8		
		ue-id-cannot-be-derived-from-network-9		
		plmn-not-allowed-11		
		tracking-area-not-allowed-12		
		roaming-not-allowed-in-this-ta-13		
		eps-service-not-allowed-in-this-plmn-14		
		no-suitable-cells-in-ta-15		
network-failure-17				
esm-failure-19				
closed-subscriber-group	enumeration	enable disable	disable	Controls the closed subscriber group for this mobility class. Default: disable

Attribute	Type	Range/Length/Pattern	Default	Description
inter-rat-ho-restriction-list	leaf-list of enumeration	eutran nrln5gs all		Radio Access Type (RAT) types for which Inter RAT Hand Over (HO) is not allowed. Valid values are: 'all' or up to 1 of the following RAT types 'eutran' and 'nrln5gs'. Maximum number of elements: 1
secondary-rat-en-dc	enumeration	standard prohibit	standard	This attribute provides the control to allow or prohibit the E-UTR-NR Dual Connectivity (EN-DC) functionality for this class of subscribers when 'nsa-en-dc' feature is enabled. MME support for EN-DC in general is configured under 'mme-cc feature nsa-en-dc'. Default: standard
user-type	enumeration	home-user visiting-user	home-user	Defines the type of the subscriber. Default: home-user

Ran Reroute

Reroute subscribers to other MME

XPath: [cna-mme-cc:/mme-cc/mobility-policy/class/access-control/ran-reroute](#)

CLI Path: [mme-cc mobility-policy class access-control ran-reroute](#)

Triggers

List of triggers to reroute subscribers to the configured MME.

XPath: [cna-mme-cc:/mme-cc/mobility-policy/class/access-control/ran-reroute/triggers](#)

CLI Path: [mme-cc mobility-policy class access-control ran-reroute triggers](#)

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	enumeration	always user-unknown dcnr-capable n1-capable		Reroute condition for subscribers. Valid values are always, user-unknown, dcnr-capable, n1-capable. The entry with trigger 'name' value 'always' is not allowed when the entries with other trigger names are already configured The entry with trigger 'name' value 'always' is not allowed when the entries with other trigger names are already configured The entry with trigger 'name' value 'always' is not allowed when the entries with other trigger names are already configured
dest-mme-group-id	uint16			MME group identifier to use as the destination of the reroute The 'dest-mme-group-id' should not be the same as configured under 'mme-cc mme-profile mme-groupid'

Service Control

Service Control Policy

XPath: [cna-mme-cc:/mme-cc/mobility-policy/class/service-control](#)

CLI Path: [mme-cc mobility-policy class service-control](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
sgs	enumeration	enable disable	enable	Control for MME to perform Combined Attach/TAU for a UE, over SGS Interface. Default: enable
force-combined-on-request	enumeration	enable disable	disable	Controls support for combined attach for the UE with data only SIMs even when the combined service is not available. Default: disable.
ims-vops-agreement	enumeration	enable disable enable-if-srvcc-subscribed	disable	Controls support for IMS voice over PS. Default: disable

Attribute	Type	Range/Length/Pattern	Default	Description
emergency-support	enumeration	enable disable	enable	Controls support for emergency voice over PS. Default: enable
voice-domain-profile-name	leafref			Name of voice domain profile Leaf Reference Path: cna-mme-cc:/mme-cc/mobility-policy/voice-domain-profile/name
rfsp-index	union	['uint16,1..256', 'Enumerations: use-subscribed-value none ']	use-subscribed-value	RAT frequency selection priority index. Default: use-subscribed-value

Ciot

Cellular Internet of Things (CIoT) parameters

XPath: [cna-mme-cc:/mme-cc/mobility-policy/class/service-control/CIoT](#)

CLI Path: [mme-cc mobility-policy class service-control CIoT](#)

Ciot Eps Optimization

Feature list for CIoT capable UEs.

XPath: [cna-mme-cc:/mme-cc/mobility-policy/class/service-control/CIoT/CIoT-Eps-Optimization](#)

CLI Path: [mme-cc mobility-policy class service-control CIoT CIoT-Eps-Optimization](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
control-plane-optimization	enumeration	enable disable	disable	Controls support of Control Plane Optimization. Default: disable
s1u-data-transfer	enumeration	enable disable	disable	Controls support of S1-U data transfer for UEs. If enabled and negotiated with a UE, it allows establishment of traditional S1-U bearers which are not subject to CIoT optimizations. Default: disable
extended-pco	enumeration	enable disable	disable	Controls support of Extended Protocol Configuration Option (PCO). Default: disable

Sgw Extended DI Buffering

Extended downlink data buffering configuration.

XPath: [cna-mme-cc:/mme-cc/mobility-policy/class/service-control/CIoT/sgw-extended-dl-buffering](#)

CLI Path: [mme-cc mobility-policy class service-control CIoT sgw-extended-dl-buffering](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
status	enumeration	enable disable	enable	Controls support of extended downlink data buffering on the gateway as indicated by the MME in a DL Buffering Duration IE, during DDN ACK messaging, as described in 3GPP TS 29.272, section 7.2.11.2. Default:enable
policy-source	enumeration	subscription mme	mme	Extended downlink data buffering policy source. Valid values are subscription and mme. Default: mme
suggested-packet-count	union	['Enumerations: none , 'uint16,1..1024']	5	Extended downlink data buffering suggested packet count for data transferred over s1-u interface. Default: 5

Purge Trigger

List of purge triggers

XPath: [cna-mme-cc:/mme-cc/mobility-policy/class/service-control/purge-trigger](#)

CLI Path: *mme-cc mobility-policy class service-control purge-trigger*

List Properties:

Keys: detach-type

Maximum number of elements: 3

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
detach-type	enumeration	oam-forced explicit implicit all		Detach type The 'detach-type' with value all is not allowed when entries with other detach types are already configured The 'detach-type' with value all is not allowed when entries with other detach types are already configured The 'detach-type' with value all is not allowed when entries with other detach types are already configured
inactive-age	union	['Enumerations: immediate ']	immediate	Time to purge. Default: immediate

Nas Policy

Non Access Stratum Policy

XPath: *cna-mme-cc:/mme-cc/mobility-policy/class/nas-policy*

CLI Path: *mme-cc mobility-policy class nas-policy*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
mm-profile-name	leafref			Name of the MM Profile Leaf Reference Path: cna-mme-cc:/mme-cc/interface/nas/mm/profile/name
paging-profile-set-name	leafref			Name of the paging profile set Leaf Reference Path: cna-mme-cc:/mme-cc/interface/nas/paging/profile-set/name
security-profile-name	leafref			Name of the Security Profile Leaf Reference Path: cna-mme-cc:/mme-cc/interface/nas/security/profile/name

Voice Domain Profile

A profile that includes RFSP indexes for voice domain and data domain preferences

XPath: *cna-mme-cc:/mme-cc/mobility-policy/voice-domain-profile*

CLI Path: *mme-cc mobility-policy voice-domain-profile*

List Properties:

Keys: name

Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	1..32		Unique profile name
dc00-cs-voice-only	union	['uint16,1..256', 'Enumerations: use-subscribed-value none ']	use-subscribed-value	Data Centric Circuit Switched Voice Only. Controls the data centric RFSP index selection. Default: use-subscribed-value
dc01-ps-voice-only	union	['uint16,1..256', 'Enumerations: use-subscribed-value none ']	use-subscribed-value	Data Centric Packet Switched Voice Only. Controls the data centric RFSP index selection. Default: use-subscribed-value

Attribute	Type	Range/Length/Pattern	Default	Description
dc10-cs-voice-preferred	union	['uint16,1..256', 'Enumerations: use-subscribed-value none ']	use-subscribed-value	Data Centric Circuit Switched Voice Preferred. Controls the data centric RFSP index selection. Default: use-subscribed-value
dc11-ps-voice-preferred	union	['uint16,1..256', 'Enumerations: use-subscribed-value none ']	use-subscribed-value	Data Centric Packet Switched Voice Preferred. Controls the data centric RFSP index selection. Default: use-subscribed-value
vc00-cs-voice-only	union	['uint16,1..256', 'Enumerations: use-subscribed-value none ']	use-subscribed-value	Voice Centric Circuit Switched Voice Only. Controls the voice centric RFSP index selection. Default: use-subscribed-value
vc01-ps-voice-only	union	['uint16,1..256', 'Enumerations: use-subscribed-value none ']	use-subscribed-value	Voice Centric Packet Switched Voice Only. Controls the voice centric RFSP index selection. Default: use-subscribed-value
vc10-cs-voice-preferred	union	['uint16,1..256', 'Enumerations: use-subscribed-value none ']	use-subscribed-value	Voice Centric Circuit Switched Voice Preferred. Controls the voice centric RFSP index selection. Default: use-subscribed-value
vc11-ps-voice-preferred	union	['uint16,1..256', 'Enumerations: use-subscribed-value none ']	use-subscribed-value	Voice Centric Packet Switched Voice Preferred. Controls the voice centric RFSP index selection. Default: use-subscribed-value

Session Policy

Session Policy

XPath: [cna-mme-cc:/mme-cc/session-policy](#)

CLI Path: [mme-cc session-policy](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
analyzer-pre-apn-selection	string	1..64		Name of the analyzer used for initial classification of the subscriber
analyzer-post-apn-selection	string	1..64		Name of the analyzer used for subscriber classification post apn-selection

Default

Session class with default values

XPath: [cna-mme-cc:/mme-cc/session-policy/default](#)

CLI Path: [mme-cc session-policy default](#)

Access Control

PDN Access Control Policy

XPath: [cna-mme-cc:/mme-cc/session-policy/default/access-control](#)

CLI Path: [mme-cc session-policy default access-control](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
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Attribute	Type	Range/Length/Pattern	Default	Description
pdn-reject	enumeration	undefined-esm-cause-0	undefined-esm-cause-0	PDN reject cause
		request-rejected-31		
		service-option-not-supported-32		
		requested-service-option-not-subscribed-33		
		multiple-pdn-connections-for-a-given-apn-not-allowed-55		

Gateway Selection Policy

Gateway Selection Policy

XPath: [cna-mme-cc:/mme-cc/session-policy/default/gateway-selection-policy](#)

CLI Path: [mme-cc session-policy default gateway-selection-policy](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
hplmn-oi-replacement-for-dns	string	([A-Za-z0-9]+)..		Defines the Access Point Name (APN) Operator Identifier that is appended to the APN Network Identifier and used to identify the PGW for home subscribers, or visiting subscribers without VPLMN access. It is a fully qualified domain name according to the Domain Name System (DNS) naming conventions and consists of three labels, that is (DNS-label).(DNS-label).gprs. Preferably in the mnc(MNC digits).mcc(MCC digits).gprs format. This is used only for the DNS query if no replacement OI is present under subscription

Dedicated Gateway

List of dedicated gateways

XPath: [cna-mme-cc:/mme-cc/session-policy/default/gateway-selection-policy/dedicated-gateway](#)

CLI Path: [mme-cc session-policy default gateway-selection-policy dedicated-gateway](#)

List Properties:

Keys: fqdn

Maximum number of elements: 2

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
fqdn	inet:domain-name			Fully qualified domain name of the gateway
gateway-type	enumeration	combo sgw pgw		Type of the gateway. Allowed values are sgw, pgw and combo, either two entries can be configured with type 'sgw' and 'pgw' or single entry with type 'combo' Either two entries can be configured with type 'sgw' and 'pgw' or single entry with type 'combo' Either two entries can be configured with type 'sgw' and 'pgw' or single entry with type 'combo'

Apn Management Policy

Access Point Name (APN) management policy

XPath: [cna-mme-cc:/mme-cc/session-policy/default/apn-management-policy](#)

CLI Path: [mme-cc session-policy default apn-management-policy](#)

Pdn Type Selection

Packet Data Network (PDN) type selection

XPath: [cna-mme-cc:/mme-cc/session-policy/default/apn-management-policy/pdn-type-selection](#)

CLI Path: [mme-cc session-policy default apn-management-policy pdn-type-selection](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
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Attribute	Type	Range/Length/Pattern	Default	Description
selection-method	enumeration	standard match-subscribed-pdn-type	standard	Defines the Packet Data Network (PDN) type selection method. Default: standard
pdn-type	enumeration	non-ip ipv4 ipv6 ipv4ipv6		PDN (Packet Data Network) type to match the subscription.
pdn-type-preference	enumeration	ipv4 ipv6	ipv4	Defines the preference of PDN type for the UE. Default: ipv4

Initial Apn Ni Selection

Initial Access Point Name Network Identifier (APN-NI) selection

XPath: [cna-mme-cc:/mme-cc/session-policy/default/apn-management-policy/initial-apn-ni-selection](#)

CLI Path: [mme-cc session-policy default apn-management-policy initial-apn-ni-selection](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
selection-method	enumeration	standard use-default-apn use-subscribed-apn use-configured-apn	standard	Defines the Access Point Name Network Identifier (APN-NI) selection method. Default: standard
subscribed-apn-ni	string	(([a-zA-Z0-9])+((- or .)[A-Za-z0-9]+)*)		APN Network Identifier to match the subscription
replacement-for-subscription-apn-ni	string	(([a-zA-Z0-9])+((- or .)[A-Za-z0-9]+)*)		Replacement APN-NI for the subscription APN-NI
configured-apn-ni	string	(([a-zA-Z0-9])+((- or .)[A-Za-z0-9]+)*)		Configured APN Network Identifier

Apn Qos

Quality of Service (QoS) parameters for the configured APN

XPath: [cna-mme-cc:/mme-cc/session-policy/default/apn-management-policy/initial-apn-ni-selection/apn-qos](#)

CLI Path: [mme-cc session-policy default apn-management-policy initial-apn-ni-selection apn-qos](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
qos-class-identifier	enumeration	qci1 qci2 qci3 qci4 qci5 qci6 qci7 qci8 qci9 qci80 qci82 qci83 qci65 qci66 qci69 qci70 qci128 qci129 qci130 qci131 qci132		QoS Class Identifier (QCI)

Attribute	Type	Range/Length/Pattern	Default	Description
		qci133		
		qci134		
		qci135		
		qci136		
		qci137		
		qci138		
		qci139		
		qci140		
		qci141		
		qci142		
		qci143		
		qci144		
		qci145		
		qci146		
		qci147		
		qci148		
		qci149		
		qci150		
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		qci191		
		qci192		
		qci193		
		qci194		
		qci195		
		qci196		
		qci197		

Attribute	Type	Range/Length/Pattern	Default	Description
		qci198		
		qci199		
		qci200		
		qci201		
		qci202		
		qci203		
		qci204		
		qci205		
		qci206		
		qci207		
		qci208		
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		qci247		
		qci248		
		qci249		
		qci250		
		qci251		
		qci252		
		qci253		
		qci254		
arp-priority	uint8	1..100		Allocation and Retention Priority (ARP) priority
pre-emption-capability	enumeration	enable disable	disable	Controls the pre-emption capability. Default: disable
pre-emption-vulnerability	enumeration	enable disable	enable	Controls the pre-emption vulnerability. Default: enable

Attribute	Type	Range/Length/Pattern	Default	Description
apn-ambr-uplink	uint32	0..4000000000	4000000000	[future] APN aggregated max bitrate for uplink in Kbps. Default: 4000000000
apn-ambr-downlink	uint32	0..4000000000	4000000000	[future] APN aggregated max bitrate for downlink in Kbps. Default: 4000000000

Initial Apn Oi Selection

Initial Access Point Name Operator Identifier (APN-OI) selection

XPath: [cna-mme-cc:/mme-cc/session-policy/default/apn-management-policy/initial-apn-oi-selection](#)

CLI Path: [mme-cc session-policy default apn-management-policy initial-apn-oi-selection](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
selection-method	enumeration	standard use-home-plmn use-configured-oi	standard	Defines the Access Point Name Operator Identifier (APN-OI) selection method. Default: standard 'vplmn-configured-oi' or 'hplmn-configured-oi' must be configured for selection-method 'use-configured-oi'
vplmn-configured-oi	string	([A-Za-z0-9]+)..		APN-OI used to identify PGW for roaming subscribers
hplmn-configured-oi	string	([A-Za-z0-9]+)..		APN-OI used to identify PGW for home or home-routed subscribers

Fallback Apn Oi Selection

Fallback Access Point Name Operator Identifier (APN-OI) selection

XPath: [cna-mme-cc:/mme-cc/session-policy/default/apn-management-policy/fallback-apn-oi-selection](#)

CLI Path: [mme-cc session-policy default apn-management-policy fallback-apn-oi-selection](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
selection-method	enumeration	none use-home-plmn	use-home-plmn	Defines the Access Point Name Operator Identifier (APN-OI) selection method. Default: none

Apn Usage

Access Point Name (APN) usage

XPath: [cna-mme-cc:/mme-cc/session-policy/default/apn-management-policy/apn-usage](#)

CLI Path: [mme-cc session-policy default apn-management-policy apn-usage](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
apn-in-nas	enumeration	requested selected	selected	Controls whether requested or selected APN will be sent in Activate EPS Default Bearer Context Request message to the UE. If the UE did not request for a specific APN, the selected APN will be sent irrespective of the value of this attribute. Default: selected
apn-to-gw	enumeration	requested selected	selected	Controls whether requested or selected APN will be sent in Create Session Request message to the GW. If the UE did not request for a specific APN, the selected APN will be sent irrespective of the value of this attribute. Default: selected

Class

Session Class

XPath: [cna-mme-cc:/mme-cc/session-policy/class](#)

CLI Path: [mme-cc session-policy class](#)

List Properties:

Keys: name
Maximum number of elements: 1000

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	1..64		Name of the session class

Access Control

PDN Access Control Policy

XPath: [cna-mme-cc:/mme-cc/session-policy/class/access-control](#)

CLI Path: [mme-cc session-policy class access-control](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
pdn-reject	enumeration	undefined-esm-cause-0	undefined-esm-cause-0	PDN reject cause
		request-rejected-31		
		service-option-not-supported-32		
		requested-service-option-not-subscribed-33		
		multiple-pdn-connections-for-a-given-apn-not-allowed-55		

Gateway Selection Policy

Gateway Selection Policy

XPath: [cna-mme-cc:/mme-cc/session-policy/class/gateway-selection-policy](#)

CLI Path: [mme-cc session-policy class gateway-selection-policy](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
hplmn-oi-replacement-for-dns	string	([A-Za-z0-9]+)..		Defines the Access Point Name (APN) Operator Identifier that is appended to the APN Network Identifier and used to identify the PGW for home subscribers, or visiting subscribers without VPLMN access. It is a fully qualified domain name according to the Domain Name System (DNS) naming conventions and consists of three labels, that is (DNS-label).(DNS-label).gprs. Preferably in the mnc(MNC digits).mcc(MCC digits).gprs format. This is used only for the DNS query if no replacement OI is present under subscription

Dedicated Gateway

List of dedicated gateways

XPath: [cna-mme-cc:/mme-cc/session-policy/class/gateway-selection-policy/dedicated-gateway](#)

CLI Path: [mme-cc session-policy class gateway-selection-policy dedicated-gateway](#)

List Properties:

Keys: fqdn
Maximum number of elements: 2

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
fqdn	inet:domain-name			Fully qualified domain name of the gateway
gateway-type	enumeration	combo		Type of the gateway. Allowed values are sgw, pgw and combo, either two entries can be configured with type 'sgw' and 'pgw' or single entry with type 'combo'
		sgw		Either two entries can be configured with type 'sgw' and 'pgw' or single entry with type 'combo'
		pgw		Either two entries can be configured with type 'sgw' and 'pgw' or single entry with type 'combo'

Apn Management Policy

Access Point Name (APN) management policy

XPath: *cna-mme-cc:/mme-cc/session-policy/class/apn-management-policy*

CLI Path: *mme-cc session-policy class apn-management-policy*

Pdn Type Selection

Packet Data Network (PDN) type selection

XPath: *cna-mme-cc:/mme-cc/session-policy/class/apn-management-policy/pdn-type-selection*

CLI Path: *mme-cc session-policy class apn-management-policy pdn-type-selection*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
selection-method	enumeration	standard match-subscribed-pdn-type	standard	Defines the Packet Data Network (PDN) type selection method. Default: standard
pdn-type	enumeration	non-ip ipv4 ipv6 ipv4ipv6		PDN (Packet Data Network) type to match the subscription.
pdn-type-preference	enumeration	ipv4 ipv6	ipv4	Defines the preference of PDN type for the UE. Default: ipv4

Initial Apn Ni Selection

Initial Access Point Name Network Identifier (APN-NI) selection

XPath: *cna-mme-cc:/mme-cc/session-policy/class/apn-management-policy/initial-apn-ni-selection*

CLI Path: *mme-cc session-policy class apn-management-policy initial-apn-ni-selection*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
selection-method	enumeration	standard use-default-apn use-subscribed-apn use-configured-apn	standard	Defines the Access Point Name Network Identifier (APN-NI) selection method. Default: standard
subscribed-apn-ni	string	(((a-zA-Z0-9))+((- or .)[A-Za-z0-9]+)*)		APN Network Identifier to match the subscription
replacement-for-subscription-apn-ni	string	(((a-zA-Z0-9))+((- or .)[A-Za-z0-9]+)*)		Replacement APN-NI for the subscription APN-NI
configured-apn-ni	string	(((a-zA-Z0-9))+((- or .)[A-Za-z0-9]+)*)		Configured APN Network Identifier

Apn Qos

Quality of Service (QoS) parameters for the configured APN

XPath: *cna-mme-cc:/mme-cc/session-policy/class/apn-management-policy/initial-apn-ni-selection/apn-qos*

CLI Path: *mme-cc session-policy class apn-management-policy initial-apn-ni-selection apn-qos*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
qos-class-identifier	enumeration	qci1 qci2 qci3 qci4 qci5		QoS Class Identifier (QCI)

Attribute	Type	Range/Length/Pattern	Default	Description
		qci6		
		qci7		
		qci8		
		qci9		
		qci80		
		qci82		
		qci83		
		qci65		
		qci66		
		qci69		
		qci70		
		qci128		
		qci129		
		qci130		
		qci131		
		qci132		
		qci133		
		qci134		
		qci135		
		qci136		
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		qci173		
		qci174		
		qci175		
		qci176		
		qci177		
		qci178		
		qci179		
		qci180		
		qci181		

Attribute	Type	Range/Length/Pattern	Default	Description
		qci182		
		qci183		
		qci184		
		qci185		
		qci186		
		qci187		
		qci188		
		qci189		
		qci190		
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		qci243		
		qci244		
		qci245		
		qci246		

Attribute	Type	Range/Length/Pattern	Default	Description
		qci247 qci248 qci249 qci250 qci251 qci252 qci253 qci254		
arp-priority	uint8	1..100		Allocation and Retention Priority (ARP) priority
pre-emption-capability	enumeration	enable disable	disable	Controls the pre-emption capability. Default: disable
pre-emption-vulnerability	enumeration	enable disable	enable	Controls the pre-emption vulnerability. Default: enable
apn-ambr-uplink	uint32	0..4000000000	4000000000	[future] APN aggregated max bitrate for uplink in Kbps. Default: 4000000000
apn-ambr-downlink	uint32	0..4000000000	4000000000	[future] APN aggregated max bitrate for downlink in Kbps. Default: 4000000000

Initial Apn Oi Selection

Initial Access Point Name Operator Identifier (APN-OI) selection

XPath: *cna-mme-cc:/mme-cc/session-policy/class/apn-management-policy/initial-apn-oi-selection*

CLI Path: *mme-cc session-policy class apn-management-policy initial-apn-oi-selection*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
selection-method	enumeration	standard use-home-plmn use-configured-oi	standard	Defines the Access Point Name Operator Identifier (APN-OI) selection method. Default: standard 'vplmn-configured-oi' or 'hplmn-configured-oi' must be configured for selection-method 'use-configured-oi'
vplmn-configured-oi	string	([A-Za-z0-9]+)..		APN-OI used to identify PGW for roaming subscribers
hplmn-configured-oi	string	([A-Za-z0-9]+)..		APN-OI used to identify PGW for home or home-routed subscribers

Fallback Apn Oi Selection

Fallback Access Point Name Operator Identifier (APN-OI) selection

XPath: *cna-mme-cc:/mme-cc/session-policy/class/apn-management-policy/fallback-apn-oi-selection*

CLI Path: *mme-cc session-policy class apn-management-policy fallback-apn-oi-selection*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
selection-method	enumeration	none use-home-plmn	use-home-plmn	Defines the Access Point Name Operator Identifier (APN-OI) selection method. Default: none

Apn Usage

Access Point Name (APN) usage

XPath: *cna-mme-cc:/mme-cc/session-policy/class/apn-management-policy/apn-usage*

CLI Path: *mme-cc session-policy class apn-management-policy apn-usage*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
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Attribute	Type	Range/Length/Pattern	Default	Description
apn-in-nas	enumeration	requested selected	selected	Controls whether requested or selected APN will be sent in Activate EPS Default Bearer Context Request message to the UE. If the UE did not request for a specific APN, the selected APN will be sent irrespective of the value of this attribute. Default: selected
apn-to-gw	enumeration	requested selected	selected	Controls whether requested or selected APN will be sent in Create Session Request message to the GW. If the UE did not request for a specific APN, the selected APN will be sent irrespective of the value of this attribute. Default: selected

R8 Qos Conversion

Used for R8 to R99 QoS mapping and for R99 to R8 QoS mapping as defined by TS 23.401 Annex E

XPath: [cna-mme-cc:/mme-cc/session-policy/r8-qos-conversion](#)

CLI Path: [mme-cc session-policy r8-qos-conversion](#)

List Properties:

Keys: qci

Maximum number of elements: 143

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
qci	uint8	1..9	65	66
traffic-class	enumeration	conversational streaming background interactive-thp1 interactive-thp2 interactive-thp3		Type of application for which the bearer service is optimized
max-sdu-size	uint16	1..1500	1500	This attribute indicates the maximum SDU size for the converted R99 QoS. The maximum SDU size is encoded as specified in TS 24.008, section 10.5.6.5. Default: 1500; Unit: bytes
residual-ber	enumeration	residualber-5hundredth residualber-1hundredth residualber-5thousandth residualber-4thousandth residualber-1thousandth residualber-1tenthousandth residualber-1hundredthousandth residualber-1millionth residualber-6hundredmillionth		This attribute indicates the undetected bit error ratio in the delivered SDUs for the selected traffic class.
delivery-of-erroneous-sdus	enumeration	no-detect err-sdus-delivered err-sdus-not-delivered	err-sdus-not-delivered	This attribute defines how erroneous SDUs are handled by the network. Default: err-sdus-not-delivered
sdu-error-ratio	enumeration	sdu-err-ratio-1hundredth sdu-err-ratio-7thousandth sdu-err-ratio-1thousandth sdu-err-ratio-1tenthousandth sdu-err-ratio-1hundredthousandth sdu-err-ratio-1tenth		This attribute indicates the fraction of Signaling Data Units (SDUs) lost or detected as erroneous.

Attribute	Type	Range/Length/Pattern	Default	Description
transfer-delay	uint16	10..4000	4000	This attribute specifies the maximum delay for the 95th percentile of the distribution of delay for all delivered Signaling Data Units (SDUs) during the lifetime of a bearer service. This attribute can take a numeric range from 10 to 4000 ms with encoding as specified in TS 24.008, section 10.5.6.5. When the traffic-class attribute is conversational the transfer-delay attribute is recommended to be set to a value between 80 to 4000 ms. When the traffic-class attribute is streaming the transfer-delay attribute is recommended to be set to a value between 250 to 4000 ms. The transfer-delay attribute is not used when the traffic-class attribute is configured to interactive or background. Default: 4000
arp-high	uint8	1..13	5	Defines the relative importance compared to other radio access bearers (RABs) for allocation and retention of the RAB; Default: 5 arp-high value must be less than arp-medium value.
arp-medium	uint8	2..14	10	Defines the relative importance compared to other radio access bearers (RABs) for allocation and retention of the RAB; Default: 10 arp-medium value must be greater than arp-high value.

Feature

Feature control

XPath: [cna-mme-cc:/mme-cc/feature](#)

CLI Path: [mme-cc feature](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
nsa-en-dc	enumeration	enable disable	disable	Controls the Non-stand-alone (NSA) feature status. Default: disable
ims-emergency	enumeration	enable disable	disable	Controls the IMS Emergency feature status. Default: disable
interworking-with-5gc-over-n26	enumeration	enable disable	disable	Controls the interworking with 5G Core (5GC) over N26 feature status. Default: disable
nb-ciot	enumeration	enable disable	disable	Controls the Narrow Band Cellular Internet of Things (NB-CIoT) feature status. Default: disable

Trace

Trace objects

XPath: [cna-mme-cc:/mme-cc/trace](#)

CLI Path: [mme-cc trace](#)

Session

Per UE tracing

XPath: [cna-mme-cc:/mme-cc/trace/session](#)

CLI Path: [mme-cc trace session](#)

List Properties:

Keys: target-type, target-value

Maximum number of elements: 100

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
target-type	enumeration	imsi		Type of target for tracing
target-value	string	([0-9]){6,15})		Value of the target

Attribute	Type	Range/Length/Pattern	Default	Description
session-id	string	[0-9]{5,6}[.][0-9a-fA-F]{6}		Identity to uniquely identify the trace session. Format: PLMN.trace_id, where PLMN is input as 5 or 6 digit BCD number and trace-id is input as a 3 byte hexadecimal number. Example: 50631.70F13A

MME S1 Configuration

The module for entities implementing cna-mme-s1.

Table of Contents

- mme-s1
 - s1
 - endpoint
 - sctp-profile

Mme S1

Top level container for mme s1 Configuration.

XPath: *cna-mme-s1:/mme-s1*

CLI Path: *mme-s1*

S1

s1 interface to 4G access (4G AN)

XPath: *cna-mme-s1:/mme-s1/s1*

CLI Path: *mme-s1 s1*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
time-to-wait	enumeration	1	60	Time that the eNodeB waits before reinitiating the exchange of application level data after a failure of the S1 setup or eNodeB configuration update procedure. Default: 60; Unit: seconds
		2		
		5		
		10		
		20		
		60		

Endpoint

SCTP endpoint for mme side of s1
Either 'ipv4-fqdn' or 'ipv6-fqdn' must be provisioned

XPath: *cna-mme-s1:/mme-s1/s1/endpoint*

CLI Path: *mme-s1 s1 endpoint*

List Properties:

Keys: name
Minimum number of elements: 1
Maximum number of elements: 1

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	1..254		unique Endpoint name ; length: 1..254
ipv4-fqdn	inet:domain-name			The IPv4 FQDN of the associated s1 endpoint. If provided it is published in the s1 Profile.
dscp-marking	string	([0-9] or [1-5][0-9] or [6][0-3]) or df or af[1-4][1-3] or cs[1-7] or ef	df	DSCP marking. Valid values are 0..63, df, af11, af12, af13, af21, af22, af23, af31, af32, af33, af41, af42, af43, cs1, cs2, cs3, cs4, cs5, cs6, cs7, ef; Default: df

Sctp Profile

SCTP parameters for s1

XPath: *cna-mme-s1:/mme-s1/s1/endpoint/sctp-profile*

CLI Path: *mme-s1 s1 endpoint sctp-profile*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
rto-initial	uint32	1..5000	500	Retransmission Time Out (RTO) initial value. Default: 500; Unit: milliseconds The 'rto-initial' value should be in the range from 'rto-min' to 'rto-max'
rto-min	uint32	1..5000	250	Retransmission Time Out (RTO) minimum value. Default: 250; Unit: milliseconds
rto-max	uint32	1..60000	2000	Retransmission Time Out (RTO) maximum value. Default: 2000; Unit: milliseconds
path-max-retrans	uint16	1..20	5	Maximum number of consecutive retransmissions over a destination transport address of a peer endpoint before it is marked as inactive. Default: 5
association-init-max-transmit	uint16	1..10	3	Maximum number of times an INIT chunk or a COOKIE ECHO chunk is retransmitted before an endpoint aborts the initialization process and closes the association. Default: 3
association-max-retrans	uint16	1..20	6	Maximum number of consecutive retransmissions to a peer before an endpoint considers that the peer is unreachable and closes the association. Default: 6
no-delay	boolean		true	Specifies no delay. Default: true
max-out-stream	uint16	1..64	11	Specifies the maximum out stream. Default: 11
max-in-stream	uint16	1..64	11	Specifies the maximum in stream. Default: 11
hb-interval	uint16	1..120	30	Specifies the heartbeat interval. Default: 30; Unit: seconds
rx-buffer-size	uint32	25000..2097152	149000	Specifies the receive buffer size. Default: 149000; Unit: bytes
tx-buffer-size	uint32	16000..5509000	149000	Specifies the transmit buffer size. Default: 149000; Unit: bytes
sack-delay	uint16	1	10..500	200
sack-frequency	uint16	1..5	2	Specifies the SCTP SACK frequency value. Disable: 1; Default: 2; Unit: packets

MME SGS Configuration

This module defines the configuration for MME SGS.

Table of Contents

- mme-sgs
 - sgs
 - local-endpoint
 - sctp-profile
 - peer
 - remote-endpoint

Mme Sgs

Top level container for MME SGS Configuration.

XPath: *cna-mme-sgs:/mme-sgs*

CLI Path: *mme-sgs*

Sgs

SGS interface to 4G access (4G AN)

XPath: *cna-mme-sgs:/mme-sgs/sgs*

CLI Path: *mme-sgs sgs*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
connection-retry-timer	uint16	1..300	5	Connection Retry Timer; Default: 5; Unit: Seconds

Local Endpoint

SCTP endpoint for MME side of SGS

Either 'ipv4-fqdn' or 'ipv6-fqdn' must be provisioned

XPath: *cna-mme-sgs:/mme-sgs/sgs/local-endpoint*

CLI Path: *mme-sgs sgs local-endpoint*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
port	uint32	1..65535	29118	SCTP Port number; Default: 29118
ipv4-fqdn	inet:domain-name			The IPv4 FQDN of the associated SGS endpoint. If provided it is published in the SGS Profile.
dscp-marking	string	(([0-9] or [1-5][0-9] or [6][0-3]) or de or af[1-4][1-3] or cs[1-7] or ef	de	DSCP marking for SGS. Valid values are 0..63, de, af11, af12, af13, af21, af22, af23, af31, af32, af33, af41, af42, af43, cs1, cs2, cs3, cs4, cs5, cs6, cs7, ef; Default: de

Sctp Profile

SCTP parameters for SGS

XPath: *cna-mme-sgs:/mme-sgs/sgs/local-endpoint/sctp-profile*

CLI Path: *mme-sgs sgs local-endpoint sctp-profile*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
rto-initial	uint32	1..5000	500	Retransmission Time Out (RTO) initial value. Default: 500; Unit: milliseconds The 'rto-initial' value should be in the range from 'rto-min' to 'rto-max'
rto-min	uint32	1..5000	250	Retransmission Time Out (RTO) minimum value. Default: 250; Unit: milliseconds
rto-max	uint32	1..60000	2000	Retransmission Time Out (RTO) maximum value. Default: 2000; Unit: milliseconds
path-max-retrans	uint16	1..20	5	Maximum number of consecutive retransmissions over a destination transport address of a peer endpoint before it is marked as inactive. Default: 5
association-init-max-transmit	uint16	1..10	3	Maximum number of times an INIT chunk or a COOKIE ECHO chunk is retransmitted before an endpoint aborts the initialization process and closes the association. Default: 3
association-max-retrans	uint16	1..20	6	Maximum number of consecutive retransmissions to a peer before an endpoint considers that the peer is unreachable and closes the association. Default: 6
no-delay	boolean		true	Specifies no delay. Default: true
max-out-stream	uint16	1..64	11	Specifies the maximum out stream. Default: 11
max-in-stream	uint16	1..64	11	Specifies the maximum in stream. Default: 11
hb-interval	uint16	1..120	30	Specifies the heartbeat interval. Default: 30; Unit: seconds
rx-buffer-size	uint32	25000..2097152	149000	Specifies the receive buffer size. Default: 149000; Unit: bytes
tx-buffer-size	uint32	16000..5509000	149000	Specifies the transmit buffer size. Default: 149000; Unit: bytes
sack-delay	uint16	1	10..500	200
sack-frequency	uint16	1..5	2	Specifies the SCTP SACK frequency value. Disable: 1; Default: 2; Unit: packets

Peer

List of peer configurations

XPath: *cna-mme-sgs:/mme-sgs/peer*

CLI Path: *mme-sgs peer*

List Properties:

Keys: vlr-name

Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
vlr-name	string	1..255		Name of the Visitor Location Register (VLR)

Remote Endpoint

SGs peer endpoint for a VLR

XPath: [cna-mme-sgs/mme-sgs/peer/remote-endpoint](#)

CLI Path: [mme-sgs peer remote-endpoint](#)

List Properties:

Keys: association-id

Maximum number of elements: 4

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
association-id	uint8			SGs VLR association ID
port	uint16	1..65535		SGs peer port number
ip-address	inet:ip-address			Primary IP address of the peer

N3IWF CC Configuration

This module defines the configuration for N3IWF Call Control.

Table of Contents

- [n3iwf](#)
 - [service-construct](#)
 - [plmn](#)
 - [plmn-list](#)
 - [plmn](#)
 - [nssai](#)
 - [s-nssai](#)
 - [s-nssai](#)
 - [tai](#)
 - [five-qi-dscp-mapping-policy](#)
 - [five-qi-dscp-mapping](#)
 - [service](#)

N3Iwf

N3IWF Call Control

XPath: [cna-n3iwf-cc/n3iwf](#)

CLI Path: [n3iwf](#)

Service Construct

Interface objects

XPath: [cna-n3iwf-cc/n3iwf/service-construct](#)

CLI Path: [n3iwf service-construct](#)

Plmn

PLMN construct

Defines the PLMN containing PLMN ID and NSSAI supported by this PLMN

XPath: [cna-n3iwf-cc/n3iwf/service-construct/plmn](#)

CLI Path: [n3iwf service-construct plmn](#)

List Properties:

Keys: name

Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		
plmn-id	string	(([0-9]{5,6}))		Supported PLMN identity; Range: numbers with 5-6 digits
nssai	leafref			name of the nssai supported by this plmn Leaf Reference Path: cna-n3iwf-cc:/n3iwf/service-construct/nssai/name

Plmn List

name the plmn-list containing the list of plmns

XPath: [cna-n3iwf-cc:/n3iwf/service-construct/plmn-list](#)

CLI Path: [n3iwf service-construct plmn-list](#)

List Properties:

Keys: name

Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		

Plmn

List of PLMNs

XPath: [cna-n3iwf-cc:/n3iwf/service-construct/plmn-list/plmn](#)

CLI Path: [n3iwf service-construct plmn-list plmn](#)

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: cna-n3iwf-cc:/n3iwf/service-construct/plmn/name

Nssai

list of S-NSSAI's supported by the N3IWF service

XPath: [cna-n3iwf-cc:/n3iwf/service-construct/nssai](#)

CLI Path: [n3iwf service-construct nssai](#)

List Properties:

Keys: name

Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		

S Nssai

list of S-NSSAI's upported by the N3IWF service

XPath: [cna-n3iwf-cc:/n3iwf/service-construct/nssai/s-nssai](#)

CLI Path: [n3iwf service-construct nssai s-nssai](#)

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: cna-n3iwf-cc:/n3iwf/service-construct/s-nssai/name

S Nssai

S-NSSAI construct
Defines the S-NSSAI containing SST, SD and DNN List

XPath: [cna-n3iwf-cc:/n3iwf/service-construct/s-nssai](#)

CLI Path: [n3iwf service-construct s-nssai](#)

List Properties:
Keys: name
Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		
slice-service-type	union	['uint8,0 or 4 .. 255', 'Enumerations: eMBB URLLC MIoT ']		Identifies the expected Network Slice behavior in terms of features and services
slice-differentiator	string	([A-Fa-f0-9]{6})		Differentiate amongst multiple Network Slices of the same Slice/Service type

Tai

tracking area identifier construct
Defines the tracking area identifier that consist of mcc, mnc and tracking are code

XPath: [cna-n3iwf-cc:/n3iwf/service-construct/tai](#)

CLI Path: [n3iwf service-construct tai](#)

List Properties:
Keys: name
Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		
tac	string	([A-Fa-f0-9]{4} or [A-Fa-f0-9]{6})		tracking are code of the tai
broadcast-plmn-list	leafref			Name of PLMN list Leaf Reference Path: cna-n3iwf-cc:/n3iwf/service-construct/plmn-list/name

Five Qi Dscp Mapping Policy

5QI to DSCP mapping policy
5QI to DSCP mapping policy

XPath: [cna-n3iwf-cc:/n3iwf/service-construct/five-qi-dscp-mapping-policy](#)

CLI Path: [n3iwf service-construct five-qi-dscp-mapping-policy](#)

List Properties:
Keys: name
Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			5QI to DSCP mapping policy name

Five Qi Dscp Mapping

5QI to DSCP mapping
5QI to DSCP mapping

XPath: *cna-n3iwf-cc:/n3iwf/service-construct/five-qi-dscp-mapping-policy/five-qi-dscp-mapping*

CLI Path: *n3iwf service-construct five-qi-dscp-mapping-policy five-qi-dscp-mapping*

List Properties:

Keys: name
Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			5QI to DSCP mapping name
five-qi	uint32	0 .. 255		5G QoS Identifier
arp-priority-level	uint32	0 .. 15	0	Priority level in Allocation and Retention Priority. Valid values are 1..15. 0 implies that this is the default configuration for 5QI/ARP priority to DSCP mapping when no other configuration for 5QI/ARP priority to DSCP mapping is defined

Attribute	Type	Range/Length/Pattern	Default	Description
dscp	enumeration	cs0		DSCP value
		dscp_01		
		dscp_02		
		dscp_03		
		dscp_04		
		dscp_05		
		dscp_06		
		dscp_07		
		cs1		
		dscp_09		
		af11		
		dscp_11		
		af12		
		dscp_13		
		af13		
		dscp_15		
		cs2		
		dscp_17		
		af21		
		dscp_19		
		af22		
		dscp_21		
		af23		
		dscp_23		
		cs3		
		dscp_25		
		af31		
		dscp_27		
		af32		
		dscp_29		
		af33		
		dscp_31		
		cs4		
		dscp_33		
		af41		
		dscp_35		
		af42		
		dscp_37		
		af43		
		dscp_39		
		cs5		
		dscp_41		
		dscp_42		
		dscp_43		
		dscp_44_va		
		dscp_45		
		dscp_46_ef		
		dscp_47		
		cs6		
		dscp_49		
		dscp_50		
		dscp_51		
		dscp_52		
		dscp_53		
		dscp_54		
		dscp_55		
		cs7		
		dscp_57		
		dscp_58		
		dscp_59		
		dscp_60		
		dscp_61		
		dscp_62		
		dscp_63		

Service

N3IWF Service

XPath: [cna-n3iwf-cc:/n3iwf/service](#)

CLI Path: [n3iwf service](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	1..128		N3IWF name
node-id	uint16			N3IWF Global Node ID
serving-tai	leafref			N3IWF serving tracking Area Identifier Leaf Reference Path: cna-n3iwf-cc:/n3iwf/service-construct/tai/name
ue-context-cache-timer	uint32	120..86400	600	N3IWF UE Context Cache Timer in seconds
home-plmn-id	string	(((0-9){5,6}))		Home PLMN ID; Range: numbers with 5-6 digits
five-qi-dscp-mapping-policy	leafref			5QI to DSCP mapping policy Leaf Reference Path: cna-n3iwf-cc:/n3iwf/service-construct/five-qi-dscp-mapping-policy/name

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N2 Endpoint

N3IWF N2 Endpoint

XPath: [cna-n3iwf-n2:/n2-endpoint](#)

CLI Path: [n2-endpoint](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
default-amf	leafref			Addition, modification or deletion of this field requires SCTP association restart for changes to take effect Leaf Reference Path: cna-n3iwf-n2:/n2-endpoint/amf/name

Sctp Profile

SCTP profile

XPath: [cna-n3iwf-n2:/n2-endpoint/sctp-profile](#)

CLI Path: [n2-endpoint sctp-profile](#)

List Properties:

Keys: name

Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			SCTP Profile Name
rto-initial	uint16	1..5000	500	Retransmission Time Out (RTO) initial value. Default: 500; Unit: milliseconds The 'rto-initial' value should be in the range from 'rto-min' to 'rto-max'
rto-min	uint16	1..5000	250	Retransmission Time Out (RTO) minimum value. Default: 250; Unit: milliseconds
rto-max	uint16	1..60000	2000	Retransmission Time Out (RTO) maximum value. Default: 2000; Unit: milliseconds
path-max-retrans	uint16	1..20	5	Maximum number of consecutive retransmissions over a destination transport address of a peer endpoint before it is marked as inactive. Default: 5

Attribute	Type	Range/Length/Pattern	Default	Description
association-init-max-transmit	uint16	1..10	3	Maximum number of times an INIT chunk or a COOKIE ECHO chunk is retransmitted before an endpoint aborts the initialization process and closes the association. Default: 3
association-max-retrans	uint16	1..20	6	Maximum number of consecutive retransmissions to a peer before an endpoint considers that the peer is unreachable and closes the association. Default: 6
no-delay	boolean		true	Specifies no delay. Default: true
max-out-stream	uint16	1..64	11	Specifies the maximum out stream. Default: 11
max-in-stream	uint16	1..64	11	Specifies the maximum in stream. Default: 11
hb-interval	uint16	1..120	30	Specifies the heartbeat interval. Default: 30; Unit: seconds
rx-buffer-size	uint32	25000..2097152	149000	Specifies the receive buffer size. Default: 149000; Unit: bytes
tx-buffer-size	uint32	16000..5509000	149000	Specifies the transmit buffer size. Default: 149000; Unit: bytes
sack-delay	uint16	1	10..500	200
sack-frequency	uint16	1..5	2	Specifies the SCTP SACK frequency value. Disable: 1; Default: 2; Unit: packets

Amf

AMF List

XPath: [cna-n3iwf-n2:/n2-endpoint/amf](#)

CLI Path: [n2-endpoint amf](#)

List Properties:

Keys: name

Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	1..254		AMF Name
ip-address	inet:ip-address			AMF IP Address. Can be IPv4/IPv6
port	uint32			AMF port number
sctp-profile	leafref			Associated SCTP Profile Leaf Reference Path: cna-n3iwf-n2:/n2-endpoint/sctp-profile/name
admin-state	enumeration	disabled enabled	disabled	Admin state of AMF. default state is "disabled".

NF INV MGR Configuration

The module for entities implementing cna-nf-inv-mgr.

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- [nf-inv-mgr](#)
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Nf Inv Mgr

Nf Inv Mgr. See 3GPP specification TS29.510 for more information

XPath: [cna-nf-inv-mgr:/nf-inv-mgr](#)

CLI Path: [nf-inv-mgr](#)

Trace

Trace objects

XPath: [cna-nf-inv-mgr:/nf-inv-mgr/trace](#)

CLI Path: [nf-inv-mgr trace](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
nf-instance-id	leaf-list of string	[0-9a-fA-F]{8}-[0-9a-fA-F]{4}-[0-9a-fA-F]{4}-[0-9a-fA-F]{4}-[0-9a-fA-F]{12}		Unique identity of an NF in the form of a uuid
nf-type	leaf-list of enumeration	AMF		Type of Network Function
		SMF		
		UDM		
		AUSF		
		UPF		
		PCF		
		NEF		
		5G_EIR		
		SMSF		
		NSSF		
		UDR		
		CHF		
		BSF		
		AF		
		LMF		
		GMLC		
		NWDAF		
		CBCF		
		N3IWF		
		PCSCF		
		SEPP		
probability	string		0.1	Probability of sampling the trace for the target; valid values are 0.0 to 1.0 The 'probability' value must be less than 1.0
default-probability	string		0.1	Probability of sampling the trace for a non-target; valid values are 0.0 to 1.0 The 'probability' value must be less than 1.0

S Nssai

Supported slices per PLMN

Value 0xFFFFFFFF is reserved and cannot be used to configure SD

XPath: [cna-nf-inv-mgr:/nf-inv-mgr/trace/s-nssai](#)

CLI Path: [nf-inv-mgr trace s-nssai](#)

List Properties:

Keys: sst, sd

Minimum number of elements: 0

Maximum number of elements: 8

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
sst	union	['uint8,0 or 4 .. 255', 'Enumerations: eMBB		Identifies the expected Network Slice behavior in terms of features and services
		URLLC		
		MIoT		
		']		

Attribute	Type	Range/Length/Pattern	Default	Description
sd	union	['string,(0x[0-9a-fA-F]{6})', 'Enumerations: not-specified ']		Differentiate amongst multiple Network Slices of the same Slice/Service type

Nf Profile Management

NF profile management Configuration. See 3GPP specification TS29.510 for more information.

XPath: [cna-nf-inv-mgr:/nf-inv-mgr/nf-profile-management](#)

CLI Path: [nf-inv-mgr nf-profile-management](#)

Authorization

NF management authorization parameters

XPath: [cna-nf-inv-mgr:/nf-inv-mgr/nf-profile-management/authorization](#)

CLI Path: [nf-inv-mgr nf-profile-management authorization](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
heart-beat	uint32	2 .. 86400		Time expected between 2 consecutive heart-beat between Nf and Nrf. Unit: second
allowed-plmns	leaf-list of string	(((0-9){5,6}))		List of allowed PLMN identifiers. Default: all Maximum number of elements: 16
oauth2	boolean		false	Indicates if oAuth2 is applicable for client registration. Default: false

Requester Nf Type

Requester Network Function Types; Override the authorization value per 'requester-nf-type'

XPath: [cna-nf-inv-mgr:/nf-inv-mgr/nf-profile-management/requester-nf-type](#)

CLI Path: [nf-inv-mgr nf-profile-management requester-nf-type](#)

List Properties:

Keys: requester-nf-type

Minimum number of elements: 0

Maximum number of elements: 20

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
requester-nf-type	enumeration	AMF		Requester Network Function Types; Override the authorization value per 'requester-nf-type'
		SMF		
		UDM		
		AUSF		
		UPF		
		PCF		
		NEF		
		5G_EIR		
		SMSF		
		NSSF		
		UDR		
		CHF		
		BSF		
		AF		
		LMF		
		GMLC		
		NWDAF		
		CBCF		
		N3IWF		
		PCSCF		
		SEPP		

Authorization

NF management authorization parameters

XPath: *cna-nf-inv-mgr:/nf-inv-mgr/nf-profile-management/requester-nf-type/authorization*

CLI Path: *nf-inv-mgr nf-profile-management requester-nf-type authorization*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
heart-beat	uint32	2 .. 86400		Time expected between 2 consecutive heart-beat between Nf and Nrf. Unit: second
allowed-plmns	leaf-list of string	(([0-9]{5,6}))		List of allowed PLMN identifiers. Default: all Maximum number of elements: 16
oauth2	boolean		false	Indicates if oAuth2 is applicable for client registration. Default: false

Subscription

NF management subscription parameters

XPath: *cna-nf-inv-mgr:/nf-inv-mgr/nf-profile-management/requester-nf-type/subscription*

CLI Path: *nf-inv-mgr nf-profile-management requester-nf-type subscription*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
validity-time	uint32	1 .. 1440		Validity time duration that the subscription will be valid for. Default duration is 15 minutes, Unit: minutes
subscr-cond-nf-type	leaf-list of enumeration	all AMF SMF UDM AUSF UPF PCF NEF 5G_EIR SMSF NSSF UDR CHF BSF AF LMF GMLC NWDAF CBCF N3IWF PCSCF SEPP		List of target NF Types that NRF allows NF Service Consumer to subscribe to changes on status of NF Instance registered in NRF. When NF Service Consumer requests to subscribe for a set of NF Instances identified by NF type (by providing NF Type in the 'subscrCond' attribute of the 'SubscriptionData' parameter). NRF will authorize the subscription when the requested NF type matches the NF type of this attribute. The 'all' value indicates that NRF allows NF Consumer to subscribe to any target NF Types. subscr-cond-nf-type can be either all or combination valid values but not both.

Attribute	Type	Range/Length/Pattern	Default	Description
subscr-cond-instance-nf-type	leaf-list of enumeration	all		List of target NF Types that NRF allows NF Service Consumer to subscribe to changes on status of NF Instance registered in NRF for specific target NF Instance. When NF Service Consumer requests to subscribe for a specific target NF Instance (by providing 'NF Instance Id' in the 'subscrCond' attribute of the 'SubscriptionData' parameter). NRF will find NF type of the requested target NF Instance from registered NFs then NRF will authorize the subscription if the target NF type matches the one of NF types of this attribute. The 'all' value indicates that NRF allows NF Consumer to subscribe to any target NF Types. subscr-cond-instance-nf-type can be either all or combination valid values but not both.
		AMF		
		SMF		
		UDM		
		AUSF		
		UPF		
		PCF		
		NEF		
		5G_EIR		
		SMSF		
		NSSF		
		UDR		
		CHF		
		BSF		
		AF		
		LMF		
		GMLC		
		NWDAF		
		CBCF		
		N3IWF		
		PCSCF		
		SEPP		

Nrf Profile

NRF profile

XPath: *cna-nf-inv-mgr:/nf-inv-mgr/nrf-profile*

CLI Path: *nf-inv-mgr nrf-profile*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
nrf-name	string	[A-Za-z0-9 '() +, -./: =?]*		NRF name communicated to the 5GAN
nf-instance-id	string	[0-9a-fA-F]{8}-[0-9a-fA-F]{4}-[0-9a-fA-F]{4}-[0-9a-fA-F]{4}-[0-9a-fA-F]{12}		Unique identity of the NRF in the form of a uuid
oauth2	enumeration	enable disable	disable	Controls the enabling and disabling of the oauth2 feature for this NRF. Default: disable

Service Resource

List of service names which supports Content/Accept encoding

XPath: *cna-nf-inv-mgr:/nf-inv-mgr/service-resource*

CLI Path: *nf-inv-mgr service-resource*

List Properties:

Keys: name

Minimum number of elements: 0

Maximum number of elements: 10

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	enumeration	nnrf-nfm-list-retrieval		service names
		nnrf-nfm-profile-retrieval		
		nnrf-nfm-subscription		
		nnrf-nfm-notification		
		nnrf-nfm-discovery		
		nnrf-nfm-registration		

Attribute	Type	Range/Length/Pattern	Default	Description
encoding-type	enumeration	gzip	gzip	Type of encoding supported; default gzip
payload-size	uint32	50..2000		Maximum payload size (before compression, if any) of the response, expressed in kilo octets.

Nrf Set

Set of NRFs for GeoRedundancy
Primary NRF where nrf-set config is present must be a peer
Own instance id must be configured as a peer

XPath: [cna-nf-inv-mgr:/nf-inv-mgr/nrf-set](#)

CLI Path: [nf-inv-mgr nrf-set](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
config-sync	boolean		true	Controls if nrf-set config information needs to be synced across the NRFs in the set. Default: true
nf-state-notify-primary-only	boolean		false	Controls if status down notification needs to be sent to NFs for which this NRF is primary. Default: false
nrf-set-primary-instance-id	string	[0-9a-fA-F]{8}-[0-9a-fA-F]{4}-[0-9a-fA-F]{4}-[0-9a-fA-F]{4}-[0-9a-fA-F]{12}		Unique identity of the primary NRF in nrf-set, in form of a uuid

Peers

List of NRFs in nrf-set
Either 'ip' or 'fqdn' must be provisioned

XPath: [cna-nf-inv-mgr:/nf-inv-mgr/nrf-set/peers](#)

CLI Path: [nf-inv-mgr nrf-set peers](#)

List Properties:
Keys: nrf-instance
Minimum number of elements: 2
Maximum number of elements: 10

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
nrf-instance	string	[0-9a-fA-F]{8}-[0-9a-fA-F]{4}-[0-9a-fA-F]{4}-[0-9a-fA-F]{4}-[0-9a-fA-F]{12}		Unique identity of the NRF in the form of a uuid
ip	union	['inet:ipv4-address;', 'inet:ipv6-address,']		Load balancer IP of peer NRF The 'ip-address' must not be '0.0.0.0' The 'ip-address' must not be '::'
fqdn	inet:domain-name			FQDN of the peer NRF
port	uint32	1..65535		Port for the peer NRF
scheme	enumeration	http https	http	Protocol scheme

NF SELECTOR Configuration

The module for entities implementing nf-selector

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Nf Selector

Network Function (NF) Selector. See 3GPP specification TS29.510 for more information

XPath: *cna-nf-selector:/nf-selector*

CLI Path: *nf-selector*

Serving Plmn

NRF Selection Manager serving PLMN Configuration

XPath: *cna-nf-selector:/nf-selector/serving-plmn*

CLI Path: *nf-selector serving-plmn*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
plmn-id-list	leaf-list of string	(((0-9){5,6}))		List of PLMN identifiers that this NRF supports Maximum number of elements: 16

Service Auth

NRF Selection Manager service authorization Configuration. See 3GPP specification TS29.510 for more information.

XPath: *cna-nf-selector:/nf-selector/service-auth*

CLI Path: *nf-selector service-auth*

Target Nfs

Target Network Functions

XPath: *cna-nf-selector:/nf-selector/service-auth/target-nfs*

CLI Path: *nf-selector service-auth target-nfs*

List Properties:

Keys: target-nf-type
Minimum number of elements: 0
Maximum number of elements: 18

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
target-nf-type	enumeration	AMF		Type of target Network Function.
		SMF		
		UDM		
		AUSF		
		NEF		
		UPF		
		PCF		
		SMSF		
		BSF		
		CHF		
		AF		
		UDR		
		LMF		
		GMLC		
		5G_EIR		
		CBCF		
		NWDAF		
		N3IWF		
		NSSF		
validity-period	uint32	10 .. 86400	360	Time period that the discovery result is considered as valid. Unit: second; Default: 360

Services

Services supported by the NF. An empty list implies all applicable services are supported

XPath: *cna-nf-selector:/nf-selector/service-auth/target-nfs/services*

CLI Path: *nf-selector service-auth target-nfs services*

List Properties:

Keys: service-name
Minimum number of elements: 0
Maximum number of elements: 15

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
service-name	string	([a-zA-Z0-9_!]{0,1})		Name of the services. The service name may include a wildcard. Examples: 'AMF'; 'a*', '*'
allowed-requester-nf-types	leaf-list of enumeration	AMF		List of allowed requester Network Function types. Default is an empty list which implies allow all Minimum number of elements: 0 Maximum number of elements: 18
		SMF		
		UDM		
		AUSF		
		NEF		
		UPF		
		PCF		
		SMSF		
		BSF		
		CHF		
		AF		
		UDR		
		LMF		
		GMLC		
		5G_EIR		
		CBCF		
		NWDAF		
		N3IWF		
		NSSF		
allowed-plmns	leaf-list of string	(((0-9){5,6}))		List of allowed target PLMN identifiers. Default is an empty list which implies allow all Maximum number of elements: 16

Attribute	Type	Range/Length/Pattern	Default	Description
allowed-nf-realms	leaf-list of string	([^[+])+		List of target Network Function realm identities allowed. Default is an empty list which implies allow all. Up to 10 realm string may be listed Maximum number of elements: 10
oauth2	boolean		false	Indicates if oauth2 is required for this service. Default: false

Allowed S Nssais

List of allowed Single Network Slice Selection Assistance Information per PLMN. Default is an empty list which implies allow all
Value FFFFFFFF is reserved and cannot be used to configure slice-differentiator

XPath: *cna-nf-selector:/nf-selector/service-auth/target-nfs/services/allowed-s-nssais*

CLI Path: *nf-selector service-auth target-nfs services allowed-s-nssais*

List Properties:

Keys: slice-name
Minimum number of elements: 0
Maximum number of elements: 8

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
slice-name	string	(a-zA-Z{0,63})		
slice-service-type	union	['uint8,0 or 4 .. 255', 'Enumerations: eMBB URLLC MIoT ']		Identifies the expected Network Slice behavior in terms of features and services
slice-differentiator	string	(([0-9a-fA-F]{2}){:[0-9a-fA-F]{2}})*?		Differentiate amongst multiple Network Slices of the same Slice/Service type

Static Nf Profile

Static configuration for NF profile. See 3GPP specification TS29.510 for more information.
Either 'ipv4-address' or 'ipv6-address' or 'fqdn' must be provisioned

XPath: *cna-nf-selector:/nf-selector/static-nf-profile*

CLI Path: *nf-selector static-nf-profile*

List Properties:

Keys: nf-instance-id
Maximum number of elements: 100

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
nf-instance-id	string	[A-Za-z0-9 '() +, - . / : = ?]*		Unique identity of the NF Instance.

Attribute	Type	Range/Length/Pattern	Default	Description
nf-type	enumeration	AMF		Type of Network Function
		SMF		
		UDM		
		AUSF		
		NEF		
		UPF		
		PCF		
		SMSF		
		BSF		
		CHF		
		AF		
		UDR		
		LMF		
		GMLC		
		5G_EIR		
		CBCF		
		NWDAF		
		N3IWF		
		PCSCF		
		SEPP		
plmn-id	string	(((0-9){5,6}))		PLMN of Location Area Identifier
fqdn	inet:domain-name			The FQDN of the NF
ipv4-address	inet:ipv4-address			IP address for NF The 'ip-address' must not be '0.0.0.0'
ipv6-address	inet:ipv6-address			IP address for NF The 'ip-address' must not be '::'
capacity	uint32	1..10		Weight for this object. Default: 10; Unit: relative
load	uint32	1..100		Dynamic load information, ranged from 0 to 100
priority	uint32	1..100		Priority (relative to other NFs of the same type).
locality	string			Location of the NF instance (e.g. geographic location, data center)

S Nssai

Supported slices per PLMN
Value FFFFFFFF is reserved and cannot be used to configure slice-differentiator

XPath: *cna-nf-selector:/nf-selector/static-nf-profile/s-nssai*

CLI Path: *nf-selector static-nf-profile s-nssai*

List Properties:
Keys: slice-name
Minimum number of elements: 0
Maximum number of elements: 8

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
slice-name	string	(a-zA-Z{0,63})		
slice-service-type	union	[uint8,0 or 4 .. 255', 'Enumerations: eMBB URLLC MIoT ']		Identifies the expected Network Slice behavior in terms of features and services
slice-differentiator	string	([a-zA-Z0-9]*)		Differentiate amongst multiple Network Slices of the same Slice/Service type

Amf Info

AMF Specific Information. See 3GPP specification TS29.510 for more information.

XPath: *cna-nf-selector:/nf-selector/static-nf-profile/amf-info*

CLI Path: *nf-selector static-nf-profile amf-info*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
amf-region-id	string	[A-Fa-f0-9]{2}		AMF region identifier
amf-set-id	string	[0-3][A-Fa-f0-9]{2}		AMF set identifier

Guami List

List of supported GUAMIs

XPath: *cna-nf-selector:/nf-selector/static-nf-profile/amf-info/guami-list*

CLI Path: *nf-selector static-nf-profile amf-info guami-list*

List Properties:

Keys: plmn-id, amf-id

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
plmn-id	string	(([0-9]{5,6}))		PLMN of Location Area Identifier
amf-id	string	(([A-Fa-f0-9]{6}))		AMF Identity

Tai List

The list of TAIs the AMF can serve

XPath: *cna-nf-selector:/nf-selector/static-nf-profile/amf-info/tai-list*

CLI Path: *nf-selector static-nf-profile amf-info tai-list*

List Properties:

Keys: plmn-id, tac

Minimum number of elements: 0

Maximum number of elements: 1000

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
plmn-id	string	(([0-9]{5,6}))		
tac	union	['string,[0-9a-fA-F]{4,6}', 'Enumerations: none']		

Tai Range List

The list of TAIs the AMF can serve

If 'tac-start' is set to none then 'tac-end' must also be set to none

XPath: *cna-nf-selector:/nf-selector/static-nf-profile/amf-info/tai-range-list*

CLI Path: *nf-selector static-nf-profile amf-info tai-range-list*

List Properties:

Keys: plmn-id, tac-start, tac-end

Minimum number of elements: 0

Maximum number of elements: 1000

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
plmn-id	string	(([0-9]{5,6}))		

Attribute	Type	Range/Length/Pattern	Default	Description
tac-start	union	['string,[0-9a-fA-F]{4,6}', 'Enumerations: none']		TAC start value
tac-end	union	['string,[0-9a-fA-F]{4,6}', 'Enumerations: none']		TAC end value

Smf Info

SMF Specific Information. See 3GPP specification TS29.510 for more information.

XPath: [cna-nf-selector:/nf-selector/static-nf-profile/smf-info](#)

CLI Path: [nf-selector static-nf-profile smf-info](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
pgw-fqdn	inet:domain-name			The FQDN of the PGW if the SMF is a combined SMF/PGW-C

S Nssai Smfinfo

Supported slices per PLMN

Value FFFFFFFF is reserved and cannot be used to configure slice-differentiator

XPath: [cna-nf-selector:/nf-selector/static-nf-profile/smf-info/s-nssai-smfinfo](#)

CLI Path: [nf-selector static-nf-profile smf-info s-nssai-smfinfo](#)

List Properties:

Keys: slice-name

Minimum number of elements: 0

Maximum number of elements: 8

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
slice-name	string	(a-zA-Z{0,63})		
slice-service-type	union	['uint8,0 or 4 .. 255', 'Enumerations: eMBB URLLC MIoT']		Identifies the expected Network Slice behavior in terms of features and services
slice-differentiator	string	([a-zA-Z0-9]*)		Differentiate amongst multiple Network Slices of the same Slice/Service type. Value FFFFFFFF is reserved and cannot be used to configure slice-differentiator.
dnn-list	leaf-list of string	0..100		Dnn supported by the SMF Minimum number of elements: 1 Maximum number of elements: 50

Udm Info

UDM Specific Information. See 3GPP specification TS29.510 for more information.

XPath: [cna-nf-selector:/nf-selector/static-nf-profile/udm-info](#)

CLI Path: [nf-selector static-nf-profile udm-info](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
group-id	string	0..20		Identity of the UDM group that is served by the UDM instance
routing-indicators	leaf-list of string	0..50		Routing indicator information Minimum number of elements: 0 Maximum number of elements: 50

Supi Range

List of ranges of SUPIs
Both 'supi-start' and 'supi-end' must be configured together
Either the 'supi-start' and 'supi-end' attributes, or the pattern attribute, shall be present.

XPath: *cna-nf-selector:/nf-selector/static-nf-profile/udm-info/supi-range*

CLI Path: *nf-selector static-nf-profile udm-info supi-range*

List Properties:
Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	0..20		
supi-start	string	([0-9]{3,15})		
supi-end	string	([0-9]{3,15})		The range value of 'supi-end must be greater than the value of 'supi-start'
pattern	string	3..100		

Ausf Info

AUSF Specific Information. See 3GPP specification TS29.510 for more information.

XPath: *cna-nf-selector:/nf-selector/static-nf-profile/ausf-info*

CLI Path: *nf-selector static-nf-profile ausf-info*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
group-id	string	0..20		Identity of the AUSF group
routing-indicators	leaf-list of string	0..50		Routing indicator information Minimum number of elements: 0 Maximum number of elements: 50

Upf Info

UPF Specific Information. See 3GPP specification TS29.510 for more information.

XPath: *cna-nf-selector:/nf-selector/static-nf-profile/upf-info*

CLI Path: *nf-selector static-nf-profile upf-info*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
smf-serving-area-list	leaf-list of string	1..20		List of SMF Serving Areas that UPF can serve. If not provided, the UPF can serve any SMF Service Area

S Nssai Upfinfo

List of parameters supported by the UPF per S-NSSAI
Value FFFFFFFF is reserved and cannot be used to configure slice-differentiator

XPath: *cna-nf-selector:/nf-selector/static-nf-profile/upf-info/s-nssai-upfinfo*

CLI Path: *nf-selector static-nf-profile upf-info s-nssai-upfinfo*

List Properties:
Keys: slice-name
Minimum number of elements: 0
Maximum number of elements: 8

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
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Attribute	Type	Range/Length/Pattern	Default	Description
slice-name	string	(a-zA-Z{0,63})		
slice-service-type	union	['uint8,0 or 4 .. 255', 'Enumerations: eMBB URLLC MIoT ']		Identifies the expected Network Slice behavior in terms of features and services
slice-differentiator	string			
		[(a-zA-Z0-9)*]		Differentiate amongst multiple Network Slices of the same Slice/Service type. Value FFFFFFFF is reserved and cannot be used to configure slice-differentiator.

Dnn Upfinfo List

List of parameters supported by the UPF per DNN

XPath: *cna-nf-selector:/nf-selector/static-nf-profile/upf-info/s-nssai-upfinfo/dnn-upfinfo-list*

CLI Path: *nf-selector static-nf-profile upf-info s-nssai-upfinfo dnn-upfinfo-list*

List Properties:

Keys: dnn
Minimum number of elements: 0
Maximum number of elements: 50

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
dnn	string	0..100		Dnn supported by the UPF per sNssai
dnai-list	leaf-list of string	0..100		List of Data network access identifiers supported by the UPF for this DNN Minimum number of elements: 0 Maximum number of elements: 50

Pcf Info

PCF Specific Information. See 3GPP specification TS29.510 for more information.

XPath: *cna-nf-selector:/nf-selector/static-nf-profile/pcf-info*

CLI Path: *nf-selector static-nf-profile pcf-info*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
dnn-list	leaf-list of string	0..100		Dnn supported by the PCF Minimum number of elements: 1 Maximum number of elements: 50

Supi Range

List of ranges of SUPIs
Both 'supi-start' and 'supi-end' must be configured together
Either the 'supi-start' and 'supi-end' attributes, or the pattern attribute, shall be present.

XPath: *cna-nf-selector:/nf-selector/static-nf-profile/pcf-info/supi-range*

CLI Path: *nf-selector static-nf-profile pcf-info supi-range*

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	0..20		
supi-start	string	[(0-9){3,15}]		

Attribute	Type	Range/Length/Pattern	Default	Description
supi-end	string	([0-9]{3,15})		The range value of 'supi-end' must be greater than the value of 'supi-start'
pattern	string	3..100		

Nf Service

NF Services Information. See 3GPP specification TS29.510 for more information.

XPath: *cna-nf-selector:/nf-selector/static-nf-profile/nf-service*

CLI Path: *nf-selector static-nf-profile nf-service*

Services

NF Services Information. See 3GPP specification TS29.510 for more information.

XPath: *cna-nf-selector:/nf-selector/static-nf-profile/nf-service/services*

CLI Path: *nf-selector static-nf-profile nf-service services*

List Properties:

Keys: service-instance-id
Minimum number of elements: 0
Maximum number of elements: 10

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
service-instance-id	string	0..255		Unique ID of the service instance within a given NF Instance
service-name	string	0..30		Name of the service instance (e.g. 'nudm-sdm')
schema	enumeration	http https		URI schema (e.g. 'http', 'https')

Version

List of supported versions

XPath: *cna-nf-selector:/nf-selector/static-nf-profile/nf-service/services/version*

CLI Path: *nf-selector static-nf-profile nf-service services version*

List Properties:

Keys: api-version-uri, api-full-version
Minimum number of elements: 1
Maximum number of elements: 10

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
api-version-uri	enumeration	v1 v2		Version of the service instance to be used in the URI for accessing the API (e.g. 'v1').
api-full-version	string	2..200		
expiry	string	\d{4}-\d{2}-\d{2}T\d{2}:\d{2}:\d{2}(\.\d+)?(Z or [+ -]\d{2}:\d{2})		

Ip Endpoint

List of ipEndPoint for each service

XPath: *cna-nf-selector:/nf-selector/static-nf-profile/nf-service/services/ip-endpoint*

CLI Path: *nf-selector static-nf-profile nf-service services ip-endpoint*

List Properties:

Keys: name

Minimum number of elements: 0
Maximum number of elements: 10

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	1..100		Name of the ipendpoint
ipv4-address	inet:ipv4-address			IP address for NF The 'ip-address' must not be '0.0.0.0'
ipv6-address	inet:ipv6-address			IP address for NF The 'ip-address' must not be '::'
protocol	string	1..7		protocol example: TCP,SCTP,UDP
port	uint32			port for the NF Services

Lmf Info

LMF Specific Information. See 3GPP specification TS29.510 for more information.

XPath: [cna-nf-selector:/nf-selector/static-nf-profile/lmf-info](#)

CLI Path: [nf-selector static-nf-profile lmf-info](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
serving-client-type-list	leaf-list of enumeration	EMERGENCY_SERVICES VALUE_ADDED_SERVICES PLMN_OPERATOR_SERVICES LAWFUL_INTERCEPT_SERVICES PLMN_OPERATOR_BROADCAST_SERVICES PLMN_OPERATOR_OM PLMN_OPERATOR_ANONYMOUS_STATISTICS PLMN_OPERATOR_TARGET_MS_SERVICE_SUPPORT		List of Serving Client Type. Absence of this means the LMF is not dedicated to serve specific client types Minimum number of elements: 0 Maximum number of elements: 8
lmf-id	string	0..36		LMF identification
serving-access-type-list	leaf-list of enumeration	threeGPP_ACCESS NON_threeGPP_ACCESS		List of Serving Access Type. If not included, it is assumed that all access types are supported Minimum number of elements: 0 Maximum number of elements: 2
serving-an-node-type-list	leaf-list of enumeration	GNB NG_GNB		List of Serving AN Node Type. If not included, it is assumed that all AN node types are supported Minimum number of elements: 0 Maximum number of elements: 2
serving-rat-type-list	leaf-list of enumeration	NR EUTRA WLAN VIRTUAL NBIOT WIRELINE WIRELINE_CABLE WIRELINE_BBF LTE-M NR_U EUTRA_U TRUSTED_N3GA TRUSTED_WLAN UTRA GERA		List of Serving RAT Type. If not included, it is assumed that all RAT types are supported Minimum number of elements: 0 Maximum number of elements: 15

Gmlc Info

GMLC Specific Information. See 3GPP specification TS29.510 for more information.

XPath: [cna-nf-selector:/nf-selector/static-nf-profile/gmlc-info](#)

CLI Path: *nf-selector static-nf-profile gmlc-info*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
serving-client-type-list	leaf-list of enumeration	EMERGENCY_SERVICES		List of Serving Client Type. Absence of this means the LMF is not dedicated to serve specific client types Minimum number of elements: 0 Maximum number of elements: 8
		VALUE_ADDED_SERVICES		
		PLMN_OPERATOR_SERVICES		
		LAWFUL_INTERCEPT_SERVICES		
		PLMN_OPERATOR_BROADCAST_SERVICES		
		PLMN_OPERATOR_OM		
		PLMN_OPERATOR_ANONYMOUS_STATISTICS		
		PLMN_OPERATOR_TARGET_MS_SERVICE_SUPPORT		

Pcscf Info

PCSCF Specific Information. See 3GPP specification TS29.510 for more information.

XPath: *cna-nf-selector:/nf-selector/static-nf-profile/pcscf-info*

CLI Path: *nf-selector static-nf-profile pcscf-info*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
dnn-list	leaf-list of string	0..100		Dnn supported by the P-CSCF Minimum number of elements: 0 Maximum number of elements: 300
access-type	leaf-list of enumeration	3GPP_ACCESS NON_3GPP_ACCESS		Access type supported by P-CSCF Minimum number of elements: 0 Maximum number of elements: 2
gm-ipv4-address	leaf-list of inet:ipv4-address			IPv4 address(es) of the P-CSCF for the Gm interface The 'ip-address' must not be '0.0.0.0'
gm-ipv6-address	leaf-list of inet:ipv6-address			IPv6 address(es) of the P-CSCF for the Gm interface The 'ip-address' must not be '::'

Trace

Trace objects

XPath: *cna-nf-selector:/nf-selector/trace*

CLI Path: *nf-selector trace*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
default-probability	string		0.0000000001	Probability of sampling the trace for a non-target; valid values are 0.0 to 1.0. Default: 0.0000000001 The 'probability' value must be less than 1.0

Targets

List of trace targets

One and only one of requester-nf-type or target-nf-type must be set

XPath: *cna-nf-selector:/nf-selector/trace/targets*

CLI Path: *nf-selector trace targets*

List Properties:

Keys: profile-name

Maximum number of elements: 100

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
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Attribute	Type	Range/Length/Pattern	Default	Description
profile-name	string			Target profile name :
target-nf-type	enumeration	AMF		Type of target Network Function
		SMF		
		UDM		
		AUSF		
		NEF		
		UPF		
		PCF		
		SMSF		
		BSF		
		CHF		
		AF		
		UDR		
		LMF		
		GMLC		
		5G_EIR		
		CBCF		
		NWDAF		
		N3IWF		
requester-nf-type	enumeration	NSSF		Type of requester Network Function
		AMF		
		SMF		
		UDM		
		AUSF		
		NEF		
		UPF		
		PCF		
		SMSF		
		BSF		
		CHF		
		AF		
		UDR		
		LMF		
		GMLC		
		5G_EIR		
		CBCF		
		NWDAF		
probability	string		1	Probability of sampling the trace for the target; valid values are 0.0 to 1.0. Default: 1
				The 'probability' value must be less than 1.0

NRF AGENT Configuration

The module for entities implementing cna-nrf-agent.

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Nrf Agent

NRF Agent Config. See 3GPP specification TS29.510v6 for more information.
One and only one 'primary' NRF must be configured when the nrf-redundancy-mode is set to primary-secondary

XPath: *cna-nrf-agent:/nrf-agent*

CLI Path: *nrf-agent*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
nrf-redundancy-mode	enumeration	multi-active primary-secondary	multi-active	NRF redundancy mode. In multi-active mode, all NRF are used equally for discovery, and all are updated with NF status. Whereas in primary-secondary mode, only the primary is used for discovery and NF status update. When heartbeat fails, the system will revert to a secondary until the heartbeat recovers based on the recovery criteria configured under the nrf-profile. Default: multi-active
failsafe-validity-timer	uint32	5..120	10	Validity time for which the discovery cache expiry is extended, when all the NRF status is down in 'primary-secondary'. Default: 10; Unit: minutes
enable-3gpp-peer-info	enumeration	true false	false	This attribute controls the addition of 3gpp-peer-info header in sbi message towards NRF'. Default: false

Static Nf Profile

Static configuration for Network Function (NF) profile. See 3GPP specification TS29.510 for more information.
The 'static-nf-profile' of nf-type 'nssf' already exists
Either 'ipv4-address' or 'ipv6-address' or 'fqdn' must be provisioned

XPath: *cna-nrf-agent:/nrf-agent/static-nf-profile*

CLI Path: *nrf-agent static-nf-profile*

List Properties:
Keys: nf-instance-id
Maximum number of elements: 20

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
nf-instance-id	string	[A-Za-z0-9 '()+,-./:=?]*		Unique identity of the NF Instance

Attribute	Type	Range/Length/Pattern	Default	Description
nf-type	enumeration	amf		Type of Network Function
		smf		
		udm		
		ausf		
		upf		
		pcf		
		nssf		
		smsf		
		bsf		
		chf		
		af		
		udr		
		lmf		
		gmlc		
		nef		
		pcscf		
		sepp		
plmn-id	string	(((0-9){5,6}))		PLMN of Location Area Identifier
fqdn	inet:domain-name			The FQDN of the NF
ipv4-address	inet:ipv4-address			IP address for NF The 'ip-address' must not be '0.0.0.0'
ipv6-address	inet:ipv6-address			IP address for NF The 'ip-address' must not be '::'
capacity	uint32	1..10		Weight for this object. Default: 10; Unit: relative
load	uint32	1..100		Dynamic load information, Range: 0 to 100
priority	uint32	1..100		Priority (relative to other NFs of the same type).
locality	string			Location of the NF instance (e.g. geographic location, data center)
nf-set-id-list	leaf-list of string	1..50		List of NF set ids
default-profile	enumeration	true false	false	Controls if nf-profile is used as fallback profile during discovery, i.e. if enabled profile is defaulted to be used only on discovery failure, otherwise will be looked in for discovery first on discovery request. Default: false The 'static-nf-profile' of nf-type 'AMF' and default-profile 'true' already exists The 'static-nf-profile' of nf-type 'SMF' and default-profile 'true' already exists The 'static-nf-profile' of nf-type 'UPF' and default-profile 'true' already exists The 'static-nf-profile' of nf-type 'UDM' and default-profile 'true' already exists The 'static-nf-profile' of nf-type 'PCF' and default-profile 'true' already exists The 'static-nf-profile' of nf-type 'AUSF' and default-profile 'true' already exists The 'static-nf-profile' of nf-type 'AF' and default-profile 'true' already exists The 'static-nf-profile' of nf-type 'BSF' and default-profile 'true' already exists The 'static-nf-profile' of nf-type 'chF' and default-profile 'true' already exists The 'static-nf-profile' of nf-type 'SMSF' and default-profile 'true' already exists The 'static-nf-profile' of nf-type 'URD' and default-profile 'true' already exists The 'static-nf-profile' of nf-type 'NEF' and default-profile 'true' already exists
nrf-instance-id	string			Nssf Specific Information. Nrf Instance Id to be used for Nssf oauth token operation.

S Nssai

Supported slices per PLMN
Value 0xFFFFFFFF is reserved and cannot be used to configure slice-differentiator

XPath: *cna-nrf-agent/nrf-agent/static-nf-profile/s-nssai*

CLI Path: *nrf-agent static-nf-profile s-nssai*

List Properties:
Keys: slice-name

Minimum number of elements: 0
Maximum number of elements: 8

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
slice-name	string	(a-zA-Z{0,63})		
slice-service-type	union	['uint8,0 or 4 .. 255', 'Enumerations: eMBB URLLC MIoT ']		Identifies the expected Network Slice behavior in terms of features and services
slice-differentiator	string	([a-fA-F0-9]*)		Differentiate amongst multiple Network Slices of the same Slice/Service type

Amf Info

AMF Specific Information. See 3GPP specification TS29.510 for more information.

XPath: *cna-nrf-agent:/nrf-agent/static-nf-profile/amf-info*

CLI Path: *nrf-agent static-nf-profile amf-info*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
amf-region-id	string	[A-Fa-f0-9]{2}		AMF region identifier
amf-set-id	string	[0-3][A-Fa-f0-9]{2}		AMF set identifier

Guami List

List of supported GUAMIs

XPath: *cna-nrf-agent:/nrf-agent/static-nf-profile/amf-info/guami-list*

CLI Path: *nrf-agent static-nf-profile amf-info guami-list*

List Properties:

Keys: plmn-id, amf-id
Minimum number of elements: 0
Maximum number of elements: 1000

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
plmn-id	string	((([0-9]{5,6})))		PLMN of Location Area Identifier
amf-id	string	0x[A-Fa-f0-9]{6}		AMF Identity (hexadecimal values of length 6 with a prefix of '0x')

Tai List

The list of TAIs the AMF can serve
Cannot configure 'tai-list' when 'tai-range-list' is already configured

XPath: *cna-nrf-agent:/nrf-agent/static-nf-profile/amf-info/tai-list*

CLI Path: *nrf-agent static-nf-profile amf-info tai-list*

List Properties:

Keys: plmn-id, tac
Minimum number of elements: 0
Maximum number of elements: 1000

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
plmn-id	string	((([0-9]{5,6})))		

Attribute	Type	Range/Length/Pattern	Default	Description
tac	string	0x[0-9a-fA-F]{4,6}		

Tai Range List

The list of TAIs the AMF can serve
Cannot configure 'tai-range-list' when 'tai-list' is already configured

XPath: *cna-nrf-agent:/nrf-agent/static-nf-profile/amf-info/tai-range-list*

CLI Path: *nrf-agent static-nf-profile amf-info tai-range-list*

List Properties:
Keys: plmn-id, tac-start, tac-end
Minimum number of elements: 0
Maximum number of elements: 1000

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
plmn-id	string	(([0-9]{5,6}))		
tac-start	string	0x[0-9a-fA-F]{4,6}		TAC start value
tac-end	string	0x[0-9a-fA-F]{4,6}		TAC end value

Smf Info

SMF Specific Information. See 3GPP specification TS29.510 for more information.

XPath: *cna-nrf-agent:/nrf-agent/static-nf-profile/smf-info*

CLI Path: *nrf-agent static-nf-profile smf-info*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
pgw-fqdn	inet:domain-name			The FQDN of the PGW if the SMF is a combined SMF/PGW-C

S Nssai

Supported slices per PLMN
Value 0xFFFFF is reserved and cannot be used to configure slice-differentiator

XPath: *cna-nrf-agent:/nrf-agent/static-nf-profile/smf-info/s-nssai*

CLI Path: *nrf-agent static-nf-profile smf-info s-nssai*

List Properties:
Keys: slice-name
Minimum number of elements: 0
Maximum number of elements: 8

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
slice-name	string	(a-zA-Z{0,63})		
slice-service-type	union	['uint8,0 or 4 .. 255', 'Enumerations: eMBB URLLC MIoT ']		Identifies the expected Network Slice behavior in terms of features and services
slice-differentiator	string	([a-fA-F0-9]*)		Differentiate amongst multiple Network Slices of the same Slice/Service type
dnn-list	leaf-list of string	0..100		Dnn supported by the SMF Minimum number of elements: 1 Maximum number of elements: 50

Udm Info

UDM Specific Information. See 3GPP specification TS29.510 for more information.

XPath: *cna-nrf-agent:/nrf-agent/static-nf-profile/udm-info*

CLI Path: *nrf-agent static-nf-profile udm-info*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
group-id	string	0..20		Identity of the UDM group that is served by the UDM instance
routing-indicators	leaf-list of string	0..50		Routing indicator information Minimum number of elements: 0 Maximum number of elements: 50

Supi Range

List of ranges of SUPIs

Both 'supi-start' and 'supi-end' must be configured together

Either the 'supi-start' and 'supi-end' attributes, or the pattern attribute, shall be present.

XPath: *cna-nrf-agent:/nrf-agent/static-nf-profile/udm-info/supi-range*

CLI Path: *nrf-agent static-nf-profile udm-info supi-range*

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	0..20		
supi-start	string	([0-9]{3,15})		
supi-end	string	([0-9]{3,15})		The range value of 'supi-end must be greater than the value of 'supi-start'
pattern	string	3..100		

Ausf Info

AUSF Specific Information. See 3GPP specification TS29.510 for more information.

XPath: *cna-nrf-agent:/nrf-agent/static-nf-profile/ausf-info*

CLI Path: *nrf-agent static-nf-profile ausf-info*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
group-id	string	0..20		Identity of the AUSF group
routing-indicators	leaf-list of string	0..50		Routing indicator information Minimum number of elements: 0 Maximum number of elements: 50

Upf Info

UPF Specific Information. See 3GPP specification TS29.510 for more information.

XPath: *cna-nrf-agent:/nrf-agent/static-nf-profile/upf-info*

CLI Path: *nrf-agent static-nf-profile upf-info*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
smf-serving-area-list	leaf-list of string	1..20		List of SMF Serving Areas the UPF can serve. If not provided, the UPF can serve any SMF service area

Attribute	Type	Range/Length/Pattern	Default	Description
iwk-eps-ind	enumeration	true false	false	Indicates whether interworking with EPS is supported by the UPF; Default: false
ipups	enumeration	true false	false	Indicates whether the UPF is configured for IPUPS(Inter-PLMN User Plane Security); Default: false

S Nssai Upfinfo

List of parameters supported by the UPF per S-NSSAI
Value 0xFFFFFFFF is reserved and cannot be used to configure slice-differentiator

XPath: *cna-nrf-agent:/nrf-agent/static-nf-profile/upf-info/s-nssai-upfinfo*

CLI Path: *nrf-agent static-nf-profile upf-info s-nssai-upfinfo*

- List Properties:**
Keys: slice-name
Minimum number of elements: 0
Maximum number of elements: 8

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
slice-name	string	(a-zA-Z{0,63})		
slice-service-type	union	[uint8,0 or 4 .. 255', 'Enumerations: eMBB URLLC MIoT ']		Identifies the expected Network Slice behavior in terms of features and services
slice-differentiator	string	([a-fA-F0-9]*)		Differentiate amongst multiple Network Slices of the same Slice/Service type

Dnn Upfinfo List

List of parameters supported by the UPF per DNN

XPath: *cna-nrf-agent:/nrf-agent/static-nf-profile/upf-info/s-nssai-upfinfo/dnn-upfinfo-list*

CLI Path: *nrf-agent static-nf-profile upf-info s-nssai-upfinfo dnn-upfinfo-list*

- List Properties:**
Keys: dnn
Minimum number of elements: 0
Maximum number of elements: 50

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
dnn	string	0..100		Dnn supported by the UPF per sNssai
dnai-list	leaf-list of string	0..100		List of Data network access identifiers supported by the UPF for this DNN Minimum number of elements: 0 Maximum number of elements: 50

Interface Upfinfo

List of User Plane interfaces configured on the UPF
Either 'ipv4-address' or 'ipv6-address' or 'fqdn' must be provisioned

XPath: *cna-nrf-agent:/nrf-agent/static-nf-profile/upf-info/interface-upfinfo*

CLI Path: *nrf-agent static-nf-profile upf-info interface-upfinfo*

- List Properties:**
Keys: interface-type
Minimum number of elements: 0
Maximum number of elements: 3

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
interface-type	string			Types of User-Plane interfaces of the UPF. Valid values are 'N3', 'N6' and 'N9' The 'interface-type' must be 'N3', 'N6 or 'N9'
fqdn	inet:domain-name			The FQDN of the NF
ipv4-address	leaf-list of inet:ipv4-address			List of IPv4 addresses for the interface The 'ipv4-address' must not be '0.0.0.0'
ipv6-address	leaf-list of inet:ipv6-address			List of IPv6 addresses for the interface The 'ipv6-address' must not be '::'
network-instance	string			Network instance associated to the User Plane interface

Pcf Info

PCF Specific Information. See 3GPP specification TS29.510 for more information.

XPath: *cna-nrf-agent:/nrf-agent/static-nf-profile/pcf-info*

CLI Path: *nrf-agent static-nf-profile pcf-info*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
dnn-list	leaf-list of string	0..100		Dnn supported by the PCF Minimum number of elements: 1 Maximum number of elements: 300

Supi Range

List of ranges of SUPIs

Both 'supi-start' and 'supi-end' must be configured together

Either the 'supi-start' and 'supi-end' attributes, or the pattern attribute, shall be present.

XPath: *cna-nrf-agent:/nrf-agent/static-nf-profile/pcf-info/supi-range*

CLI Path: *nrf-agent static-nf-profile pcf-info supi-range*

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	0..20		
supi-start	string	([0-9]{3,15})		
supi-end	string	([0-9]{3,15})		The range value of 'supi-end must be greater than the value of 'supi-start'
pattern	string	3..100		

Nf Service

NF Services Information. See 3GPP specification TS29.510 for more information.

XPath: *cna-nrf-agent:/nrf-agent/static-nf-profile/nf-service*

CLI Path: *nrf-agent static-nf-profile nf-service*

Services

NF Services Information. See 3GPP specification TS29.510 for more information.

XPath: *cna-nrf-agent:/nrf-agent/static-nf-profile/nf-service/services*

CLI Path: *nrf-agent static-nf-profile nf-service services*

List Properties:

Keys: service-instance-id
Minimum number of elements: 0
Maximum number of elements: 10

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
service-instance-id	string	0..255		Unique ID of the service instance within a given NF Instance
service-name	string	0..30		Name of the service instance (e.g. 'nudm-sdm')
schema	enumeration	http https		Protocol schema

Version

List of supported versions

XPath: *cna-nrf-agent:/nrf-agent/static-nf-profile/nf-service/services/version*

CLI Path: *nrf-agent static-nf-profile nf-service services version*

List Properties:

Keys: api-version-uri, api-full-version
Minimum number of elements: 0
Maximum number of elements: 10

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
api-version-uri	enumeration	v1 v2		Version of the service instance to be used in the URI for accessing the API (e.g. 'v1').
api-full-version	string	2..200		Full version number of the API as specified in subclause 4.3.1 of 3GPP TS 29.501.
expiry	string	\d{4}-\d{2}-\d{2}T\d{2}:\d{2}:\d{2}(\.\d+)? (Z or [+ -]\d{2}:\d{2})		Expiry date and time of the NF service. This represents the planned retirement date as specified in 3GPP TS 29.501

Ip Endpoint

List of ipEndPoint for each service

XPath: *cna-nrf-agent:/nrf-agent/static-nf-profile/nf-service/services/ip-endpoint*

CLI Path: *nrf-agent static-nf-profile nf-service services ip-endpoint*

List Properties:

Keys: name
Minimum number of elements: 0
Maximum number of elements: 10

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	1..100		Name of the ipendpoint
ipv4-address	inet:ipv4-address			IP address for NF The 'ip-address' must not be '0.0.0.0'
ipv6-address	inet:ipv6-address			IP address for NF The 'ip-address' must not be '::'
protocol	string	1..7		protocol example: TCP,SCTP,UDP
port	uint32			port for the NF Services

Lmf Info

LMF Specific Information. See 3GPP specification TS29.510 for more information.

XPath: *cna-nrf-agent:/nrf-agent/static-nf-profile/lmf-info***CLI Path:** *nrf-agent static-nf-profile lmf-info*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
serving-client-type-list	leaf-list of enumeration	EMERGENCY_SERVICES VALUE_ADDED_SERVICES PLMN_OPERATOR_SERVICES LAWFUL_INTERCEPT_SERVICES PLMN_OPERATOR_BROADCAST_SERVICES PLMN_OPERATOR_OM PLMN_OPERATOR_ANONYMOUS_STATISTICS PLMN_OPERATOR_TARGET_MS_SERVICE_SUPPORT		List of Serving Client Type. Absence of this means the LMF is not dedicated to serve specific client types Minimum number of elements: 0 Maximum number of elements: 8
lmf-id	string	0..36		LMF identification
serving-access-type-list	leaf-list of enumeration	threeGPP_ACCESS NON_threeGPP_ACCESS		List of Serving Access Type. If not included, it is assumed that all access types are supported Minimum number of elements: 0 Maximum number of elements: 2
serving-an-node-type-list	leaf-list of enumeration	GNB NG_GNB		List of Serving AN Node Type. If not included, it is assumed that all AN node types are supported Minimum number of elements: 0 Maximum number of elements: 2
serving-rat-type-list	leaf-list of enumeration	NR EUTRA WLAN VIRTUAL NBIOT WIRELINE WIRELINE_CABLE WIRELINE_BBF LTE-M NR_U EUTRA_U TRUSTED_N3GA TRUSTED_WLAN UTRA GERA		List of Serving RAT Type. If not included, it is assumed that all RAT types are supported Minimum number of elements: 0 Maximum number of elements: 15

Gmlc Info

GMLC Specific Information. See 3GPP specification TS29.510 for more information.

XPath: *cna-nrf-agent:/nrf-agent/static-nf-profile/gmlc-info***CLI Path:** *nrf-agent static-nf-profile gmlc-info*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
serving-client-type-list	leaf-list of enumeration	EMERGENCY_SERVICES VALUE_ADDED_SERVICES PLMN_OPERATOR_SERVICES LAWFUL_INTERCEPT_SERVICES PLMN_OPERATOR_BROADCAST_SERVICES PLMN_OPERATOR_OM PLMN_OPERATOR_ANONYMOUS_STATISTICS PLMN_OPERATOR_TARGET_MS_SERVICE_SUPPORT		List of Serving Client Type. Absence of this means the LMF is not dedicated to serve specific client types Minimum number of elements: 0 Maximum number of elements: 8

Nef Info

NEF Specific Information. See 3GPP specification TS29.510 for more information.

XPath: *cna-nrf-agent:/nrf-agent/static-nf-profile/nef-info*

CLI Path: *nrf-agent static-nf-profile nef-info*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
nef-id	string			Identity of the NEF

Pfd Data

List of Application IDs and/or AF IDs managed by a given NEF Instance
Atleast one nnef-pfdmanagemnet service needs to be present under nf-service for configuring pfdData

XPath: *cna-nrf-agent:/nrf-agent/static-nf-profile/nef-info/pfd-data*

CLI Path: *nrf-agent static-nf-profile nef-info pfd-data*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
app-ids	leaf-list of string			List of Application Ids
af-ids	leaf-list of string			List of AF Ids

Pcscf Info

PCSCF Specific Information. See 3GPP specification TS29.510 for more information.

XPath: *cna-nrf-agent:/nrf-agent/static-nf-profile/pcscf-info*

CLI Path: *nrf-agent static-nf-profile pcscf-info*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
dnn-list	leaf-list of string	0..100		Dnn supported by the P-CSCF Minimum number of elements: 1 Maximum number of elements: 300
access-type	leaf-list of enumeration	3GPP_ACCESS NON_3GPP_ACCESS		Access type supported by P-CSCF Minimum number of elements: 0 Maximum number of elements: 2
gm-ipv4-address	leaf-list of inet:ipv4-address			IPv4 address(es) of the P-CSCF for the Gm interface The 'ip-address' must not be '0.0.0.0'
gm-ipv6-address	leaf-list of inet:ipv6-address			IPv6 address(es) of the P-CSCF for the Gm interface The 'ip-address' must not be '::'

Default Notification Subscriptions

Notification endpoints for different notification types
When notification-type is N1-MESSAGE, n1-message-class must be set
When notification-type is N2-INFORMATION, n2-information-class must be set

XPath: *cna-nrf-agent:/nrf-agent/static-nf-profile/default-notification-subscriptions*

CLI Path: *nrf-agent static-nf-profile default-notification-subscriptions*

List Properties:
Keys: notification-type
Minimum number of elements: 0
Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
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Attribute	Type	Range/Length/Pattern	Default	Description
notification-type	enumeration	N1_MESSAGES N2_INFORMATION LOCATION_NOTIFICATION DATA_REMOVAL_NOTIFICATION DATA_CHANGE_NOTIFICATION LOCATION_UPDATE_NOTIFICATION NSSAA_REAUTH_NOTIFICATION NSSAA_REVOC_NOTIFICATION		Type of notification for which the corresponding callback URI is provided
callback-uri	string			A default notification endpoint to be used by a NF Service Producer towards an NF Service Consumer that has not registered explicitly a callback URI in the NF Service Producer
n1-message-class	enumeration	SM LPP SMS UPDP LCS 5GMM		If the notification type is N1_MESSAGES, this IE shall be present and shall identify the class of N1 messages to be notified
n2-information-class	enumeration	SM NRPPa PWS PWS-BCAL PWS-RF RAN V2Xkk		If the notification type is N2_INFORMATION, this IE shall be present and shall identify the class of N2 information to be notified
version-list	leaf-list of string	(\vd{1})		API versions (for example: v1) supported for the default notification type Minimum number of elements: 0 Maximum number of elements: 5
binding	string			Contain the value of the Binding Indication for the default subscription notification

Nrf Profile

Configuration for NF Repository Function (NRF) profile. See 3GPP specification TS29.510 for more information.

Priority must be provisioned when the role is set to secondary

XPath: *cna-nrf-agent/nrf-agent/nrf-profile*

CLI Path: *nrf-agent nrf-profile*

List Properties:

Keys: nrf-instance-id

Maximum number of elements: 20

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
nrf-instance-id	string	[A-Za-z0-9 '() +, -./:=?]*		Unique identity of the NRF Instance
role	enumeration	primary secondary		Role of the NRF. This is only applicable when nrf-redundancy-mode is set to primary-secondary. Role must be provisioned when the nrf-redundancy-mode is set to primary-secondary
failover-errorcode	leaf-list of uint32	400..599		HTTP error codes (4xx to 5xx) to consider for detecting NRF failure and trigger failover, a max 30 error codes can be configured. Default: '429','460','500','501','502','503','504','508'; To operate on the entire list, use square brackets, but to operate on individual leafs in the list, omit the square brackets. Minimum number of elements: 0 Maximum number of elements: 30
priority	uint32	1..20		Priority of the secondary NRF. Lower value is the higher priority
heartbeat-interval	uint32	60..360	120	Heartbeat duration interval value in seconds. Default: 120

Attribute	Type	Range/Length/Pattern	Default	Description
subscription-validity-timer	uint32	1..31680	31680	Validity time which the subscription is desired to be kept alive. Default: 31680; Unit: minutes The 'subscription-validity-time' must not be '0'
nrf-oauth-support	enumeration	enable disable	disable	Controls the enabling and disabling of the oauth2 feature for this NRF; When this flag is enabled nrf-agent gets the access token for the NRF services which are 'nnrf-nfm', 'nnrf-disc' during the registration process, and uses the token to access the given NRF services for subsequent management and discovery procedure. Default: disable
encoding-type	enumeration	gzip	gzip	Type of encoding supported as defined by 3GPP TS 29.510 section 6.1.2.2.3, as expressed in the http header. Default: gzip
compress-threshold-size	uint8	0..50	16	Maximum payload size allowed to send to NRF, expressed in kilo octets. Payloads that exceed this size will be automatically compressed if the NRF's capability supports compression as indicated by the OPTIONS API as defined in by 3GPP TS 29.510 section 6.1.3.2.3.2. Default: 16. Value 0 means compression is disabled.

Plmn Info

Supported PLMNs and S-NSSAIs information

XPath: *cna-nrf-agent:/nrf-agent/nrf-profile/plmn-info*

CLI Path: *nrf-agent nrf-profile plmn-info*

List Properties:

Keys: plmn-id
Minimum number of elements: 1
Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
plmn-id	string	(([0-9]{5,6}))		PLMN of Location Area Identifier

S Nssai

Supported slices per PLMN

XPath: *cna-nrf-agent:/nrf-agent/nrf-profile/plmn-info/s-nssai*

CLI Path: *nrf-agent nrf-profile plmn-info s-nssai*

List Properties:

Keys: slice-name
Minimum number of elements: 0
Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
slice-name	string	(a-zA-Z{0,63})		
slice-service-type	union	['uint8,0 or 4 .. 255', 'Enumerations: eMBB URLLC MIoT ']		Identifies the expected Network Slice behavior in terms of features and services
slice-differentiator	string	([a-fA-F0-9]*)		Differentiate amongst multiple Network Slices of the same Slice/Service type

Services

NRF Services Information. See 3GPP specification TS29.510 for more information.
service-heartbeat-interval must be provisioned when the nrf-redundancy-mode is set to 'primary-secondary'
recovery-count must be provisioned when the role is set to primary and the service-type is 'nnrf-nfm' or 'nnrf-disc'
failover-count must be provisioned when the nrf-redundancy-mode is set to 'primary-secondary' and the service-type is 'nnrf-nfm' or 'nnrf-disc'
Either 'ipv4-address' or 'ipv6-address' or 'fqdn' must be provisioned

XPath: *cna-nrf-agent:/nrf-agent/nrf-profile/services***CLI Path:** *nrf-agent nrf-profile services***List Properties:**

Keys: service-type

Minimum number of elements: 1

Maximum number of elements: 3

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
service-type	enumeration	nnrf-nfm nnrf-disc nnrf-accesstoken		Service that the NRF will be used for
dscp-marking	string	([0-9] or [1-5][0-9] or [6][0-3]) or df or af[1-4] [1-3] or cs[1-7] or ef	df	DSCP marking, Default: df; Valid values are: 0..63, df (DSCP-0), af11 (DSCP-10), af12 (DSCP-12), af13 (DSCP-14), af21 (DSCP-18), af22 (DSCP-20), af23 (DSCP-22), af31 (DSCP-26), af32 (DSCP-28), af33 (DSCP-30), af41 (DSCP-34), af42 (DSCP-36), af43 (DSCP-38), cs1 (DSCP-8), cs2 (DSCP-16), cs3 (DSCP-24), cs4 (DSCP-32), cs5 (DSCP-40), cs6 (DSCP-48), cs7 (DSCP-56), ef (DSCP-46)
service-heartbeat-interval	uint32	30..120		Keep-alive message per service interval in seconds; If it is configured for service 'nnrf-nfm' then NRF level failover is indicated. The service-heartbeat-interval must be less than the heartbeat-interval
recovery-count	uint32	1..3		Number of successful service keep-alive responses received before moving back to Primary NRF. This is only applicable to primary NRF.
failover-count	uint32	1..3		Number of monitoring failure responses received from the primary NRF before failover to secondary NRF
spec-version	enumeration	16.5.0	16.5.0	3GPP TS 29.510 version supported for the service. Default: 16.5.0
fqdn	inet:domain-name			The FQDN of the NRF service
ipv4-address	inet:ipv4-address			IP address for NRF service The 'ip-address' must not be '0.0.0.0'
ipv6-address	inet:ipv6-address			IP address for NRF service The 'ip-address' must not be '::'
port	uint32		80	Port for the NRF Service. Default: 80
schema	enumeration	http https	http	Protocol schema

Service Heartbeat Uri

URI of the API to be used in the service heartbeat message. This is only applicable when the nrf-redundancy-mode is set to 'primary-secondary'

XPath: *cna-nrf-agent:/nrf-agent/nrf-profile/services/service-heartbeat-uri***CLI Path:** *nrf-agent nrf-profile services service-heartbeat-uri*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
http-verb	enumeration	GET PATCH POST PUT		HTTP verb to be used in the service heartbeat message. If nothing is specified, the default verb for the service 'nnrf-nfm' or 'nnrf-disc' is 'GET'.
api-version	enumeration	v1 v2		Version of the service instance to be used in the URI
api-specific-path	string			API specific resource URI part. The specific resource must be compatible to the service-type of this service. If nothing is specified, the default URI for the service 'nnrf-nfm' or 'nnrf-disc' is 'nf-instances'.

Oauth Profile

Oauth profile related parameters; As defined by OAuth 2.0 authorization framework in IETF RFC 6749.

XPath: *cna-nrf-agent:/nrf-agent/nrf-profile/oauth-profile*

CLI Path: *nrf-agent nrf-profile oauth-profile*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
algorithm	enumeration	HS256 RS256		Type of algorithm supported; Value: 'HS256' or 'RS256' If 'algorithm' is set to 'HS256' then 'secret-key' must be configured. If 'algorithm' is set to 'RS256' then and 'public-key' must be configured.
secret-key	string			Secret key for validating JWT tokens in Base64 url safe encoded string; preferably 256-bit key; This attribute must be configured if the algorithm is set to 'HS256'.
public-key	string			Public key for validating JWT tokens in Base64 url safe encoded string; This attribute must be configured if the algorithm is set to 'RS256'.

Trace

NRF Agent Jaeger Tracing

XPath: *cna-nrf-agent:/nrf-agent/trace*

CLI Path: *nrf-agent trace*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
default-probability	string		0.0000000001	Probability of sampling the trace for a non-target; valid values are 0.0 to 1.0. Default: 0.0000000001 The 'probability' value must be less than 1.0

Targets

List of trace targets
One and only one of target-nrf-instance-id or service-type must be set

XPath: *cna-nrf-agent:/nrf-agent/trace/targets*

CLI Path: *nrf-agent trace targets*

List Properties:

Keys: profile-name
Maximum number of elements: 100

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
profile-name	string			Target profile name
target-nrf-instance-id	string	[A-Za-z0-9 '() +, -./: = ?]*		Unique identity of the NRF Instance
service-type	enumeration	nf-register nf-deregister nf-update-ex-heartbeat nf-update-heartbeat nf-status-subscribe nf-status-notify nf-status-unsubscribe access-token		Type of service provided by NRF NF management
probability	string		1	Probability of sampling the trace for the target; valid values are 0.0 to 1.0. Default: 1 The 'probability' value must be less than 1.0

NRF MONITOR Configuration

The module for entities implementing nrf-monitor

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- [nrf-monitor](#)
 - [critical-service-frequent-restart](#)
 - [deregistration-time](#)
 - [critical-service-readiness-probe](#)

Nrf Monitor

Network Function (NF) Monitor. See 3GPP specification TS29.510 for more information

XPath: [cna-nrf-monitor:/nrf-monitor](#)

CLI Path: [nrf-monitor](#)

Critical Service Frequent Restart

Configure critical service frequent Restart monitoring feature
status-check-interval should be always greater than stable-interval
critical-flip-count should be always greater than threshold-flip-count

XPath: [cna-nrf-monitor:/nrf-monitor/critical-service-frequent-restart](#)

CLI Path: [nrf-monitor critical-service-frequent-restart](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
monitoring	string	enabled or disabled	disabled	Feature enabled or disabled
stable-interval	uint16	1..3600	120	The minimum runtime in seconds for a critical service to be considered as stable
status-check-interval	uint16	1..3600	180	The duration in seconds after which the critical service status should be rechecked.
threshold-flip-count	uint16	1..1000	3	Number of times a critical service is found unstable after which alarm/alert is generated
critical-flip-count	uint16	1..1000	20	Number of times a critical service is found unstable after which vnf restart is triggered

Deregistration Time

Duration(in Secs) for which Monitor waits for a Critical Service to become available

XPath: [cna-nrf-monitor:/nrf-monitor/deregistration-time](#)

CLI Path: [nrf-monitor deregistration-time](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
value	uint16	1..3600	300	Duration(in Secs) for which Monitor waits for a Critical Service to become available

Critical Service Readiness Probe

Configure Critical Service Readiness monitoring feature
critical-failure-count should be always greater than threshold-failure-count

XPath: [cna-nrf-monitor:/nrf-monitor/critical-service-readiness-probe](#)

CLI Path: [nrf-monitor critical-service-readiness-probe](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
monitoring	enumeration	disabled enabled	disabled	Feature enabled or disabled
check-interval	uint16	60..36000	300	The duration in seconds after which the CriticalService status should be rechecked.
threshold-failure-count	uint16	1..20000	5	Number of times a CriticalService is found NotReady after which Alarm/Alert is generated

Attribute	Type	Range/Length/Pattern	Default	Description
critical-failure-count	uint16	1..20000	10	Number of times a CriticalService is found NotReady after which VNFRestart is triggered

NSSF MONITOR Configuration

This module defines the configuration for NSSF Monitor.

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- nssf-monitor
 - load-profile
 - critical-service-frequent-restart
 - deregistration-time
 - critical-service-readiness-probe

Nssf Monitor

NSSF Monitor interface object

XPath: *cna-nssf-monitor:/nssf-monitor*

CLI Path: *nssf-monitor*

Load Profile

Configure NSSF Load Profile

XPath: *cna-nssf-monitor:/nssf-monitor/load-profile*

CLI Path: *nssf-monitor load-profile*

List Properties:

Keys: load-profile

Maximum number of elements: 1

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
load-profile	string			NSSF Load Profile Name
reporting-frequency-interval	uint64		120	Frequency interval(seconds) at which the NSSF load should be reported to NRF. Default 120s
reporting-frequency-change-factor	uint64	1..100	20	Load change factor(+/- load difference) which should result in reporting the NSSF load to the NRF. Range 1..100. Default 20

Critical Service Frequent Restart

Configure critical service frequent Restart monitoring feature

status-check-interval should be always greater than stable-interval

critical-flip-count should be always greater than threshold-flip-count

XPath: *cna-nssf-monitor:/nssf-monitor/critical-service-frequent-restart*

CLI Path: *nssf-monitor critical-service-frequent-restart*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
monitoring	enumeration	disabled enabled	disabled	Feature enabled or disabled
stable-interval	uint16	1..3600	120	The minimum runtime in seconds for a critical service to be considered as stable
status-check-interval	uint16	1..3600	180	The duration in seconds after which the critical service status should be rechecked.
threshold-flip-count	uint16	1..1000	3	Number of times a critical service is found unstable after which alarm/alert is generated

Attribute	Type	Range/Length/Pattern	Default	Description
critical-flip-count	uint16	1..1000	20	Number of times a critical service is found unstable after which vnf restart is triggered

Deregistration Time

Duration(in Secs) for which Monitor waits for a Critical Service to become available

XPath: [cna-nssf-monitor:/nssf-monitor/deregistration-time](#)

CLI Path: [nssf-monitor deregistration-time](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
value	uint16	1..3600	300	Duration(in Secs) for which Monitor waits for a Critical Service to become available

Critical Service Readiness Probe

Configure Critical Service Readiness monitoring feature

critical-failure-count should be always greater than threshold-failure-count

XPath: [cna-nssf-monitor:/nssf-monitor/critical-service-readiness-probe](#)

CLI Path: [nssf-monitor critical-service-readiness-probe](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
monitoring	enumeration	disabled enabled	disabled	Feature enabled or disabled
check-interval	uint16	60..36000	300	The duration in seconds after which the CriticalService status should be rechecked.
threshold-failure-count	uint16	1..20000	5	Number of times a CriticalService is found NotReady after which Alarm/Alert is generated
critical-failure-count	uint16	1..20000	10	Number of times a CriticalService is found NotReady after which VNFRestart is triggered

OAUTH TOKEN Configuration

The module for entities implementing oauth-token generation

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- [oauth-token](#)
 - [nf-type](#)
 - [oauth](#)
 - [nf-instance-id](#)
 - [oauth](#)

Oauth Token

Network Function (NF) Access Token Service; See 3GPP specification TS 29.510 for more information

XPath: [cna-oauth-token:/oauth-token](#)

CLI Path: [oauth-token](#)

Nf Type

Network Function types

XPath: [cna-oauth-token:/oauth-token/nf-type](#)

CLI Path: [oauth-token nf-type](#)

List Properties:

Keys: nf-type

Minimum number of elements: 0
Maximum number of elements: 20

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
nf-type	enumeration	AMF		Network Function types
		SMF		
		UDM		
		AUSF		
		UPF		
		PCF		
		NEF		
		5G_EIR		
		SMSF		
		NSSF		
		UDR		
		CHF		
		BSF		
		AF		
		LMF		
		GMLC		
		NWDAF		
		CBCF		
		N3IWF		
		NRF		
		PCSCF		
		SEPP		

Oauth

OAuth related parameters; As defined by OAuth 2.0 authorization framework in IETF RFC 6749.

XPath: *cna-oauth-token:/oauth-token/nf-type/oauth*

CLI Path: *oauth-token nf-type oauth*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
secret-key	string			Secret key for signing JWT tokens in Base64 url safe encoded string; preferably 256-bit key
private-key	string			Private key for signing JWT tokens in Base64 url safe encoded string
public-key	string			Public key for signing JWT tokens in Base64 url safe encoded string
expiry-time	uint32	5 .. 86400	3600	Time for which a new token is valid relative to the time it was created; Unit: seconds
algorithm	enumeration	HS256 RS256		Type of algorithm supported. If 'algorithm' is set to 'HS256' then 'secret-key' must be configured. If 'algorithm' is set to 'HS256' then 'private-key' and 'public-key' must not be configured. If 'algorithm' is set to 'RS256' then 'private-key' and 'public-key' must be configured.
token-type	enumeration	JWT	JWT	Type of token supported; JWT is JSON Web Token as defined by RFC 7519. 3GPP mandates use of JWT. Default: JWT

Nf Instance Id

Unique identity of the NF Instance; Override the oauth value per 'nf-instance-id'

XPath: *cna-oauth-token:/oauth-token/nf-instance-id*

CLI Path: *oauth-token nf-instance-id*

List Properties:
Keys: nf-instance-id

Minimum number of elements: 0
Maximum number of elements: 1000

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
nf-instance-id	string			Unique identity of the NF instance

Oauth

OAuth related parameters; As defined by OAuth 2.0 authorization framework in IETF RFC 6749.

XPath: *cna-oauth-token:/oauth-token/nf-instance-id/oauth*

CLI Path: *oauth-token nf-instance-id oauth*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
secret-key	string			Secret key for signing JWT tokens in Base64 url safe encoded string; preferably 256-bit key
private-key	string			Private key for signing JWT tokens in Base64 url safe encoded string
public-key	string			Public key for signing JWT tokens in Base64 url safe encoded string
expiry-time	uint32	5 .. 86400	3600	Time for which a new token is valid relative to the time it was created; Unit: seconds
algorithm	enumeration	HS256 RS256		Type of algorithm supported. If 'algorithm' is set to 'HS256' then 'secret-key' must be configured. If 'algorithm' is set to 'HS256' then 'private-key' and 'public-key' must not be configured. If 'algorithm' is set to 'RS256' then 'private-key' and 'public-key' must be configured.
token-type	enumeration	JWT	JWT	Type of token supported; JWT is JSON Web Token as defined by RFC 7519. 3GPP mandates use of JWT. Default: JWT

PFCP CP Configuration

The module for entities implementing cna-pfcp-cp.

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PFCP UP Configuration

The module for entities implementing cna-pfcp-up.

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PFCP Configuration

The module for entities implementing cna-pfcp.

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- pfcp
 - service
 - peer
 - profile
 - trace
 - trace-keys

Pfcp

PFCP

XPath: *cna-pfcp:/pfcp*

CLI Path: *pfcp*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
admin-state	enumeration	disabled enabled	enabled	admin state for pfcp association service

Association

PFCP association config - contains list of peers for which association is initiated

XPath: *cna-pfcp:/pfcp/service*

CLI Path: *pfcp association*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
fqdn	string			Fully Qualified Domain Name of the PFCP-UPF
default-profile	leafref			default PFCP profile - used if peer has no profile Leaf Reference Path: <i>cna-pfcp:/pfcp/profile/name</i>
pfcp-set-id	string			PFCP SMF Set ID is an FQDN representing the SMF Set
bundling-support	enumeration	disabled enabled	enabled	PFCP Bundling feature support

Peer

PFCP Node - Contains the Peer-IP, UE-pool list, and UPResourceInfo List
must configure ipv4 or ipv6 addresses

XPath: *cna-pfcp:/pfcp/peer*

CLI Path: *pfcp peer*

List Properties:

Keys: node-name

Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
node-name	string			peer node name
fqdn	string			Peer FQDN
port	uint16		8805	Peer Port
ipv4-address	inet:ipv4-address			Peer IPV4 address
ipv6-address	inet:ipv6-address			Peer IPV6 address
pfcp-profile	leafref			PFCP profile Leaf Reference Path: <i>cna-pfcp:/pfcp/profile/name</i>

Profile

PFCP Profile - contains timers/counts used by PFCP service
at least one profile must be configured, and attached to the association

XPath: *cna-pfcp:/pfcp/profile*

CLI Path: *pfcp profile*

List Properties:

Keys: name

Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
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Attribute	Type	Range/Length/Pattern	Default	Description
name	string			Profile name
pfcf-association-retry-interval	uint32	20000 .. 3600000	120000	PFCE Association Retry Interval - amount of time to wait before re-initiating Association (ms). NOTE: If the configured timer value is less than net.netfilter.nf_conntrack_udp_timeout_stream value, the retry duration will be used as (net.netfilter.nf_conntrack_udp_timeout_stream + 10 seconds)
pfcf-session-response-timeout	uint32	100 .. 3600000	2400	PFCE Session Response Timeout Value - amount of time to wait before retransmitting a session level message (ms)
pfcf-session-retransmit-count	uint32	1 .. 5	2	PFCE Session Retransmit Count - number of session level messages retransmissions to make before declaring timeout
pfcf-node-response-timeout	uint32	100 .. 3600000	2000	PFCE Node Response Timeout Value - amount of time to wait before retransmitting a node level message (ms)
pfcf-node-retransmit-count	uint32	1 .. 5	3	PFCE Node Retransmit Count - number of node level messages retransmissions to make before declaring a timeout
pfcf-heartbeat-periodic-interval	uint32	300 .. 3600000	60000	PFCE HeartBeat Periodic Interval - amount of time to wait before restrasmitting a heartbeat request (ms)
pfcf-heartbeat-down-threshold	uint32	0 .. 300	0	PFCE HeartBeat Down Threshold - amount of pfcf-heartbeat-periodic-intervals after which the association will be restarted, 0 means retry heartbeat indefinitely, operator can set it accordingly.
pfcf-guard-time-interval	uint32	300 .. 3600000	120000	PFCE Guard Time Interval - amount of time to wait before tearing down Association (ms)
dscp-marking	string	([0-9] or [1-5][0-9] or [6][0-3]) or df or af[1-4][1-3] or cs[1-7] or ef	df	DSCP marking, Default: df; Valid values are: 0..63, df (DSCP-0), af11 (DSCP-10), af12 (DSCP-12), af13 (DSCP-14), af21 (DSCP-18), af22 (DSCP-20), af23 (DSCP-22), af31 (DSCP-26), af32 (DSCP-28), af33 (DSCP-30), af41 (DSCP-34), af42 (DSCP-36), af43 (DSCP-38), cs1 (DSCP-8), cs2 (DSCP-16), cs3 (DSCP-24), cs4 (DSCP-32), cs5 (DSCP-40), cs6 (DSCP-48), cs7 (DSCP-56), ef (DSCP-46)
pfcf-bundling-interval	uint32	5 .. 1000	10	PFCE Bundling Interval - amount of time to wait before sending bundled messages to peer(milli second)
pfcf-bundling-buffer-size	uint32	100 .. 32000	1472	PFCE Bundle Buffer Size - maximum buffer size of bundled messages(byte)
pfcf-bundling-message-count	uint32	1 .. 64	16	PFCE Bundle Message Count - maximum number of PFCE message in bundled messages(count)

Trace

PFCE Tracing Configuration

XPath: *cna-pfcf:/pfcf/trace*

CLI Path: *pfcf trace*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
trace-keys-admin-state	enumeration	disabled enabled	enabled	Enable or Disable Tracing Keys

Attribute	Type	Range/Length/Pattern	Default	Description
verbose	enumeration	disabled enabled	disabled	PFCP Verbose Tracing
probability	string		0	Probability of sampling the trace for PFCP The 'probability' value must be less than or equal to 1.0 The 'probability' value must be greater than or equal to 0

Trace Keys

Name
Tracing Key List
Atleast one of procedure-type or peer-id should be configured for the trace keys

XPath: [cna-pfcp/pfcp/trace/trace-keys](#)
CLI Path: [pfcp trace trace-keys](#)

List Properties:
Keys: name
Maximum number of elements: 128

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			Tracing Key Name
priority	uint32	1 .. 255	255	trace-keys priority from 1 to 255
probability	string		0	Probability of sampling the trace for PFCP trace-key The 'probability' value must be less than or equal to 1.0 The 'probability' value must be greater than or equal to 0
procedure-type	enumeration	association-setup association-update association-release association-heartbeat session-establishment session-modification session-delete session-report pfd-management		PFCP procedure type to match traces
peer-id	string			peer's IP address or FQDN value to match traces

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PPE Configuration

The module for entities implementing cna-ppe

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Upf

Affirmed UPF Packet Processing Engine

XPath: [cna-ppe:/upf](#)

CLI Path: [upf](#)

Service Construct

Affirmed UPF Service-Construct

XPath: [cna-ppe:/upf/service-construct](#)

CLI Path: [upf service-construct](#)

Packet Filter

Packet Filter: match on all fields
ue-port-end must be greater than or equal to ue-port-start
ue-port-end must be greater than or equal to ue-port-start

XPath: [cna-ppe:/upf/service-construct/packet-filter](#)

CLI Path: [upf service-construct packet-filter](#)

List Properties:
Keys: name
Maximum number of elements: 65535

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			
ip-version	enumeration	v4_only v6_only v4_or_v6	v4_or_v6	IP version

Attribute	Type	Range/Length/Pattern	Default	Description
ue-ip-v4-address-start	inet:ipv4-address		0.0.0.0	user equipment IPv4 address range start (inclusive)
ue-ip-v4-address-end	inet:ipv4-address		0.0.0.0	user equipment IPv4 address range end (inclusive)
network-ip-v4-address-start	inet:ipv4-address		0.0.0.0	network equipment IPv4 address range start (inclusive)
network-ip-v4-address-end	inet:ipv4-address		0.0.0.0	network equipment IPv4 address range end (inclusive)
ue-ip-v6-address-start	inet:ipv6-address		::	user equipment IPv6 address range start (inclusive)
ue-ip-v6-address-end	inet:ipv6-address		::	user equipment IPv6 address range end (inclusive)
network-ip-v6-address-start	inet:ipv6-address		::	network IPv6 address range start (inclusive)
network-ip-v6-address-end	inet:ipv6-address		::	network IPv6 address range end (inclusive)
ue-port-start	uint32	0 .. 65535	0	user equipment port range start value (valid for protocol = TCP (6), UDP (17), SCTP (132), only)
ue-port-end	uint32	0 .. 65535	65535	user equipment port range end value (valid for protocol = TCP (6), UDP (17), SCTP (132), only)
network-port-start	uint32	0 .. 65535	0	network equipment port range start value (valid for protocol = TCP (6), UDP (17), SCTP (132), only)
network-port-end	uint32	0 .. 65535	65535	network equipment port range end value (valid for protocol = TCP (6), UDP (17), SCTP (132), only)
protocol	uint32	0 .. 255	255	ip protocol values (e.g., all: 255, ICMP: 1, TCP: 6, UDP: 17, SCTP: 132)
initiator	enumeration	ue network both	both	match against ue or network traffic
min-uplink-ttl	uint32	0 .. 255	0	minimum value of the connection TTL(Time-To-Live)
max-uplink-ttl	uint32	0 .. 255	255	maximum value of the connection TTL(Time-To-Live)

Http Rule Group

HTTP Rule Group: match on any rule

XPath: *cna-ppe:/upf/service-construct/http-rule-group*

CLI Path: *upf service-construct http-rule-group*

List Properties:

Keys: name

Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			Content HTTP Rule Group Name - string

Rule

must enter host-port-start and host-port-end together to have a valid range
host-port-end must be >= host-port-start

XPath: *cna-ppe:/upf/service-construct/http-rule-group/rule*

CLI Path: *upf service-construct http-rule-group rule*

List Properties:

Keys: name

Minimum number of elements: 1
Maximum number of elements: 512

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			Rule name
priority	uint32	1 .. 4096		Priority
request-method	enumeration	GET HEAD POST PUT DELETE TRACE OPTIONS CONNECT		Specify a string for matching against HTTP request's method field
response-status-code	enumeration	100 101 200 201 202 203 204 205 206 300 301 302 303 304 305 306 307 400 401 402 403 404 405 406 407 408 409 410 411 412 413 414 415 416 417 500 501 502 503 504 505		Specify a status code for matching against HTTP response's status line
host-port-start	uint32	0 .. 65535		Specify a port start value for matching against HTTP request host port
host-port-end	uint32	0 .. 65535		Specify a port end value for matching against HTTP request host port

Content Type

Specify a string for matching against HTTP request's content-type header

XPath: [cna-ppe:/upf/service-construct/http-rule-group/rule/content-type](#)

CLI Path: *upf service-construct http-rule-group rule content-type*

Matches (choice)

XPath: *cna-ppe:/upf/service-construct/http-rule-group/rule/content-type*

CLI Path: *upf service-construct http-rule-group rule content-type*

Case : begins-with

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
begins-with	string			Specify a string that begins-with to match against incoming domain Choice: This begins-with attribute is mutually exclusive with attributes [['ends-with'], ['contains'], ['regex'], ['exact']]

Case : ends-with

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
ends-with	string			Specify a string that ends-with to match against incoming domain Choice: This ends-with attribute is mutually exclusive with attributes [['begins-with'], ['contains'], ['regex'], ['exact']]

Case : contains

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
contains	string			Specify a string that contains to match against incoming domain Choice: This contains attribute is mutually exclusive with attributes [['begins-with'], ['ends-with'], ['regex'], ['exact']]

Case : regex

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
regex	string			Specify a regex to match against incoming domain Choice: This regex attribute is mutually exclusive with attributes [['begins-with'], ['ends-with'], ['contains'], ['exact']]

Case : exact

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
exact	string			Specify an exact string to match against incoming domain Choice: This exact attribute is mutually exclusive with attributes [['begins-with'], ['ends-with'], ['contains'], ['regex']]

Host

Specify a string for matching against HTTP request's host header

XPath: *cna-ppe:/upf/service-construct/http-rule-group/rule/host*

CLI Path: *upf service-construct http-rule-group rule host*

Matches (choice)

XPath: [cna-ppe:/upf/service-construct/http-rule-group/rule/host](#)

CLI Path: [upf service-construct http-rule-group rule host](#)

Case : begins-with

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
begins-with	string			Specify a string that begins-with to match against incoming domain Choice: This begins-with attribute is mutually exclusive with attributes [['ends-with'], ['contains'], ['regex'], ['exact']]

Case : ends-with

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
ends-with	string			Specify a string that ends-with to match against incoming domain Choice: This ends-with attribute is mutually exclusive with attributes [['begins-with'], ['contains'], ['regex'], ['exact']]

Case : contains

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
contains	string			Specify a string that contains to match against incoming domain Choice: This contains attribute is mutually exclusive with attributes [['begins-with'], ['ends-with'], ['regex'], ['exact']]

Case : regex

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
regex	string			Specify a regex to match against incoming domain Choice: This regex attribute is mutually exclusive with attributes [['begins-with'], ['ends-with'], ['contains'], ['exact']]

Case : exact

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
exact	string			Specify an exact string to match against incoming domain Choice: This exact attribute is mutually exclusive with attributes [['begins-with'], ['ends-with'], ['contains'], ['regex']]

User Agent

Specify a string for matching against HTTP request's user-agent header

XPath: [cna-ppe:/upf/service-construct/http-rule-group/rule/user-agent](#)

CLI Path: [upf service-construct http-rule-group rule user-agent](#)

Matches (choice)

XPath: [cna-ppe:/upf/service-construct/http-rule-group/rule/user-agent](#)

CLI Path: [upf service-construct http-rule-group rule user-agent](#)

Case : begins-with

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
begins-with	string			Specify a string that begins-with to match against incoming domain Choice: This begins-with attribute is mutually exclusive with attributes [['ends-with'], ['contains'], ['regex'], ['exact']]

Case : ends-with

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
ends-with	string			Specify a string that ends-with to match against incoming domain Choice: This ends-with attribute is mutually exclusive with attributes [['begins-with'], ['contains'], ['regex'], ['exact']]

Case : contains

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
contains	string			Specify a string that contains to match against incoming domain Choice: This contains attribute is mutually exclusive with attributes [['begins-with'], ['ends-with'], ['regex'], ['exact']]

Case : regex

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
regex	string			Specify a regex to match against incoming domain Choice: This regex attribute is mutually exclusive with attributes [['begins-with'], ['ends-with'], ['contains'], ['exact']]

Case : exact

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
exact	string			Specify an exact string to match against incoming domain Choice: This exact attribute is mutually exclusive with attributes [['begins-with'], ['ends-with'], ['contains'], ['regex']]

Request Uri

Specify a string for matching against HTTP request's URI field

XPath: *cna-ppe:/upf/service-construct/http-rule-group/rule/request-uri*

CLI Path: *upf service-construct http-rule-group rule request-uri*

Matches (choice)

XPath: *cna-ppe:/upf/service-construct/http-rule-group/rule/request-uri*

CLI Path: *upf service-construct http-rule-group rule request-uri*

Case : begins-with

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
begins-with	string			Specify a string that begins-with to match against incoming domain Choice: This begins-with attribute is mutually exclusive with attributes [['ends-with'], ['contains'], ['regex'], ['exact']]

Case : ends-with

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
ends-with	string			Specify a string that ends-with to match against incoming domain Choice: This ends-with attribute is mutually exclusive with attributes [['begins-with'], ['contains'], ['regex'], ['exact']]

Case : contains

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
contains	string			Specify a string that contains to match against incoming domain Choice: This contains attribute is mutually exclusive with attributes [['begins-with'], ['ends-with'], ['regex'], ['exact']]

Case : regex

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
regex	string			Specify a regex to match against incoming domain Choice: This regex attribute is mutually exclusive with attributes [['begins-with'], ['ends-with'], ['contains'], ['exact']]

Case : exact

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
exact	string			Specify an exact string to match against incoming domain Choice: This exact attribute is mutually exclusive with attributes [['begins-with'], ['ends-with'], ['contains'], ['regex']]

Referer

Specify a string for matching against HTTP request's Referer field

XPath: *cna-ppe:/upf/service-construct/http-rule-group/rule/referer*

CLI Path: *upf service-construct http-rule-group rule referer*

Matches (choice)

XPath: *cna-ppe:/upf/service-construct/http-rule-group/rule/referer*

CLI Path: *upf service-construct http-rule-group rule referer*

Case : begins-with

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
-----------	------	----------------------	---------	-------------

Attribute	Type	Range/Length/Pattern	Default	Description
begins-with	string			Specify a string that begins-with to match against incoming domain Choice: This begins-with attribute is mutually exclusive with attributes [['ends-with'], ['contains'], ['regex'], ['exact']]

Case : ends-with

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
ends-with	string			Specify a string that ends-with to match against incoming domain Choice: This ends-with attribute is mutually exclusive with attributes [['begins-with'], ['contains'], ['regex'], ['exact']]

Case : contains

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
contains	string			Specify a string that contains to match against incoming domain Choice: This contains attribute is mutually exclusive with attributes [['begins-with'], ['ends-with'], ['regex'], ['exact']]

Case : regex

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
regex	string			Specify a regex to match against incoming domain Choice: This regex attribute is mutually exclusive with attributes [['begins-with'], ['ends-with'], ['contains'], ['exact']]

Case : exact

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
exact	string			Specify an exact string to match against incoming domain Choice: This exact attribute is mutually exclusive with attributes [['begins-with'], ['ends-with'], ['contains'], ['regex']]

Accept

Specify a string for matching against HTTP request's Accept header

XPath: [cna-ppe:/upf/service-construct/http-rule-group/rule/accept](#)

CLI Path: [upf service-construct http-rule-group rule accept](#)

Matches (choice)

XPath: [cna-ppe:/upf/service-construct/http-rule-group/rule/accept](#)

CLI Path: [upf service-construct http-rule-group rule accept](#)

Case : begins-with

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
-----------	------	----------------------	---------	-------------

Attribute	Type	Range/Length/Pattern	Default	Description
begins-with	string			Specify a string that begins-with to match against incoming domain Choice: This begins-with attribute is mutually exclusive with attributes [['ends-with'], ['contains'], ['regex'], ['exact']]

Case : ends-with

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
ends-with	string			Specify a string that ends-with to match against incoming domain Choice: This ends-with attribute is mutually exclusive with attributes [['begins-with'], ['contains'], ['regex'], ['exact']]

Case : contains

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
contains	string			Specify a string that contains to match against incoming domain Choice: This contains attribute is mutually exclusive with attributes [['begins-with'], ['ends-with'], ['regex'], ['exact']]

Case : regex

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
regex	string			Specify a regex to match against incoming domain Choice: This regex attribute is mutually exclusive with attributes [['begins-with'], ['ends-with'], ['contains'], ['exact']]

Case : exact

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
exact	string			Specify an exact string to match against incoming domain Choice: This exact attribute is mutually exclusive with attributes [['begins-with'], ['ends-with'], ['contains'], ['regex']]

Media

Specify media types for matching against response payload

XPath: [cna-ppe:/upf/service-construct/http-rule-group/rule/media](#)

CLI Path: [upf service-construct http-rule-group rule media](#)

Mediachoice (choice)

XPath: [cna-ppe:/upf/service-construct/http-rule-group/rule/media](#)

CLI Path: [upf service-construct http-rule-group rule media](#)

Case : all-video

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
-----------	------	----------------------	---------	-------------

Attribute	Type	Range/Length/Pattern	Default	Description
all-video	empty			Specify all video type for matching against response payload
Choice: This all-video attribute is mutually exclusive with attributes [['progressive-video'], ['abr-video']]				

Case : progressive-video

Progressive Video

Specify progressive video types(FLV, MP4, WebM) for matching against the response payload

Choice: This progressive-video attribute is mutually exclusive with attributes [['all-video'], ['abr-video']]

XPath: *cna-ppe:/upf/service-construct/http-rule-group/rule/media/progressive-video*

CLI Path: *upf service-construct http-rule-group rule media progressive-video*

Progressivevideochoice (choice)

Choice: This progressiveVideoChoice attribute is mutually exclusive with attributes [['all-video'], ['abr-video']]

XPath: *cna-ppe:/upf/service-construct/http-rule-group/rule/media/progressive-video*

*****CLI Path:** *upf service-construct http-rule-group rule media progressive-video ****

Case : progressive-video-all

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
progressive-video-all	empty			Choice: This progressive-video-all attribute is mutually exclusive with attributes [['all-video'], ['progressive-video'], ['abr-video'], ['progressive-video-type']]

Case : progressive-video-type

Progressive Video Type

Choice: This progressive-video-type attribute is mutually exclusive with attributes [['all-video'], ['progressive-video'], ['abr-video'], ['progressive-video-all']]

XPath: *cna-ppe:/upf/service-construct/http-rule-group/rule/media/progressive-video/progressive-video-type*

CLI Path: *upf service-construct http-rule-group rule media progressive-video progressive-video-type*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
flv	empty			Specify FLV for matching against the response payload
Choice: This flv attribute is mutually exclusive with attributes [['all-video'], ['progressive-video'], ['abr-video'], ['progressive-video-all']]				
mp4	empty			Specify MP4 for matching against the response payload
Choice: This mp4 attribute is mutually exclusive with attributes [['all-video'], ['progressive-video'], ['abr-video'], ['progressive-video-all']]				
webm	empty			Specify WebM for matching against the response payload
Choice: This webm attribute is mutually exclusive with attributes [['all-video'], ['progressive-video'], ['abr-video'], ['progressive-video-all']]				

Case : abr-video

Abr Video

Specify all ABR video types(HLS, DASH, HDS) for matching against the response payload

XPath: *cna-ppe:/upf/service-construct/http-rule-group/rule/media/abr-video*

CLI Path: *upf service-construct http-rule-group rule media abr-video*

Abrvideochoice (choice)

XPath: *cna-ppe:/upf/service-construct/http-rule-group/rule/media/abr-video*

CLI Path: *upf service-construct http-rule-group rule media abr-video*

Case : abr-all

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
abr-all	empty			Choice: This abr-all attribute is mutually exclusive with attributes [['abr-video-type']]

Case : abr-video-type

Abr Video Type

Choice: This abr-video-type attribute is mutually exclusive with attributes [['abr-all']]

XPath: *cna-ppe:/upf/service-construct/http-rule-group/rule/media/abr-video/abr-video-type*

CLI Path: *upf service-construct http-rule-group rule media abr-video abr-video-type*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
hls	empty			Specify HLS for matching against the response payload Choice: This hls attribute is mutually exclusive with attributes [['abr-all']]
dash	empty			Specify DASH for matching against the response payload Choice: This dash attribute is mutually exclusive with attributes [['abr-all']]
hds	empty			Specify HDS for matching against the response payload Choice: This hds attribute is mutually exclusive with attributes [['abr-all']]

Response Header

Configure HTTP response header matching rule

XPath: *cna-ppe:/upf/service-construct/http-rule-group/rule/response-header*

CLI Path: *upf service-construct http-rule-group rule response-header*

List Properties:

Keys: header-name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
header-name	string			Specify a string as the header name for matching against HTTP response

Header Value

Specify a string as the header value for matching against HTTP response

XPath: *cna-ppe:/upf/service-construct/http-rule-group/rule/response-header/header-value*

CLI Path: *upf service-construct http-rule-group rule response-header header-value*

Matches (choice)

XPath: *cna-ppe:/upf/service-construct/http-rule-group/rule/response-header/header-value*

CLI Path: *upf service-construct http-rule-group rule response-header header-value*

Case : begins-with

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
begins-with	string			Specify a string that begins-with to match against incoming domain Choice: This begins-with attribute is mutually exclusive with attributes [['ends-with'], ['contains'], ['regex'], ['exact']]

Case : ends-with

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
ends-with	string			Specify a string that ends-with to match against incoming domain Choice: This ends-with attribute is mutually exclusive with attributes [['begins-with'], ['contains'], ['regex'], ['exact']]

Case : contains

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
contains	string			Specify a string that contains to match against incoming domain Choice: This contains attribute is mutually exclusive with attributes [['begins-with'], ['ends-with'], ['regex'], ['exact']]

Case : regex

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
regex	string			Specify a regex to match against incoming domain Choice: This regex attribute is mutually exclusive with attributes [['begins-with'], ['ends-with'], ['contains'], ['exact']]

Case : exact

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
exact	string			Specify an exact string to match against incoming domain Choice: This exact attribute is mutually exclusive with attributes [['begins-with'], ['ends-with'], ['contains'], ['regex']]

Domain Rule Group

Domain Rule Group: match on any rule

XPath: *cna-ppe:/upf/service-construct/domain-rule-group*

CLI Path: *upf service-construct domain-rule-group*

List Properties:

Keys: name

Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			Domain Rule Group Name - string
dns-snoop	enumeration	disabled enabled	enabled	Specify DNS snoop needs to be enabled for this domain rule

Rule

Domain Rule: match on all fields

XPath: *cna-ppe:/upf/service-construct/domain-rule-group/rule*

CLI Path: *upf service-construct domain-rule-group rule*

List Properties:

Keys: name

Minimum number of elements: 1

Maximum number of elements: 512

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			Rule name

Domain Name

Specify a domain name to match against

XPath: *cna-ppe:/upf/service-construct/domain-rule-group/rule/domain-name*

CLI Path: *upf service-construct domain-rule-group rule domain-name*

Matches (choice)

XPath: *cna-ppe:/upf/service-construct/domain-rule-group/rule/domain-name*

CLI Path: *upf service-construct domain-rule-group rule domain-name*

Case : begins-with

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
begins-with	string			Specify a string that begins-with to match against incoming domain Choice: This begins-with attribute is mutually exclusive with attributes [['ends-with'], ['contains'], ['regex'], ['exact']]

Case : ends-with

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
ends-with	string			Specify a string that ends-with to match against incoming domain Choice: This ends-with attribute is mutually exclusive with attributes [['begins-with'], ['contains'], ['regex'], ['exact']]

Case : contains

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
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Attribute	Type	Range/Length/Pattern	Default	Description
contains	string			Specify a string that contains to match against incoming domain Choice: This contains attribute is mutually exclusive with attributes [['begins-with'], ['ends-with'], ['regex'], ['exact']]

Case : regex

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
regex	string			Specify a regex to match against incoming domain Choice: This regex attribute is mutually exclusive with attributes [['begins-with'], ['ends-with'], ['contains'], ['exact']]

Case : exact

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
exact	string			Specify an exact string to match against incoming domain Choice: This exact attribute is mutually exclusive with attributes [['begins-with'], ['ends-with'], ['contains'], ['regex']]

Tls Rule Group

TLS Rule Group: match on any rule

XPath: *cna-ppe:/upf/service-construct/tls-rule-group*

CLI Path: *upf service-construct tls-rule-group*

List Properties:

Keys: name

Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			TLS Rule Group Name - string

Rule

TLS Rule: match on all fields

XPath: *cna-ppe:/upf/service-construct/tls-rule-group/rule*

CLI Path: *upf service-construct tls-rule-group rule*

List Properties:

Keys: name

Minimum number of elements: 1

Maximum number of elements: 512

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			

Sni

Specify a Server Name Indication string for matching against the incoming SNI

XPath: *cna-ppe:/upf/service-construct/tls-rule-group/rule/sni*

CLI Path: *upf service-construct tls-rule-group rule sni*

Matches (choice)

XPath: `cna-ppe:/upf/service-construct/tls-rule-group/rule/sni`

CLI Path: `upf service-construct tls-rule-group rule sni`

Case : begins-with

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
begins-with	string			Specify a string that begins-with to match against incoming domain Choice: This begins-with attribute is mutually exclusive with attributes [['ends-with'], ['contains'], ['regex'], ['exact']]

Case : ends-with

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
ends-with	string			Specify a string that ends-with to match against incoming domain Choice: This ends-with attribute is mutually exclusive with attributes [['begins-with'], ['contains'], ['regex'], ['exact']]

Case : contains

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
contains	string			Specify a string that contains to match against incoming domain Choice: This contains attribute is mutually exclusive with attributes [['begins-with'], ['ends-with'], ['regex'], ['exact']]

Case : regex

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
regex	string			Specify a regex to match against incoming domain Choice: This regex attribute is mutually exclusive with attributes [['begins-with'], ['ends-with'], ['contains'], ['exact']]

Case : exact

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
exact	string			Specify an exact string to match against incoming domain Choice: This exact attribute is mutually exclusive with attributes [['begins-with'], ['ends-with'], ['contains'], ['regex']]

Server Common Name

Specify a Common Name string for matching against the incoming Common Name

XPath: `cna-ppe:/upf/service-construct/tls-rule-group/rule/server-common-name`

CLI Path: `upf service-construct tls-rule-group rule server-common-name`

Matches (choice)

XPath: `cna-ppe:/upf/service-construct/tls-rule-group/rule/server-common-name`

CLI Path: [upf service-construct tls-rule-group rule server-common-name](#)

Case : begins-with

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
begins-with	string			Specify a string that begins-with to match against incoming domain Choice: This begins-with attribute is mutually exclusive with attributes [['ends-with'], ['contains'], ['regex'], ['exact']]

Case : ends-with

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
ends-with	string			Specify a string that ends-with to match against incoming domain Choice: This ends-with attribute is mutually exclusive with attributes [['begins-with'], ['contains'], ['regex'], ['exact']]

Case : contains

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
contains	string			Specify a string that contains to match against incoming domain Choice: This contains attribute is mutually exclusive with attributes [['begins-with'], ['ends-with'], ['regex'], ['exact']]

Case : regex

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
regex	string			Specify a regex to match against incoming domain Choice: This regex attribute is mutually exclusive with attributes [['begins-with'], ['ends-with'], ['contains'], ['exact']]

Case : exact

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
exact	string			Specify an exact string to match against incoming domain Choice: This exact attribute is mutually exclusive with attributes [['begins-with'], ['ends-with'], ['contains'], ['regex']]

San

Specify a Subject Alternative Name string for matching against the incoming SAN

XPath: [cna-ppe:/upf/service-construct/tls-rule-group/rule/san](#)

CLI Path: [upf service-construct tls-rule-group rule san](#)

Matches (choice)

XPath: [cna-ppe:/upf/service-construct/tls-rule-group/rule/san](#)

CLI Path: [upf service-construct tls-rule-group rule san](#)

Case : begins-with

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
begins-with	string			Specify a string that begins-with to match against incoming domain Choice: This begins-with attribute is mutually exclusive with attributes [['ends-with'], ['contains'], ['regex'], ['exact']]

Case : ends-with

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
ends-with	string			Specify a string that ends-with to match against incoming domain Choice: This ends-with attribute is mutually exclusive with attributes [['begins-with'], ['contains'], ['regex'], ['exact']]

Case : contains

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
contains	string			Specify a string that contains to match against incoming domain Choice: This contains attribute is mutually exclusive with attributes [['begins-with'], ['ends-with'], ['regex'], ['exact']]

Case : regex

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
regex	string			Specify a regex to match against incoming domain Choice: This regex attribute is mutually exclusive with attributes [['begins-with'], ['ends-with'], ['contains'], ['exact']]

Case : exact

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
exact	string			Specify an exact string to match against incoming domain Choice: This exact attribute is mutually exclusive with attributes [['begins-with'], ['ends-with'], ['contains'], ['regex']]

Acl Rule

acl-rule-description: match on any packet-filter

XPath: *cna-ppe:/upf/service-construct/acl-rule*

CLI Path: *upf service-construct acl-rule*

List Properties:

Keys: name

Maximum number of elements: 1000

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			
priority	uint32	1 .. 4096		Priority

Attribute	Type	Range/Length/Pattern	Default	Description
action	enumeration	deny allow	deny	allow/deny

Packet Filter

Name of a Packet Filter associated with this ACL rule

XPath: [cna-ppe:/upf/service-construct/acl-rule/packet-filter](#)

CLI Path: [upf service-construct acl-rule packet-filter](#)

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: cna-ppe:/upf/service-construct/packet-filter/name

Acl Rule Group

Configure ACL rule group

XPath: [cna-ppe:/upf/service-construct/acl-rule-group](#)

CLI Path: [upf service-construct acl-rule-group](#)

List Properties:

Keys: name

Maximum number of elements: 1000

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			ACL rule group name

Aclrule

List of ACL rules for this rule group

XPath: [cna-ppe:/upf/service-construct/acl-rule-group/aclRule](#)

CLI Path: [upf service-construct acl-rule-group aclRule](#)

List Properties:

Keys: name

Minimum number of elements: 0

Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: cna-ppe:/upf/service-construct/acl-rule/name

Security Policy

Configure security-policy

XPath: [cna-ppe:/upf/service-construct/security-policy](#)

CLI Path: [upf service-construct security-policy](#)

List Properties:

Keys: name

Maximum number of elements: 1024

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			
acl-rule-group	leafref			Name of an acl-rule-group to attach Leaf Reference Path: cna-ppe:/upf/service-construct/acl-rule-group/name

Network Context

Network-Context instance.
Configure network context object

XPath: [cna-ppe:/upf/service-construct/network-context](#)

CLI Path: [upf service-construct network-context](#)

List Properties:
Keys: name
Maximum number of elements: 4096

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
tcp-mss-adjust-ipv4	uint32	0 .. 1800	0	TCP MSS Adjustment for IPv4 to override the MSS in TCP connections
tcp-mss-adjust-ipv6	uint32	0 .. 1800	0	TCP MSS Adjustment for IPv6 to override the MSS in TCP connections
name	string	1..64		Name of this network-context. This name must be unique.
type	bits			Type of this network-context. Possible values are {N3}, {N6}, {N4}, {N9} and {NWU}.
header-sanity-check	enumeration	false true	false	Checks for packet header consistency when this flag is set to true
ignore-notify-bgp-peer-state-change	boolean		false	Ignore BGP Peer State Change notification from this network-context
security-policy	leafref			Name of security-policy to attach to this network-context Leaf Reference Path: cna-ppe:/upf/service-construct/security-policy/name

Nat Pool

Configure nat pool, Warning!!! name of the nat-pool must be unique across network-context
If the nat mode is One to One, the value of max-ports-per-session and max-port-alloc-retries cannot be changed

XPath: [cna-ppe:/upf/service-construct/network-context/nat-pool](#)

CLI Path: [upf service-construct network-context nat-pool](#)

List Properties:
Keys: name
Maximum number of elements: 4096

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			Name of this nat-pool, Warning!!! This name must be unique across network-context.
nat-mode	enumeration	Many_to_One One_to_One	Many_to_One	Setting that controls whether NAT or NAPT is applied for subscribers using this NAT pool - Many_to_One or One_to_One
nat-allocation-policy	enumeration	statistical guaranteed	statistical	Policy that controls when and how NAT resources are assigned to UEs when using this NAT pool
nat-filter	enumeration	not_set address_port_dependent	address_port_dependent	Nat filter type for this nat pool

Attribute	Type	Range/Length/Pattern	Default	Description
max-ports-per-session	uint32	0..64511	0	Number of ports allowed per UE. Default 0 implies unlimited ports per UE.
l3am-nat-ips-per-ppe	uint32	1 .. max	1	Number of ip's per ppe to be allocated
max-port-alloc-retries	uint32	0..64511	0	Maximum number of retries to find nat port for allocation. Default 0 implies max possible retries.
max-users-per-nat-ip	uint32	2..10000		The maximum number of UE/APNs that can be assigned to a public IP - if not configured, there is no limit enforced. Value is only valid for nat-mode Many_to_One
nat-ip-release-timer	uint32	0..3600		Timer value (in seconds) to age out and release inactive NAT address allocations from a Statistical NAT pool for a subscriber session, If nat-ip-release-timer configured with Zero(0) NAT IP will be released back to free pool when all the active connections of a subscriber goes away, If nat-ip-release-timer configured with Postive Value(in Seconds) NAT IP will be released back to free pool after n seconds when last connection of a subscriber goes away, If not configured NAT IP is not released from session to the pool even though all connections are aged out and it released to pool after session deletion
associated-vrf-network-context	leafref			If the NAT Pool is used for sessions anchored in a VRF network-context, the vrf network-context name we need to mention here - This config is needed only in the case when number of NAT addresses is less than number of PPEs and in the VRF network-context small UE Pool is configured which is not anchored on all the PPEs. Leaf Reference Path: cna-ppe:/upf/service-construct/network-context/name
group	leafref			Name of the group the NAT Pool is classified into - for resource management Leaf Reference Path: cna-ppe:/upf/service-construct/network-context/ip-pool/group
nat-port-chunk-size	uint32	8	16	32

Ip Pool

IP Pool configuration, Warning!!! name of the ip-pool must be unique across network-context
Must specify NAT Pool when ip-pool-type is 'NAT_IP_Pool'

XPath: [cna-ppe:/upf/service-construct/network-context/ip-pool](#)

CLI Path: [upf service-construct network-context ip-pool](#)

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	1..64		Name of this ip-pool, Warning!!! This name must be unique across network-context.
admin-state	enumeration	disabled enabled maintenance	enabled	Administrative State(enabled
type	enumeration	UE_IP_Pool NAT_IP_Pool Remote_UE_IP_Pool	UE_IP_Pool	Type of the IP pool
range-start	inet:ip-address			Starting value for IP address of this IP pool
range-end	inet:ip-address			Ending value for IP address of this IP pool

Attribute	Type	Range/Length/Pattern	Default	Description
route-summary-prefix-length	uint32		0	If advertise-pool-aggregate is true, this attribute controls whether the pool's address range is advertised as a set of routes that exactly cover the address range, or by a single route that encompasses the pool range but may include addresses that are outside the pool range. If route-summary-prefix-length is not configured, the exact address range is advertised. If route-summary-prefix-length is configured, a single route will be advertised consisting of the starting address and this prefix length.
advertise	boolean		true	Advertise PPE-specific reachability for addresses in this pool. UE_IP_Pool and NAT-IP_Pool addresses will be advertised in pool chunks, Remote_UE_IP_Pool addresses will be individually advertised as host routes.
advertise-pool-aggregate	boolean		false	Advertise reachability for the entire address range of this pool. Each PPE will advertise the same set of routes.
route-tag-aggregate	uint32	0 .. max	0	If aggregate route(s) are generated for this pool, the route-tag value will be attached to each route passed to BGP. BGP can apply policy to these routes by matching the route-tag. A route-tag value does not have to be unique across ip-pools.
route-tag	uint32	0 .. max	0	For route(s) generated for I3rm chunk or host route of this pool, this route-tag value will be attached to each route passed to BGP. BGP can apply policy to these routes by matching the route-tag. A route-tag value does not have to be unique across ip-pools.
I3am-chunk-size	uint32	0 .. max	0	Size of the UE IP pool chunks that will be used in resource blocks for Layer 3 Affinity mode - for resource management with multiple PPE instances. When set to 0, chunk size is auto calculated
group	string			Name of the group the UE IP Pool is classified into - for resource management
nat-pool	leafref			NAT pool associated with an IP sub-pool of type NAT-IP-Pool Leaf Reference Path: cna-ppe:/upf/service-construct/network-context/nat-pool/name

Serving Area

Serving-area instance.
Configure serving area object

XPath: [cna-ppe:/upf/service-construct/serving-area](#)

CLI Path: [upf service-construct serving-area](#)

List Properties:
Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	1..64		Name of serving-area. This name must be unique

Serving Area List

Serving-area-list instance.
Configure serving area list object

XPath: [cna-ppe:/upf/service-construct/serving-area-list](#)

CLI Path: [upf service-construct serving-area-list](#)

List Properties:
Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	1..64		Name of serving-area-list. This name must be unique

Serving Area

Name of serving-area

XPath: [cna-ppe:/upf/service-construct/serving-area-list/serving-area](#)

CLI Path: *upf service-construct serving-area-list serving-area*

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: cna-ppe:/upf/service-construct/serving-area/name

Tdef

Affirmed UPF Service-Construct Tdef objects

XPath: *cna-ppe:/upf/service-construct/tdef*

CLI Path: *upf service-construct tdef*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
cert-expiry-days	uint32	10 .. 600	30	Configure Number of days before which warning/critical alerts are raised for TDEF Proxy Certs

Enrichment Profile

Configure Content Provider Enrichment Profile

XPath: *cna-ppe:/upf/service-construct/tdef/enrichment-profile*

CLI Path: *upf service-construct tdef enrichment-profile*

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			Enrichment Profile Name - string
index	uint32	1 .. max		Unique Index

Header Name

Add the Header Name

XPath: *cna-ppe:/upf/service-construct/tdef/enrichment-profile/headerString*

CLI Path: *upf service-construct tdef enrichment-profile header-name*

List Properties:

Keys: headerString

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
header-string	string	1..64		

Headervalue (choice)

XPath: *cna-ppe:/upf/service-construct/tdef/enrichment-profile/headerString*

CLI Path: *upf service-construct tdef enrichment-profile header-name*

Case : containerValue

Value

Custom value to include for the header.

Choice: This value attribute is mutually exclusive with attributes [['ratType'], ['chargingCharacteristic'], ['servingNetworkPlmnid'], ['uli'], ['msisdn'], ['imsi'], ['imei'], ['httpProtocol'], ['sgwlpAddr'], ['sgwlpv6Addr'], ['originalApn'], ['mappedApn'], ['subscriberId'], ['resellerId'], ['timezone']]

XPath: *cna-ppe:/upf/service-construct/tdef/enrichment-profile/headerString/containerValue*

CLI Path: *upf service-construct tdef enrichment-profile header-name value*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
value	string	1..255		Choice: This value attribute is mutually exclusive with attributes [['ratType'], ['chargingCharacteristic'], ['servingNetworkPlmnid'], ['uli'], ['msisdn'], ['imsi'], ['imei'], ['httpProtocol'], ['sgwlpAddr'], ['sgwlpv6Addr'], ['originalApn'], ['mappedApn'], ['subscriberId'], ['resellerId'], ['timezone']]

Case : ratType

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
				Include Radio Access Type (2G, 3G or 4G) in the header
rat-type	empty			Choice: This rat-type attribute is mutually exclusive with attributes [['containerValue'], ['chargingCharacteristic'], ['servingNetworkPlmnid'], ['uli'], ['msisdn'], ['imsi'], ['imei'], ['httpProtocol'], ['sgwlpAddr'], ['sgwlpv6Addr'], ['originalApn'], ['mappedApn'], ['subscriberId'], ['resellerId'], ['timezone']]

Case : chargingCharacteristic

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
				Include Charging Characteristic in the header
charging-characteristic	empty			Choice: This charging-characteristic attribute is mutually exclusive with attributes [['containerValue'], ['ratType'], ['servingNetworkPlmnid'], ['uli'], ['msisdn'], ['imsi'], ['imei'], ['httpProtocol'], ['sgwlpAddr'], ['sgwlpv6Addr'], ['originalApn'], ['mappedApn'], ['subscriberId'], ['resellerId'], ['timezone']]

Case : servingNetworkPlmnid

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
				Include Serving Network PLMNID in the header
serving-network-plmnid	empty			Choice: This serving-network-plmnid attribute is mutually exclusive with attributes [['containerValue'], ['ratType'], ['chargingCharacteristic'], ['uli'], ['msisdn'], ['imsi'], ['imei'], ['httpProtocol'], ['sgwlpAddr'], ['sgwlpv6Addr'], ['originalApn'], ['mappedApn'], ['subscriberId'], ['resellerId'], ['timezone']]

Case : uli

User Location Info

Include User Location Information in the header

Choice: This user-location-info attribute is mutually exclusive with attributes [['containerValue'], ['ratType'], ['chargingCharacteristic'], ['servingNetworkPlmnid'], ['msisdn'], ['imsi'], ['imei'], ['httpProtocol'], ['sgwlpAddr'], ['sgwlpv6Addr'], ['originalApn'], ['mappedApn'], ['subscriberId'], ['resellerId'], ['timezone']]

XPath: *cna-ppe:/upf/service-construct/tdef/enrichment-profile/headerString/uli*

CLI Path: *upf service-construct tdef enrichment-profile header-name user-location-info*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
				Include Cell Global Information in the header
cgi	empty			Choice: This cgi attribute is mutually exclusive with attributes [['containerValue'], ['ratType'], ['chargingCharacteristic'], ['servingNetworkPlmnid'], ['msisdn'], ['imsi'], ['imei'], ['httpProtocol'], ['sgwlpAddr'], ['sgwlpv6Addr'], ['originalApn'], ['mappedApn'], ['subscriberId'], ['resellerId'], ['timezone']]]

Case : msisdn

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
				Include MSISDN in the header
msisdn	empty			Choice: This msisdn attribute is mutually exclusive with attributes [['containerValue'], ['ratType'], ['chargingCharacteristic'], ['servingNetworkPlmnid'], ['uli'], ['imsi'], ['imei'], ['httpProtocol'], ['sgwlpAddr'], ['sgwlpv6Addr'], ['originalApn'], ['mappedApn'], ['subscriberId'], ['resellerId'], ['timezone']]]

Case : imsi

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
				Include IMSI in the header
imsi	empty			Choice: This imsi attribute is mutually exclusive with attributes [['containerValue'], ['ratType'], ['chargingCharacteristic'], ['servingNetworkPlmnid'], ['uli'], ['msisdn'], ['imei'], ['httpProtocol'], ['sgwlpAddr'], ['sgwlpv6Addr'], ['originalApn'], ['mappedApn'], ['subscriberId'], ['resellerId'], ['timezone']]]

Case : imei

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
				Include IMEI in the header
imei	empty			Choice: This imei attribute is mutually exclusive with attributes [['containerValue'], ['ratType'], ['chargingCharacteristic'], ['servingNetworkPlmnid'], ['uli'], ['msisdn'], ['imsi'], ['httpProtocol'], ['sgwlpAddr'], ['sgwlpv6Addr'], ['originalApn'], ['mappedApn'], ['subscriberId'], ['resellerId'], ['timezone']]]

Case : httpProtocol

Case : sgwlpAddr

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
				Include Serving Gateway IPv4 Address in the header
serving-gw-ip-addr	empty			Choice: This serving-gw-ip-addr attribute is mutually exclusive with attributes [['containerValue'], ['ratType'], ['chargingCharacteristic'], ['servingNetworkPlmnid'], ['uli'], ['msisdn'], ['imsi'], ['imei'], ['httpProtocol'], ['sgwlpv6Addr'], ['originalApn'], ['mappedApn'], ['subscriberId'], ['resellerId'], ['timezone']]]

Case : sgwlpv6Addr

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
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Attribute	Type	Range/Length/Pattern	Default	Description
serving-gw-ipv6-addr	empty			Include Serving Gateway IPv6 Address in the header Choice: This <code>serving-gw-ipv6-addr</code> attribute is mutually exclusive with attributes <code>['containerValue']</code>, <code>['ratType']</code>, <code>['chargingCharacteristic']</code>, <code>['servingNetworkPlmnid']</code>, <code>['uli']</code>, <code>['msisdn']</code>, <code>['imsi']</code>, <code>['imei']</code>, <code>['httpProtocol']</code>, <code>['sgwlpAddr']</code>, <code>['originalApn']</code>, <code>['mappedApn']</code>, <code>['subscriberId']</code>, <code>['resellerId']</code>, <code>['timezone']</code>

Case : originalApn

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
original-apn-name	empty			Include original APN name in the header Choice: This <code>original-apn-name</code> attribute is mutually exclusive with attributes <code>['containerValue']</code>, <code>['ratType']</code>, <code>['chargingCharacteristic']</code>, <code>['servingNetworkPlmnid']</code>, <code>['uli']</code>, <code>['msisdn']</code>, <code>['imsi']</code>, <code>['imei']</code>, <code>['httpProtocol']</code>, <code>['sgwlpAddr']</code>, <code>['sgwlpv6Addr']</code>, <code>['mappedApn']</code>, <code>['subscriberId']</code>, <code>['resellerId']</code>, <code>['timezone']</code>

Case : mappedApn

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
mapped-apn-name	empty			Include mapped APN name in the header Choice: This <code>mapped-apn-name</code> attribute is mutually exclusive with attributes <code>['containerValue']</code>, <code>['ratType']</code>, <code>['chargingCharacteristic']</code>, <code>['servingNetworkPlmnid']</code>, <code>['uli']</code>, <code>['msisdn']</code>, <code>['imsi']</code>, <code>['imei']</code>, <code>['httpProtocol']</code>, <code>['sgwlpAddr']</code>, <code>['sgwlpv6Addr']</code>, <code>['originalApn']</code>, <code>['subscriberId']</code>, <code>['resellerId']</code>, <code>['timezone']</code>

Case : subscriberId

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
subscriber-id	empty			Include subscriber id in the header Choice: This <code>subscriber-id</code> attribute is mutually exclusive with attributes <code>['containerValue']</code>, <code>['ratType']</code>, <code>['chargingCharacteristic']</code>, <code>['servingNetworkPlmnid']</code>, <code>['uli']</code>, <code>['msisdn']</code>, <code>['imsi']</code>, <code>['imei']</code>, <code>['httpProtocol']</code>, <code>['sgwlpAddr']</code>, <code>['sgwlpv6Addr']</code>, <code>['originalApn']</code>, <code>['mappedApn']</code>, <code>['resellerId']</code>, <code>['timezone']</code>

Case : resellerId

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
reseller-id	empty			Include reseller ID in the header Choice: This <code>reseller-id</code> attribute is mutually exclusive with attributes <code>['containerValue']</code>, <code>['ratType']</code>, <code>['chargingCharacteristic']</code>, <code>['servingNetworkPlmnid']</code>, <code>['uli']</code>, <code>['msisdn']</code>, <code>['imsi']</code>, <code>['imei']</code>, <code>['httpProtocol']</code>, <code>['sgwlpAddr']</code>, <code>['sgwlpv6Addr']</code>, <code>['originalApn']</code>, <code>['mappedApn']</code>, <code>['subscriberId']</code>, <code>['timezone']</code>

Case : timezone

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
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Attribute	Type	Range/Length/Pattern	Default	Description
				Include time zone in the header
time-zone	empty			Choice: This time-zone attribute is mutually exclusive with attributes [['containerValue'], ['ratType'], ['chargingCharacteristic'], ['servingNetworkPlmnid'], ['uli'], ['msisdn'], ['imsi'], ['imei'], ['httpProtocol'], ['sgwlpAddr'], ['sgwlpv6Addr'], ['originalApn'], ['mappedApn'], ['subscriberId'], ['resellerId']]]

Encryption Profile

Configure Content Provider Encryption Profile

XPath: *cna-ppe:/upf/service-construct/tdef/encryption-profile*

CLI Path: *upf service-construct tdef encryption-profile*

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			Encryption Profile Name - string
index	uint32	1 .. max		Unique Index
admin-state	enumeration	disabled enabled	enabled	Enable/Disable encryption profile
encoding	enumeration	base64 hex	hex	Select an encoding format for encrypted information

EncryptionAlgorithm

Algorithm used for encryption

XPath: *cna-ppe:/upf/service-construct/tdef/encryption-profile/encryptionAlgorithm*

CLI Path: *upf service-construct tdef encryption-profile encryptionAlgorithm*

EncryptionAlgorithmchoice (choice)

XPath: *cna-ppe:/upf/service-construct/tdef/encryption-profile/encryptionAlgorithm*

***CLI Path: *upf service-construct tdef encryption-profile encryptionAlgorithm* ***

Case : AES128

Aes 128 Cbc

Encrypted by Advanced Encryption Standard 128-bit Cipher Block Chaining

Choice: This aes-128-cbc attribute is mutually exclusive with attributes [['AES192'], ['AES256']]

XPath: *cna-ppe:/upf/service-construct/tdef/encryption-profile/encryptionAlgorithm/AES128*

CLI Path: *upf service-construct tdef encryption-profile encryptionAlgorithm aes-128-cbc*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
key	string			Key for AES encryption, please input HEX format without space
				Choice: This key attribute is mutually exclusive with attributes [['AES192'], ['AES256']]
iv	string			IV for AES128 encryption, please input HEX format without space
				Choice: This iv attribute is mutually exclusive with attributes [['AES192'], ['AES256']]

Case : AES192

Aes 192 Cbc

Encrypted by Advanced Encryption Standard 192-bit Cipher Block Chaining

Choice: This aes-192-cbc attribute is mutually exclusive with attributes [['AES128'], ['AES256']]

XPath: *cna-ppe:/upf/service-construct/tdef/encryption-profile/encryptionAlgorithm/AES192*

CLI Path: *upf service-construct tdef encryption-profile encryptionAlgorithm aes-192-cbc*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
key	string			Key for AES encryption, please input HEX format without space
				Choice: This key attribute is mutually exclusive with attributes [['AES128'], ['AES256']]
iv	string			IV for AES192 encryption, please input HEX format without space
				Choice: This iv attribute is mutually exclusive with attributes [['AES128'], ['AES256']]

Case : AES256

Aes 256 Cbc

Encrypted by Advanced Encryption Standard 256-bit Cipher Block Chaining

Choice: This aes-256-cbc attribute is mutually exclusive with attributes [['AES128'], ['AES192']]

XPath: *cna-ppe:/upf/service-construct/tdef/encryption-profile/encryptionAlgorithm/AES256*

CLI Path: *upf service-construct tdef encryption-profile encryptionAlgorithm aes-256-cbc*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
key	string			Key for AES encryption, please input HEX format without space
				Choice: This key attribute is mutually exclusive with attributes [['AES128'], ['AES192']]
iv	string			IV for AES256 encryption, please input HEX format without space
				Choice: This iv attribute is mutually exclusive with attributes [['AES128'], ['AES192']]

Tcp Optimization Profile

Configure TCP optimization profile

XPath: *cna-ppe:/upf/service-construct/tdef/tcp-optimization-profile*

CLI Path: *upf service-construct tdef tcp-optimization-profile*

List Properties:

Keys: name

Maximum number of elements: 8192

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			TCP Optimization Profile Name - string
profile-id	uint32	1..8192		TCP Optimization Profile Unique ID
mss-v4	uint32	88..1500	1460	Configure IPv4 TCP maximum segment size in bytes
mss-v6	uint32	88..1500	1440	Configure IPv6 TCP maximum segment size in bytes
fast-retransmit-dupacks	uint32	1..255	3	Configure TCP number of duplicate acks that will trigger fast retransmit

Attribute	Type	Range/Length/Pattern	Default	Description
congestion-control	enumeration	cubic vegas reno htcp cdg chd dctcp hd bbr	reno	Configure TCP congestion control method
send-buffer-size	uint32	4096 .. 33554432		Configure TCP send buffer size in bytes
recv-buffer-size	uint32	4096 .. 33554432		Configure TCP receive buffer size in bytes
initial-congestion-window	uint32	1..16000	10	Configure TCP initial congestion window in units of mss
slow-start	enumeration	enabled disabled	disabled	Enable/Disable TCP slow start after idle option
nagle-algorithm	enumeration	enabled disabled	disabled	Enable/Disable TCP Nagle algorithm
buffer-auto-tuning	enumeration	enabled disabled	enabled	Enable/Disable TCP buffer auto-tuning
min-congestion-window	uint32	1..16000	1	Configure Minimum value Congestion Window can drop to in units of mss
min-ssthresh	uint32	0..32000000	0	Configure Minimum value Slow-start threshold can drop to in units of bytes
intelligent-buffering	enumeration	enabled disabled	disabled	Configure intelligent server-side buffering Must enable recv-buffer-auto-tuning first
syn-passthrough	enumeration	enabled disabled	disabled	Enable/Disable SYN passthrough option

Http Header Enrichment

Configure HTTP Header Enrichment

XPath: [cna-ppe:/upf/service-construct/tdef/http-header-enrichment](#)

CLI Path: [upf service-construct tdef http-header-enrichment](#)

List Properties:

Keys: name

Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			Header Enrichment List Name - string

Request

Http Header Enrichment supports for request packets

XPath: [cna-ppe:/upf/service-construct/tdef/http-header-enrichment/request](#)

CLI Path: [upf service-construct tdef http-header-enrichment request](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
encryption-profile	leafref			Encryption Profile Leaf Reference Path: cna-ppe:/upf/service-construct/tdef/encryption-profile/name

Add Header

Add a new HTTP header field name string

XPath: *cna-ppe:/upf/service-construct/tdef/http-header-enrichment/request/add-header*

CLI Path: *upf service-construct tdef http-header-enrichment request add-header*

List Properties:

Keys: header-string

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
header-string	string	1..64		
encrypted	enumeration	enabled disabled	disabled	Enable/Disable Encryption
value	string	1..255		Value to include for the header.

Remove Header

Remove a HTTP header field name string

XPath: *cna-ppe:/upf/service-construct/tdef/http-header-enrichment/request/remove-header*

CLI Path: *upf service-construct tdef http-header-enrichment request remove-header*

List Properties:

Keys: header-string

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
header-string	string	1..64		

Rewrite Header

Rewrite a HTTP header field name string

XPath: *cna-ppe:/upf/service-construct/tdef/http-header-enrichment/request/rewrite-header*

CLI Path: *upf service-construct tdef http-header-enrichment request rewrite-header*

List Properties:

Keys: header-string

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
header-string	string	1..64		

Rewrite Option (choice)

XPath: *cna-ppe:/upf/service-construct/tdef/http-header-enrichment/request/rewrite-header*

CLI Path: *upf service-construct tdef http-header-enrichment request rewrite-header*

Case : replace-header

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
replace-header	string	1..64		Replace a HTTP header field Choice: This replace-header attribute is mutually exclusive with attributes [['replace-value'], ['append-value']]

Case : replace-value

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
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Attribute	Type	Range/Length/Pattern	Default	Description
replace-value	string	1 .. 255		Replace header value with a new value Choice: This replace-value attribute is mutually exclusive with attributes [['replace-header'], ['append-value']]

Case : append-value

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
append-value	string	1..255		Append a value to an existed header Choice: This append-value attribute is mutually exclusive with attributes [['replace-header'], ['replace-value']]

Response

Http Header Enrichment supports for response packets

XPath: [cna-ppe:/upf/service-construct/tdef/http-header-enrichment/response](#)

CLI Path: [upf service-construct tdef http-header-enrichment response](#)

Add Header

Add a new HTTP header field name string

XPath: [cna-ppe:/upf/service-construct/tdef/http-header-enrichment/response/add-header](#)

CLI Path: [upf service-construct tdef http-header-enrichment response add-header](#)

List Properties:

Keys: header-string

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
header-string	string	1..64		
value	string	1..255		Value to include for the header.

Remove Header

Remove a HTTP header field name string

XPath: [cna-ppe:/upf/service-construct/tdef/http-header-enrichment/response/remove-header](#)

CLI Path: [upf service-construct tdef http-header-enrichment response remove-header](#)

List Properties:

Keys: header-string

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
header-string	string	1..64		

Rewrite Header

Rewrite a HTTP header field name string

XPath: [cna-ppe:/upf/service-construct/tdef/http-header-enrichment/response/rewrite-header](#)

CLI Path: [upf service-construct tdef http-header-enrichment response rewrite-header](#)

List Properties:

Keys: header-string

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
header-string	string	1..64		

Rewrite Option (choice)

XPath: *cna-ppe:/upf/service-construct/tdef/http-header-enrichment/response/rewrite-header*

CLI Path: *upf service-construct tdef http-header-enrichment response rewrite-header*

Case : replace-header

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
replace-header	string	1..64		Replace a HTTP header field Choice: This replace-header attribute is mutually exclusive with attributes [['replace-value'], ['append-value']]

Case : replace-value

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
replace-value	string	1 .. 255		Replace header value with a new value Choice: This replace-value attribute is mutually exclusive with attributes [['replace-header'], ['append-value']]

Case : append-value

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
append-value	string	1..255		Append a value to an existed header Choice: This append-value attribute is mutually exclusive with attributes [['replace-header'], ['replace-value']]

Http Url Rewrite Profile

Configure HTTP URL Rewrite Profile

XPath: *cna-ppe:/upf/service-construct/tdef/http-url-rewrite-profile*

CLI Path: *upf service-construct tdef http-url-rewrite-profile*

List Properties:

Keys: name

Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			Url Rewrite List Name - string

Rewrite Rule

XPath: *cna-ppe:/upf/service-construct/tdef/http-url-rewrite-profile/rewrite-rule*

CLI Path: *upf service-construct tdef http-url-rewrite-profile rewrite-rule*

List Properties:

Keys: name

Minimum number of elements: 1

Maximum number of elements: 512

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			Rule name
priority	uint32	1 .. 4096		Priority
request-url	string			Configure regular expression pattern to match request URL
substitution-url	string			Configure substitution URL to replace matched request URL

Load Balancer

Configure Load Balancer

XPath: *cna-ppe:/upf/service-construct/tdef/load-balancer*

CLI Path: *upf service-construct tdef load-balancer*

List Properties:

Keys: name
Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			Load Balancer group name - string

Server

XPath: *cna-ppe:/upf/service-construct/tdef/load-balancer/server*

CLI Path: *upf service-construct tdef load-balancer server*

List Properties:

Keys: name
Minimum number of elements: 1
Maximum number of elements: 512

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			Server name
admin-state	enumeration	disabled enabled	enabled	admin state: enable/disable profile
port	uint32	1 .. 65535		Configure Server port
weight	uint32	1 .. 255	1	Configure Server weight
backup	enumeration	false true	false	Configure Server as a backup

Destination (choice)

XPath: *cna-ppe:/upf/service-construct/tdef/load-balancer/server*

***CLI Path: *upf service-construct tdef load-balancer server* ***

Case : ip-address

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
ip-address	inet:ip-address			Configure Server IP address Choice: This ip-address attribute is mutually exclusive with attributes [['host-name']]

Case : host-name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
host-name	string			Configure Server Host name
Choice: This host-name attribute is mutually exclusive with attributes [['ip-address']]				

Tdef Tls Termination Policy

TLS termination policy

XPath: cna-ppe:/upf/service-construct/tdef/tdef-tls-termination-policy

CLI Path: upf service-construct tdef tdef-tls-termination-policy

List Properties:
Keys: name
Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			

Site

TLS virtual server name list

XPath: cna-ppe:/upf/service-construct/tdef/tdef-tls-termination-policy/site

CLI Path: upf service-construct tdef tdef-tls-termination-policy site

List Properties:
Keys: name
Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			TLS virtual server name
certificate	string	1 .. 255		TLS server certificate file name
key	string	1 .. 255		TLS server key file name
key-passphrase	string			TLS server key file passphrase

Ca Certificate List

Configure CA certificate List

XPath: cna-ppe:/upf/service-construct/tdef/tdef-ca-certificate-list

CLI Path: upf service-construct tdef ca-certificate-list

List Properties:
Keys: name
Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			CA certificate List Name - string

Ca Certificate

Configure CA certificate File

XPath: cna-ppe:/upf/service-construct/tdef/tdef-ca-certificate-list/ca-certificate

CLI Path: upf service-construct tdef ca-certificate-list ca-certificate

List Properties:
Keys: name

Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			CA certificate filename - string

Tdef Https Profile

HTTPS service profile

XPath: [cna-ppe:/upf/service-construct/tdef/tdef-https-profile](#)

CLI Path: [upf service-construct tdef tdef-https-profile](#)

List Properties:

Keys: name

Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			
admin-state	enumeration	disabled enabled	enabled	admin state: enable/disable profile

Tls Termination Policy

Name of TLS termination policy

XPath: [cna-ppe:/upf/service-construct/tdef/tdef-https-profile/tls-termination-policy](#)

CLI Path: [upf service-construct tdef tdef-https-profile tls-termination-policy](#)

List Properties:

Keys: name

Maximum number of elements: 1

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: cna-ppe:/upf/service-construct/tdef/tdef-tls-termination-policy/name

Ca Certificate List

Name of CA certificate List

XPath: [cna-ppe:/upf/service-construct/tdef/tdef-https-profile/ca-certificate-list](#)

CLI Path: [upf service-construct tdef tdef-https-profile ca-certificate-list](#)

List Properties:

Keys: name

Maximum number of elements: 1

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: cna-ppe:/upf/service-construct/tdef/tdef-ca-certificate-list/name

Tdef Proxy Policy

Configure TDEF Proxy Policy

XPath: [cna-ppe:/upf/service-construct/tdef/tdef-proxy-policy](#)

CLI Path: [upf service-construct tdef tdef-proxy-policy](#)

List Properties:

Keys: name
Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			TDEF Proxy Policy - Name
priority	uint32	1 .. 4096		Priority
protocol	enumeration	http https tcp udp terminated-https	http	Service Protocol (http/https/tcp/udp)
proxy-bypass	enumeration	disabled enabled	disabled	Enable/disable proxy bypass
proxy-bypass-on-overload	enumeration	disabled enabled	disabled	Enable/disable bypass when proxy overload-control is in place
dns-resolution	enumeration	disabled enabled	disabled	DNS resolution to be used for lookup destination
compress-force	enumeration	disabled enabled	disabled	Force on-the-fly HTTP compression for all client requests
compress-inline	enumeration	disabled enabled	disabled	Enable on-the-fly HTTP compression for supported client requests
compress-min-size	uint32		512	The minimum response size in bytes required for compression to take effect compress-min-size must be lesser than compress-min-size
compress-max-size	uint32		2000000	The maximum response size in bytes allowed for compression to take effect compress-max-size must be greater than compress-min-size
http-header-enrichment	leafref			Configure the Header Enrichment Profile to use for HTTP packets Leaf Reference Path: cna-ppe:/upf/service-construct/tdef/http-header-enrichment/name
http-url-rewrite-profile	leafref			Configure the URL Rewrite Profile to use for HTTP packets Leaf Reference Path: cna-ppe:/upf/service-construct/tdef/http-url-rewrite-profile/name
load-balancer	leafref			Configure the load-balancer Leaf Reference Path: cna-ppe:/upf/service-construct/tdef/load-balancer/name
https-profile	leafref			HTTPS profile asociated with this policy Leaf Reference Path: cna-ppe:/upf/service-construct/tdef/tdef-https-profile/name
web-api	enumeration	api-ship redirect-ship simple-entitlement secure-entitlement		Configure the Web-API termination script and protocol

Tdef Video Opt Policy

Configure TDEF Video-Optimization Policy

XPath: [cna-ppe:/upf/service-construct/tdef/tdef-video-opt-policy](#)

CLI Path: [upf service-construct tdef tdef-video-opt-policy](#)

List Properties:

Keys: name
Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
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Attribute	Type	Range/Length/Pattern	Default	Description
name	string			TDEF Video Optimization Policy - Name

Shaping

XPath: *cna-ppe:/upf/service-construct/tdef/tdef-video-opt-policy/shaping*

CLI Path: *upf service-construct tdef tdef-video-opt-policy shaping*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
admin-state	enumeration	disabled enabled	enabled	Enable/Disable TDEF Video-Optimization Policy
send-interval	uint32	40 .. 500	100	Configure Shaping send-interval in seconds
udp-congestion-control	enumeration	head-drop tail-drop	head-drop	Configure Udp congestion control
flow-aggregation	enumeration	disabled enabled	disabled	Configure shaping flow-aggregation
http-segment-threshold	uint32	1 .. 10000000	200000	Configure Http Segment Threshold in bytes
threshold-bytes	uint32	1 .. 20000000	750000	Configure Threshold in bytes for shaping
threshold-duration	uint32	1 .. 90	2	Configure Threshold Duration in seconds for shaping
upstream-buffering-limit	uint32	64 .. 32000	8000	Configure the maximum data to store when processing upstream data, units KB
downstream-buffering-limit	uint32	64 .. 32000	8000	Configure the maximum data to store when processing downstream data, units KB

Burst

burst-size must be less than or equal to 2 gbytes

XPath: *cna-ppe:/upf/service-construct/tdef/tdef-video-opt-policy/shaping/burst*

CLI Path: *upf service-construct tdef tdef-video-opt-policy shaping burst*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
initial-burst-size	uint32	0 .. 2000000000	100	Configure Shaping Initial Burst Size
initial-burst-size-unit	enumeration	bytes kbytes mbytes gbytes	kbytes	burst-size-units: bytes/kbytes/mbytes/gbytes

Bitrate

bitrate must be greater than or equal to 128 kbps and less than or equal to 2 gbps

XPath: *cna-ppe:/upf/service-construct/tdef/tdef-video-opt-policy/shaping/bitrate*

CLI Path: *upf service-construct tdef tdef-video-opt-policy shaping bitrate*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
bitrate	uint32	1 .. 2000000000	1500	Configure Shaping Bitrate
bitrate-unit	enumeration	bps kbps mbps gbps	kbps	bitrate-units: bps/kbps/mbps/gbps

Dns Client Profile

Configure TDEF DNS Client Profile

XPath: *cna-ppe:/upf/service-construct/tdef/dns-client-profile*

CLI Path: *upf service-construct tdef dns-client-profile*

List Properties:

Keys: name

Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			TDEF DNS Client Profile - Name
profile-id	uint32	1 .. max		Unique ID
query-attempt	uint32	1 .. 5	2	The number of times the resolver will send a query to servers before giving up and returning an error
query-timeout	uint32	1 .. 10	5	The amount of time in seconds the resolver will wait for a response from a server before retrying the query via a different server
use-extension-dns	enumeration	false true	false	Use extension mechanisms for DNS
edns0-udp-packet-size	uint32	512 .. 4096	1280	UDP packet size to use for EDNS0 (default is 1280)
transport-protocol	enumeration	udp-with-tcp-fallback tcp-only	udp-with-tcp-fallback	Configure transport layer protocol used to connect to dns server

Attribute	Type	Range/Length/Pattern	Default	Description
dscp-value	enumeration	dscp_0_be	dscp_0_be	DiffServ Code Point value with which to mark DNS IP packets
		dscp_01		
		dscp_02		
		dscp_03		
		dscp_04		
		dscp_05		
		dscp_06		
		dscp_07		
		dscp_08		
		dscp_09		
		dscp_10_af11		
		dscp_11		
		dscp_12_af12		
		dscp_13		
		dscp_14_af13		
		dscp_15		
		dscp_16		
		dscp_17		
		dscp_18_af21		
		dscp_19		
		dscp_20_af22		
		dscp_21		
		dscp_22_af23		
		dscp_23		
		dscp_24		
		dscp_25		
		dscp_26_af31		
		dscp_27		
		dscp_28_af32		
		dscp_29		
		dscp_30_af33		
		dscp_31		
		dscp_32		
		dscp_33		
		dscp_34_af41		
		dscp_35		
		dscp_36_af42		
		dscp_37		
		dscp_38_af43		
		dscp_39		
		dscp_40		
		dscp_41		
		dscp_42		
		dscp_43		
		dscp_44_va		
		dscp_45		
		dscp_46_ef		
		dscp_47		
		dscp_48		
		dscp_49		
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		dscp_61		
		dscp_62		
		dscp_63		

Attribute	Type	Range/Length/Pattern	Default	Description
dns-cache-aging-seconds	uint32		0	DNS cache stale record aging time in seconds - 0 disables aging

Server Address

Configuration of dns server IP addresses

XPath: *cna-ppe:/upf/service-construct/tdef/dns-client-profile/dnsServer*

CLI Path: *upf service-construct tdef dns-client-profile server-address*

List Properties:

Keys: name

Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
ip-address	inet:ip-address			IPv4 or IPv6 DNS server IP address
udp-port	uint32	1 .. 65535	53	Configure dns server udp port, must be same for all servers within the network-context
tcp-port	uint32	1 .. 65535	53	Configure dns server tcp port, must be same for all servers within the network-context

Network Context

Configuration of network context

XPath: *cna-ppe:/upf/service-construct/tdef/dns-client-profile/netContext*

CLI Path: *upf service-construct tdef dns-client-profile network-context*

List Properties:

Keys: name

Maximum number of elements: 1

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
network-context	string			Network context to be used

Loopback Ipv4

Configuration of ipv4 loopback IP address

XPath: *cna-ppe:/upf/service-construct/tdef/dns-client-profile/loopbackIpv4*

CLI Path: *upf service-construct tdef dns-client-profile loopback-ipv4*

List Properties:

Keys: name

Maximum number of elements: 1

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
loopback	string			IPv4 IPv4 loopback to be used as local IP address

Loopback Ipv6

Configuration of ipv6 loopback IP address

XPath: *cna-ppe:/upf/service-construct/tdef/dns-client-profile/loopbackIpv6*

CLI Path: *upf service-construct tdef dns-client-profile loopback-ipv6*

List Properties:

Keys: name

Maximum number of elements: 1

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
loopback	string			IPv4 IPv6 loopback to be used as local IP address

Explicit Proxy Profile

Configure Explicit proxy address profile

XPath: *cna-ppe:/upf/service-construct/tdef/explicit-proxy-profile*

CLI Path: *upf service-construct tdef explicit-proxy-profile*

List Properties:

Keys: name
Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			Content Explicit Proxy profile name - string

Explicit Proxy Networkcontext

Configure Explicit proxy network context and address list

XPath: *cna-ppe:/upf/service-construct/tdef/explicit-proxy-profile/explicitProxyNetworkContext*

CLI Path: *upf service-construct tdef explicit-proxy-profile explicit-proxy-networkcontext*

List Properties:

Keys: name
Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
network-context	string			Configure network-context to be used for explicit proxy address list

Explicit Proxy Address

Configure Explicit proxy address

XPath: *cna-ppe:/upf/service-construct/tdef/explicit-proxy-profile/explicitProxyNetworkContext/explicitProxyAddress*

CLI Path: *upf service-construct tdef explicit-proxy-profile explicit-proxy-networkcontext explicit-proxy-address*

List Properties:

Keys: name
Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			Content Explicit Proxy Address - string

Address

Configure Explicit Proxy address

XPath: *cna-ppe:/upf/service-construct/tdef/explicit-proxy-profile/explicitProxyNetworkContext/explicitProxyAddress/address*

CLI Path: *upf service-construct tdef explicit-proxy-profile explicit-proxy-networkcontext explicit-proxy-address address*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
address	inet:ip-address			Configure Explicit Proxy IP address
port	uint32	0 .. 65535		Configure Explicit Proxy IP port

Dpi Rule List

Dpi-rule-list: Match any dpi rule to divert packet to DPI
Defines dpi-rule-list for TVS video-opt-rule-group

XPath: *cna-ppe:/upf/service-construct/tdef/dpi-rule-list*

CLI Path: *upf service-construct tdef dpi-rule-list*

List Properties:

Keys: name
Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			DPI rule list Name - string

Rule

DPI rule list: match on all of the fields that are specified

XPath: *cna-ppe:/upf/service-construct/tdef/dpi-rule-list/rule*

CLI Path: *upf service-construct tdef dpi-rule-list rule*

List Properties:

Keys: name
Minimum number of elements: 1
Maximum number of elements: 512

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			Rule name
protocol	enumeration	tcp udp both	both	Apply TVS on matching protocol (tcp/udp/both)
port	uint32	1 .. 65535		Apply TVS on matching port
priority	uint32	1 .. 4096		Priority

Edr Profile

Configure TDEF EDR Profile

XPath: *cna-ppe:/upf/service-construct/tdef/edr-profile*

CLI Path: *upf service-construct tdef edr-profile*

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			EDR Profile Name - string
time-info	enumeration	disabled enabled	disabled	Enable/Disable time info
subscriber-info	enumeration	disabled enabled	disabled	Enable/Disable subscriber info
connection-info	enumeration	disabled enabled	disabled	Enable/Disable connection info
http-request-info	enumeration	disabled enabled	disabled	Enable/Disable HTTP request info
http-response-info	enumeration	disabled enabled	disabled	Enable/Disable HTTP response info

Attribute	Type	Range/Length/Pattern	Default	Description
compression-info	enumeration	disabled enabled	disabled	Enable/Disable HTTP compression info
video-info	enumeration	disabled enabled	disabled	Enable/Disable video info
tcp-splicing-info	enumeration	disabled enabled	disabled	Enable/Disable TCP splicing info
tcp-client-connection-info	enumeration	disabled enabled	disabled	Enable/Disable TCP client connection info
tcp-server-connection-info	enumeration	disabled enabled	disabled	Enable/Disable TCP server connection info
udp-proxy-info	enumeration	disabled enabled	disabled	Enable/Disable UDP Proxy info
dpi-info	enumeration	disabled enabled	disabled	Enable/Disable DPI Info
protocol-application-summary	enumeration	disabled enabled	disabled	Enable/Disable Protocol Application Summary
protocol-application-summary-log-interval	uint32	1 .. 3600	5	Configure the Protocol Application Summary log interval in seconds

Proxy Rule Group

Configure TDEF Proxy Rule Group

XPath: [cna-ppe:/upf/service-construct/tdef/proxy-rule-group](#)

CLI Path: [upf service-construct tdef proxy-rule-group](#)

List Properties:

Keys: name

Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			Content TDEF Proxy Rule Group Name - string
admin-state	enumeration	disabled enabled	enabled	Enable/Disable TDEF Proxy Rule Group
priority	uint32	1 .. 4096		priority - unique
dns-client-profile	leafref			Configure the DNS Client Profile to use Leaf Reference Path: cna-ppe:/upf/service-construct/tdef/dns-client-profile/name
explicit-proxy-profile	leafref			Configure Explicit Proxy Profile to use Leaf Reference Path: cna-ppe:/upf/service-construct/tdef/explicit-proxy-profile/name
edr-profile	leafref			Configure EDR Profile to use Leaf Reference Path: cna-ppe:/upf/service-construct/tdef/edr-profile/name

Pdr

List of PDR's for proxy-rule-group

XPath: [cna-ppe:/upf/service-construct/tdef/proxy-rule-group/pdr](#)

CLI Path: [upf service-construct tdef proxy-rule-group pdr](#)

List Properties:

Keys: name

Minimum number of elements: 0

Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: cna-ppe:/upf/service-construct/pdr/name

Loopback Pool

Configure loopback pools for the network-context

XPath: [cna-ppe:/upf/service-construct/tdef/proxy-rule-group/loopback-pool](#)

CLI Path: [upf service-construct tdef proxy-rule-group loopback-pool](#)

List Properties:

Keys: name
Minimum number of elements: 0
Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			

Video Opt Rule Group

Configure Video-Optimization rule-group

XPath: [cna-ppe:/upf/service-construct/tdef/video-opt-rule-group](#)

CLI Path: [upf service-construct tdef video-opt-rule-group](#)

List Properties:

Keys: name
Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			Video-Optimization rule-group Name - string
admin-state	enumeration	disabled enabled	enabled	Enable/Disable Video-Optimization rule-group
opting-mode	enumeration	opt_in opt_out	opt_out	ABR Opting mode Opt_in/ Opt_out
edr-profile	leafref			Configure EDR Profile to use Leaf Reference Path: cna-ppe:/upf/service-construct/tdef/edr-profile/name
tvsv	enumeration	disabled enabled	disabled	Enable/Disable transparent-video-shaping(tvs) for video-opt-rule-group

Pdr

List of Pdr's for video-opt-rule-group

XPath: [cna-ppe:/upf/service-construct/tdef/video-opt-rule-group/pdr](#)

CLI Path: [upf service-construct tdef video-opt-rule-group pdr](#)

List Properties:

Keys: name
Minimum number of elements: 0
Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: cna-ppe:/upf/service-construct/pdr/name

Dpi Rule List

Configure dpi-rule-list

XPath: *cna-ppe:/upf/service-construct/tdef/video-opt-rule-group/dpi-rule-list*

CLI Path: *upf service-construct tdef video-opt-rule-group dpi-rule-list*

List Properties:

Keys: name
Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: <i>cna-ppe:/upf/service-construct/tdef/dpi-rule-list/name</i>

Entitlement Profile

Configure Content Provider Entitlement Profile

XPath: *cna-ppe:/upf/service-construct/tdef/entitlement-profile*

CLI Path: *upf service-construct tdef entitlement-profile*

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			Entitlement Profile Name - string
uri	string	(/.*)		URI of the service (the uri should always begin with a forward slash)
vsa	string			Entitlement Vendor Specific Attribute Value

Subscriber Tracing Profile

Configure Subscriber Tracing Profile

XPath: *cna-ppe:/upf/service-construct/tdef/subscriber-tracing-profile*

CLI Path: *upf service-construct tdef subscriber-tracing-profile*

List Properties:

Keys: name
Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			Subscriber Tracing Profile Name - string
default-event-logging	enumeration	none failure notice success trace dump	none	Configure default event logging level

Subscriber

XPath: *cna-ppe:/upf/service-construct/tdef/subscriber-tracing-profile/subscriber*

CLI Path: *upf service-construct tdef subscriber-tracing-profile subscriber*

List Properties:

Keys: subscriber-name
Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
subscriber-name	string			Subscriber Name - string
event-logging	enumeration	none failure notice success trace dump	none	Configure event logging level

Filter Key

XPath: *cna-ppe:/upf/service-construct/tdef/subscriber-tracing-profile/subscriber/filter-key*

CLI Path: *upf service-construct tdef subscriber-tracing-profile subscriber filter-key*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
supi	uint64			Configure SUPI value

Packet Capture

XPath: *cna-ppe:/upf/service-construct/tdef/subscriber-tracing-profile/subscriber/packet-capture*

CLI Path: *upf service-construct tdef subscriber-tracing-profile subscriber packet-capture*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
interface	enumeration	client server both		Configure Packet Capture Interface
duration	uint64			Configure Packet Capture Duration in seconds
max-packets	uint64			Configure maximum number of packets to capture
file-name	string			Configure the filename to store captured packets

Tvs Subscriber Tracing Profile

Configure Subscriber Tracing Profile for TVS

XPath: *cna-ppe:/upf/service-construct/tdef/tvs-subscriber-tracing-profile*

CLI Path: *upf service-construct tdef tvs-subscriber-tracing-profile*

List Properties:

Keys: name

Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			TVS Subscriber Tracing Profile Name - string
default-event-logging	enumeration	none error warning info debug	none	Configure default event logging level

Subscriber

XPath: *cna-ppe:/upf/service-construct/tdef/tvs-subscriber-tracing-profile/subscriber*

CLI Path: *upf service-construct tdef tvs-subscriber-tracing-profile subscriber*

List Properties:

Keys: subscriber-name

Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
subscriber-name	string			Subscriber Name - string

Filter Key

XPath: *cna-ppe:/upf/service-construct/tdef/tvs-subscriber-tracing-profile/subscriber/filter-key*

CLI Path: *upf service-construct tdef tvs-subscriber-tracing-profile subscriber filter-key*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
supi	uint64			Configure SUPI value

Protocol Application Rule Group

Protocol Application Rule Group: match on any rule in the rule group
Defines a layer 7 protocol and application rule group

XPath: *cna-ppe:/upf/service-construct/protocol-application-rule-group*

CLI Path: *upf service-construct protocol-application-rule-group*

List Properties:

Keys: name

Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			Protocol Application Rule Group Name - string

Rule

protocol Application Rule Group Rule: match on all of the fields that are specified

XPath: *cna-ppe:/upf/service-construct/protocol-application-rule-group/rule*

CLI Path: *upf service-construct protocol-application-rule-group rule*

List Properties:

Keys: name

Minimum number of elements: 1

Maximum number of elements: 512

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			Rule name

Application Group

match the application group specified

XPath: *cna-ppe:/upf/service-construct/protocol-application-rule-group/rule/application-group*

CLI Path: *upf service-construct protocol-application-rule-group rule application-group*

List Properties:

Keys: name

Maximum number of elements: 1

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
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Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path:

Application

match the application specified

XPath: *cna-ppe:/upf/service-construct/protocol-application-rule-group/rule/application-group/application*

CLI Path: *upf service-construct protocol-application-rule-group rule application-group application*

List Properties:

Keys: name
Maximum number of elements: 1

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path:

Flow Attribute

match the flow attribute specified

XPath: *cna-ppe:/upf/service-construct/protocol-application-rule-group/rule/flow-attribute*

CLI Path: *upf service-construct protocol-application-rule-group rule flow-attribute*

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path:

Subscriber

match operating-system and the tethered attribute when specified

XPath: *cna-ppe:/upf/service-construct/protocol-application-rule-group/rule/subscriber*

CLI Path: *upf service-construct protocol-application-rule-group rule subscriber*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
operating-system	leafref			Leaf Reference Path:
tethered	enumeration	false true		

Flow

match operating-system and the tethered attribute when specified
Please specify both tdcj-lower-limit and tdcj-upper-limit. Configuring only one limit is invalid

XPath: *cna-ppe:/upf/service-construct/protocol-application-rule-group/rule/flow*

CLI Path: *upf service-construct protocol-application-rule-group rule flow*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
operating-system	leafref			Leaf Reference Path:

Attribute	Type	Range/Length/Pattern	Default	Description
tethered	enumeration	false true		
tdci-lower-limit	uint32			match tethering detection confidence index(tdci) lower limit tdci-lower-limit must be greater than 0
tdci-upper-limit	uint32			match tethering detection confidence index(tdci) upper limit For tdci configurations, flow tethered true must be configured Please specify upper limit greater than lower limit

Protocol Group

match the protocol group specified

XPath: *cna-ppe:/upf/service-construct/protocol-application-rule-group/rule/protocol-group*

CLI Path: *upf service-construct protocol-application-rule-group rule protocol-group*

List Properties:

Keys: name

Maximum number of elements: 1

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path:

Protocol

match the protocol specified

XPath: *cna-ppe:/upf/service-construct/protocol-application-rule-group/rule/protocol-group/protocol*

CLI Path: *upf service-construct protocol-application-rule-group rule protocol-group protocol*

List Properties:

Keys: name

Maximum number of elements: 1

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path:

Attribute

XPath: *cna-ppe:/upf/service-construct/protocol-application-rule-group/rule/protocol-group/protocol/attribute*

CLI Path: *upf service-construct protocol-application-rule-group rule protocol-group protocol attribute*

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path:

Pfd

packet-flow-description: match on any packet-filter AND http-rule-group AND domain-rule-group

XPath: *cna-ppe:/upf/service-construct/pfd*

CLI Path: *upf service-construct pfd*

List Properties:

Keys: name

Maximum number of elements: 4000

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			
admin-state	enumeration	disabled enabled	enabled	admin state: enabled/disable pfd
priority	uint32	1 .. 4096		Priority
service-data-flow-id	uint32	1 .. max		Mandatory numerical identifier for this pfd-rule
http-rule-group	leafref			Name of a Http-Rule-Group associated with this PFD Leaf Reference Path: cna-ppe:/upf/service-construct/http-rule-group/name
domain-rule-group	leafref			Name of a Domain-Rule-Group associated with this PFD Leaf Reference Path: cna-ppe:/upf/service-construct/domain-rule-group/name
tls-rule-group	leafref			Name of a Tls-Rule-Group associated with this PFD Leaf Reference Path: cna-ppe:/upf/service-construct/tls-rule-group/name

Attribute	Type	Range/Length/Pattern	Default	Description
dscp	enumeration	dscp_0_be		ip dscp values (e.g., dscp_02, dscp_10)
		dscp_01		
		dscp_02		
		dscp_03		
		dscp_04		
		dscp_05		
		dscp_06		
		dscp_07		
		dscp_08		
		dscp_09		
		dscp_10_af11		
		dscp_11		
		dscp_12_af12		
		dscp_13		
		dscp_14_af13		
		dscp_15		
		dscp_16		
		dscp_17		
		dscp_18_af21		
		dscp_19		
		dscp_20_af22		
		dscp_21		
		dscp_22_af23		
		dscp_23		
		dscp_24		
		dscp_25		
		dscp_26_af31		
		dscp_27		
		dscp_28_af32		
		dscp_29		
		dscp_30_af33		
		dscp_31		
		dscp_32		
		dscp_33		
		dscp_34_af41		
		dscp_35		
		dscp_36_af42		
		dscp_37		
		dscp_38_af43		
		dscp_39		
		dscp_40		
		dscp_41		
		dscp_42		
		dscp_43		
		dscp_44_va		
		dscp_45		
		dscp_46_ef		
		dscp_47		
		dscp_48		
		dscp_49		
		dscp_50		
		dscp_51		
		dscp_52		
		dscp_53		
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		dscp_55		
		dscp_56		
		dscp_57		
		dscp_58		
		dscp_59		
		dscp_60		
		dscp_61		
		dscp_62		
		dscp_63		

Attribute	Type	Range/Length/Pattern	Default	Description
protocol-application-rule-group	leafref			Name of a protocol-application-rule-group associated with this PFD Leaf Reference Path: cna-ppe:/upf/service-construct/protocol-application-rule-group/name

Packet Filter

Name of a Packet Filter associated with this PFD

XPath: [cna-ppe:/upf/service-construct/pfd/packet-filter](#)

CLI Path: [upf service-construct pfd packet-filter](#)

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: cna-ppe:/upf/service-construct/packet-filter/name

Firewall Service Policy

Configure firewall service policy

nat-pool must be specified when action is 'nat'

nat-pool must not be specified when action is not 'nat'

ipsec-natt-assist-nat-pool must be specified when alg is 'ipsecnattassist'

ipsec-natt-assist-nat-pool must not be specified when alg is not 'ipsecnattassist'

XPath: [cna-ppe:/upf/service-construct/firewall-service-policy](#)

CLI Path: [upf service-construct firewall-service-policy](#)

List Properties:

Keys: name

Maximum number of elements: 4096

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			Firewall service policy name
priority	uint32	1 .. 4096		priority - unique
nat-pool	leafref			Leaf Reference Path: cna-ppe:/upf/service-construct/network-context/nat-pool/name
ipsec-natt-assist-nat-pool	leafref			NAT 1:1 pool name. This pool will help assist with ipsec nat-traversal Leaf Reference Path: cna-ppe:/upf/service-construct/network-context/nat-pool/name
action	enumeration	nat pass drop	nat	Action to take for this subscriber-firewall flow
alg	enumeration	none tftp ftp rtsp pptp ipsecnattassist sip	none	ALG application

Flow Idle Timer Profile

Configure flow idle timer profile

XPath: [cna-ppe:/upf/service-construct/flow-idle-timer-profile](#)

CLI Path: [upf service-construct flow-idle-timer-profile](#)

List Properties:

Keys: name
Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			Flow idle timeout profile name
default-tcp-flow-idle-timeout	uint32	10 .. 3600	45	Configure tcp flow idle timeout in seconds, Warning!!! 3600 is a long timeout of about 1 hour, use it judiciously
default-udp-flow-idle-timeout	uint32	10 .. 3600	30	Configure udp flow idle timeout in seconds, Warning!!! 3600 is a long timeout of about 1 hour, use it judiciously
default-non-tcp-udp-flow-idle-timeout	uint32	10 .. 3600	45	Configure non tcp/udp flow idle timeout in seconds, Warning!!! 3600 is a long timeout of about 1 hour, use it judiciously
default-tvs-flow-idle-timeout	uint32	10 .. 3600	45	Configure tvs flow idle timeout in seconds.
default-tcp-flow-fast-age-timeout	uint32	10 .. 3600		Configure tcp flow fast age timeout in seconds.
default-udp-flow-fast-age-timeout	uint32	10 .. 3600		Configure udp flow fast age timeout in seconds.

Flow Idle Timer Rule

Configure flow idle timer rule
The total number of flow-idle-timer-rule elements must not exceed 10,000

XPath: *cna-ppe:/upf/service-construct/flow-idle-timer-profile/flow-idle-timer-rule*

CLI Path: *upf service-construct flow-idle-timer-profile flow-idle-timer-rule*

List Properties:

Keys: name
Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			Flow idle timeout rule name
tcp-flow-idle-timeout	uint32	10 .. 3600	45	Configure tcp flow idle timeout in seconds, Warning!!! 3600 is a long timeout of about 1 hour, use it judiciously
udp-flow-idle-timeout	uint32	10 .. 3600	30	Configure udp flow idle timeout in seconds, Warning!!! 3600 is a long timeout of about 1 hour, use it judiciously
non-tcp-udp-flow-idle-timeout	uint32	10 .. 3600	45	Configure non tcp/udp flow idle timeout in seconds, Warning!!! 3600 is a long timeout of about 1 hour, use it judiciously
tvs-flow-idle-timeout	uint32	10 .. 3600	45	Configure tvs flow idle timeout in seconds.
tcp-flow-fast-age-timeout	uint32	10 .. 3600		Configure tcp flow fast age timeout in seconds.
udp-flow-fast-age-timeout	uint32	10 .. 3600		Configure udp flow fast age timeout in seconds.

Idle Timeout Packet Filter

Packet Filter: match on all fields
ue-port-end must be greater than or equal to ue-port-start
ue-port-end must be greater than or equal to ue-port-start
The total number of packet-filter elements must not exceed 10,000

XPath: *cna-ppe:/upf/service-construct/flow-idle-timer-profile/flow-idle-timer-rule/packet-filter*

CLI Path: *upf service-construct flow-idle-timer-profile flow-idle-timer-rule idle-timeout-packet-filter*

List Properties:

Keys: name

Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			
ip-version	enumeration	v4_only v6_only v4_or_v6	v4_or_v6	IP version
ue-ip-v4-address-start	inet:ipv4-address		0.0.0.0	user equipment IPv4 address range start (inclusive)
ue-ip-v4-address-end	inet:ipv4-address		0.0.0.0	user equipment IPv4 address range end (inclusive)
network-ip-v4-address-start	inet:ipv4-address		0.0.0.0	network equipment IPv4 address range start (inclusive)
network-ip-v4-address-end	inet:ipv4-address		0.0.0.0	network equipment IPv4 address range end (inclusive)
ue-ip-v6-address-start	inet:ipv6-address		::	user equipment IPv6 address range start (inclusive)
ue-ip-v6-address-end	inet:ipv6-address		::	user equipment IPv6 address range end (inclusive)
network-ip-v6-address-start	inet:ipv6-address		::	network IPv6 address range start (inclusive)
network-ip-v6-address-end	inet:ipv6-address		::	network IPv6 address range end (inclusive)
ue-port-start	uint32	0 .. 65535	0	user equipment port range start value (valid for protocol = TCP (6), UDP (17), SCTP (132), only)
ue-port-end	uint32	0 .. 65535	65535	user equipment port range end value (valid for protocol = TCP (6), UDP (17), SCTP (132), only)
network-port-start	uint32	0 .. 65535	0	network equipment port range start value (valid for protocol = TCP (6), UDP (17), SCTP (132), only)
network-port-end	uint32	0 .. 65535	65535	network equipment port range end value (valid for protocol = TCP (6), UDP (17), SCTP (132), only)
protocol	uint32	0 .. 255	255	ip protocol values (e.g., all: 255, ICMP: 1, TCP: 6, UDP: 17, SCTP: 132)
initiator	enumeration	ue network both	both	match against ue or network traffic

Pdr

Packet Detection Rule

XPath: *cna-ppe:/upf/service-construct/pdr***CLI Path:** *upf service-construct pdr***List Properties:**

Keys: name

Maximum number of elements: 4000

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			Packet Detection Rule - string
pdr-id	uint32	1 .. 65535		Unique ID
priority	uint32	1 .. 4096		Priority
application-id	string			application identifier used in PCC rule to refer a pre-configured rule (this one) in UPF

Attribute	Type	Range/Length/Pattern	Default	Description
proxy-dpi-notifications	enumeration	disabled enabled	disabled	Enable/disable Proxy DPI notifications
tdef-video-opt-policy	leafref			Configure associated tdef-video-opt-policy Leaf Reference Path: cna-ppe:/upf/service-construct/tdef/tdef-video-opt-policy/name
tdef-proxy-policy	leafref			Configure associated tdef-proxy-policy Leaf Reference Path: cna-ppe:/upf/service-construct/tdef/tdef-proxy-policy/name
tdef-client-tcp-opt-profile	leafref			Configure associated tdef tcp-optimization-profile for client connections Leaf Reference Path: cna-ppe:/upf/service-construct/tdef/tcp-optimization-profile/name
tdef-server-tcp-opt-profile	leafref			Configure associated tdef tcp-optimization-profile for server connections Leaf Reference Path: cna-ppe:/upf/service-construct/tdef/tcp-optimization-profile/name
firewall-service-policy	leafref			Configure associated firewall-service-policy Leaf Reference Path: cna-ppe:/upf/service-construct/firewall-service-policy/name
flow-idle-timer-profile	leafref			Flow idle timer profile applicable for this PDR Leaf Reference Path: cna-ppe:/upf/service-construct/flow-idle-timer-profile/name

Pfd

List of PFD's for this PDR

XPath: [cna-ppe:/upf/service-construct/pdr/pfd](#)

CLI Path: [upf service-construct pdr pfd](#)

List Properties:

Keys: name

Minimum number of elements: 0

Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: cna-ppe:/upf/service-construct/pfd/name

Firewall Rule Group

Configure firewall rule group

XPath: [cna-ppe:/upf/service-construct/firewall-rule-group](#)

CLI Path: [upf service-construct firewall-rule-group](#)

List Properties:

Keys: name

Maximum number of elements: 4096

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			Firewall rule group

Pdr

List of pdr's for firewall rule group

XPath: [cna-ppe:/upf/service-construct/firewall-rule-group/pdr](#)

CLI Path: [upf service-construct firewall-rule-group pdr](#)

List Properties:

Keys: name
Minimum number of elements: 0
Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: cna-ppe:/upf/service-construct/pdr/name

Udp Tunnel Profile

Configure udp tunnel profile

XPath: [cna-ppe:/upf/service-construct/udp-tunnel-profile](#)

CLI Path: [upf service-construct udp-tunnel-profile](#)

List Properties:

Keys: name
Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			Flow idle timeout profile name
ipv4-loopback-template-name	string	1..64		Name of the ipv4 Loopback Template to be used for UDP Tunnel.
ipv6-loopback-template-name	string	1..64		Name of the ipv6 Loopback Template to be used for UDP Tunnel.
udp-port	uint32	1 .. 65535		UDP Tunnel source port.

Service Profile

Configure Service Profile

XPath: [cna-ppe:/upf/service-construct/service-profile](#)

CLI Path: [upf service-construct service-profile](#)

List Properties:

Keys: name
Maximum number of elements: 4096

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			Service Profile Name - string
tdef-proxy-rule-group	leafref			TDEF Proxy Rule Group applicable for this service-profile Leaf Reference Path: cna-ppe:/upf/service-construct/tdef/proxy-rule-group/name
firewall-rule-group	leafref			Firewall rule group applicable for this service-profile Leaf Reference Path: cna-ppe:/upf/service-construct/firewall-rule-group/name
non-blocking-urr	enumeration	false true	false	set URR to be in non-blocking mode. This flag dictates the behaviour of all URR's provisioned on the session. If 'enabled', packets will not be dropped when the quota expires instead they will be let allowed to go and counted.
flow-idle-timer-profile	leafref			Flow idle timer profile applicable for this service-profile Leaf Reference Path: cna-ppe:/upf/service-construct/flow-idle-timer-profile/name
udp-tunnel-profile	leafref			UDP Tunnel profile applicable for this service-profile Leaf Reference Path: cna-ppe:/upf/service-construct/udp-tunnel-profile/name
max-connections	uint32	1 .. 65535	1024	Enter the maximum number of allowed connections for a single session. Enter the maximum number of allowed connections for a single session. Range [1 .. 65535]

Attribute	Type	Range/Length/Pattern	Default	Description
legacy-sdfilter-format	enumeration	false true	false	Enable legacy SDFFilter format. Default is disabled If this feature flag is enabled UPF will not swap the source and destination in the SDF Filter for ACCESS PDRs

Tdef Video Rule Group

Configure TDEF Video Rule Group

XPath: *cna-ppe:/upf/service-construct/service-profile/tdef-video-rule-group*

CLI Path: *upf service-construct service-profile tdef-video-rule-group*

List Properties:

Keys: name

Minimum number of elements: 0

Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: <i>cna-ppe:/upf/service-construct/tdef/video-opt-rule-group/name</i>

Dpi Profile

tune dpi related various knobs for a subscriber

XPath: *cna-ppe:/upf/service-construct/service-profile/dpi-profile*

CLI Path: *upf service-construct service-profile dpi-profile*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
force-dpi	enumeration	false true	false	Enable force-dpi. default is disabled Enable force-dpi.
dpi-pre-detection-max-packets	uint32		0	allow max number of packets for DPI pre protocol and application detection
dpi-pre-detection-max-bytes	uint32		0	allow max number of bytes for DPI pre protocol and application detection

Dnai

Data Network Access Identifier.

Data Network Access Identifier

XPath: *cna-ppe:/upf/service-construct/dnai*

CLI Path: *upf service-construct dnai*

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	1..64		Data Network Access Identifier name. This name must be unique

Dnai List

Data Network Access Identifier List.

Configure DNAI list object

XPath: *cna-ppe:/upf/service-construct/dnai-list*

CLI Path: *upf service-construct dnai-list*

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	1..64		Data Network Access Identifier List name. This name must be unique

Dnai

Data-Network-Access-Identifier name

XPath: [cna-ppe:/upf/service-construct/dnai-list/dnai](#)

CLI Path: [upf service-construct dnai-list dnai](#)

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: cna-ppe:/upf/service-construct/dnai/name

Dnn

data network name construct

Defines the data network name

Network-context must be configured when ip pool list is specified

XPath: [cna-ppe:/upf/service-construct/dnn](#)

CLI Path: [upf service-construct dnn](#)

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z0-9{0,63})		
network-context	string	(a-zA-Z{1,64})		Network-Context to be associated with the DNN
pdu-session-type	enumeration	IPv4 IPv6 IPv4V6		Type of UE-IP address supported for this DNN
service-profile	leafref			List of service profiles to be associated with the DNN Leaf Reference Path: cna-ppe:/upf/service-construct/service-profile/name
dnai-list	leafref			Data Network Access Identifier List name. This name must be unique Leaf Reference Path: cna-ppe:/upf/service-construct/dnai-list/name
allow-ue-hairpin	enumeration	false true	false	Allow UE to UE communication

Ipv6 Ra Param

ipv6 RA parameters

XPath: [cna-ppe:/upf/service-construct/dnn/ipv6-ra-param](#)

CLI Path: [upf service-construct dnn ipv6-ra-param](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
mtu	uint32	1280 .. 1800	1400	MTU for IPV6 RA Packet

Attribute	Type	Range/Length/Pattern	Default	Description
router-lifetime	uint32	1 .. 65535	1800	Router-lifetime for IPV6 RA Packet

Ipv4 Ue Pool

List of IPv4 UE Pool to be associated with the DNN

XPath: [cna-ppe:/upf/service-construct/dnn/ipv4-ue-pool](#)

CLI Path: [upf service-construct dnn ipv4-ue-pool](#)

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: cna-ppe:/upf/service-construct/network-context/ip-pool/name

Ipv6 Ue Pool

List of IPv6 UE Pool to be associated with the DNN

XPath: [cna-ppe:/upf/service-construct/dnn/ipv6-ue-pool](#)

CLI Path: [upf service-construct dnn ipv6-ue-pool](#)

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: cna-ppe:/upf/service-construct/network-context/ip-pool/name

Dnn List

list of data network names supported by the upf service

XPath: [cna-ppe:/upf/service-construct/dnn-list](#)

CLI Path: [upf service-construct dnn-list](#)

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{1,64})		

Dnn

data network name supported by the upf service

XPath: [cna-ppe:/upf/service-construct/dnn-list/dnn](#)

CLI Path: [upf service-construct dnn-list dnn](#)

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: cna-ppe:/upf/service-construct/dnn/name

S Nssai

S-NSSAI construct
Defines the S-NSSAI containing SST, SD and DNN List

XPath: [cna-ppe:/upf/service-construct/s-nssai](#)

CLI Path: [upf service-construct s-nssai](#)

List Properties:
Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{1,64})		
slice-service-type	union	['uint8,0 or 4 .. 255', 'Enumerations: eMBB URLLC MIoT ']		Identifies the expected Network Slice behavior in terms of features and services
slice-differentiator	string	([A-Fa-f0-9]{6})		Differentiate amongst multiple Network Slices of the same Slice/Service type
dnn-list	leafref			name of the list of DNNs those are supported by this s-nssai Leaf Reference Path: cna-ppe:/upf/service-construct/dnn-list/name

Nssai

list of S-NSSAI's supported by the upf service

XPath: [cna-ppe:/upf/service-construct/nssai](#)

CLI Path: [upf service-construct nssai](#)

List Properties:
Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{1,64})		

S Nssai

list of S-NSSAI's upported by the upf service

XPath: [cna-ppe:/upf/service-construct/nssai/s-nssai](#)

CLI Path: [upf service-construct nssai s-nssai](#)

List Properties:
Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: cna-ppe:/upf/service-construct/s-nssai/name

Plmn

PLMN construct
Defines the plmn identier and nssai list

XPath: [cna-ppe:/upf/service-construct/plmn](#)

CLI Path: [upf service-construct plmn](#)

List Properties:

Keys: plmn-id

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
plmn-id	string	(([0-9]{5,6}))		Unique PLMN identifier; Range: numbers with 5-6 digits
nssai	leafref			name of the nssai supported by this plmn Leaf Reference Path: cna-ppe:/upf/service-construct/nssai/name

Plmn List

name the plmn-list containing the list of plmns

XPath: [cna-ppe:/upf/service-construct/plmn-list](#)

CLI Path: [upf service-construct plmn-list](#)

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{1,64})		

Plmn

name of the plmn

XPath: [cna-ppe:/upf/service-construct/plmn-list/plmn](#)

CLI Path: [upf service-construct plmn-list plmn](#)

List Properties:

Keys: plmn-id

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
plmn-id	leafref			Leaf Reference Path: cna-ppe:/upf/service-construct/plmn/plmn-id

Tai

Tracking area identifier construct

Defines the tracking area identifier that consist of mcc, mnc and tracking are code

XPath: [cna-ppe:/upf/service-construct/tai](#)

CLI Path: [upf service-construct tai](#)

List Properties:

Keys: name

Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		
plmn-id	string	(([0-9]{5,6}))		Unique PLMN identifier; Range: numbers with 5-6 digits
tac	string	([A-Fa-f0-9]{4}) or [A-Fa-f0-9]{6})		tracking are code of the tai

Tai List

tracking area identifiers list supported by the UPF service

XPath: [cna-ppe:/upf/service-construct/tai-list](#)

CLI Path: *upf service-construct tai-list*

List Properties:

Keys: name
Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		
dnai-list	leafref			name of the dnai-list of DNAs those are used for local traffic steering Leaf Reference Path: <i>cna-ppe:/upf/service-construct/dnai-list/name</i>

Tai

tracking area identifiers list supported by the UPF service

XPath: *cna-ppe:/upf/service-construct/tai-list/tai*

CLI Path: *upf service-construct tai-list tai*

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: <i>cna-ppe:/upf/service-construct/tai/name</i>

Tac Range

tracking area code range construct
Defines the tracking area code range
Either start-tac and end-tac or pattern should be configured
Both start-tac and end-tac should be configured

XPath: *cna-ppe:/upf/service-construct/tac-range*

CLI Path: *upf service-construct tac-range*

List Properties:

Keys: name
Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		
start-tac	string	([A-Fa-f0-9]{4} or [A-Fa-f0-9]{6})		
end-tac	string	([A-Fa-f0-9]{4} or [A-Fa-f0-9]{6})		
pattern	string			

Tac Range List

tracking area codes range list supported by the SMF service

XPath: *cna-ppe:/upf/service-construct/tac-range-list*

CLI Path: *upf service-construct tac-range-list*

List Properties:

Keys: name
Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
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Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		

Tac Range

tracking area codes range list supported by the SMF service

XPath: [cna-ppe:/upf/service-construct/tac-range-list/tac-range](#)

CLI Path: [upf service-construct tac-range-list tac-range](#)

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: cna-ppe:/upf/service-construct/tac-range/name

Tai Range

tracking area identifiers range construct
Defines the tracking area identifiers range

XPath: [cna-ppe:/upf/service-construct/tai-range](#)

CLI Path: [upf service-construct tai-range](#)

List Properties:

Keys: name

Maximum number of elements: 32

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		
plmn-id	string	(([0-9]{5,6}))		Unique PLMN identifier; Range: numbers with 5-6 digits
tac-range-list	leafref			Tracking Area Code Range List Name Leaf Reference Path: cna-ppe:/upf/service-construct/tac-range-list/name
nid	string	([a-zA-Z0-9]{0,63})		NID related to the TacRange

Tai Range List

tracking area identifiers range list supported by the SMF service

XPath: [cna-ppe:/upf/service-construct/tai-range-list](#)

CLI Path: [upf service-construct tai-range-list](#)

List Properties:

Keys: name

Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		

Tai Range

tracking area identifiers range list supported by the SMF service

XPath: [cna-ppe:/upf/service-construct/tai-range-list/tai-range](#)

CLI Path: [upf service-construct tai-range-list tai-range](#)

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: cna-ppe:/upf/service-construct/tai-range/name

Upf Info

UPF Info - Data specific for UPF

UPF Info - Data specific for UPF

XPath: [cna-ppe:/upf/service-construct/upf-info](#)

CLI Path: [upf service-construct upf-info](#)

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			UPFInfo name, should be unique
admin-state	enumeration	disabled enabled	enabled	Admin State, if disabled it is not published to NRF
ipups-support	enumeration	false true	false	UPF support for Inter PLMN User Plane Security feature
priority	uint32	1 .. 4096		Priority for NF selection, lower value has higher priority
iwk-eps-ind	enumeration	false true	true	UPF support for interworking with EPS - Published only for primary-upf-info
ueip-addr-ind	enumeration	false true	true	Indicates UPF support for UE-IP allocation - Published only for primary-upf-info
smf-serving-area	leafref			SMF serving area served by the UPF - Published only for primary-upf-info Leaf Reference Path: cna-ppe:/upf/service-construct/serving-area-list/name
tai-list	leafref			List of TAIs served by the UPF Leaf Reference Path: cna-ppe:/upf/service-construct/tai-list/name
tai-range-list	leafref			List of TAI Ranges served by the UPF Leaf Reference Path: cna-ppe:/upf/service-construct/tai-range-list/name

Upf Info List

List of upf-info published in upfInfoList - Only priority and tai-list will be published

List of upf-info published in upfInfoList - Only priority and tai-list will be published

XPath: [cna-ppe:/upf/service-construct/upf-info-list](#)

CLI Path: [upf service-construct upf-info-list](#)

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			UPFInfo List name, should be unique

Upf Info

List of upf-info to be published by the UPF service in upfInfoList

XPath: [cna-ppe:/upf/service-construct/upf-info-list/upf-info](#)

CLI Path: [upf service-construct upf-info-list upf-info](#)

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: cna-ppe:/upf/service-construct/upf-info/name

Edr Profile

Configure Edr Profile

XPath: [cna-ppe:/upf/service-construct/edr-profile](#)

CLI Path: [upf service-construct edr-profile](#)

List Properties:

Keys: name

Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			Edr Profile Name - string
start-flow-edr-generation	enumeration	false true	false	Enable or Disable START type Flow Edr generation Enable or Disable START type Flow Edr generation
interim-flow-edr-generation	enumeration	false true	false	Enable or Disable INTERIM type Flow Edr generation Enable or Disable INTERIM type Flow Edr generation
stop-flow-edr-generation	enumeration	false true	false	Enable or Disable STOP type Flow Edr generation Enable or Disable STOP type Flow Edr generation
interim-edr-interval	uint16	10 .. 360	60	time interval between two successive interim edrs in seconds time interval between two successive interim edrs in seconds
optimized-interim-records	enumeration	false true	true	List of EDR types for which interim record optimization is enabled. if enabled, interim record size is optimized by skipping fields that did not change from last EDR and incremental stats List of EDR types for which interim record optimization is enabled. if enabled, interim record size is optimized by skipping fields that did not change from last EDR and incremental stats
start-session-edr-generation	enumeration	false true	false	Enable or Disable START type Session Edr generation Enable or Disable START type Session Edr generation
stop-session-edr-generation	enumeration	false true	false	Enable or Disable STOP type Session Edr generation Enable or Disable STOP type Session Edr generation
session-record-nat-info	enumeration	false true	true	Enable or Disable including NAT Binding Info (when applicable) in Session Record Enable or Disable including NAT Binding Info (when applicable) in Session Record

Li Profile

Affirmed UPF LI-Profile

XPath: [cna-ppe:/upf/service-construct/li-profile](#)

CLI Path: [upf service-construct li-profile](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
ipv4-loopback-name	string	1..64		Name of the pod-scope ipv4 Loopback Template to be used for LI on this UPF.
ipv6-loopback-name	string	1..64		Name of the pod-scope ipv6 Loopback Template to be used for LI on this UPF.
start-port	uint16	1..65535	2000	Start port value to be used for LI on this UPF.
end-port	uint16	1..65535	65535	End port value to be used for LI on this UPF.
x2-tcp-connections-per-destination	uint32	1..128	1	Number of TCP connections established for each X2 destination on this UPF.

Attribute	Type	Range/Length/Pattern	Default	Description
x3-tcp-connections-per-destination	uint32	1..128	1	Number of TCP connections established for each X3 destination on this UPF.

Ppe Session Management Profile

Affirmed UPF SessionManager-Profile

XPath: *cna-ppe:/upf/service-construct/ppe-session-management-profile*

CLI Path: *upf service-construct ppe-session-management-profile*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
proxy-service	enumeration	disabled enabled	enabled	Enable/disable proxy services, default is enabled
guaranteed-nat	enumeration	disabled enabled	enabled	Enable/disable Guaranteed NAT functionality, default is enabled
session-establishment-timeout	uint32	1000 .. 10000	2000	Session Establishment Request Timer, specified in milliseconds. Range [1000-10000], default 2000
session-modification-timeout	uint32	100 .. 10000	2000	Session Modification Request Timer, specified in milliseconds. Range [100-10000], default 2000
session-deletion-timeout	uint32	1000 .. 10000	2000	Session Deletion Request Timer, specified in milliseconds. Range [1000-10000], default 2000
standby-session-establishment-timeout	uint32	1000 .. 10000	5000	Standby Session Establishment Request Timer, specified in milliseconds. Range [1000-10000], default 5000
standby-session-modification-timeout	uint32	1000 .. 10000	3000	Standby Session Modification Request Timer, specified in milliseconds. Range [1000-10000], default 3000
standby-session-deletion-timeout	uint32	1000 .. 10000	3000	Standby Session Deletion Request Timer, specified in milliseconds. Range [1000-10000], default 3000
session-report-timeout	uint32	1000 .. 10000	3000	Session Report Timer, specified in milliseconds. Range [1000-10000], default 3000
session-report-retry-upon-timeout	enumeration	disabled enabled	disabled	Retry to different SMF in the SMF Set upon timeout from PFCP-U, default disabled
session-restore-timeout	uint32	1000 .. 10000	5000	Session Restore Timer, specified in milliseconds. Range [1000-10000], default 5000
vpp-retry-timer	uint32	50 .. 2000	500	VPP Retry Timer for PFCP sessions, specified in milliseconds. Range [50-2000], default 500
vpp-session-modification-retry-timer	uint32	50 .. 2000	500	VPP Retry Timer for PFCP session modification, specified in milliseconds. Range [50-2000], default 500
session-restore-batch-size	uint32	1 .. 100	10	Session Restore batch size, Range [1-100], default 1
session-idle-timeout	uint32	60 .. 604800	60	Session Idle Timer, specified in seconds. Range [60-604800], default 600
clear-sessions-for-pfcp-assoc-down	enumeration	disabled enabled	enabled	Clear all sessions on UPF when PFCP association is down. Default is enabled
pfcp-message-batch-timer-interval	uint32	10 .. 10000	10	PFCP message batch Timer interval, specified in milliseconds. Range [10-10000], default 10
pfcp-message-batch-size	uint32	1 .. 100	10	PFCP message batch size, Range [1-100], default 10
best-effort-partial-db-update	enumeration	disabled enabled	enabled	Do not wait on DB response for Partial Modify Session
wait-for-proxy-create-response	enumeration	disabled enabled	enabled	Enable/disable waiting for proxy create session response, default is enabled
dnn-usage-update-timer	uint32	5 .. 3600	60	DNN usage update timer. Range [5 - 3600], default 60
dnn-usage-fast-update-timer	uint32	1 .. 1800	5	DNN usage fast update timer. Range [1 - 1800], default 5

Attribute	Type	Range/Length/Pattern	Default	Description
dnn-crankback-throttle-time	uint32	5 .. 5000	20	Minimum time between DNN usage updates triggered by crankback events. Range [5 - 5000] milliseconds, default 20 milliseconds
sm-load-update-timer	uint32	5 .. 3600	60	SM load update timer. Range [5 - 3600], default 60
sm-load-fast-update-timer	uint32	1 .. 1800	5	SM load fast update timer. Range [1 - 1800], default 5
session-clear-pacer-timer-interval	uint32	1 .. 1800	10	Session Clear Pacer each invocation time interval. Range [1 - 1800], default 10
session-clear-pacer-count-per-interval	uint32	1 .. 1000	10	Number of max Session to be cleared during each Session Clear Pacer invocation. Range [1 - 1000], default 10
standby-vpp-retry-timer	uint32	500 .. 2000	1000	VPP Retry Timer for sessions created as standby, specified in milliseconds. Range [500-2000], default 1000
sm-bundle-allrb-msg	enumeration	disabled enabled	enabled	Enable/Disable bundling of resourceblock messages from session manager to VPP to support UE2UE hairpinning. Default is enabled

Vprobe Profile

Affirmed UPF VPROBE-Profile

XPath: [cna-ppe:/upf/service-construct/vprobe-profile](#)

CLI Path: [upf service-construct vprobe-profile](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
statistics	string	1..64		The EDR statistics in vprobe agent process.
vpp-statistics	string	1..64		The EDR statistics in vpp process.

Load Reporting Profile

Configure parameters for UPF Load Reporting to NRF

XPath: [cna-ppe:/upf/service-construct/load-reporting-profile](#)

CLI Path: [upf service-construct load-reporting-profile](#)

List Properties:

Keys: name

Maximum number of elements: 1

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			Name of the load-reporting-profile
cpuLoad	enumeration	disabled enabled	enabled	Enable/disable cpu-load reporting
sessionLoad	enumeration	disabled enabled	disabled	Enable/disable session-load reporting

Flow Rate Limit Profile

Configure the connection creation rate limiting profile. NOTE: If you are not binding the profile to a DNN or NC then it is advisable to configure only 1 profile that acts globally across all NCs and DNNs supported

XPath: [cna-ppe:/upf/service-construct/flow-rate-limit-profile](#)

CLI Path: [upf service-construct flow-rate-limit-profile](#)

List Properties:

Keys: name

Minimum number of elements: 0

Maximum number of elements: 1

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			Flow Rate Limit profile name
new-flow-creation-rate-limit-ul	uint32	0 .. 20000000000	0	The flow creation rate on the uplink direction, indicates connections per second
new-flow-creation-rate-limit-dl	uint32	0 .. 20000000000	0	The flow creation rate on the downlink direction, indicates connections per second

Service

UPF Service Configuration

XPath: [cna-ppe:/upf/service](#)

CLI Path: [upf service](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
admin-state	enumeration	disabled enabled	enabled	Admin state of the UPF Service. default state is "Enabled".
instance-id	string			Unique Instance Id of the UPF Service.
upf-fqdn	string			Fully Qualified Domain Name of the UPF Service.
priority	uint32	0 .. 65535		Priority of the UPF Service.
capacity	uint32	0 .. 65535	10000	Capacity of the UPF Service.
locality	string			Location of the UPF to be registered with NRF
smf-serving-area	leafref			Serving-Area-list selected from "/upf/service-construct/serving-area-list" list Leaf Reference Path: cna-ppe:/upf/service-construct/serving-area-list/name
home-plmn-list	leafref			Home-plmn list selected from "/upf/service-construct/plmn-list" list Leaf Reference Path: cna-ppe:/upf/service-construct/plmn-list/name
allowed-visited-plmn-list	leafref			Allowed visited-plmn list selected from "/upf/service-construct/plmn-list" list Leaf Reference Path: cna-ppe:/upf/service-construct/plmn-list/name
allowed-nssai	leafref			Allowed S-NSSAI list selected from "/upf/service-construct/nssai" list Leaf Reference Path: cna-ppe:/upf/service-construct/nssai/name
fteid-allocation	enumeration	disabled enabled	enabled	Control for local FTEID allocation on UPF. default state is "Enabled".
fsm-logging	enumeration	DEBUG_1 DEBUG_2 DEBUG_3	DEBUG_1	Log level for FSM logging, default is debug1
standby-fsm-logging	enumeration	DEBUG_1 DEBUG_2 DEBUG_3	DEBUG_1	Log level for standby FSM, default is debug1
ucd-infra-logging	enumeration	DEBUG_1 DEBUG_2 DEBUG_3	DEBUG_1	Log level for UCD infrastructure, default is debug1
cfg-logging	enumeration	DEBUG_1 DEBUG_2 DEBUG_3	DEBUG_1	Log level for config logging, default is debug1
primary-upf-info	leafref			Primary upf-info to be used for NRF registration of UPF Leaf Reference Path: cna-ppe:/upf/service-construct/upf-info/name
upf-info-list	leafref			upf-info-list to be used for NRF registration of UPF Leaf Reference Path: cna-ppe:/upf/service-construct/upf-info-list/name
malloc-trim-interval	uint16	0 .. 65535	0	time interval for malloc trim in minute time interval for malloc trim in minute

Attribute	Type	Range/Length/Pattern	Default	Description
ppe-single-flow-anchor-core	enumeration	disabled enabled	disabled	enable or disable anchoring flow on a single core
max-dl-buffers-per-session	uint32	16 .. 65535	16384	Enter the maximum number of downlink packets to buffer per session. Enter the maximum number of downlink packets to buffer per session. Range [16 .. 65535]
max-dl-buffers-per-ppe	uint32	4096 .. 1000000	500000	Enter the maximum number of downlink packets to buffer per ppe. Enter the maximum number of downlink packets to buffer per ppe. Range [4096 .. 1000000]
max-dl-buffering-hold-time	uint64	25 .. 26784000	25	Enter the maximum DL buffering hold time, it is applied only when CP not configures buffering duration. Enter the value in seconds. Range [25 .. 26784000]
uepcap-copy-interval-time	uint16	5 .. 30	10	time interval for uepcap copy in seconds time interval for uepcap copy in seconds
uepcap-file-chunk-size	uint16	1 .. 100	10	default file chunk size in MB for UE packet capture default file chunk size in MB for UE packet capture
ucd-initial-sync-timer	uint32	0 .. 1000	50	This timer specifies the interval at which server communicates records to clients through UCD initially This timer specifies the interval at which server communicates records to clients through UCD initially.
default-service-profile	leafref			Configure Default Service Profile Leaf Reference Path: cna-ppe:/upf/service-construct/service-profile/name
default-edr-profile	leafref			Configure Default EDR Profile Leaf Reference Path: cna-ppe:/upf/service-construct/edr-profile/name

Network Context

list of Network-Contexts - selected from "/upf/service-construct/network-context" list

XPath: [cna-ppe:/upf/service/network-context](#)

CLI Path: [upf service network-context](#)

List Properties:
Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: cna-ppe:/upf/service-construct/network-context/name

Tdef Proxy

Configure Content TDEF Proxy Service

XPath: [cna-ppe:/upf/service/tdef-proxy](#)

CLI Path: [upf service tdef-proxy](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
admin-state	enumeration	disabled enabled	disabled	Enable/Disable TDEF Proxy Service
http-client-idle-timeout	uint32	10 .. 600	60	Configure HTTP client idle timeout in seconds
tcp-client-idle-timeout	uint32	10 .. 600	60	Configure TCP client idle timeout in seconds
udp-client-idle-timeout	uint32	10 .. 600	60	Configure UDP client idle timeout in seconds

Attribute	Type	Range/Length/Pattern	Default	Description
http-fast-open	enumeration	disabled enabled		Turn on/off HTTP server connection fast open
tcp-fast-open	enumeration	disabled enabled		Turn on/off TCP server connection fast open
network-context	string			Configure network-context for tdef-proxy
subscriber-tracing-profile	leafref			Configure Subscriber Tracing Profile Leaf Reference Path: cna-ppe:/upf/service-construct/tdef/subscriber-tracing-profile/name
dscp-marking	enumeration	disabled enabled	enabled	Enable/Disable DSCP Marking for Server-Side Connections
loopback-ip-family	enumeration	ipv4 ipv6 dual-stack transparent	dual-stack	Configure Proxy Loopback IP Family

Overload Control

Configure OverLoad Control

XPath: [cna-ppe:/upf/service/tdef-proxy/overload-control](#)

CLI Path: [upf service tdef-proxy overload-control](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
admin-state	enumeration	disabled enabled	disabled	Enable/disable overload control
minor-cpu-threshold	uint32	66 .. 75	70	Threshold of minor overload
major-cpu-threshold	uint32	76 .. 85	80	Threshold of major overload
critical-cpu-threshold	uint32	86 .. 100	90	Threshold of critical overload
minor-bypass-percent	uint32	0 .. 100	20	Bypass rate of new connections if reaching minor threshold
major-bypass-percent	uint32	0 .. 100	40	Bypass rate of new connections if reaching major threshold
critical-bypass-percent	uint32	0 .. 100	60	Bypass rate of new connections if reaching critical threshold

Tdef Video Optimization

Configure TDEF Video-optimization

XPath: [cna-ppe:/upf/service/tdef-video-optimization](#)

CLI Path: [upf service tdef-video-optimization](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
admin-state	enumeration	disabled enabled	disabled	Enable/Disable TDEF Video-optimization Service

Tdef Ship

Configure TDEF SHIP Encryption and Header Enrichment

XPath: [cna-ppe:/upf/service/tdef-ship](#)

CLI Path: [upf service tdef-ship](#)

Content Provider

Content Provider Information

XPath: [cna-ppe:/upf/service/tdef-ship/content-provider](#)

CLI Path: *upf service tdef-ship content-provider*

List Properties:

Keys: content-provider-domain
Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
domain	string			Content Provider Domain
timeout	int32		1440	Request expire time for the content provider
consent-id	string	[A-Za-z0-9*]+	*	Consent ID for the content provider - string
http-redirect-code	int32		303	Content provider HTTP request success code Success code must be 301, 302 or 303
enrichment-profile	leafref			Leaf Reference Path: cna-ppe:/upf/service-construct/tdef/enrichment-profile/name
encryption-profile	leafref			Leaf Reference Path: cna-ppe:/upf/service-construct/tdef/encryption-profile/name

Tdef Entitlement

Configure TDEF Entitlement information

XPath: *cna-ppe:/upf/service/tdef-entitlement*

CLI Path: *upf service tdef-entitlement*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
bypass-entitlement-validation	enumeration	disabled enabled	disabled	Enable/Disable Bypassing Entitlement Validation

Entitlement Service

Entitlement Service Domain

XPath: *cna-ppe:/upf/service/tdef-entitlement/entitlement-service*

CLI Path: *upf service tdef-entitlement entitlement-service*

List Properties:

Keys: entitlement-service-domain
Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
entitlement-service-domain	string			Entitlement
entitlement-profile	leafref			Leaf Reference Path: cna-ppe:/upf/service-construct/tdef/entitlement-profile/name

Tdef Tcp Optimization

Configure TDEF Global TCP Parameters

XPath: *cna-ppe:/upf/service/tdef-tcp-optimization*

CLI Path: *upf service tdef-tcp-optimization*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
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Attribute	Type	Range/Length/Pattern	Default	Description
auto-tuning-recv-buffer-size-default	uint32	4096 .. 33554432	1048576	Configure auto-tuning range for TCP receive buffer size
auto-tuning-recv-buffer-size-max	uint32	4096 .. 33554432	33554432	Configure auto-tuning range for TCP receive buffer size
auto-tuning-send-buffer-size-default	uint32	4096 .. 33554432	1048576	Configure auto-tuning range for TCP send buffer size
auto-tuning-send-buffer-size-max	uint32	4096 .. 33554432	33554432	Configure auto-tuning range for TCP send buffer size
delay-ack-admin-state	enumeration	disabled enabled	enabled	Configure admin state to enable or disable for TCP Delayed ACK
delay-ack-time	uint32	1 .. 500	100	Configure TCP Delayed ACK to try and piggyback it onto a data packet
max-syn-backlog	uint32	16384 .. 1048576	32768	Configure TCP max syn backlog.
newreno-beta	uint32	1 .. 100	50	Configure newreno beta value, only applicable to reno CCA
proportional-rate-reduction	enumeration	disabled enabled	enabled	Configure TCP Proportional Rate Reduction admin state
prp-conservative	enumeration	disabled enabled	disabled	Configure conservative Proportional Rate Reduction
tcp-abc-l-value	uint32	1 .. 65535	256	Configure the max CWND increment during slow-start to this number of segments
tcp-ecn	enumeration	disable-ecn enable-with-request-ecn-on-outgoing enable-without-request-ecn-on-outgoing	disable-ecn	Configure TCP ECN Options.
tcp-sack-admin-state	enumeration	disabled enabled	enabled	Configure TCP SACK options
tcp-sack-maxholes	uint32	128 .. 3000000	1024	Configure TCP SACK options.
tcp-sack-global-maxholes	uint32	65536 .. 3000000	3000000	Configure TCP SACK options.
server-path-mtu-discovery	enumeration	disabled enabled	enabled	Configure Server-Side Path MTU Discovery
server-path-blackhole-detection	enumeration	disabled enabled	disabled	Configure Server-Side Path Blackhole detection
server-path-blackhole-detection-v4-mss	uint16	100 .. 65535	1200	Configure Server-Side Path Blackhole detection v4 mss
server-path-blackhole-detection-v6-mss	uint16	100 .. 65535	1220	Configure Server-Side Path Blackhole detection v6 mss

Tdef Tvs

TDEF-Transparent Video Shaping
TDEF-Transparent Video Shaping

XPath: *cna-ppe:/upf/service/tdef-tvs*

CLI Path: *upf service tdef-tvs*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
admin-state	enumeration	disabled enabled	disabled	Admin state of the UPF Service. default state is "Disabled".
threshold-bypass	enumeration	disabled enabled	disabled	Admin state of the threshold-bypass feature of the TVS service. Default state is "Disabled".

Attribute	Type	Range/Length/Pattern	Default	Description
burst-detect-ratio	uint32	0 .. 200	0	Ratio in percentage that determines the burst detection threshold. Default value is 0 as disabled.
subscriber-tracing-profile	leafref			Configure Subscriber Tracing Profile Leaf Reference Path: cna-ppe:/upf/service-construct/tdef/tvs-subscriber-tracing-profile/name
edr-profile	leafref			Configure EDR Profile Leaf Reference Path: cna-ppe:/upf/service-construct/tdef/edr-profile/name

Dpi

dpi control

XPath: [cna-ppe:/upf/service/dpi-detection](#)

CLI Path: [upf service dpi](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
tethering-detection-packet-limit	uint32	3..max	64	The maximum number of packets of a flow to be analysed for tethering detection, default is 64. The minimum value that can be configured is 3

Tethering Quality Index

Set the value for tethering quality index

Set the value for tethering quality index

XPath: [cna-ppe:/upf/service/dpi-detection/tethering-quality-index](#)

CLI Path: [upf service dpi tethering-quality-index](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
value	uint32	0 .. 100	70	Percentage value for Quality Index (Confidence Index) to mark connection as tethered 1..100

Tethering Quality Index Result Weight

Set the weight percentage for different tethering method's quality index

Set the weight percentage for different tethering method's quality index

XPath: [cna-ppe:/upf/service/dpi-detection/tethering-quality-index-result-weight](#)

CLI Path: [upf service dpi tethering-quality-index-result-weight](#)

Percentage

Set the weight percentage for tethering detection algorithms

Set the weight percentage for tethering detection algorithm

XPath: [cna-ppe:/upf/service/dpi-detection/tethering-quality-index-result-weight/percentage](#)

CLI Path: [upf service dpi tethering-quality-index-result-weight percentage](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
ttl	uint32	0 .. 100	80	Percentage value for Quality Index (Confidence Index) weight for ttl method
user-agent	uint32	0 .. 100	99	Percentage value for Quality Index (Confidence Index) weight for user-agent method
tcp-fingerprinting	uint32	0 .. 100	100	Percentage value for Quality Index (Confidence Index) weight for tcp-fingerprinting
gquic-user-agent	uint32	0 .. 100	100	Percentage value for Quality Index (Confidence Index) weight for gquic-user-agent

Tethering Detection Algorithm

Control tethering detection algorithms

Control tethering detection algorithms

XPath: *cna-ppe:/upf/service/dpi-detection/tethering-detection-algorithm*

CLI Path: *upf service dpi tethering-detection-algorithm*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
ttd	enumeration	disabled enabled report-only	enabled	Admin state for running tethering detection algorithm using ttl Admin state for running tethering detection algorithm using ttl
user-agent	enumeration	disabled enabled report-only	enabled	Admin state for running tethering detection algorithm using user-agent Admin state for running tethering detection algorithm using user-agent
tcp-fingerprinting	enumeration	disabled enabled report-only	disabled	Admin state for running tethering detection algorithm using tcp fingerprinting - Changes to this object do not take effect until the next POD restart Admin state for running tethering detection algorithm using tcp fingerprinting - Changes to this object do not take effect until the next POD restart
gquic-user-agent	enumeration	disabled enabled report-only	disabled	Admin state for running tethering detection algorithm using gquic useragent - Changes to this object do not take effect until the next POD restart Admin state for running tethering detection algorithm using gquic useragent - Changes to this object do not take effect until the next POD restart
ttd-connection-tracking	enumeration	disabled enabled report-only	enabled	Admin state for running tethering detection algorithm using ttl by taking connection tracking into consideration Admin state for running tethering detection algorithm using ttl by taking connection tracking into consideration
tdci	enumeration	disabled enabled report-only	disabled	Admin state for running tethering detection algorithm using tddi Admin state for running tethering detection algorithm using tddi

Tethering Tddi Weight

Set the weight for different tethering method's confidence index

Set the weight for different tethering method's confidence index

XPath: *cna-ppe:/upf/service/dpi-detection/tethering-tddi-weight*

CLI Path: *upf service dpi tethering-tddi-weight*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
ttd	uint32	0 .. 200	100	Tethering Detection Confidence Index weight in percentage for ttd method
user-agent	uint32	0 .. 200	0	Tethering Detection Confidence Index weight in percentage for user-agent method
tcp-fingerprinting	uint32	0 .. 200	70	Tethering Detection Confidence Index weight in percentage for tcp-fingerprinting
gquic-user-agent	uint32	0 .. 200	100	Tethering Detection Confidence Index weight in percentage for gquic-user-agent

Proxy Dpi

Configure Proxy DPI packet-limit and attribute information

XPath: *cna-ppe:/upf/service/dpi-detection/proxydpi*

CLI Path: *upf service dpi proxy-dpi*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
dpi-packet-count-with-proxy	uint32	0..max	50	Set the max packet count for dpi processing when proxy is enabled,0 is for all packets

Flow Attribute

Please specify flow-attributes to match the proxy flow

XPath: *cna-ppe:/upf/service/dpi-detection/proxydpi/flowattributes*

CLI Path: *upf service dpi proxy-dpi flow-attribute*

List Properties:

Keys: flowAttrType

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
flowAttrType	enumeration	video_download		
		video		
		video_stream		

Flow Attribute Detection

Configuration option to tune the flow-attribute detection

XPath: *cna-ppe:/upf/service/dpi-detection/attributeDetection*

CLI Path: *upf service dpi flow-attribute-detection*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
packet-limit	uint32	10..max	50	The maximum number of packets to detect flow-attribute of a dpi flow
first-attribute-match	enumeration	disabled	disabled	Consider only first attribute of a flow for rule match
		enabled		

Control Flow

Configure DPI Control Application/Protocol Information

XPath: *cna-ppe:/upf/service/dpi-detection/controlApps*

CLI Path: *upf service dpi control-flow*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
skip-attr-check	enumeration	disabled enabled	disabled	Admin state to enable/disable attribute check for marking control connections in Control Flow Feature

Optimize Dpi

Optimize DPI rules evaluation based on configuration

XPath: *cna-ppe:/upf/service/dpi-detection/optimizeDpi*

CLI Path: *upf service dpi optimize-dpi*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
match-on-configured-parg	enumeration	disabled enabled	enabled	Evaluate parg rules configured may match to classification event

Dns Snoop

DNS Snooping for traffic classification

XPath: *cna-ppe:/upf/service/dpi-detection/dnsSnooping*

CLI Path: *upf service dpi dns-snoop*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
enableDnsSnooping	enumeration	disabled enabled	disabled	Enable DNS Snooping for traffic classification

Attribute	Type	Range/Length/Pattern	Default	Description
maximum-snooping-capacity	uint32	1000..10000	10000	The maximum number of servers to snoop from authorized dns servers
maximum-snooping-per-drg	uint32	100..10000	100	The maximum number of servers to snoop per drg rule

Dns Ip Address

The list of IP addresses of authorized DNS Servers

XPath: [cna-ppe:/upf/service/dpi-detection/dnsSnooping/dnsSnoopingServerList](#)

CLI Path: [upf service dpi dns-snoop dns-ip-address](#)

List Properties:
Keys: dnsServerIp
Maximum number of elements: 20

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
dnsServerIp	inet:ip-address			IP v4/v6 addresses of authorized DNS Server Please enter valid ipv6-address Please enter valid ipv4-address

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The module for entities implementing cna-radiusagent.

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Radiusagent

Radius-Agent interface object

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ROUTING Configuration

The module for cna-routing.

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Routing

Routing Service configuration
Routing Service handles the deployment of all routing protocols and its infrastructure in this NF.
Value should be between 30s to 1800s, or 0 to disable

XPath: *cna-routing:/routing*

CLI Path: *routing*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
rib-logging-enabled	empty			Enable Rib logging
bgp-logging-enabled	empty			Enable Bgp logging
unsolicit-na-refresh-time	uint32		60	Set Unsolicit Neighbor Advertisement refresh time in seconds (range: <30-1800>, default: 60s, 0 to disable)
fib-sync-time	uint32	100..65535	100	VPP Fib Synchronous timer (ms). Recommend multiples of 100ms for every multiple of 50 of fib-sync-batch-size. Default: 100ms
fib-sync-batch-size	uint32	1..65535	50	Fib Batch size in an interval of (fib-sync-time) ms. Recommend multiples of 50 for every multiple of 100ms of fib-sync-time. Default: 50

Label Allocation Mode

Configure label-allocation-mode (default: label-per-vrf)

XPath: *cna-routing:/routing/labelAllocationMode*

CLI Path: *routing label-allocation-mode*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
label-per-prefix	empty			Set mpls-label allocation mode to label-per-prefix

Network Context

Configure the Network Contexts supported by this NF and this acts as shadow of Network Context configured in Interface Manager
Configure the Network Contexts supported by this NF. Each Network Context represents an entire IPv4 address space and an entire IPv6 address space. The address spaces of all Network Contexts are disjoint

XPath: *cna-routing:/routing/network-context*

CLI Path: *routing network-context*

List Properties:

Keys: nc-name
Maximum number of elements: 4096

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
nc-name	string	(a-zA-Z{1,64})		Network Context name - type is <alphanumeric, '_' is allowed>
pod-type	enumeration	ulb		Specify Pod-type if this network-context needs to be used for a specific type of pod - No need to configure for IPPE or PPE
vrf	empty			Specify a Vrf Vrf Name for Routing
bfd-all-static-enable	empty			Enable BFD sessions for all IPv4/IPv6 static routes in this context
bfd-sh-profile	leafref			BFD profile to be used for Single-Hop BFD sessions under this context, if not configured default timers (tx-interval = 300, rx-interval = 300, multiplier = 3) will be used Leaf Reference Path:

Vrf Name

Configure and manage VRFs to associate to VR
Vrf Name for Routing

XPath: *cna-routing:/routing/network-context/vrf-name*

CLI Path: *routing network-context vrf-name*

List Properties:

Keys: vrf-nc-name
Maximum number of elements: 4096

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
vrf-nc-name	leafref			VRF name - type is <alphanumeric, '_' is allowed> Leaf Reference Path:

Ip Route

Configuration of static IP routes
Configuration of Static routes

XPath: *cna-routing:/routing/network-context/ip-route*

CLI Path: *routing network-context ip-route*

List Properties:

Keys: dest, length, nexthop

Maximum number of elements: 4096

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
dest	inet:ip-address			IP address of the route destination IP address of the route destination
length	uint32			IP prefix length of the route destination Prefix length of destination Address
next-hop	inet:ip-address			IP Address of NextHop Gateway IP Address of nexthop gateway
preference	uint32	1..255		Distance metric for this route <1-255> Distance metric for this route
tag	uint32			Specify a tag for this route <1-4294967295> Tag Value for making Policy Decision

Bgp

Manage BGP4 routing

XPath: [cna-routing:/routing/network-context/bgp](#)

CLI Path: [routing network-context bgp](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
local-as	inet:as-number			Autonomous System Number for this BGP router
router-id	string	((([0-9] or [1-9][0-9] or 1[0-9][0-9] or 2[0-4][0-9] or 25[0-5]).){3}([0-9] or [1-9][0-9] or 1[0-9][0-9] or 2[0-4][0-9] or 25[0-5]))		Router ID
bgp-up-notification-to-l3am	enumeration	on-single-session-up on-all-session-up	on-single-session-up	bgp-up-notification will be sent to l3am:

Graceful Restart

Configuration of BGP Graceful Restart Timers

XPath: [cna-routing:/routing/network-context/bgp/bgpGracefulRestartParams](#)

CLI Path: [routing network-context bgp graceful-restart](#)

List Properties:

Keys: graceRestartEnable

Maximum number of elements: 1

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
graceRestartEnable	enumeration	enable		
restart-time	uint32	1 .. 3600	90	Restart Time [1..3600 in secs]
stalepath-time	uint32	1 .. 3600	360	Stale Path Time [1..3600 in secs]

Neighbor

Configure and manage BGP neighbors

XPath: [cna-routing:/routing/network-context/bgp/neighbor](#)

CLI Path: [routing network-context bgp neighbor](#)

List Properties:

Keys: remoteAddr

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
neighbor-addr	inet:ip-address			IP address of neighbor
remote-as	inet:as-number			Autonomous System of remote neighbor
fall-over-bfd	empty			Enable Bi-directional Forwarding Detection

Ipv4 Ucast Af

IPv4 Address Family

XPath: [cna-routing:/routing/network-context/bgp/ipv4Ucast](#)

CLI Path: [routing network-context bgp ipv4-ucast-af](#)

Neighbor

Manage ipv4 unicast address family features on a neighbor peering

XPath: [cna-routing:/routing/network-context/bgp/ipv4Ucast/afNeighbor](#)

CLI Path: [routing network-context bgp ipv4-ucast-af neighbor](#)

List Properties:

Keys: remoteAddr

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
activate	empty			Activate address family on peering.
nexthop-self	empty			Set router as next-hop for all routes advertised to peer
neighbor-addr	leafref			IP address of neighbor Leaf Reference Path:

Capabilities

Configure the capabilities of this neighbor

XPath: [cna-routing:/routing/network-context/bgp/ipv4Ucast/afNeighbor/capabilities](#)

CLI Path: [routing network-context bgp ipv4-ucast-af neighbor capabilities](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
graceful-restart	empty			Advertise support of graceful restart

Route Map

Apply route-map filtering to in(bound) and/or out(bound) update messages carried by this peering

XPath: [cna-routing:/routing/network-context/bgp/ipv4Ucast/afNeighbor/afNeighborRouteMap](#)

CLI Path: [routing network-context bgp ipv4-ucast-af neighbor route-map](#)

List Properties:

Keys: direction

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
route-map-name	leafref			Name of a route map Leaf Reference Path:
direction	enumeration	in out		Specify filter direction

Redistribute

Redistribution configuration

XPath: *cna-routing:/routing/network-context/bgp/ipv4Ucast/redistribute*

CLI Path: *routing network-context bgp ipv4-ucast-af redistribute*

List Properties:

Keys: protocol

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
protocol	enumeration	connected static		Protocol type to redistribute into BGP

Ipv6 Ucast Af

IPv6 Address Family

XPath: *cna-routing:/routing/network-context/bgp/ipv6Ucast*

CLI Path: *routing network-context bgp ipv6-ucast-af*

Neighbor

Manage ipv6 unicast address family features on a neighbor peering

XPath: *cna-routing:/routing/network-context/bgp/ipv6Ucast/afNeighbor*

CLI Path: *routing network-context bgp ipv6-ucast-af neighbor*

List Properties:

Keys: remoteAddr

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
activate	empty			Activate address family on peering.
nexthop-self	empty			Set router as next-hop for all routes advertised to peer
neighbor-addr	leafref			IP address of neighbor Leaf Reference Path:

Capabilities

Configure the capabilities of this neighbor

XPath: *cna-routing:/routing/network-context/bgp/ipv6Ucast/afNeighbor/capabilities*

CLI Path: *routing network-context bgp ipv6-ucast-af neighbor capabilities*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
graceful-restart	empty			Advertise support of graceful restart

Route Map

Apply route-map filtering to in(bound) and/or out(bound) update messages carried by this peering

XPath: *cna-routing:/routing/network-context/bgp/ipv6Ucast/afNeighbor/afNeighborRouteMap*

CLI Path: *routing network-context bgp ipv6-ucast-af neighbor route-map*

List Properties:

Keys: direction

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
route-map-name	leafref			Name of a route map Leaf Reference Path:

Attribute	Type	Range/Length/Pattern	Default	Description
direction	enumeration	in out		Specify filter direction

Redistribute

Redistribution configuration

XPath: [cna-routing:/routing/network-context/bgp/ipv6Ucast/redistribute](#)

CLI Path: [routing network-context bgp ipv6-ucast-af redistribute](#)

List Properties:

Keys: protocol

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
protocol	enumeration	connected static		Protocol type to redistribute into BGP

Vpnv4 Ucast Af

VPN-IPv4 Address Family

XPath: [cna-routing:/routing/network-context/bgp/vpnv4Ucast](#)

CLI Path: [routing network-context bgp vpnv4-ucast-af](#)

Neighbor

Manage vpn-v4 unicast address family features on a neighbor peering

XPath: [cna-routing:/routing/network-context/bgp/vpnv4Ucast/afNeighbor](#)

CLI Path: [routing network-context bgp vpnv4-ucast-af neighbor](#)

List Properties:

Keys: remoteAddr

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
activate	empty			Activate address family on peering.
nexthop-self	empty			Set router as next-hop for all routes advertised to peer
neighbor-addr	leafref			IP address of neighbor Leaf Reference Path:
send-community	enumeration	standard extended both		Configure type of community support - standard, extended or both. Default "extended"

Capabilities

Configure the capabilities of this neighbor

XPath: [cna-routing:/routing/network-context/bgp/vpnv4Ucast/afNeighbor/capabilities](#)

CLI Path: [routing network-context bgp vpnv4-ucast-af neighbor capabilities](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
graceful-restart	empty			Advertise support of graceful restart

Route Map

Apply route-map filtering to in(bound) and/or out(bound) update messages carried by this peering

XPath: [cna-routing:/routing/network-context/bgp/vpnv4Ucast/afNeighbor/afNeighborRouteMap](#)

CLI Path: *routing network-context bgp vpnv4-ucast-af neighbor route-map*

List Properties:

Keys: direction

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
route-map-name	leafref			Name of a route map Leaf Reference Path:
direction	enumeration	in out		Specify filter direction

Vpnv6 Ucast Af

VPN-IPv6 Address Family

XPath: *cna-routing:/routing/network-context/bgp/vpnv6Ucast*

CLI Path: *routing network-context bgp vpnv6-ucast-af*

Neighbor

Manage vpn-v6 unicast address family features on a neighbor peering

XPath: *cna-routing:/routing/network-context/bgp/vpnv6Ucast/afNeighbor*

CLI Path: *routing network-context bgp vpnv6-ucast-af neighbor*

List Properties:

Keys: remoteAddr

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
activate	empty			Activate address family on peering.
nexthop-self	empty			Set router as next-hop for all routes advertised to peer
neighbor-addr	leafref			IP address of neighbor Leaf Reference Path:
send-community	enumeration	standard extended both		Configure type of community support - standard, extended or both. Default "extended"

Capabilities

Configure the capabilities of this neighbor

XPath: *cna-routing:/routing/network-context/bgp/vpnv6Ucast/afNeighbor/capabilities*

CLI Path: *routing network-context bgp vpnv6-ucast-af neighbor capabilities*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
graceful-restart	empty			Advertise support of graceful restart

Route Map

Apply route-map filtering to in(bound) and/or out(bound) update messages carried by this peering

XPath: *cna-routing:/routing/network-context/bgp/vpnv6Ucast/afNeighbor/afNeighborRouteMap*

CLI Path: *routing network-context bgp vpnv6-ucast-af neighbor route-map*

List Properties:

Keys: direction

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
route-map-name	leafref			Name of a route map Leaf Reference Path:
direction	enumeration	in out		Specify filter direction

Rt Constrain Af

Route Target Constrain Address Family

XPath: [cna-routing:/routing/network-context/bgp/rtcAf](#)

CLI Path: [routing network-context bgp rt-constrain-af](#)

Neighbor

Manage Route Target Constrain address family features on a neighbor peering

XPath: [cna-routing:/routing/network-context/bgp/rtcAf/afNeighbor](#)

CLI Path: [routing network-context bgp rt-constrain-af neighbor](#)

List Properties:

Keys: remoteAddr

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
neighbor-addr	leafref			IP address of neighbor Leaf Reference Path:
activate	empty			Activate address family on peering.

Vrf

Configure and manage VRFs

Vrf Name for Routing

XPath: [cna-routing:/routing/network-context/bgp/vrf](#)

CLI Path: [routing network-context bgp vrf](#)

List Properties:

Keys: vrf-nc-name

Maximum number of elements: 4096

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
vrf-nc-name	leafref			VRF name - type is <alphanumeric, '_' is allowed> Leaf Reference Path:

Attribute	Type	Range/Length/Pattern	Default	Description
route-distinguisher	string	((([1-9] or [1-9][0-9] or [1-9][0-9][0-9] or [1-9][0-9][0-9][0-9] or [1-5][0-9][0-9][0-9][0-9] or 6[0-4][0-9][0-9][0-9] or 65[0-4][0-9][0-9] or 655[0-2][0-9] or 6553[0-5]);([1-9] or [1-9][0-9] or [1-9][0-9][0-9] or [1-9][0-9][0-9][0-9] or [1-9][0-9][0-9][0-9][0-9] or [1-9][0-9][0-9][0-9][0-9] or [1-9][0-9][0-9][0-9][0-9][0-9] or [1-9][0-9][0-9][0-9][0-9][0-9] or [1-9][0-9][0-9][0-9][0-9][0-9][0-9] or [1-9][0-9][0-9][0-9][0-9][0-9][0-9][0-9] or [1-3][0-9][0-9][0-9][0-9][0-9][0-9][0-9][0-9] or 4[0-1][0-9][0-9][0-9][0-9][0-9][0-9][0-9] or 42[0-8][0-9][0-9][0-9][0-9][0-9][0-9][0-9] or 429[0-3][0-9][0-9][0-9][0-9][0-9][0-9] or 4294[0-8][0-9][0-9][0-9][0-9][0-9][0-9] or 42949[0-5][0-9][0-9][0-9][0-9][0-9] or 429496[0-6][0-9][0-9][0-9][0-9] or 4294967[0-1][0-9][0-9][0-9] or 42949672[0-8][0-9] or 429496729[0-5])) or ((([0-9] or [1-9][0-9] or 1[0-9][0-9] or 2[0-4][0-9] or 25[0-5]);{3}([0-9] or [1-9][0-9] or 1[0-9][0-9] or 2[0-4][0-9] or 25[0-5]);([1-9] or [1-9][0-9] or [1-9][0-9][0-9] or [1-9][0-9][0-9][0-9] or [1-5][0-9][0-9][0-9][0-9] or 6[0-4][0-9][0-9][0-9] or 65[0-4][0-9][0-9] or 655[0-2][0-9] or 6553[0-5])) or ((([1-9] or [1-9][0-9] or [1-9][0-9][0-9] or [1-9][0-9][0-9][0-9] or [1-9][0-9][0-9][0-9][0-9] or [1-9][0-9][0-9][0-9][0-9][0-9] or [1-9][0-9][0-9][0-9][0-9][0-9][0-9] or [1-9][0-9][0-9][0-9][0-9][0-9][0-9][0-9] or [1-3][0-9][0-9][0-9][0-9][0-9][0-9][0-9] or 4[0-1][0-9][0-9][0-9][0-9][0-9][0-9][0-9] or 42[0-8][0-9][0-9][0-9][0-9][0-9][0-9] or 429[0-3][0-9][0-9][0-9][0-9][0-9][0-9] or 4294[0-8][0-9][0-9][0-9][0-9][0-9] or 42949[0-5][0-9][0-9][0-9][0-9] or 429496[0-6][0-9][0-9][0-9][0-9] or 4294967[0-1][0-9][0-9][0-9] or 42949672[0-8][0-9] or 429496729[0-5]);([1-9] or [1-9][0-9] or [1-9][0-9][0-9] or [1-9][0-9][0-9][0-9] or [1-5][0-9][0-9][0-9][0-9] or 6[0-4][0-9][0-9][0-9] or 65[0-4][0-9][0-9] or 655[0-2][0-9] or 6553[0-5]))		Route Distinguisher - ':' or '::'. Cannot be modified once VRF has been created
mpls-exp	uint32	0..7		

Route Target

Route Target configuration

XPath: [cna-routing:/routing/network-context/bgp/vrf/routeTarget](#)

CLI Path: [routing network-context bgp vrf route-target](#)

List Properties:

Keys: direction, value

Maximum number of elements: unbounded

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
direction	enumeration	import export both		Filter direction - 'import', 'export' or 'both'
value	string	((([1-9] or [1-9][0-9] or [1-9][0-9][0-9] or [1-9][0-9][0-9][0-9] or [1-5][0-9][0-9][0-9][0-9] or 6[0-4][0-9][0-9][0-9] or 65[0-4][0-9][0-9] or 655[0-2][0-9] or 6553[0-5]);([1-9] or [1-9][0-9] or [1-9][0-9][0-9] or [1-9][0-9][0-9][0-9] or [1-9][0-9][0-9][0-9][0-9] or [1-9][0-9][0-9][0-9][0-9][0-9] or [1-9][0-9][0-9][0-9][0-9][0-9][0-9] or [1-9][0-9][0-9][0-9][0-9][0-9][0-9][0-9] or [1-3][0-9][0-9][0-9][0-9][0-9][0-9][0-9] or 4[0-1][0-9][0-9][0-9][0-9][0-9][0-9][0-9] or 42[0-8][0-9][0-9][0-9][0-9][0-9][0-9] or 429[0-3][0-9][0-9][0-9][0-9][0-9][0-9] or 4294[0-8][0-9][0-9][0-9][0-9][0-9][0-9] or 42949[0-5][0-9][0-9][0-9][0-9] or 429496[0-6][0-9][0-9][0-9][0-9] or 4294967[0-1][0-9][0-9][0-9] or 42949672[0-8][0-9] or 429496729[0-5])) or ((([0-9] or [1-9][0-9] or 1[0-9][0-9] or 2[0-4][0-9] or 25[0-5]);{3}([0-9] or [1-9][0-9] or 1[0-9][0-9] or 2[0-4][0-9] or 25[0-5]);([1-9] or [1-9][0-9] or [1-9][0-9][0-9] or [1-9][0-9][0-9][0-9] or [1-5][0-9][0-9][0-9][0-9] or 6[0-4][0-9][0-9][0-9] or 65[0-4][0-9][0-9] or 655[0-2][0-9] or 6553[0-5])) or ((([1-9] or [1-9][0-9] or [1-9][0-9][0-9] or [1-9][0-9][0-9][0-9] or [1-9][0-9][0-9][0-9][0-9] or [1-9][0-9][0-9][0-9][0-9][0-9] or [1-9][0-9][0-9][0-9][0-9][0-9][0-9] or [1-9][0-9][0-9][0-9][0-9][0-9][0-9][0-9] or [1-3][0-9][0-9][0-9][0-9][0-9][0-9][0-9] or 4[0-1][0-9][0-9][0-9][0-9][0-9][0-9][0-9] or 42[0-8][0-9][0-9][0-9][0-9][0-9][0-9] or 429[0-3][0-9][0-9][0-9][0-9][0-9][0-9] or 4294[0-8][0-9][0-9][0-9][0-9][0-9] or 42949[0-5][0-9][0-9][0-9][0-9] or 429496[0-6][0-9][0-9][0-9][0-9] or 4294967[0-1][0-9][0-9][0-9] or 42949672[0-8][0-9] or 429496729[0-5]);([1-9] or [1-9][0-9] or [1-9][0-9][0-9] or [1-9][0-9][0-9][0-9] or [1-5][0-9][0-9][0-9][0-9] or 6[0-4][0-9][0-9][0-9] or 65[0-4][0-9][0-9] or 655[0-2][0-9] or 6553[0-5]))		Route Target - ':' or '::'

Import Map

Route map to filter routes imported into VRF. Applied after import route targets.(Needs hard clear of bgp sessions to take effect)

XPath: [cna-routing:/routing/network-context/bgp/vrf/importMap](#)

CLI Path: [routing network-context bgp vrf import-map](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Name of Route map filter. Leaf Reference Path:
ignore-route-target-import	empty			Ignore Route Target Import and exclusively use Import-Map to filter routes. (Needs hard clear of bgp sessions to take effect)

Ipv4 Ucast Af

IPv4 Address Family

XPath: [cna-routing:/routing/network-context/bgp/vrf/ipv4Ucast](#)

CLI Path: [routing network-context bgp vrf ipv4-ucast-af](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
mpls-exp	uint32	0..7		

Redistribute

Redistribution configuration

XPath: [cna-routing:/routing/network-context/bgp/vrf/ipv4Ucast/redistribute](#)

CLI Path: [routing network-context bgp vrf ipv4-ucast-af redistribute](#)

List Properties:

Keys: protocol

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
protocol	enumeration	connected static		Protocol type to redistribute into BGP

Ipv6 Ucast Af

IPv6 Address Family

XPath: [cna-routing:/routing/network-context/bgp/vrf/ipv6Ucast](#)

CLI Path: [routing network-context bgp vrf ipv6-ucast-af](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
mpls-exp	uint32	0..7		

Redistribute

Redistribution configuration

XPath: [cna-routing:/routing/network-context/bgp/vrf/ipv6Ucast/redistribute](#)

CLI Path: [routing network-context bgp vrf ipv6-ucast-af redistribute](#)

List Properties:

Keys: protocol

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
protocol	enumeration	connected static		Protocol type to redistribute into BGP

Bfd Session Profile

Configure BFD parameters

XPath: *cna-routing:/routing/bfdProfile*

CLI Path: *routing bfd-session-profile*

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{1,64})		BFD Profile name - type is <alphanumeric, '_' is allowed>
tx-interval	uint32	100 .. 4294967		
rx-interval	uint32	100 .. 4294967		
multiplier	uint32	1 .. 255		

Policy

Manage routing policies

XPath: *cna-routing:/routing/policy*

CLI Path: *routing policy*

Prefix List

Configure an ipv4 prefix list

XPath: *cna-routing:/routing/policy/prefixList*

CLI Path: *routing policy prefix-list*

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{1,64})		Ipv4 Prefix list name - type is <alphanumeric, '_' is allowed>
description	string	(a-zA-Z{1,64})		Ipv4 Prefix list description - type is <alphanumeric, '_' is allowed>

Line

<1-2147483647>; Sequence Number
Ipv4 prefix list entry; sequence number <1-2147483647>, action, ipv4 address, le and ge ranges <1..32>
ge should be greater than prefix-length and less than le
le should be greater than prefix-length and ge

XPath: *cna-routing:/routing/policy/prefixList/line*

CLI Path: *routing policy prefix-list line*

List Properties:

Keys: number

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
sequence-number	uint32	1 .. 2147483647		
action	enumeration	deny permit		Action:
address	inet:ip-address			Configure the ip of the prefix
prefix-length	uint32	0 .. 32		Configure the prefix length
le	uint32	0 .. 32	0	LE range type <0..32> (Value must be greater than prefix-length)
ge	uint32	0 .. 32	0	GE range type <0..32> (Value must be greater than prefix-length)

Prefix List Ipv6

Configure an ipv6 prefix list

XPath: [cna-routing:/routing/policy/prefixListIpv6](#)

CLI Path: [routing policy prefix-list-ipv6](#)

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{1,64})		Ipv6 Prefix list name - type is <alphanumeric, '_' is allowed>
description	string	(a-zA-Z{1,64})		Ipv6 Prefix list description - type is <alphanumeric, '_' is allowed>

Line

<1-2147483647>; Sequence Number

Ipv6 prefix list entry; sequence number <1-2147483647>, action, ipv6 address, le and ge ranges <1..128>

ge should be greater than prefix-length and less than le

le should be greater than prefix-length and ge

XPath: [cna-routing:/routing/policy/prefixListIpv6/line](#)

CLI Path: [routing policy prefix-list-ipv6 line](#)

List Properties:

Keys: number

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
sequence-number	uint32	1 .. 2147483647		
action	enumeration	deny permit		Action:
address	inet:ip-address			Configure the ip of the prefix
prefix-length	uint32	0 .. 128		Configure the prefix length
le	uint32	0 .. 128	0	LE range type <0..128> (Value must be greater than prefix-length)
ge	uint32	0 .. 128	0	GE range type <0..128> (Value must be greater than prefix-length)

Comm List

Configure a Community List <1-99> to match BGP Standard Community Attributes

Configure a Community List <1-99> to match BGP Standard Community Attributes

XPath: [cna-routing:/routing/policy/commList](#)

CLI Path: [routing policy comm-list](#)

List Properties:

Keys: id

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
id	uint32	1 .. 99		
description	string			

Line

community num | AN:NN | local-AS | no-advertise | no-export

Community List Entry <1-99>: community num | AN:NN | local-AS | no-advertise | no-export, action

XPath: [cna-routing:/routing/policy/commList/line](#)

CLI Path: [routing policy comm-list line](#)

List Properties:

Keys: expr

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
action	enumeration	deny permit		Action:
expr	string	((internet) or (local-AS) or (no-advertise) or (no-export) or (\d+) or ((\d+:\d+) or (\d+.\d+.\d+.\d+)(\d+:\d+) or (\d+)))((internet) or (local-AS) or (no-advertise) or (no-export) or ((\d+:\d+) or (\d+.\d+.\d+.\d+)(\d+:\d+) or (\d+)))		

Comm List Ext

Configure an Extended Community List <100-199> to match BGP Extended Community Attributes

Configure an Extended Community List <100-199> to match BGP Extended Community Attributes

XPath: [cna-routing:/routing/policy/commListExt](#)

CLI Path: [routing policy comm-list-ext](#)

List Properties:

Keys: id

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
id	uint32	100 .. 199		
description	string			

Line

Regular Expression

Extended Community List Entry <100-199>: action, regular expression

XPath: [cna-routing:/routing/policy/commListExt/line](#)

CLI Path: [routing policy comm-list-ext line](#)

List Properties:

Keys: expr

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
action	enumeration	deny permit		Action:
expr	string			

Comm List Word

Configure a named Community List to match BGP Standard Community Attributes

Configure a named Community List to match BGP Standard Community Attributes

XPath: [cna-routing:/routing/policy/commListWord](#)

CLI Path: [routing policy comm-list-word](#)

List Properties:

Keys: id

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
id	string			
description	string			

Line

community num | AN:NN | local-AS | no-advertise | no-export
Community List Entry : community num | AN:NN | local-AS | no-advertise | no-export, action

XPath: [cna-routing:/routing/policy/commListWord/line](#)

CLI Path: [routing policy comm-list-word line](#)

List Properties:
Keys: expr

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
action	enumeration	deny permit		Action:
expr	string	((internet) or (local-AS) or (no-advertise) or (no-export) or (\d+) or ((\d+:\d+) or (\d+.\d+.\d+.\d+)(\d+:\d+) or (\d+)))((internet) or (local-AS) or (no-advertise) or (no-export) or ((\d+:\d+) or (\d+.\d+.\d+.\d+)(\d+:\d+) or (\d+)))*		

Comm List Word Ext

Configure a Named Extended Community List to match BGP Extended Community Attributes
Configure a Named Extended Community List to match BGP Extended Community Attributes

XPath: [cna-routing:/routing/policy/commListWordExt](#)

CLI Path: [routing policy comm-list-word-ext](#)

List Properties:
Keys: id

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
id	string			
description	string			

Line

Regular Expression
Extended Community List Entry : action, regular expression

XPath: [cna-routing:/routing/policy/commListWordExt/line](#)

CLI Path: [routing policy comm-list-word-ext line](#)

List Properties:
Keys: expr

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
action	enumeration	deny permit		Action:
expr	string			

Route Map

Create route-map

XPath: [cna-routing:/routing/policy/routeMap](#)

CLI Path: [routing policy route-map](#)

List Properties:
Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{1,64})		Route-map tag - type is <alphanumeric, '_' is allowed>
continue	empty			Enable/continue matching all route-map match rules (significant when >1 match rules present)

Routemaprule

Create route-map rule

XPath: [cna-routing:/routing/policy/routeMap/routeMapRule](#)

CLI Path: [routing policy route-map routeMapRule](#)

List Properties:

Keys: action, number

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
action	enumeration	deny permit		
number	uint32	1 .. 65535		Sequence number

Matchtable

Defines the match clauses for this route map entry

XPath: [cna-routing:/routing/policy/routeMap/routeMapRule/matchTable](#)

CLI Path: [routing policy route-map routeMapRule matchTable](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
match-ip-address-prefix-list	leafref			IPv4 prefix list match by name Leaf Reference Path:
match-ipv6-address-prefix-list	leafref			IPv6 prefix list match by name Leaf Reference Path:
match-ip-next-hop	inet:ipv4-address			Match ipv4 address of next hop (Only applicable to route map in)
match-ipv6-next-hop	inet:ipv6-address			Match ipv6 address of next hop (Only applicable to route map in)
match-tag	uint32			Match tag of route
match-metric	uint32			Match metric of route

Match Community List Types

Different community list match types

XPath: [cna-routing:/routing/policy/routeMap/routeMapRule/matchTable/containerCommListType](#)

CLI Path: [routing policy route-map routeMapRule matchTable match-community-list-types](#)

Commlistreftype (choice)

XPath: [cna-routing:/routing/policy/routeMap/routeMapRule/matchTable/containerCommListType](#)

***CLI Path: routing policy route-map routeMapRule matchTable match-community-list-types ***

Case : commList

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
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Attribute	Type	Range/Length/Pattern	Default	Description
match-comm-list	leafref			Match BGP community list by ID <1-99> Choice: This match-comm-list attribute is mutually exclusive with attributes [['commListExt'], ['commListWord'], ['commListWordExt']] Leaf Reference Path: cna-routing:/routing/policy/commList/id

Case : commListExt

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
match-comm-list-ext	leafref			Match BGP community list extended by ID <100-199> Choice: This match-comm-list-ext attribute is mutually exclusive with attributes [['commList'], ['commListWord'], ['commListWordExt']] Leaf Reference Path: cna-routing:/routing/policy/commListExt/id

Case : commListWord

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
match-comm-list-word	leafref			Match BGP community list by name Choice: This match-comm-list-word attribute is mutually exclusive with attributes [['commList'], ['commListExt'], ['commListWordExt']] Leaf Reference Path: cna-routing:/routing/policy/commListWord/id

Case : commListWordExt

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
match-comm-list-word-ext	leafref			Match BGP community list extended by name Choice: This match-comm-list-word-ext attribute is mutually exclusive with attributes [['commList'], ['commListExt'], ['commListWord']] Leaf Reference Path: cna-routing:/routing/policy/commListWordExt/id

Settable

Defines the set clauses for this route map entry

XPath: [cna-routing:/routing/policy/routeMap/routeMapRule/setTable](#)

CLI Path: [routing policy route-map routeMapRule setTable](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
set-ip-next-hop	inet:ipv4-address			Set ipv4 address of next hop
set-ipv6-next-hop	inet:ipv6-address			Set ipv6 address of next hop
set-vpnv4-next-hop	inet:ipv4-address			Set the next hop address for vpnv4 routes
set-local-preference	uint32			Set BGP local preference path attribute
set-tag	uint32			Set tag value for destination routing protocol

Set Community

BGP community attribute

XPath: [cna-routing:/routing/policy/routeMap/routeMapRule/setTable/setCommunity](#)

CLI Path: [routing policy route-map routeMapRule setTable set-community](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
additive	empty			Add to the existing community

Community Choice (choice)

XPath: [cna-routing:/routing/policy/routeMap/routeMapRule/setTable/setCommunity](#)

CLI Path: [routing policy route-map routeMapRule setTable set-community](#)

Case : community-number-case

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
community-number	uint32	1..4294967295		Choice: This community-number attribute is mutually exclusive with attributes [['well-known-communities']]
community-number-alt-format	string	((\d+:\d+) or ((\d+:\d+) (\d+:\d+)) or ((\d+:\d+) (\d+:\d+) (\d+:\d+)) or ((\d+:\d+) (\d+:\d+) (\d+:\d+) (\d+:\d+)) or ((\d+:\d+) (\d+:\d+) (\d+:\d+)))		Choice: This community-number-alt-format attribute is mutually exclusive with attributes [['well-known-communities']]

Case : well-known-community-case

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
well-known-communities	enumeration	internet local-AS no-advertise no-export none		Choice: This well-known-communities attribute is mutually exclusive with attributes [['community-number', 'community-number-alt-format']]

Set Ext Community

Configure a Named Extended Community

Configure a Named Extended Community

XPath: [cna-routing:/routing/policy/routeMap/routeMapRule/setTable/setExtCommunity](#)

CLI Path: [routing policy route-map routeMapRule setTable set-ext-community](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
rt	string	((\d+:\d+) or \d+.\d+.\d+.\d+:\d+)(\d+:\d+) or (\d+)))		
soo	string	((\d+:\d+) or \d+.\d+.\d+.\d+:\d+)(\d+:\d+) or (\d+)))		

SLICE INV MGR Configuration

The module for entities implementing cna-slice-inv-mgr.

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 - [network-slice-instance-list](#)
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- [network-subslice-instance-list](#)
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- [serving-area-list](#)
 - [network-subslice-instance-list](#)
- [serving-nrf-list](#)
- [amf-set-list](#)

Slice Inv Mgr

Slice Inv Mgr. See 3GPP specification TS29.510 for more information

XPath: [cna-slice-inv-mgr:/slice-inv-mgr](#)

CLI Path: [slice-inv-mgr](#)

Network Slice Instance List

Network slice instance list

XPath: [cna-slice-inv-mgr:/slice-inv-mgr/network-slice-instance-list](#)

CLI Path: [slice-inv-mgr network-slice-instance-list](#)

List Properties:
Keys: slice-instance
Minimum number of elements: 0
Maximum number of elements: 1000

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
slice-instance	string	[A-Za-z0-9 '() +, -./:=?]*		Unique identity of the Network Slice Instance
serving-nrf	leafref			Network Function Repository Function (NRF) that provides services to this network slice instance Leaf Reference Path: cna-slice-inv-mgr:/slice-inv-mgr/serving-nrf-list/nrf-name
serving-amf-set-id	leafref			Access and Mobility Management Function (AMF) set that serves this network slice instance Leaf Reference Path: cna-slice-inv-mgr:/slice-inv-mgr/amf-set-list/amf-set-id

Network Subslice Instance List

Network Subslice Instance list

XPath: [cna-slice-inv-mgr:/slice-inv-mgr/network-slice-instance-list/network-subslice-instance-list](#)

CLI Path: [slice-inv-mgr network-slice-instance-list network-subslice-instance-list](#)

List Properties:
Keys: subslice-instance
Minimum number of elements: 1
Maximum number of elements: 128

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
subslice-instance	leafref			Unique identity of the Network Subslice Instance Leaf Reference Path: cna-slice-inv-mgr:/slice-inv-mgr/network-subslice-instance-list/subslice-instance

Network Subslice Instance List

Network Subslice Instance list

XPath: [cna-slice-inv-mgr:/slice-inv-mgr/network-subslice-instance-list](#)

CLI Path: [slice-inv-mgr network-subslice-instance-list](#)

List Properties:
Keys: subslice-instance
Minimum number of elements: 0
Maximum number of elements: 1000

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
subslice-instance	string	[A-Za-z0-9 '()+,.-:/:=?]*		Unique identity of the Network Subslice Instance

S Nssai List

S-NSSAI list

Value 0xFFFFFFFF is reserved and cannot be used to configure SD

XPath: *cna-slice-inv-mgr:/slice-inv-mgr/network-subslice-instance-list/s-nssai-list*

CLI Path: *slice-inv-mgr network-subslice-instance-list s-nssai-list*

List Properties:

Keys: sst, sd

Minimum number of elements: 1

Maximum number of elements: 128

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
sst	union	['uint8,0 or 4 .. 255', 'Enumerations: eMBB URLLC MIoT']		Identifies the expected Network Slice behavior in terms of features and services
sd	union	['string,(0x[0-9a-fA-F]{6})', 'Enumerations: not-specified']		Differentiate amongst multiple Network Slices of the same Slice/Service type

Serving Area List

Serving areas of the network subslice instances

If 'tac-low' is set to none then 'tac-high' must also be set to none

XPath: *cna-slice-inv-mgr:/slice-inv-mgr/serving-area-list*

CLI Path: *slice-inv-mgr serving-area-list*

List Properties:

Keys: plmn-id, tac-low, tac-high

Minimum number of elements: 0

Maximum number of elements: 128000

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
plmn-id	string	((([0-9]{5,6}))		PLMN ID
tac-low	union	['string,0x[0-9a-f]{4,6}', 'Enumerations: none']		TAC start value
tac-high	union	['string,0x[0-9a-f]{4,6}', 'Enumerations: none']		TAC end value

Network Subslice Instance List

list of network-subslice-instance identity

XPath: *cna-slice-inv-mgr:/slice-inv-mgr/serving-area-list/network-subslice-instance-list*

CLI Path: *slice-inv-mgr serving-area-list network-subslice-instance-list*

List Properties:

Keys: subslice-instance

Minimum number of elements: 1

Maximum number of elements: 32

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
subslice-instance	leafref			Leaf Reference Path: cna-slice-inv-mgr:/slice-inv-mgr/network-subslice-instance-list/subslice-instance

Serving Nrf List

List of Network Function Repository Functions (NRF)s that provide services to the network slices defined by this network

XPath: [cna-slice-inv-mgr:/slice-inv-mgr/serving-nrf-list](#)

CLI Path: [slice-inv-mgr serving-nrf-list](#)

- List Properties:**
Keys: nrf-name
Minimum number of elements: 0
Maximum number of elements: 100

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
nrf-name	string			NRF unique name string
nf-discovery-uri	string			API URI of NRF NF Discovery services as defined in the 3GPP TS 29.510 section 6.2.1
nf-management-uri	string			API URI of NRF NF Management services as defined in the 3GPP TS 29.510 section 6.1.1
access-token-uri	string			API URI of NRF Access Token services as defined in the 3GPP TS 29.510 section 6.3.2

Amf Set List

List of Access and Mobility Management Function (AMF) set data

XPath: [cna-slice-inv-mgr:/slice-inv-mgr/amf-set-list](#)

CLI Path: [slice-inv-mgr amf-set-list](#)

- List Properties:**
Keys: amf-set-id
Minimum number of elements: 0
Maximum number of elements: 100

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
amf-set-id	string	([0-9]{3}-[0-9]{2,3}-[A-Fa-f0-9]{2}-[0-3][A-Fa-f0-9]{2})		AMF set identifier which includes PLMN ID (MCC-MNC), AMF region ID (8 bit) and AMF set ID (10 bit)
nrf-nf-discovery-uri	string			API URI of NRF NF Discovery services as defined in the 3GPP TS 29.510 section 6.2.1
nrf-nf-management-uri	string			API URI of NRF NF Management services as defined in the 3GPP TS 29.510 section 6.1.1
nrf-access-token-uri	string			API URI of NRF Access Token services as defined in the 3GPP TS 29.510 section 6.3.2
amf-id-list	leaf-list of string	[0-9a-fA-F]{8}-[0-9a-fA-F]{4}-[0-9a-fA-F]{4}-[0-9a-fA-F]{4}-[0-9a-fA-F]{12}		List of serving AMF instance identifiers Maximum number of elements: 10

SLICE SELECTOR Configuration

The module for entities implementing slice-selector

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Slice Selector

Network Slice Selection Services. See 3GPP specification TS29.531 for more information

XPath: [cna-slice-selector:/slice-selector](#)

CLI Path: [slice-selector](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
analyzer-name	string	([0-9a-zA-Z_-]*)		Analyzer policy instance name
nrf-registration-enable	boolean		true	Enable NSSF profile and service registration with the NRF; When the value changes from true to false, NF will be deregistered from the NRF and federation will be restarted. Default: true

Restricted Tracking Areas

A list of restricted serving tracking areas
If 'tac-low' is set to none then 'tac-high' must also be set to none

XPath: [cna-slice-selector:/slice-selector/restricted-tracking-areas](#)

CLI Path: [slice-selector restricted-tracking-areas](#)

List Properties:
Keys: plmn-id, tac-low, tac-high
Minimum number of elements: 0
Maximum number of elements: 10000

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
plmn-id	string	(((0-9){5,6}))		PLMN ID
tac-low	union	['string,0x[0-9a-fA-F]{4,6}', 'Enumerations: none']		TAC start value
tac-high	union	['string,0x[0-9a-fA-F]{4,6}', 'Enumerations: none']		TAC end value

Restricted S Nssai List

A list of restricted S-NSSAI for a specific tracking area
Value 0xFFFFFFFF is reserved and cannot be used to configure SD

XPath: [cna-slice-selector:/slice-selector/restricted-tracking-areas/restricted-s-nssai-list](#)

CLI Path: [slice-selector restricted-tracking-areas restricted-s-nssai-list](#)

List Properties:
Keys: sst, sd
Minimum number of elements: 1
Maximum number of elements: 128

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
sst	union	['uint8,0 or 4 .. 255', 'Enumerations: eMBB URLLC MlIoT ']		Identifies the expected Network Slice behavior in terms of features and services
sd	union	['string,(0x[0-9a-fA-F]{6})', 'Enumerations: not-specified ']		Differentiate amongst multiple Network Slices of the same Slice/Service type

Registration And Uecu Policy

List of UE Registration and UE Configuration Update policies in the slice selector

At least one 'local-config-s-nssais' entry must be configured when 'policy-action' is set to use-local-config-nssai-as-allowed.

XPath: [cna-slice-selector:/slice-selector/registration-policy](#)

CLI Path: [slice-selector registration-and-uecu-policy](#)

List Properties:

Keys: policy-name

Minimum number of elements: 0

Maximum number of elements: 10000

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
policy-name	union	['string,([0-9a-zA-Z_-])**', 'Enumerations: default ']		Unique identity of the selection policy
policy-action	enumeration	match-subscribed-with-requested-nssai use-local-config-nssai-as-allowed		Policy action
allowed-nssai-max-size	uint8	1..8	8	The maximum number of Allowed S-NSSAI allowed. Default: 8
amf-reroute-type	enumeration	rerouteNotAllowed useAmfSetId useAmfIdList	rerouteNotAllowed	Specifies which AMF reroute data to use when AMF reroute is required. Default: rerouteNotAllowed
include-nsiInformation	boolean		false	Controls the inclusion of the NsiInformation IE in the AllowedSnsai IE in the Slice Selection response message as defined in the 3GPP TS 29.531 section 6.1.6.2.5. Default: false

Local Config S Nssais

List of local configured S-NSSAIs

Value 0xFFFFFFFF is reserved and cannot be used to configure SD

XPath: [cna-slice-selector:/slice-selector/registration-policy/local-config-s-nssais](#)

CLI Path: [slice-selector registration-and-uecu-policy local-config-s-nssais](#)

List Properties:

Keys: sst, sd

Maximum number of elements: 8

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
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Attribute	Type	Range/Length/Pattern	Default	Description
sst	union	['uint8,0 or 4 .. 255', 'Enumerations: eMBB URLLC MIoT']		SST (Slice Service Type) value for S-NSSAI
sd	union	['string,(0x[0-9a-fA-F]{6})', 'Enumerations: not-specified']		SD (Slice Differentiator) value for S-NSSAI

Pdu Establishment Policy

List of PDU session establishment policies in the slice selector

XPath: [cna-slice-selector:/slice-selector/pdu-establishment-policy](#)

CLI Path: [slice-selector pdu-establishment-policy](#)

List Properties:

Keys: policy-name

Minimum number of elements: 0

Maximum number of elements: 10000

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
policy-name	union	['string,([0-9a-zA-Z_-]*)', 'Enumerations: default']		Unique identity of the policy
nsi-id	string			Network Slice Instance Identifier (NSI) selected for the PDU establishment service
include-nsi-id	boolean		true	Controls the inclusion of the NSI ID in the NsiInformation IE in the response message as defined in the 3GPP TS 29.531 section 6.1.6.2.7

Nssf Profile

NSSF profile

XPath: [cna-slice-selector:/slice-selector/nssf-profile](#)

CLI Path: [slice-selector nssf-profile](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
nf-instance-id	string	[0-9a-fA-F]{8}-[0-9a-fA-F]{4}-[0-9a-fA-F]{4}-[0-9a-fA-F]{4}-[0-9a-fA-F]{12}		Unique identity of the NSSF in the form of a uuid
heart-beat-timer	uint32	10 .. 86400	30	Time expected between 2 consecutive heart-beat between NSSF and NRF. Uint: second; Default: 30
plmn-list	leaf-list of string	(([0-9]{5,6}))		Supported Public Land Mobile Network (PLMN); Range: numbers with 5-6 digits. If not provided, PLMNID(s) of the PLMN of the NRF are assumed for the NSSF. Minimum number of elements: 0 Maximum number of elements: 16
inter-plmn-fqdn	inet:domain-name			An FQDN that is used for inter PLMN routing
priority	uint32	0..65535	0	Priority (relative to other NFs of the same type).Lower the value has the higher priority. Default: 0
capacity	uint32	0..65535	65535	A weight relative to other NSSF instances. Higher the value has the higher capacity. Default: 65535

Attribute	Type	Range/Length/Pattern	Default	Description
locality	string	1..64		Location of the NSSF

Services

NSSF Services Information. See 3GPP specification TS29.531 for more information.

XPath: [cna-slice-selector:/slice-selector/nssf-profile/services](#)

CLI Path: [slice-selector nssf-profile services](#)

List Properties:

Keys: service-instance-id
Minimum number of elements: 1
Maximum number of elements: 2

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
service-instance-id	string	1..64		Unique ID of the service instance within a given NF Instance
service-name	enumeration	nssf-nssaiavailability nssf-nsseselection		Name of service instance that the NRF will be used for
version	enumeration	16.4.0	16.4.0	3GPP TS 29.531 version supported for the service. Default: 16.4.0

Trace

Trace objects

XPath: [cna-slice-selector:/slice-selector/trace](#)

CLI Path: [slice-selector trace](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
default-probability	string		0.0000000001	Probability of sampling the trace for a non-target; valid values are 0.0 to 1.0. Default: 0.0000000001 The 'probability' value must be less than 1.0

Targets

List of trace targets
One and only one of nf-instance-id or service-type or tracking-area or s-nssai must be set

XPath: [cna-slice-selector:/slice-selector/trace/targets](#)

CLI Path: [slice-selector trace targets](#)

List Properties:

Keys: profile-name
Maximum number of elements: 100

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
profile-name	string			Target profile name
nf-instance-id	string	[0-9a-fA-F]{8}-[0-9a-fA-F]{4}-[0-9a-fA-F]{4}-[0-9a-fA-F]{4}-[0-9a-fA-F]{12}		Unique identity of an NF in the form of a uuid
service-type	enumeration	sliceSelectUeReg sliceSelectPduEst nssaiAvailUpdate nssaiAvailSub sliceSelectUeConfigUpdate		Type of service provided by NSSF 'nssaiAvailSub' is not supported

Attribute	Type	Range/Length/Pattern	Default	Description
probability	string		1	Probability of sampling the trace for the target; valid values are 0.0 to 1.0. Default: 1 The 'probability' value must be less than 1.0

Tracking Area

Tracking area. This target applies to the Slice Selection service only.

XPath: [cna-slice-selector:/slice-selector/trace/targets/tracking-area](#)

CLI Path: [slice-selector trace targets tracking-area](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
plmn-id	string	(([0-9]{5,6}))		PLMN ID
tac	string	0x[0-9a-fA-F]{4,6}		Tracking Area Code (TAC); Valid values are hexadecimal values of length 4 or 6 prefix with '0x'

S Nssai

Single Network Slice Selection Assistance Information. This target applies to the Slice Selection service only.
Value 0xFFFFFFFF is reserved and cannot be used to configure SD

XPath: [cna-slice-selector:/slice-selector/trace/targets/s-nssai](#)

CLI Path: [slice-selector trace targets s-nssai](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
sst	union	['uint8,0 or 4 .. 255', 'Enumerations: eMBB URLLC MIoT']		SST (Slice Service Type) value for S-NSSAI
sd	union	['string,(0x[0-9a-fA-F]{6})', 'Enumerations: not-specified']		SD (Slice Differentiator) value for S-NSSAI

Service Resource

List of service names which supports Content/Accept encoding

XPath: [cna-slice-selector:/slice-selector/service-resource](#)

CLI Path: [slice-selector service-resource](#)

List Properties:

Keys: name

Minimum number of elements: 0

Maximum number of elements: 1

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	enumeration	nssf-nssaiavailability		service names
encoding-type	enumeration	gzip	gzip	Type of encoding supported; default gzip
payload-size	uint32	50..2000		Maximum payload size (before compression, if any), expressed in kilo octets.

Nssf Event Subscription

NSSF event subscription parameter

XPath: [cna-slice-selector:/slice-selector/nssf-event-subscription](#)

CLI Path: [slice-selector nssf-event-subscription](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
validity-time	uint32	1..34560		Validity time duration that the subscription will be valid for if the client did not specify the subscription expiry time in the NSSF event subscription request. If this is not configured and the client did not specify the expiry time in the request, then the subscription will never expire. Unit: minute

SMFMONITOR Configuration

The module for entities implementing cna-smfmonitor.

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Smfmonitor

SMF Monitor interface object

XPath: *cna-smfmonitor:/smfmonitor*

CLI Path: *smfmonitor*

Load Profile

Configure SMF Load Profile

XPath: *cna-smfmonitor:/smfmonitor/load-profile*

CLI Path: *smfmonitor load-profile*

List Properties:

Keys: load-profile

Maximum number of elements: 1

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
load-profile	string			SMF Load Profile Name
reporting-frequency-interval	uint64	60..36000	120	Frequence interval(seconds) at which the SMF load should be reported to NRF
reporting-frequency-change-factor	uint64			Load change factor(+/- load difference) which should result in reporting the SMF load to the NRF

Threshold Level

Configuration of the NF load threshold level parameters

XPath: *cna-smfmonitor:/smfmonitor/load-profile/threshold-level*

CLI Path: *smfmonitor load-profile threshold-level*

Minor

Configuration of the NF load minor threshold level parameters

XPath: *cna-smfmonitor:/smfmonitor/load-profile/threshold-level/minor*

CLI Path: *smfmonitor load-profile threshold-level minor*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
enter-load-value	uint8	1..100		The value of NF load from which the given threshold level will be applied. Both enter-load-value and exit-load-value must be configured
exit-load-value	uint8	1..100		The value of NF load below which the applied threshold level will be removed. Both enter-load-value and exit-load-value must be configured

Major

Configuration of the NF load major threshold level parameters

XPath: *cna-smfmonitor:/smfmonitor/load-profile/threshold-level/major*

CLI Path: *smfmonitor load-profile threshold-level major*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
enter-load-value	uint8	1..100		The value of NF load from which the given threshold level will be applied. Both enter-load-value and exit-load-value must be configured
exit-load-value	uint8	1..100		The value of NF load below which the applied threshold level will be removed. Both enter-load-value and exit-load-value must be configured

Critical

Configuration of the NF load critical threshold level parameters

XPath: *cna-smfmonitor:/smfmonitor/load-profile/threshold-level/critical*

CLI Path: *smfmonitor load-profile threshold-level critical*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
enter-load-value	uint8	1..100		The value of NF load from which the given threshold level will be applied. Both enter-load-value and exit-load-value must be configured
exit-load-value	uint8	1..100		The value of NF load below which the applied threshold level will be removed. Both enter-load-value and exit-load-value must be configured

Service

SmfMonitor service related configuration

XPath: *cna-smfmonitor:/smfmonitor/service*

CLI Path: *smfmonitor service*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
smf_gauge_query_interval	uint64		300	Time interval in seconds for smf gauge queries
nfprofile-update-on-dnn-ip-pool-exhaust	enumeration	disabled enabled	disabled	Enable/Disable NFProfile Update on DNN IP Pool Exhaustion
nfprofile-update-on-pfcp-connection-flip	enumeration	disabled enabled	disabled	Enable/Disable NFProfile Update on PFCP Connection Flip

Critical Service Frequent Restart

Configure Critical Service frequent Restart monitoring feature
status-check-interval should be always greater than stable-interval
when monitoring is enabled, critical-flip-count should not be 0
when monitoring is enabled, threshold-flip-count should not be 0
critical-flip-count should be always greater than threshold-flip-count

XPath: *cna-smfmonitor:/smfmonitor/critical-service-frequent-restart*

CLI Path: *smfmonitor critical-service-frequent-restart*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
monitoring	enumeration	disabled enabled	disabled	Feature enabled or disabled
stable-interval	uint16	1..3600	120	The minimum runtime in seconds for a CriticalService to be considered as Stable
status-check-interval	uint16	1..3600	180	The duration in seconds after which the CriticalService status should be rechecked.
threshold-flip-count	uint16	1..1000	3	Number of times a CriticalService is found unstable after which Alarm/Alert is generated
critical-flip-count	uint16	1..1000	20	Number of times a CriticalService is found unstable after which VNFRestart is triggered

Deregistration Time

Duration(in Secs) for which Monitor waits for a Critical Service to become available

XPath: *cna-smfmonitor:/smfmonitor/deregistration-time*

CLI Path: *smfmonitor deregistration-time*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
value	uint16	1..3600	300	Duration(in Secs) for which Monitor waits for a Critical Service to become available

Critical Service Readiness Probe

Configure Critical Service Readiness monitoring feature
critical-failure-count should be always greater than threshold-failure-count

XPath: *cna-smfmonitor:/smfmonitor/critical-service-readiness-probe*

CLI Path: *smfmonitor critical-service-readiness-probe*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
monitoring	enumeration	disabled enabled	disabled	Feature enabled or disabled
check-interval	uint16	60..36000	300	The duration in seconds after which the CriticalService status should be rechecked.
threshold-failure-count	uint16	1..20000	5	Number of times a CriticalService is found NotReady after which Alarm/Alert is generated
critical-failure-count	uint16	1..20000	10	Number of times a CriticalService is found NotReady after which VNFRestart is triggered

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The module for entities implementing cna-smf

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Smf

Affirmed SMF Call-Control

XPath: *cna-smf:/smf*

CLI Path: *smf*

Service Construct

Affirmed Call-Control Service Construct

XPath: *cna-smf:/smf/service-construct*

CLI Path: *smf service-construct*

Paging Profile

paging profile construct
paging profile

XPath: *cna-smf:/smf/service-construct/paging-profile*

CLI Path: *smf service-construct paging-profile*

List Properties:

Keys: name
Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		
page-timeout	uint32	1000 .. 30000	10000	Page Timeout Value in milliseconds - the length of time to wait for a UE to be paged before discarding buffered data
max-downlink-packets-to-buffer	uint32	1 .. 20	5	Maximum number of downlink data packets to buffer per flow when UE is being paged

Attribute	Type	Range/Length/Pattern	Default	Description
arp-priority-level-threshold	uint32	1 .. 15	4	Flows with ARP Priority Level values equal to or less than this threshold value will be marked for priority use
downlink-packet-buffering-capability	boolean		false	Indicates if SMF supports buffering data packets when UE is being paged
max-downlink-packets-to-buffer-for-extended-buffering	uint32	1 .. 20	10	Maximum number of downlink data packets to buffer per flow for extended buffering when UE is being paged
downlink-buffering-duration-for-extended-buffering	uint32	10 .. 1080000	15	The length of time in seconds to buffer packets when UE is being paged

Qos Flow

qos flow service

Configure qos-flow object

uplink-mbr/downlink-mbr and uplink-gbr/downlink-gbr must be greater than zero for GBR qos flow

non-dynamic non-gbr 5qi must be 5,6,7,8,9, 69, 70, 79, 80

non-dynamic non-critical-gbr 5qi must be 1, 2, 3, 4, 65, 66, 67, 75

non-dynamic critical gbr 5qi must be 82, 83, 84, 85

XPath: [cna-smf:/smf/service-construct/qos-flow](#)

CLI Path: [smf service-construct qos-flow](#)

List Properties:

Keys: name

Maximum number of elements: 4096

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			
five-qi	uint32	0 .. 255		Quality-of-service Identifier
arp-priority-level	uint32	1 .. 15		allocation & retention priority: higher values are preferred
arp-preemption-capability	enumeration	NOT_PREEMPT MAY_PREEMPT		
arp-preemption-vulnerability	enumeration	NOT_PREEMPTABLE MAY_PREEMPTABLE		
non-gbr-flow	boolean		false	indicates if this is non-gbr qos-flow
critical-gbr-type	boolean		false	delay critical gbr type
priority-level	uint32	1 .. 127		priority [1-127]
measurement-unit	enumeration	kbits mbps gbps	kbits	rate-measurement-units: kbits/mbps/gbps
uplink-mbr	uint32	1 .. 4294967295		uplink maximum bit rate
downlink-mbr	uint32	1 .. 4294967295		downlink maximum bit rate
uplink-gbr	uint32	1 .. 4294967295		uplink guaranteed bit rate
downlink-gbr	uint32	1 .. 4294967295		downlink guaranteed bit rate
averaging-window	uint32	1 .. 4095		averaging window [1-4095] in ms
max-burst	uint32	1 .. 4095		uplink maximum burst, measurement-units: bytes[1-4095]
max-packet-loss-rate-dl	uint32			downlink maximum packet loss rate
max-packet-loss-rate-ul	uint32			up-link maximum packet loss rate
reflective-qos	boolean		false	reflective qos config option

Subscribed Qos

SubscribedDefaultQos for the subscriber

Configure qos-flow object

XPath: *cna-smf:/smf/service-construct/subscribed-qos*

CLI Path: *smf service-construct subscribed-qos*

List Properties:

Keys: name
Maximum number of elements: 4096

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			
five-qi	uint32	0 .. 255		5qi: Quality-of-service Identifier
arp-priority-level	uint32	1 .. 15		allocation & retention priority: higher values are preferred
arp-preemption-capability	enumeration	NOT_PREEMPT MAY_PREEMPT		
arp-preemption-vulnerability	enumeration	NOT_PREEMPTABLE MAY_PREEMPTABLE		
priority-level	uint32	1 .. 127		5QIPriorityLevel: priority [1-127]

Pcc Usage Monitoring Data

usage-monitoring data associated to predefined/static pcc-rules and session rules
Configure usage monitoring data object
Either volume-threshold or volume-threshold-uplink or volume-threshold-downlink must be set to non-zero value

XPath: *cna-smf:/smf/service-construct/pcc-usage-monitoring-data*

CLI Path: *smf service-construct pcc-usage-monitoring-data*

List Properties:

Keys: name
Maximum number of elements: 4096

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			
usage-monitoring-id	string			usage-monitoring id
volume-threshold	uint64	0.. max		Total volume threshold in bytes
volume-threshold-uplink	uint64	0.. max		Total uplink volume threshold in bytes
volume-threshold-downlink	uint64	0.. max		Total downlink volume threshold in bytes

Session Usage Exclude Pcc Rule Ids

Pcc Rule name to be excluded from session level usage monitoring

XPath: *cna-smf:/smf/service-construct/pcc-usage-monitoring-data/session-usage-exclude-pcc-rule-ids*

CLI Path: *smf service-construct pcc-usage-monitoring-data session-usage-exclude-pcc-rule-ids*

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			

Session Ambr

configure session-ambr service construct

XPath: *cna-smf:/smf/service-construct/session-ambr*

CLI Path: *smf service-construct session-ambr*

List Properties:

Keys: name
Maximum number of elements: 512

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			
measurement-unit	enumeration	kbps mbps gbps	kbps	rate-measurement-units: kbps/mbps/gbps
uplink-ambr	uint32			per session Uplink Aggregate Maximum Bit Rate
downlink-ambr	uint32			per session Downlink Aggregate Maximum Bit Rate

Default Qos

configure session level default qos attributes construct

XPath: [cna-smf:/smf/service-construct/default-qos](#)

CLI Path: [smf service-construct default-qos](#)

List Properties:

Keys: name
Maximum number of elements: 512

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			
qos-flow	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/qos-flow/name

Session Rule

configure session level default qos attributes construct

XPath: [cna-smf:/smf/service-construct/session-rule](#)

CLI Path: [smf service-construct session-rule](#)

List Properties:

Keys: name
Maximum number of elements: 512

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			
usage-monitoring-data	leafref			usage monitoring data - for session level monitoring Leaf Reference Path: cna-smf:/smf/service-construct/pcc-usage-monitoring-data/name

Session Ambr

session-ambr - selected from session ambr list

XPath: [cna-smf:/smf/service-construct/session-rule/session-ambr](#)

CLI Path: [smf service-construct session-rule session-ambr](#)

List Properties:

Keys: name
Minimum number of elements: 1
Maximum number of elements: 1

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/session-ambr/name

Default Qos

default-qos - selected from default-qos list

XPath: [cna-smf:/smf/service-construct/session-rule/default-qos](#)

CLI Path: [smf service-construct session-rule default-qos](#)

List Properties:

Keys: name
Minimum number of elements: 1
Maximum number of elements: 1

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/default-qos/name

Inbound Roamer Qos Profile

qos profile for inbound roamers

XPath: [cna-smf:/smf/service-construct/inbound-roamer-qos-profile](#)

CLI Path: [smf service-construct inbound-roamer-qos-profile](#)

List Properties:

Keys: name
Maximum number of elements: 16000

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			
default-session-rule	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/session-rule/name

Allowed Qos Flow

qos flow with specified 5qi that is allowed to be created for inbound roamers

XPath: [cna-smf:/smf/service-construct/inbound-roamer-qos-profile/allowed-qos-flow](#)

CLI Path: [smf service-construct inbound-roamer-qos-profile allowed-qos-flow](#)

List Properties:

Keys: name
Maximum number of elements: 512

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/qos-flow/name

Home 5Qi To Qos Flow Map

inbound roamer's home 5qi value

XPath: [cna-smf:/smf/service-construct/inbound-roamer-qos-profile/home-5qi-to-qos-flow-map](#)

CLI Path: [smf service-construct inbound-roamer-qos-profile home-5qi-to-qos-flow-map](#)

List Properties:

Keys: fiveQi

Maximum number of elements: 512

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
five-qi	uint32	0 .. 255		
mapped-qos-flow	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/qos-flow/name

Pcscf Table

Configure a list of IPV4 and IPV6 P-CSCF Addresses

XPath: [cna-smf:/smf/service-construct/pcscf-table](#)

CLI Path: [smf service-construct pcscf-table](#)

List Properties:

Keys: name
Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			P-CSCF Table name
user-description	string			Description of P-CSCF Table
address-allocation-mode	enumeration	round-robin weighted-round-robin	round-robin	Algorithm to use when allocating a pair of P-CSCF Addresses from the P-CSCF table

Ip V4 List

List of IP V4 Address pairs

XPath: [cna-smf:/smf/service-construct/pcscf-table/ip-v4-list](#)

CLI Path: [smf service-construct pcscf-table ip-v4-list](#)

List Properties:

Keys: name
Maximum number of elements: 128

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			IP4 Address Pair Name
primary-ipv4-address	inet:ipv4-address			Primary P-CSCF server IP V4 Address
secondary-ipv4-address	inet:ipv4-address			Secondary P-CSCF server IP V4 Address
tertiary-ipv4-address	inet:ipv4-address			Tertiary P-CSCF server IP V4 Address
weight	uint32	0 .. 100	1	weight allocated for this pair

Ip V6 List

List of IP V6 Address pairs

XPath: [cna-smf:/smf/service-construct/pcscf-table/ip-v6-list](#)

CLI Path: [smf service-construct pcscf-table ip-v6-list](#)

List Properties:

Keys: name
Maximum number of elements: 128

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			IPV6 Address Pair Name

Attribute	Type	Range/Length/Pattern	Default	Description
primary-ipv6-address	inet:ipv6-address			Primary P-CSCF server IP V6 Address
secondary-ipv6-address	inet:ipv6-address			Secondary P-CSCF server IP V6 Address
tertiary-ipv6-address	inet:ipv6-address			Tertiary P-CSCF server IP V6 Address
weight	uint32	0 .. 100	1	weight allocated for this pair

Dnn

data network name construct

Defines the data network name

XPath: [cna-smf:/smf/service-construct/dnn](#)

CLI Path: [smf service-construct dnn](#)

List Properties:

Keys: name

Maximum number of elements: 5120

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z0-9{0,63})		
admin-state	enumeration	maintenance_mode enabled	enabled	Admin state for the dnn that can in be either enabled or maintenance-mode
allowed-dnn-selection-mode	enumeration	non-subscribed-dnn-allowed non-subscribed-dnn-blocked	non-subscribed-dnn-allowed	Policy to authorize dnn selection mode during session setup - enum list
dnn-restriction	uint32	0 .. 4	0	Restriction value that determines whether subscriber can establish bearers with other DNNs. Value ranges from 0-4, zero being least restrictive
pcscf-table	leafref			List of P-CSCF addresses Leaf Reference Path: cna-smf:/smf/service-construct/pcscf-table/name
pco-ipv4-link-mtu-size	uint32	576 .. 9216	1430	MTU Size for a Non-Roaming PDU Session to be sent in the PCO IE
pco-ipv4-link-mtu-size-roaming	uint32	576 .. 1800	1430	MTU Size for a Roaming PDU Session to be sent in the PCO IE
is-emergency-dnn	boolean		false	Indicates if the DNN is an emergency DNN
emergency-session-inactivity-timer-duration	uint32	1 .. 600	15	duration of the inactivity timer for emergency calls which specifies the time in seconds SMF/PGW-C waits to tear down emergency session after emergency dedicated bearer is released
default-user-name	string			Default User Name at DNN Level if user name is not present in PCO
default-password	string			Default Password at DNN Level if Password is not present in PCO
override-pco-option	boolean		false	Enable the capability to override pco username and password

Dns Server Configuration

dns server configuration per apn

XPath: [cna-smf:/smf/service-construct/dnn/dns-server-configuration](#)

CLI Path: [smf service-construct dnn dns-server-configuration](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
primary-ipv4-address	inet:ipv4-address			primary dns server IP V4 Address
secondary-ipv4-address	inet:ipv4-address			secondary dns server IP V4 Address
primary-ipv6-address	inet:ipv6-address			primary dns server IP V6 Address

Attribute	Type	Range/Length/Pattern	Default	Description
secondary-ipv6-address	inet:ipv6-address			secondary dns server IP V6 Address

Dnn List

list of data network names supported by the SMF service

XPath: [cna-smf:/smf/service-construct/dnn-list](#)

CLI Path: [smf service-construct dnn-list](#)

List Properties:

Keys: name

Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		

Dnn

data network name supported by the SMF service

XPath: [cna-smf:/smf/service-construct/dnn-list/dnn](#)

CLI Path: [smf service-construct dnn-list dnn](#)

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/dnn/name

S Nssai

S-NSSAI construct

Defines the S-NSSAI containing SST, SD and DNN List

XPath: [cna-smf:/smf/service-construct/s-nssai](#)

CLI Path: [smf service-construct s-nssai](#)

List Properties:

Keys: name

Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		
slice-service-type	union	['uint8,0 or 4 .. 255', 'Enumerations: eMBB URLLC MIoT ']		Identifies the expected Network Slice behavior in terms of features and services
slice-differentiator	string	([A-Za-z0-9]{6})		Differentiate amongst multiple Network Slices of the same Slice/Service type
dnn-list	leafref			name of the list of DNNs those are supported by this s-nssai Leaf Reference Path: cna-smf:/smf/service-construct/dnn-list/name

Nssai

list of S-NSSAI's supported by the SMF service

XPath: [cna-smf:/smf/service-construct/nssai](#)

CLI Path: [smf service-construct nssai](#)

List Properties:

Keys: name

Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		

S Nssai

list of S-NSSAI's upported by the SMF service

XPath: [cna-smf:/smf/service-construct/nssai/s-nssai](#)

CLI Path: [smf service-construct nssai s-nssai](#)

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/s-nssai/name

Dnai

data network access identifier

Identifier of a user plane access to DN where applications are deployed

XPath: [cna-smf:/smf/service-construct/dnai](#)

CLI Path: [smf service-construct dnai](#)

List Properties:

Keys: name

Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		

Dnai List

list of DNAs supported by the SMF service

XPath: [cna-smf:/smf/service-construct/dnai-list](#)

CLI Path: [smf service-construct dnai-list](#)

List Properties:

Keys: name

Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		

Dnai

data network access identifier name supported by the SMF service

XPath: [cna-smf:/smf/service-construct/dnai-list/dnai](#)

CLI Path: *smf service-construct dnai-list dnai*

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/dnai/name

Plmn

PLMN construct
Defines the plmn identier and nssai list

XPath: *cna-smf:/smf/service-construct/plmn*

CLI Path: *smf service-construct plmn*

List Properties:

Keys: plmn-id

Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
plmn-id	string	(([0-9]{5,6}))		Unique PLMN identifier; Range: numbers with 5-6 digits
nssai	leafref			name of the nssai supported by this plmn Leaf Reference Path: cna-smf:/smf/service-construct/nssai/name

Plmn List

name the plmn-list containing the list of plmn

XPath: *cna-smf:/smf/service-construct/plmn-list*

CLI Path: *smf service-construct plmn-list*

List Properties:

Keys: name

Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		

Plmn

name of the plmn

XPath: *cna-smf:/smf/service-construct/plmn-list/plmn*

CLI Path: *smf service-construct plmn-list plmn*

List Properties:

Keys: plmn-id

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
plmn-id	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/plmn/plmn-id

Tai

tracking area identifier construct
Defines the tracking area identifier that consist of mcc, mnc and tracking are code

XPath: [cna-smf:/smf/service-construct/tai](#)

CLI Path: [smf service-construct tai](#)

List Properties:

Keys: name
Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		
plmn-id	string	(([0-9]{5,6}))		Unique PLMN identifier; Range: numbers with 5-6 digits
tac	string	([A-Fa-f0-9]{4}) or [A-Fa-f0-9]{6})		tracking are code of the tai

Tai List

tracking area identifiers list supported by the SMF service

XPath: [cna-smf:/smf/service-construct/tai-list](#)

CLI Path: [smf service-construct tai-list](#)

List Properties:

Keys: name
Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		
dnai-list	leafref			name of the dnai-list of DNAs those are used for local traffic steering Leaf Reference Path: cna-smf:/smf/service-construct/dnai-list/name

Tai

tracking area identifiers list supported by the SMF service

XPath: [cna-smf:/smf/service-construct/tai-list/tai](#)

CLI Path: [smf service-construct tai-list tai](#)

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/tai/name

Tac Range

tracking area code range construct
Defines the tracking area code range
Either start-tac and end-tac or pattern should be configured
Both start-tac and end-tac should be configured

XPath: [cna-smf:/smf/service-construct/tac-range](#)

CLI Path: [smf service-construct tac-range](#)

List Properties:

Keys: name
Maximum number of elements: 1024

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
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Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		
start-tac	string	([A-Fa-f0-9]{4} or [A-Fa-f0-9]{6})		
end-tac	string	([A-Fa-f0-9]{4} or [A-Fa-f0-9]{6})		
pattern	string			

Tac Range List

tracking area codes range list supported by the SMF service
must enter at least one tac-range

XPath: [cna-smf:/smf/service-construct/tac-range-list](#)

CLI Path: [smf service-construct tac-range-list](#)

List Properties:

Keys: name
Maximum number of elements: 1024

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		

Tac Range

tracking area codes range list supported by the SMF service

XPath: [cna-smf:/smf/service-construct/tac-range-list/tac-range](#)

CLI Path: [smf service-construct tac-range-list tac-range](#)

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/tac-range/name

Tai Range

tracking area identifiers range construct
Defines the tracking area identifiers range
must enter at least one tac-range-list

XPath: [cna-smf:/smf/service-construct/tai-range](#)

CLI Path: [smf service-construct tai-range](#)

List Properties:

Keys: name
Maximum number of elements: 1024

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		
plmn-id	string	(([0-9]{5,6}))		Unique PLMN identifier; Range: numbers with 5-6 digits

Tac Range List

Tracking Area Code Range List Name

XPath: [cna-smf:/smf/service-construct/tai-range/tac-range-list](#)

CLI Path: [smf service-construct tai-range tac-range-list](#)

List Properties:

Keys: name
Minimum number of elements: 1
Maximum number of elements: 1024

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/tai-range-list/name

Tai Range List

tracking area identifiers range list supported by the SMF service
must enter at least one tai-range

XPath: [cna-smf:/smf/service-construct/tai-range-list](#)

CLI Path: [smf service-construct tai-range-list](#)

List Properties:

Keys: name
Maximum number of elements: 1024

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		
dnai-list	leafref			name of the dnai-list of DNAs those are used for local traffic steering Leaf Reference Path: cna-smf:/smf/service-construct/dnai-list/name

Tai Range

tracking area identifiers range list supported by the SMF service

XPath: [cna-smf:/smf/service-construct/tai-range-list/tai-range](#)

CLI Path: [smf service-construct tai-range-list tai-range](#)

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/tai-range/name

Supported Features

supported features construct
supported features

XPath: [cna-smf:/smf/service-construct/supported-features](#)

CLI Path: [smf service-construct supported-features](#)

List Properties:

Keys: name
Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		
load-control	enumeration	disabled enabled	disabled	

Attribute	Type	Range/Length/Pattern	Default	Description
overload-control	enumeration	disabled enabled	disabled	
vsmf-support	enumeration	disabled enabled	disabled	Indicates if the SMF can act as a V-SMF for inbound roamers

Packet Filter

Packet Filter construct

Defines the packet filter configuration identified via 5 tuples mainly source-ip-address, destination-ip-address, source-port, destination-port and protocol fields
source-port-start != 0 is allowed only for protocols 255, 6, 17, 132
source-port-end != 65535 is allowed only for protocols 255, 6, 17, 132
source-port-end must be greater than or equal to source-port-start
destination-port-start != 0 is allowed only for protocols 255, 6, 17, 132
destination-port-end != 65535 is allowed only for protocols 255, 6, 17, 132
destination-port-end must be greater than or equal to destination-port-start

XPath: [cna-smf:/smf/service-construct/packet-filter](#)

CLI Path: [smf service-construct packet-filter](#)

List Properties:

Keys: name
Maximum number of elements: 16000

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			
ip-version	enumeration	v4-only v6-only	v4-only	IP version
source-ipv4	inet:ipv4-address		0.0.0.0	source IPv4 address (inclusive)
source-ipv4-mask	uint32	0..32	0	source IPv4 address mask, 0 signifies for 'don't care'
source-ipv6	inet:ipv6-address		::	source IPv6 address(inclusive)
source-ipv6-prefix-length	uint32	0..128	0	source IPv6 prefix length, 0 signifies 'don't care'
destination-ipv4	inet:ipv4-address		0.0.0.0	destination IPv4 address (inclusive)
destination-ipv4-mask	uint32	0..32	0	destination IPv4 address mask, 0 signifies 'don't care'
destination-ipv6	inet:ipv6-address		::	destination IPv6 address (inclusive)
destination-ipv6-prefix-length	uint32	0..128	0	destination IPv6 prefix length, 0 signifies 'don't care'
source-port-start	inet:port-number		0	source port range start value (valid for protocol = TCP (6), UDP (17), SCTP (132), only)
source-port-end	inet:port-number		65535	source port range end value (valid for protocol = TCP (6), UDP (17), SCTP (132), only)
destination-port-start	inet:port-number		0	destination port range start value (valid for protocol = TCP (6), UDP (17), SCTP (132), only)
destination-port-end	inet:port-number		65535	destination port range end value (valid for protocol = TCP (6), UDP (17), SCTP (132), only)
protocol	uint32	0 .. 255	255	ip protocol values (e.g., all: 255, ICMP: 1, TCP: 6, UDP: 17, SCTP: 132)
direction	enumeration	ue network both	both	ue initiated or network initiated connection
flow-label	string			ipv6 flow label header
tos-traffic-class	uint32	0 .. 255		ipv4 type-of-service / ipv6 traffic class info

Attribute	Type	Range/Length/Pattern	Default	Description
tos-traffic-class-mask	uint32	0 .. 255		ipv4 type-of-service mask / ipv6 traffic class mask info

Pcc Charging Data

charging instance associated to predefined/static pcc-rule

Configure charging instance associated to predefined/static pcc-rule

both online and offline cannot be set to enabled simultaneously

both online and offline cannot be set to disabled simultaneously

reporting-level need to be configured

service-id and rating-group-id need to be filled since reporting level is serviceLevel

rating-group-id need to be filled since reporting level is ratingGroupLevel

XPath: *cna-smf:/smf/service-construct/pcc-charging-data*

CLI Path: *smf service-construct pcc-charging-data*

List Properties:

Keys: name

Maximum number of elements: 4096

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			
offline	enumeration	disabled enabled		indicates offline charging is applicable to PCC rule
online	enumeration	disabled enabled		indicates online charging is applicable to PCC rule
measurement-method	enumeration	time volume timeAndVolume event	volume	measurement method [time/volume/both/event]
rating-group-id	uint32			rating-group id
service-id	uint32			service id
sponsor-id	string			sponsor id
afcharging-id	uint32			af charging identifier
reporting-level	enumeration	serviceLevel ratingGroupLevel sponsoredLevel		policy control reporting level

Traffic Steering Policy

traffic steering policy instance

Configure traffic steering policy object

Atleast one of tunnel-endpoint-ipv4-address, tunnel-endpoint-ipv6-address should be configured

XPath: *cna-smf:/smf/service-construct/traffic-steering-policy*

CLI Path: *smf service-construct traffic-steering-policy*

List Properties:

Keys: id

Maximum number of elements: 4096

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
id	string	(a-zA-Z{0,63})		
tunnel-endpoint-ipv4-address	inet:ipv4-address			ip v4 address
tunnel-endpoint-ipv6-address	inet:ipv6-address			ip v6 address

Attribute	Type	Range/Length/Pattern	Default	Description
tunnel-endpoint-port	inet:port-number		0	port number tunnel-endpoint-port should be configured

Route To Location

Route to location primitives

XPath: [cna-smf:/smf/service-construct/route-to-location](#)

CLI Path: [smf service-construct route-to-location](#)

List Properties:

Keys: dnai
Maximum number of elements: 4192

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
dnai	string	(a-zA-Z{0,63})		
traffic-steering-policy-id	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/traffic-steering-policy/id

Redirect Information

redirect information instance
Configure redirect information object

XPath: [cna-smf:/smf/service-construct/redirect-information](#)

CLI Path: [smf service-construct redirect-information](#)

List Properties:

Keys: id
Maximum number of elements: 4096

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
id	string	(a-zA-Z{0,63})		
redirect-enabled	boolean		true	To enable or disable redirect-enabled
redirect-address-type	enumeration	ipv4 ipv6 url sip-uri		Indicates the type of redirect address
redirect-server-address	string			Indicates the address of the redirect server

Traffic Control

traffic control to support traffic steering

XPath: [cna-smf:/smf/service-construct/traffic-control](#)

CLI Path: [smf service-construct traffic-control](#)

List Properties:

Keys: name
Maximum number of elements: 4096

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			

Attribute	Type	Range/Length/Pattern	Default	Description
flow-status	enumeration	ENABLED-UPLINK ENABLED-DOWNLINK ENABLED DISABLED	ENABLED	indicates the flow status
redirect-info	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/redirect-information/id
mute-notif	boolean		false	application start or stop notification to be muted
traffic-steering-policy-id-ul	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/traffic-steering-policy/id
traffic-steering-policy-id-dl	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/traffic-steering-policy/id

Route To Location

XPath: [cna-smf:/smf/service-construct/traffic-control/route-to-location](#)

CLI Path: [smf service-construct traffic-control route-to-location](#)

List Properties:

Keys: name

Maximum number of elements: 4096

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/route-to-location/dnai

Pcc Rule

pcc rule construct to support policy and charging primitives

packet-filter count should not exceed 15 when install-packet-filters-on-ue is enabled

both packet-filter and application id cannot be configured together

XPath: [cna-smf:/smf/service-construct/pcc-rule](#)

CLI Path: [smf service-construct pcc-rule](#)

List Properties:

Keys: name

Maximum number of elements: 16000

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			
precedence	uint32	1 .. 99999		specifies the order in which this pcc rule can be applied relative to other pcc-rules
application-id	string			reference to the configured application detection filter
rule-type	enumeration	non-tdef-type predefined-tdef-type	non-tdef-type	pcc rule type
install-packet-filters-on-ue	enumeration	disabled enabled	disabled	Enable/Disable installing packet filters for this pcc rule on UE

Packet Filter

list of packet filters - selected from packet filter list

For predefined-tdef-type Rule, packet-filter is not configurable

XPath: [cna-smf:/smf/service-construct/pcc-rule/packet-filter](#)

CLI Path: *smf service-construct pcc-rule packet-filter*

List Properties:

Keys: name
Maximum number of elements: 512

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/packet-filter/name

Qos Flow

qos flow service reference - selected from qos flow list
For predefined-tdef-type Rule, qos-flow is not configurable

XPath: *cna-smf:/smf/service-construct/pcc-rule/qos-flow*

CLI Path: *smf service-construct pcc-rule qos-flow*

List Properties:

Keys: name
Maximum number of elements: 1

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/qos-flow/name

Charging Data

charging-data corresponds to pcc rule
For predefined-tdef-type Rule, charging-data is not configurable

XPath: *cna-smf:/smf/service-construct/pcc-rule/charging-data*

CLI Path: *smf service-construct pcc-rule charging-data*

List Properties:

Keys: name
Maximum number of elements: 1

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/pcc-charging-data/name

Usage Monitoring Data

usage monitoring data - selected from usage monitoring list
For predefined-tdef-type Rule, usage-monitoring-data is not configurable

XPath: *cna-smf:/smf/service-construct/pcc-rule/usage-monitoring-data*

CLI Path: *smf service-construct pcc-rule usage-monitoring-data*

List Properties:

Keys: name
Maximum number of elements: 1

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/pcc-usage-monitoring-data/name

Traffic Control Instance

traffic control reference - selected from traffic control list
For predefined-tdef-type Rule, traffic-control-instance is not configurable

XPath: [cna-smf:/smf/service-construct/pcc-rule/traffic-control-instance](#)

CLI Path: [smf service-construct pcc-rule traffic-control-instance](#)

List Properties:
Keys: name
Maximum number of elements: 1

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/traffic-control/name

Pcc Rule Base

pcc rule base construct to support list of pcc rules construct

XPath: [cna-smf:/smf/service-construct/pcc-rule-base](#)

CLI Path: [smf service-construct pcc-rule-base](#)

List Properties:
Keys: name
Maximum number of elements: 4096

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			
rule-type	enumeration	non-tdef-type predefined-tdef-type	non-tdef-type	pcc rule type

Pcc Rule

list of pcc rules - selected from pcc rules
For predefined-tdef-type Rule, pcc-rule is not configurable

XPath: [cna-smf:/smf/service-construct/pcc-rule-base/pcc-rule](#)

CLI Path: [smf service-construct pcc-rule-base pcc-rule](#)

List Properties:
Keys: name
Maximum number of elements: 4096

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/pcc-rule/name

Dscp Mapping Policy

dscp mapping policy
dscp mapping policy

XPath: [cna-smf:/smf/service-construct/dscp-mapping-policy](#)

CLI Path: [smf service-construct dscp-mapping-policy](#)

List Properties:
Keys: name
Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
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Attribute	Type	Range/Length/Pattern	Default	Description
name	string			
default-dscp	enumeration	cs0 dscp_01 dscp_02 dscp_03 dscp_04 dscp_05 dscp_06 dscp_07 cs1 dscp_09 af11 dscp_11 af12 dscp_13 af13 dscp_15 cs2 dscp_17 af21 dscp_19 af22 dscp_21 af23 dscp_23 cs3 dscp_25 af31 dscp_27 af32 dscp_29 af33 dscp_31 cs4 dscp_33 af41 dscp_35 af42 dscp_37 af43 dscp_39 cs5 dscp_41 dscp_42 dscp_43 dscp_44_va dscp_45 dscp_46_ef dscp_47 cs6 dscp_49 dscp_50 dscp_51 dscp_52 dscp_53 dscp_54 dscp_55 cs7 dscp_57 dscp_58 dscp_59 dscp_60 dscp_61		default dscp is used when 5qi isn't configured in five-qi-dscp-mapp

Attribute	Type	Range/Length/Pattern	Default	Description
		dscp_62		
		dscp_63		
default-ppi	uint32	0 .. 7		default ppi is used when dscp value isn't configured in dscp-ppi-mapp

Five Qi Dscp Map

5qi to dscp mapping table
Configure 5qi to dscp mapping

XPath: *cna-smf:/smf/service-construct/dscp-mapping-policy/five-qi-dscp-map*

CLI Path: *smf service-construct dscp-mapping-policy five-qi-dscp-map*

List Properties:
Keys: five-qi-dscp-mapping-instance
Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
five-qi-dscp-mapping-instance	string			five-qi-dscp-mapping instance name
five-qi	uint32	0 .. 255		5G QoS Identifier
arp-priority-level	uint32	1 .. 15		arp priority level

Attribute	Type	Range/Length/Pattern	Default	Description
dscp	enumeration	cs0		dscp value
		dscp_01		
		dscp_02		
		dscp_03		
		dscp_04		
		dscp_05		
		dscp_06		
		dscp_07		
		cs1		
		dscp_09		
		af11		
		dscp_11		
		af12		
		dscp_13		
		af13		
		dscp_15		
		cs2		
		dscp_17		
		af21		
		dscp_19		
		af22		
		dscp_21		
		af23		
		dscp_23		
		cs3		
		dscp_25		
		af31		
		dscp_27		
		af32		
		dscp_29		
		af33		
		dscp_31		
		cs4		
		dscp_33		
		af41		
		dscp_35		
		af42		
		dscp_37		
		af43		
		dscp_39		
		cs5		
		dscp_41		
		dscp_42		
		dscp_43		
		dscp_44_va		
		dscp_45		
		dscp_46_ef		
		dscp_47		
		cs6		
		dscp_49		
		dscp_50		
		dscp_51		
		dscp_52		
		dscp_53		
		dscp_54		
		dscp_55		
		cs7		
		dscp_57		
		dscp_58		
		dscp_59		
		dscp_60		
		dscp_61		
		dscp_62		
		dscp_63		

Dscp Ppi Map

dscp to ppi mapping
Configure dscp to ppi mapping

XPath: *cna-smf:/smf/service-construct/dscp-mapping-policy/dscp-ppi-map*

CLI Path: *smf service-construct dscp-mapping-policy dscp-ppi-map*

List Properties:
Keys: dscp
Maximum number of elements: 128

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
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Attribute	Type	Range/Length/Pattern	Default	Description
dscp	enumeration	cs0		dscp value
		dscp_01		
		dscp_02		
		dscp_03		
		dscp_04		
		dscp_05		
		dscp_06		
		dscp_07		
		cs1		
		dscp_09		
		af11		
		dscp_11		
		af12		
		dscp_13		
		af13		
		dscp_15		
		cs2		
		dscp_17		
		af21		
		dscp_19		
		af22		
		dscp_21		
		af23		
		dscp_23		
		cs3		
		dscp_25		
		af31		
		dscp_27		
		af32		
		dscp_29		
		af33		
		dscp_31		
		cs4		
		dscp_33		
		af41		
		dscp_35		
		af42		
		dscp_37		
		af43		
		dscp_39		
		cs5		
		dscp_41		
		dscp_42		
		dscp_43		
		dscp_44_va		
		dscp_45		
		dscp_46_ef		
		dscp_47		
		cs6		
		dscp_49		
		dscp_50		
		dscp_51		
		dscp_52		
		dscp_53		
		dscp_54		
		dscp_55		
		cs7		
		dscp_57		
		dscp_58		
		dscp_59		
		dscp_60		
		dscp_61		
		dscp_62		
		dscp_63		

Attribute	Type	Range/Length/Pattern	Default	Description
ppi	uint32	0 .. 7		paging policy indicator

Edr Profile

EDR profile construct
EDR profile

XPath: *cna-smf:/smf/service-construct/edr-profile*

CLI Path: *smf service-construct edr-profile*

List Properties:
Keys: name
Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		EDR Profile Name

Event Type

Event type list
Event type list

XPath: *cna-smf:/smf/service-construct/edr-profile/event-type*

CLI Path: *smf service-construct edr-profile event-type*

List Properties:
Keys: type
Minimum number of elements: 1
Maximum number of elements: 4

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
type	enumeration	establishment release modification		Event type
outcome-types	leaf-list of enumeration	success failure		Outcome types Minimum number of elements: 1

Overload Control Profile

SMF level Overload Profile
Overload Control Profile

XPath: *cna-smf:/smf/service-construct/overload-control-profile*

CLI Path: *smf service-construct overload-control-profile*

List Properties:
Keys: name
Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			

Overload Level

SMF Overload Level table
Configure mapping of system Overload Level to overload control parameters

XPath: *cna-smf:/smf/service-construct/overload-control-profile/overload-level*

CLI Path: *smf service-construct overload-control-profile overload-level*

List Properties:

Keys: overload-level-value

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
overload-level-value	enumeration	minor-overload major-overload critical-overload		overload level value
oci-period-of-validity	uint32	0	30 .. 7200	0
oci-reduction-metric	uint32	0	1 .. 100	0
min-exempt-http-message-priority	uint32	0 .. 31	31	Minimum HTTP message priority value above which messages will be exempt from overload control
min-exempt-gtp-message-priority	uint32	0 .. 15	15	Minimum GTP message priority value above which messages will be exempt from overload control

Overload Action Profile

Overload Control Profile for actions per subscriber group
Overload Control Profile for actions per subscriber group

XPath: [cna-smf:/smf/service-construct/overload-action-profile](#)

CLI Path: [smf service-construct overload-action-profile](#)

List Properties:

Keys: name

Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			

Overload Level

SMF Overload Level table
Configure mapping of system Overload Level to overload control parameters

XPath: [cna-smf:/smf/service-construct/overload-action-profile/overload-level](#)

CLI Path: [smf service-construct overload-action-profile overload-level](#)

List Properties:

Keys: overload-level-value

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
overload-level-value	enumeration	minor-overload major-overload critical-overload		overload level value
backoff-time-duration	uint32	30 .. 7200	300	Backoff time for UE during overload
overload-actions	leaf-list of enumeration	reject-create-sm-context reject-ue-init-update-sm-context reject-create-session-request reject-pcf-init-policy-update reject-new-bearers-qosflows		Overload actions Minimum number of elements: 1

Arp Policy

ARP Policy
ARP Policy

XPath: [cna-smf:/smf/service-construct/arp-policy](#)

CLI Path: *smf service-construct arp-policy*

List Properties:

Keys: name
Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			

Arp Priority Level

ARP Priority Level mapping table
Configure mapping of ARP Priority Level to message handling

XPath: *cna-smf:/smf/service-construct/arp-policy/arp-priority-level*

CLI Path: *smf service-construct arp-policy arp-priority-level*

List Properties:

Keys: arp-priority-level-value
Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
arp-priority-level-value	uint32	1 .. 15		arp priority level value
http-message-priority	uint32	0 .. 31		http message priority value
pfcf-message-priority	uint32	0 .. 15		pfcf message priority value
gtp-message-priority	uint32	0 .. 15		gtp message priority value
overload-action-profile	leafref			Leaf Reference Path: <i>cna-smf:/smf/service-construct/overload-action-profile/name</i>

In Band Radius Profile

in-band radius profile construct
in-band Radius profile

XPath: *cna-smf:/smf/service-construct/in-band-radius-profile*

CLI Path: *smf service-construct in-band-radius-profile*

List Properties:

Keys: name
Maximum number of elements: 1024

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		in-band Radius Profile Name
in-band-tunnel-dnn-ni	string	(a-zA-Z0-9{0,63})		NI portion of dnn name to be used to find network instance of DN-AAA

Upf Peer

UPF to create tunnel with, to access DN-AAA server for in-band Radius
UPF to create tunnel with, to access DN-AAA server for in-band Radius

XPath: *cna-smf:/smf/service-construct/in-band-radius-profile/upf-peer*

CLI Path: *smf service-construct in-band-radius-profile upf-peer*

List Properties:

Keys: name
Maximum number of elements: 2

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		
peer-ip-address	inet:ipv4-address			UPF peer IPv4 Address
role	enumeration	primary secondary	primary	DN-AAA UPF Type: primary / secondary
radius-client-ip-address	inet:ipv4-address			Radius client IPv4 Address to be used to send accounting requests to the UPF
n4-u-instance	string	(a-zA-Z0-9{0,63})		N4-U network instance name
n6-instance	string	(a-zA-Z0-9{0,63})		N6 network instance name
admin-state	enumeration	disabled enabled	enabled	Admin state of this Pfcps-peer/UPF connection for DN-AAA

Radius Server Group

Radius Server Group construct

Radius Server Group profile

XPath: *cna-smf:/smf/service-construct/radius-server-group*

CLI Path: *smf service-construct radius-server-group*

List Properties:

Keys: name

Maximum number of elements: 1024

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		Radius Server Group Name
server-group-type	enumeration	NotApplicable Authorization Accounting	NotApplicable	Server Group Type (NotApplicable/Authorization/Accounting)

Attribute	Type	Range/Length/Pattern	Default	Description
dscp-value	enumeration	cs0	cs0	DiffServ Code Point value with which to mark all IP packets on this Server Group
		dscp_01		
		dscp_02		
		dscp_03		
		dscp_04		
		dscp_05		
		dscp_06		
		dscp_07		
		cs1		
		dscp_09		
		af11		
		dscp_11		
		af12		
		dscp_13		
		af13		
		dscp_15		
		cs2		
		dscp_17		
		af21		
		dscp_19		
		af22		
		dscp_21		
		af23		
		dscp_23		
		cs3		
		dscp_25		
		af31		
		dscp_27		
		af32		
		dscp_29		
		af33		
		dscp_31		
		cs4		
		dscp_33		
		af41		
		dscp_35		
		af42		
		dscp_37		
		af43		
		dscp_39		
		cs5		
		dscp_41		
		dscp_42		
		dscp_43		
		dscp_44_va		
		dscp_45		
		dscp_46_ef		
		dscp_47		
		cs6		
		dscp_49		
		dscp_50		
		dscp_51		
		dscp_52		
		dscp_53		
		dscp_54		
		dscp_55		
		cs7		
		dscp_57		
		dscp_58		
		dscp_59		
		dscp_60		
		dscp_61		
		dscp_62		
		dscp_63		

Attribute	Type	Range/Length/Pattern	Default	Description
peer-retry-mode	enumeration	Alternate Sequential	Alternate	Peer Retry Mode ([Alternate]/Sequential/alternate-with-priority)
in-band-profile	leafref			in-band profile - use in-band tunnels of this profile to send and receive Radius requests and responses via UPF to and from AAA servers in DN. Leaf Reference Path: cna-smf:/smf/service-construct/in-band-radius-profile/name

Peer

Configure peer radius server attributes

XPath: [cna-smf:/smf/service-construct/radius-server-group/peer](#)

CLI Path: [smf service-construct radius-server-group peer](#)

List Properties:

Keys: name

Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		Peer radius client name
admin-state	enumeration	disabled enabled	disabled	Admin state of this peer connection
secret-key	string			Secret key string shared for a connection
remote-ip-address	inet:ipv4-address			Remote (Radius server) IP address
remote-port	uint32	1000..65535	1812	Remote (Radius server) UDP port number
max-num-response-timeout	uint32	1..5	3	Number of connection attempts
server-type	enumeration	primary	primary	Peer Server Type Primary / Secondary
server-resp-timeout-value	uint32	1..5	1	Server response timeout value in seconds

Radius Authentication Profile

Radius Authentication profile construct

Radius Authentication profile

XPath: [cna-smf:/smf/service-construct/radius-authentication-profile](#)

CLI Path: [smf service-construct radius-authentication-profile](#)

List Properties:

Keys: name

Maximum number of elements: 1024

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		Radius Authentication Profile Name
nas-port-type	enumeration	async virtual wireless-other	wireless-other	Choose the NAS-Port-Type to be sent to remote peer
server-group	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/radius-server-group/name
nas-ip-address	inet:ipv4-address			The Nas-Ip-Address to be sent in the radius authentication requests

Event Trigger Template

Configure Default Qosflow Charging Trigger Definitions

XPath: [cna-smf:/smf/service-construct/event-trigger-template](#)

CLI Path: [smf service-construct event-trigger-template](#)

List Properties:

Keys: name
Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		

Trigger Type

XPath: [cna-smf:/smf/service-construct/event-trigger-template/trigger-type](#)

CLI Path: [smf service-construct event-trigger-template trigger-type](#)

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	enumeration	plmn-change-event rat-change-event user-location-change-event ue-timezone-change-event qos-change-event session-ambr-change-event serving-node-change-event		
maximum-events	uint32	1 .. 20	1	

Radius Accounting Profile

Radius Accounting profile construct
Radius Accounting profile

XPath: [cna-smf:/smf/service-construct/radius-accounting-profile](#)

CLI Path: [smf service-construct radius-accounting-profile](#)

List Properties:

Keys: name
Maximum number of elements: 1024

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		Radius Accounting Profile Name
nas-port-type	enumeration	async virtual wireless-other	wireless- other	Choose the NAS-Port-Type to be sent to remote peer
ignore-radius-failure	boolean		false	Continue by ignoring failure from Radius - true/[false]
server-group	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/radius-server-group/name
wait-for-accounting-response	boolean		false	Wait for Accounting Response to continue processing - true/[false]
accounting-type	enumeration	qos-flow-based session-based	qos-flow-based	Radius Accounting type - [qos-flow-based]/session-based - qos-flow-based for qosflow based accounting, session-based for session based accounting
nas-ip-address	inet:ipv4-address			The Nas-Ip-Address to be sent in the accounting requests

Attribute	Type	Range/Length/Pattern	Default	Description
interim-update-interval	uint32	0	60..86400	0
data-volume-limit	uint32	0	10..500000	0
event-trigger-template	leafref			associated to event trigger template Leaf Reference Path: cna-smf:/smf/service-construct/event-trigger-template/name

Reporting Profile

Reporting profile construct
Reporting profile

XPath: [cna-smf:/smf/service-construct/reporting-profile](#)

CLI Path: [smf service-construct reporting-profile](#)

List Properties:

Keys: name
Maximum number of elements: 500

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		

Policy

Reporting Profile for Policy

XPath: [cna-smf:/smf/service-construct/reporting-profile/policy](#)

CLI Path: [smf service-construct reporting-profile policy](#)

Dnn

DNN Reporting

XPath: [cna-smf:/smf/service-construct/reporting-profile/policy/dnn](#)

CLI Path: [smf service-construct reporting-profile policy dnn](#)

Discovery

Reporting for Discovery Message

XPath: [cna-smf:/smf/service-construct/reporting-profile/policy/dnn/discovery](#)

CLI Path: [smf service-construct reporting-profile policy dnn discovery](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
dnn-type	enumeration	incoming matched current	current	DNN to be reported during discovery

Messaging

Reporting for Signaling Message

XPath: [cna-smf:/smf/service-construct/reporting-profile/policy/dnn/messaging](#)

CLI Path: [smf service-construct reporting-profile policy dnn messaging](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
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Attribute	Type	Range/Length/Pattern	Default	Description
dnn-type	enumeration	incoming matched current	current	DNN to be reported during signaling messaging

Custom

smf custom datatype and value to be sent to PCF in Policy Create message

XPath: [cna-smf:/smf/service-construct/reporting-profile/policy/custom](#)

CLI Path: [smf service-construct reporting-profile policy custom](#)

List Properties:

Keys: datatype

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
datatype	string			
value	string			

Charging

Reporting Profile for Charging

XPath: [cna-smf:/smf/service-construct/reporting-profile/charging](#)

CLI Path: [smf service-construct reporting-profile charging](#)

Dnn

DNN Reporting

XPath: [cna-smf:/smf/service-construct/reporting-profile/charging/dnn](#)

CLI Path: [smf service-construct reporting-profile charging dnn](#)

Discovery

Reporting for Discovery Message

XPath: [cna-smf:/smf/service-construct/reporting-profile/charging/dnn/discovery](#)

CLI Path: [smf service-construct reporting-profile charging dnn discovery](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
dnn-type	enumeration	incoming matched current	current	DNN to be reported during discovery

Messaging

Reporting for Signaling Message

XPath: [cna-smf:/smf/service-construct/reporting-profile/charging/dnn/messaging](#)

CLI Path: [smf service-construct reporting-profile charging dnn messaging](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
dnn-type	enumeration	incoming matched current	current	DNN to be reported during signaling messaging

Custom

smf custom datatype and value to be sent to charging interface

XPath: *cna-smf:/smf/service-construct/reporting-profile/charging/custom*

CLI Path: *smf service-construct reporting-profile charging custom*

List Properties:

Keys: datatype

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
datatype	string			
value	string			

Upf

Reporting Profile for UPF

XPath: *cna-smf:/smf/service-construct/reporting-profile/upf*

CLI Path: *smf service-construct reporting-profile upf*

Dnn

DNN Reporting

XPath: *cna-smf:/smf/service-construct/reporting-profile/upf/dnn*

CLI Path: *smf service-construct reporting-profile upf dnn*

Discovery

Reporting for Discovery Message

XPath: *cna-smf:/smf/service-construct/reporting-profile/upf/dnn/discovery*

CLI Path: *smf service-construct reporting-profile upf dnn discovery*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
dnn-type	enumeration	incoming	current	DNN to be reported during discovery
		matched		
		current		

Messaging

Reporting for Signaling Message

XPath: *cna-smf:/smf/service-construct/reporting-profile/upf/dnn/messaging*

CLI Path: *smf service-construct reporting-profile upf dnn messaging*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
dnn-type	enumeration	incoming	current	DNN to be reported during signaling messaging
		matched		
		current		

Custom

smf custom datatype and value to be sent to UPF interface

XPath: *cna-smf:/smf/service-construct/reporting-profile/upf/custom*

CLI Path: *smf service-construct reporting-profile upf custom*

List Properties:

Keys: datatype

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
datatype	string			
value	string			

Pcscf

Reporting Profile for PCSCF

XPath: *cna-smf:/smf/service-construct/reporting-profile/pcscf*

CLI Path: *smf service-construct reporting-profile pcscf*

Dnn

DNN Reporting

XPath: *cna-smf:/smf/service-construct/reporting-profile/pcscf/dnn*

CLI Path: *smf service-construct reporting-profile pcscf dnn*

Discovery

Reporting for Discovery Message

XPath: *cna-smf:/smf/service-construct/reporting-profile/pcscf/dnn/discovery*

CLI Path: *smf service-construct reporting-profile pcscf dnn discovery*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
dnn-type	enumeration	incoming matched current	current	DNN to be reported during discovery

Messaging

Reporting for Signaling Message

XPath: *cna-smf:/smf/service-construct/reporting-profile/pcscf/dnn/messaging*

CLI Path: *smf service-construct reporting-profile pcscf dnn messaging*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
dnn-type	enumeration	incoming matched current	current	DNN to be reported during signaling messaging

Custom

smf custom datatype and value to be sent to PCSCF interface

XPath: *cna-smf:/smf/service-construct/reporting-profile/pcscf/custom*

CLI Path: *smf service-construct reporting-profile pcscf custom*

List Properties:

Keys: datatype

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
datatype	string			
value	string			

Udm

Reporting Profile for UDM

XPath: *cna-smf:/smf/service-construct/reporting-profile/udm*

CLI Path: *smf service-construct reporting-profile udm*

Dnn

DNN Reporting

XPath: *cna-smf:/smf/service-construct/reporting-profile/udm/dnn*

CLI Path: *smf service-construct reporting-profile udm dnn*

Messaging

Reporting for Signaling Message

XPath: *cna-smf:/smf/service-construct/reporting-profile/udm/dnn/messaging*

CLI Path: *smf service-construct reporting-profile udm dnn messaging*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
dnn-type	enumeration	incoming matched current	current	DNN to be reported during signaling messaging

Custom

smf custom datatype and value to be sent to UDM interface

XPath: *cna-smf:/smf/service-construct/reporting-profile/udm/custom*

CLI Path: *smf service-construct reporting-profile udm custom*

List Properties:

Keys: datatype

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
datatype	string			
value	string			

Radius

Reporting Profile for RADIUS

XPath: *cna-smf:/smf/service-construct/reporting-profile/radius*

CLI Path: *smf service-construct reporting-profile radius*

Dnn

DNN Reporting

XPath: *cna-smf:/smf/service-construct/reporting-profile/radius/dnn*

CLI Path: *smf service-construct reporting-profile radius dnn*

Messaging

Reporting for Signaling Message

XPath: *cna-smf:/smf/service-construct/reporting-profile/radius/dnn/messaging*

CLI Path: *smf service-construct reporting-profile radius dnn messaging*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
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Attribute	Type	Range/Length/Pattern	Default	Description
dnn-type	enumeration	incoming matched current	current	DNN to be reported during signaling messaging

Custom

smf custom datatype and value to be sent to RADIUS interface

XPath: [cna-smf:/smf/service-construct/reporting-profile/radius/custom](#)

CLI Path: [smf service-construct reporting-profile radius custom](#)

List Properties:

Keys: datatype

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
datatype	string			
value	string			

Subscriber Management Profile

subscriber management profile to support subscriber data

XPath: [cna-smf:/smf/service-construct/subscriber-management-profile](#)

CLI Path: [smf service-construct subscriber-management-profile](#)

List Properties:

Keys: name

Maximum number of elements: 1024

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		
snssai	leafref			name of the nssai supported for this subscriber Leaf Reference Path: cna-smf:/smf/service-construct/s-nssai/name
default-ssc-mode	enumeration	SSC_MODE_1 SSC_MODE_2 SSC_MODE_3		Configure SSC_MODE_1 or SSC_MODE_2 or SSC_MODE_3
ssc-mode-3-pdu-session-address-lifetime	uint32		90	lifetime of old PDU session anchor for ssc mode 3, in unit of seconds
default-pdu-session-type	enumeration	IPV4 IPV6 IPV4V6		Configure IPV4 or IPV6 or IPV4V6
session-ambr	leafref			name of the session-ambr supported for this subscriber Leaf Reference Path: cna-smf:/smf/service-construct/session-ambr/name
fiveq-qos-profile	leafref			name of the subscribed-qos supported for this subscriber Leaf Reference Path: cna-smf:/smf/service-construct/subscribed-qos/name
iwk-eps-ind	boolean		false	Interworking with EPS Indication
dscp-mapping-policy	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/dscp-mapping-policy/name
paging-profile	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/paging-profile/name
always-on	enumeration	DISABLE ENABLE_ON_REQUEST ENABLE_ALWAYS	DISABLE	Always on grant policy for sessions

Attribute	Type	Range/Length/Pattern	Default	Description
inactivity-timer-duration	uint32	1 .. 2160000	900	Inactivity timer duration for sessions in seconds[1-2160000]
edr-profile-name	leafref			Name of the EDR Profile Leaf Reference Path: cna-smf:/smf/service-construct/edr-profile/name
arp-policy	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/arp-policy/name
default-overload-action-profile	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/overload-action-profile/name
session-idle-timer	uint32	0 .. 2160000	14400	Idle timer duration for sessions in seconds[0-2160000]
mapped-dnn	string	(a-zA-Z{0,63})		Target DNN for DNN Mapping
radius-authentication-profile	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/radius-authentication-profile/name
radius-accounting-profile	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/radius-accounting-profile/name
enable-upf-activity-report	boolean		false	Enable activity report from UPF to SMF for 5G always-on and 4G sessions after sessions have gone inactive based on inactivity-timer-duration
reporting-profile	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/reporting-profile/name
pseudo-supi	string	(a-zA-Z{0,63})		value used to fill supi attribute in CreateSmPolicy Request to PCF for emergency SIMless session
ue-ip-allocation-mode	enumeration	local radius	local	The UE IP allocation mode (Local, Radius.) to use.

Allowed Ssc Modes

list of allowed-ssc-modes for the the subscriber

XPath: [cna-smf:/smf/service-construct/subscriber-management-profile/allowed-ssc-modes](#)

CLI Path: [smf service-construct subscriber-management-profile allowed-ssc-modes](#)

List Properties:

Keys: sscmode
Minimum number of elements: 0
Maximum number of elements: 6

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
sscmode	enumeration	SSC_MODE_1 SSC_MODE_2 SSC_MODE_3		

Allowed Pdu Session Types

list of allowed-pdu-session-types for the the subscriber

XPath: [cna-smf:/smf/service-construct/subscriber-management-profile/allowed-pdu-session-types](#)

CLI Path: [smf service-construct subscriber-management-profile allowed-pdu-session-types](#)

List Properties:

Keys: pdusessiontype
Minimum number of elements: 0
Maximum number of elements: 5

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
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Attribute	Type	Range/Length/Pattern	Default	Description
pdusessiontype	enumeration	IPV4		
		IPV6		
		IPV4V6		

Qos Flow Inactivity Timer Map

QosFlow(5qi+arp) to inactivity timer mapping table
Configure QosFlow i.e. 5qi+arp(key) to inactivity-timer mapping

XPath: *cna-smf:/smf/service-construct/subscriber-management-profile/qos-flow-inactivity-timer-map*

CLI Path: *smf service-construct subscriber-management-profile qos-flow-inactivity-timer-map*

List Properties:
Keys: qos-flow-inactivity-timer-instance
Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
qos-flow-inactivity-timer-instance	string			qos-flow-inactivity-timer-instance name
five-qi	uint32	0 .. 255		5G QoS Identifier
arp-priority-level	uint32	1 .. 15		arp priority level
qos-flow-inactivity-timer	uint32	0 .. 2160000	5	Inactivity timer duration for 5qi+arp_pl combination in seconds[0-2160000]
direction	enumeration	BIDIRECTIONAL UPLINK DOWNLINK	BIDIRECTIONAL	direction of user packets to monitor inactivity This field allows user to configure direction of user packets to monitor inactivity for the configured qos-flow-inactivity-timer value

Dynamic Pcsf Parameters

Dynamic PCSCF related parameters

XPath: *cna-smf:/smf/service-construct/subscriber-management-profile/dynamic-pcsf-parameters*

CLI Path: *smf service-construct subscriber-management-profile dynamic-pcsf-parameters*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
dynamic-pcsf-discovery	boolean		false	Dynamic PCSCF Discovery
pcsaf-load-threshold	uint32	0 .. 100	80	PCSCF load threshold value
use-local-config-on-discovery-failure	boolean		true	Use local configured PCSCF Address on dynamic discovery failure

Pcf Response Code Template

Configure error codes from pcf

XPath: *cna-smf:/smf/service-construct/pcf-response-code-template*

CLI Path: *smf service-construct pcf-response-code-template*

List Properties:
Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			

Pcf Response Code

Configure pcf response code

XPath: [cna-smf:/smf/service-construct/pcf-response-code-template/pcf-response-code](#)

CLI Path: [smf service-construct pcf-response-code-template pcf-response-code](#)

List Properties:

Keys: pcf-message-response-code

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
pcf-message-response-code	uint16			Configure response code

Default Subscriber Policy Profile

Default Policy profile to be used in case of error in policy create response code
must enter at least one pcc-rule-base or pcc-rule or session-rule

XPath: [cna-smf:/smf/service-construct/default-subscriber-policy-profile](#)

CLI Path: [smf service-construct default-subscriber-policy-profile](#)

List Properties:

Keys: name

Maximum number of elements: 4192

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		
reflective-qos-timer	int32	0 .. 92160		reflective qos timer value in seconds

Session Rule

session rule to apply default qos and ambr

XPath: [cna-smf:/smf/service-construct/default-subscriber-policy-profile/session-rule](#)

CLI Path: [smf service-construct default-subscriber-policy-profile session-rule](#)

List Properties:

Keys: name

Maximum number of elements: 1

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/session-rule/name

Pcc Rule Base

list of pcc rule-bases to be associated

XPath: [cna-smf:/smf/service-construct/default-subscriber-policy-profile/pcc-rule-base](#)

CLI Path: [smf service-construct default-subscriber-policy-profile pcc-rule-base](#)

List Properties:

Keys: name

Maximum number of elements: 64

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/pcc-rule-base/name

Pcc Rule

list of pcc rules to be associated

XPath: *cna-smf:/smf/service-construct/default-subscriber-policy-profile/pcc-rule*

CLI Path: *smf service-construct default-subscriber-policy-profile pcc-rule*

List Properties:

Keys: name
Maximum number of elements: 64

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: <i>cna-smf:/smf/service-construct/pcc-rule/name</i>

Dynamic Five Qi Qos Char Profile

Configure Dynamic 5QI QoS Characteristics Profile

XPath: *cna-smf:/smf/service-construct/dynamic-five-qi-qos-char-profile*

CLI Path: *smf service-construct dynamic-five-qi-qos-char-profile*

List Properties:

Keys: name
Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		

Dynamic Five Qi Qos Char List

List of different 5QI QoS Characteristics list
Configure dynamic 5QI QoS Parameters
averaging-window must be > 0
max-data-burst-vol must be > 0

XPath: *cna-smf:/smf/service-construct/dynamic-five-qi-qos-char-profile/dynamic-five-qi-qos-char-list*

CLI Path: *smf service-construct dynamic-five-qi-qos-char-profile dynamic-five-qi-qos-char-list*

List Properties:

Keys: five-qi
Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
five-qi	uint32	10 .. 64	68	71 .. 74
qos-resource-type	enumeration	NON_GBR NON_CRITICAL_GBR CRITICAL_GBR		Resource Type [NON_GBR/NON_CRITICAL_GBR/CRITICAL_GBR]
five-qi-priority-level	uint32	1 .. 127		priority [1-127]
packet-delay-budget	uint32	1 .. 4294967295		packet delay budget in ms
packet-error-rate	string	(([0-9]E-[0-9]))		packet error rate[4E-6, 1E-2]
averaging-window	uint32	1 .. 4095		averaging window [1-4095] in ms averaging-window must be > 0
max-data-burst-vol	uint32	1 .. 4095		maximum data burst volume [1-4095] in bytes max-data-burst-vol must be > 0

Gx Response Code Template

Configure error codes from gx

XPath: *cna-smf:/smf/service-construct/gx-response-code-template*

CLI Path: [smf service-construct gx-response-code-template](#)

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			

Gx Response Code

Configure gx response code

XPath: [cna-smf:/smf/service-construct/gx-response-code-template/gx-response-code](#)

CLI Path: [smf service-construct gx-response-code-template gx-response-code](#)

List Properties:

Keys: gx-message-response-code

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
gx-message-response-code	uint16			Configure response code

Gx Policy Profile

Gx Policy profile containing Gx specific settings

XPath: [cna-smf:/smf/service-construct/gx-policy-profile](#)

CLI Path: [smf service-construct gx-policy-profile](#)

List Properties:

Keys: name

Maximum number of elements: 4192

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		
dest-hostname	string	1..128		destination host name
dest-realm	string	1..128		destination realm name
origin-hostname	string	1..128		origin host name
origin-realm	string	1..128		origin realm name
diam-group-name	string	1..128		target diameter service name from which diameter pod is selected
gx-response-code-template	leafref			Template to configure gx response codes which are considered as okay to continue smf session Leaf Reference Path: cna-smf:/smf/service-construct/gx-response-code-template/name

Subscriber Policy Profile

Policy profile to support policy and charging primitives

XPath: [cna-smf:/smf/service-construct/subscriber-policy-profile](#)

CLI Path: [smf service-construct subscriber-policy-profile](#)

List Properties:

Keys: name

Maximum number of elements: 4192

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
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Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		
pcf-response-code-template-for-session-continue	leafref			template to configure PCF response codes which are considered as okay to continue smf session Leaf Reference Path: cna-smf:/smf/service-construct/pcf-response-code-template/name
default-subscriber-policy-profile	leafref			default policy profile to be applied in case of error code in policy create response Leaf Reference Path: cna-smf:/smf/service-construct/default-subscriber-policy-profile/name
pcf-rejoin-time	uint32	0	90 .. 86400	900
pcf-down-session-hold-time	uint32	0	180 .. 172800	0
tdef-header-enrichment	enumeration	disabled enabled	disabled	send tdef header enrichment in session create and modify
supported-features	leaf-list of enumeration	TSC ADC UMC NetLoc RanNasCause PRA RANSupportInfo SessionRuleErrorHandling		List of Supported Features for PCF Interface
dynamic-five-qi-qos-char-profile	leafref			dynamic 5QI qos characteristics list profile name Leaf Reference Path: cna-smf:/smf/service-construct/dynamic-five-qi-qos-char-profile/name
pcf-interface-admin-state	enumeration	disabled enabled	enabled	admin-state for pcf interface to enable or disable dynamic pcc
use-udm-sgw-qos-no-pcf	boolean		false	indicate to use UDM or SGW qos when pcf interface is not available for any session
inbound-roamer-qos-profile	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/inbound-roamer-qos-profile/name
policy-interface-type	enumeration	PCF Gx	PCF	indicates which policy interface to be used in this policy profile either Gx or PCF - default PCF
gx-policy-profile	leafref			indicates the Gx policy profile attached in this subscriber policy profile Leaf Reference Path: cna-smf:/smf/service-construct/gx-policy-profile/name
eps-r99-priority-mapping-high	uint32	1 .. 13	5	High priority used for mapping between eps priority level and r99 ARP
eps-r99-priority-mapping-medium	uint32	2 .. 14	10	Medium priority used for mapping between eps priority level and r99 ARP value Medium priority value must be greater than High priority value
pcf-response-code-template-for-failover	leafref			template to configure PCF response codes to failover to secondary PCF in case of primary PCF fails Leaf Reference Path: cna-smf:/smf/service-construct/pcf-response-code-template/name

Session Rule

session rule to apply default qos and ambr

XPath: [cna-smf:/smf/service-construct/subscriber-policy-profile/session-rule](#)

CLI Path: [smf service-construct subscriber-policy-profile session-rule](#)

List Properties:
Keys: name
Minimum number of elements: 1
Maximum number of elements: 1

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/session-rule/name

Pcc Rule Base

list of pcc rules to be associated

XPath: [cna-smf:/smf/service-construct/subscriber-policy-profile/pcc-rule-base](#)
CLI Path: [smf service-construct subscriber-policy-profile pcc-rule-base](#)

List Properties:
Keys: name
Maximum number of elements: 64

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/pcc-rule-base/name

Pcc Rule

list of pcc rules to be associated

XPath: [cna-smf:/smf/service-construct/subscriber-policy-profile/pcc-rule](#)
CLI Path: [smf service-construct subscriber-policy-profile pcc-rule](#)

List Properties:
Keys: name
Maximum number of elements: 64

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/pcc-rule/name

Charging Trigger Template

Configure Default Subscriber Charging Trigger Definitions

XPath: [cna-smf:/smf/service-construct/charging-trigger-template](#)
CLI Path: [smf service-construct charging-trigger-template](#)

List Properties:
Keys: name
Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		

Trigger Type

volume-limit-value need to be configured since trigger volume-limit is enabled

XPath: [cna-smf:/smf/service-construct/charging-trigger-template/trigger-type](#)

CLI Path: *smf service-construct charging-trigger-template trigger-type*

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	enumeration	plmn-change rat-change user-location-change ue-timezone-change qos-change volume-limit session-ambr-change		
trigger-enabled	boolean		false	
trigger-category	enumeration	immediate-report deferred-report	immediate-report	
volume-limit-value	uint64			volume limit(threshold) bytes for session level volume reporting volume-limit-value must be > 0

Charging Msg Level Result Code Action Template

Configure error codes from chf and the actions to be performed on the session

XPath: *cna-smf:/smf/service-construct/charging-msg-level-result-code-action-template*

CLI Path: *smf service-construct charging-msg-level-result-code-action-template*

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			
default-action	enumeration	continue terminate		

Http Response Status Code

XPath: *cna-smf:/smf/service-construct/charging-msg-level-result-code-action-template/http-response-status-code*

CLI Path: *smf service-construct charging-msg-level-result-code-action-template http-response-status-code*

List Properties:

Keys: status-code

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
status-code	uint16			Configure http status code
default-action	enumeration	continue terminate		

Application Error

Configure application error codes and the actions

XPath: *cna-smf:/smf/service-construct/charging-msg-level-result-code-action-template/http-response-status-code/application-error*

CLI Path: *smf service-construct charging-msg-level-result-code-action-template http-response-status-code application-error*

List Properties:

Keys: app-error-code

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
app-error-code	enumeration	charging-failed charging-not-applicable user-unknown end-user-request-denied end-user-request-rejected istio-cfg-agent-reg-failure		
action	enumeration	continue terminate		

Online Charging Attributes List

Configure Default Online Charging Parameters

default-quota-volume-quota-threshold must be less than the default-quota-volume-quota-total

XPath: [cna-smf:/smf/service-construct/online-charging-attributes-list](#)

CLI Path: [smf service-construct online-charging-attributes-list](#)

List Properties:

Keys: name

Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		
default-quota-volume-quota-total	uint64	10000 .. max	1000000	Default Volume Quota(bytes) in both Uplink Downlink Directions
default-quota-volume-quota-threshold	uint64	1000 .. max	100000	Default Volume Quota(bytes) threshold
default-quota-max-num-default-quota	uint32	0 .. 1000	0	Max Number of Times a Session can be assigned Default Quota Zero means Unlimited Number of Default Quota Allocation
default-quota-validity-time	uint32	0 .. 7200	1800	Validity Time(seconds) of the Quota
chf-default-validity-time	uint32	60 .. 7200		Default quota validity time if VT is missing in the CHF response
chf-default-quota-holding-time	uint32	60 .. 1800		Default QHT if QHT is missing in the CHF response

Charging Rg Level Result Code Action Template

Configure list of expected RG level error result codes from CHF and any action to be taken

redirect-server-address-type should be configured if the action is redirect

redirect-server-address should be configured if the action is redirect

XPath: [cna-smf:/smf/service-construct/charging-rg-level-result-code-action-template](#)

CLI Path: [smf service-construct charging-rg-level-result-code-action-template](#)

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			
default-action	enumeration	continue terminate-service redirect		
redirect-server-address-type	enumeration	ipv4 ipv6 url		
redirect-server-address	string			

Rg Result Code

redirect-server-address-type should be configured if the action is redirect

redirect-server-address should be configured if the action is redirect

XPath: *cna-smf:/smf/service-construct/charging-rg-level-result-code-action-template/rg-result-code*

CLI Path: *smf service-construct charging-rg-level-result-code-action-template rg-result-code*

List Properties:

Keys: result-code

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
result-code	enumeration	end-user-service-denied quota-management-not-applicable quota-limit-reached end-user-service-rejected rating-failed		Configure Rg level result-code
action	enumeration	continue terminate-service redirect		Configure action to be performed
redirect-server-address-type	enumeration	ipv4 ipv6 url		
redirect-server-address	string			

Chf Profile

Configure Chf Profile Parameters

both online and offline cannot be set to enabled simultaneously

both online and offline cannot be set to disabled simultaneously

XPath: *cna-smf:/smf/service-construct/chf-profile*

CLI Path: *smf service-construct chf-profile*

List Properties:

Keys: name

Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		
online-enabled	boolean		false	default online charging enabled/disabled at PDU Session
offline-enabled	boolean		true	default offline charging enabled/disabled at PDU Session
usage-reporting-level	enumeration	serviceLevel ratingGroupLevel sponsoredLevel	ratingGroupLevel	default usage reporting-level when not provided in a policy charging data
charging-agent-profile	string			associated to charging agent profile name maintained by charging-agent
chf-session-failover-support	enumeration	failover-not-supported failover-supported	failover-not-supported	alternative CHF server support for ongoing session
max-number-of-charge-condition-change	uint32	0 .. 1000	0	max number of changes in charging condition to send explicit trigger
chf-time-limit-trigger-as-periodic-timer	boolean		true	use chf time limit trigger as periodic timer
periodic-reporting-timer	uint32	0 .. 86400	0	measurement period for generating periodic reports(in seconds)

Attribute	Type	Range/Length/Pattern	Default	Description
create-chf-session-always	boolean		false	Always create CHF Session even there are no rules in session with charging
trigger-template	leafref			associated to chargeable events trigger template Leaf Reference Path: cna-smf:/smf/service-construct/charging-trigger-template/name
online-charging-attributes-list-name	leafref			Online Charging Attributes List Name Leaf Reference Path: cna-smf:/smf/service-construct/online-charging-attributes-list/name
charging-initial-msg-result-code-action-template	leafref			During create sessions, actions to be taken on the session based on the CHF returned error-code Leaf Reference Path: cna-smf:/smf/service-construct/charging-msg-level-result-code-action-template/name
charging-update-msg-result-code-action-template	leafref			During update sessions, actions to be taken on the session based on the CHF returned error-code Leaf Reference Path: cna-smf:/smf/service-construct/charging-msg-level-result-code-action-template/name
online-charging-initial-rg-level-result-code-action-template	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/charging-rg-level-result-code-action-template/name
online-charging-update-rg-level-result-code-action-template	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/charging-rg-level-result-code-action-template/name
chf-server-discovery-through-nrf	boolean			enable chf server discovery through nrf when PCF does not provide
reflective-qos-timer	int32	0 .. 92160		reflective qos timer value in seconds
chf-interface-admin-state	enumeration	disabled enabled	enabled	admin-state for chf interface to enable or disable charging
secondary-rat-container-threshold	uint32	1 .. 100	10	secondary rat container threshold after which usage is reported to CHF
secondary-rat-reporting	enumeration	disabled enabled	disabled	enable reporting of secondary rat usage
always-send-uli-in-usage-container	boolean		true	always send ULI in usage container irrespective of ULI change trigger

Cdr Profile

Configure Cdr Profile Parameters

XPath: [cna-smf:/smf/service-construct/cdr-profile](#)

CLI Path: [smf service-construct cdr-profile](#)

List Properties:

Keys: name

Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		
cdr-report-level	enumeration	bearer	bearer	Choose the CDR reporting level. By default CDRs are generated at bearer (qosflow) level.
cdr-mode	enumeration	stand-alone co-located	stand-alone	Type of SGW CDR's to be generated: stand-alone

Attribute	Type	Range/Length/Pattern	Default	Description
interim-updates	enumeration	disabled enabled	disabled	Enable or Disable Interim Updates to Offline Charging Server
interim-update-interval	uint32	0	90..21600	1800
data-volume-limit	uint32	0	1..500000	0
event-trigger-template	leafref			associated to events trigger template Leaf Reference Path: cna-smf:/smf/service-construct/event-trigger-template/name
num-service-containers	uint32	1 .. 20	1	number of service containers to be reached to generate a new interim CDR
num-charge-condition-change	uint32	0 .. 20	0	number of charge condition changes to be reached to generate a new interim CDR

Local

local CDR

XPath: [cna-smf:/smf/service-construct/cdr-profile/local](#)

CLI Path: [smf service-construct cdr-profile local](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
asn1-cdr-version	enumeration	9.0 10.0		3gpp 32.298 version type:[10.0
encode-format	enumeration	csv asn1 both		csv or asn1 format
cdr-agent-profile	string	(a-zA-Z{0,63})		

Subscriber Charging Profile

Configure Subscriber Charging Parameters

XPath: [cna-smf:/smf/service-construct/subscriber-charging-profile](#)

CLI Path: [smf service-construct subscriber-charging-profile](#)

List Properties:

Keys: name

Maximum number of elements: 256

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		
cdr-profile	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/cdr-profile/name
chf-profile	leafref			chf profile Leaf Reference Path: cna-smf:/smf/service-construct/chf-profile/name

Network Instance Profile

Network-Instance profile to provide network-instances for N3, N6 and N9 interfaces

XPath: [cna-smf:/smf/service-construct/network-instance-profile](#)

CLI Path: [smf service-construct network-instance-profile](#)

List Properties:

Keys: name

Maximum number of elements: 4192

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		
n3-network-instance	string			n3-network-instance name
n6-network-instance	string			n6-network-instance name
n9-network-instance	string			n9-network-instance name
s5s8u-s2bu-network-instance	string			s5s8u-s2bu-network-instance name

Neighbour Discovery Profile

neighbour profile construct
neighbour discovery profile

XPath: *cna-smf:/smf/service-construct/neighbour-discovery-profile*

CLI Path: *smf service-construct neighbour-discovery-profile*

List Properties:

Keys: name
Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		
ra-mtu	uint32	1280 .. 1800	1400	MTU for ipv6 Neighbor Discovery Router Advertisements
retransmit-timer	uint32	1 .. 3600	1	The time in seconds between retransmitted Neighbor Solicitation messages
router-lifetime	uint32	1 .. 65535	1800	The lifetime associated with the default router in units of seconds
reachable-time	uint32	0 .. 3600000	0	The time, in milliseconds, a neighbor is considered reachable after receiving a reachability confirmation. 0 indicates unspecified
number-of-unsolicited-ra	uint32	0 .. 6	3	The number of unsolicited router advertisements the MCC would send when a UE attaches to the network. 0 indicates to send continuously spaced by ra-interval
unsolicited-ra-interval	uint32	1 .. 1800	3	Unsolicited Router Advertisement interval in units of seconds

Service Area

service area construct
Defines the service area name that consist tai-list

XPath: *cna-smf:/smf/service-construct/service-area*

CLI Path: *smf service-construct service-area*

List Properties:

Keys: name
Maximum number of elements: 1024

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			

Tai List

Tracking Area List Name

XPath: *cna-smf:/smf/service-construct/service-area/tai-list*

CLI Path: *smf service-construct service-area tai-list*

List Properties:

Keys: name
Maximum number of elements: 1024

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/tai-list/name
priority	uint32	0 .. 65535		Priority associated with the TAI list

Tai Range List

Tracking Area Range List Name

XPath: [cna-smf:/smf/service-construct/service-area/tai-range-list](#)

CLI Path: [smf service-construct service-area tai-range-list](#)

List Properties:

Keys: name

Maximum number of elements: 1024

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/tai-range-list/name
priority	uint32	0 .. 65535		Priority associated with the TAI Range list

Service Area List

service area list supported by the SMF service

XPath: [cna-smf:/smf/service-construct/service-area-list](#)

CLI Path: [smf service-construct service-area-list](#)

List Properties:

Keys: name

Maximum number of elements: 1024

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		

Service Area

service area supported by the SMF service

XPath: [cna-smf:/smf/service-construct/service-area-list/service-area](#)

CLI Path: [smf service-construct service-area-list service-area](#)

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/service-area/name

Nfset Nf Response Code Template

Configure error codes from nfsetNF

XPath: [cna-smf:/smf/service-construct/nfset-nf-response-code-template](#)

CLI Path: [smf service-construct nfset-nf-response-code-template](#)

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			

Nfset Nf Response Code

Configure nfsetNF response code

XPath: [cna-smf:/smf/service-construct/nfset-nf-response-code-template/nfset-nf-response-code](#)

CLI Path: [smf service-construct nfset-nf-response-code-template nfset-nf-response-code](#)

List Properties:

Keys: nfset-nf-message-response-code

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
nfset-nf-message-response-code	uint16			Configure response code

Nf Service Profile

NF Service Profile

Profile to control SMF interactions with various NF Services

XPath: [cna-smf:/smf/service-construct/nf-service-profile](#)

CLI Path: [smf service-construct nf-service-profile](#)

List Properties:

Keys: name

Maximum number of elements: 16

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			

Nf Service Name

NF Service table

Configure properties of various NF services that SMF interacts with
if oauth-required, then admin-state should not be disabled

XPath: [cna-smf:/smf/service-construct/nf-service-profile/nf-service-name](#)

CLI Path: [smf service-construct nf-service-profile nf-service-name](#)

List Properties:

Keys: nf-service-type

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
nf-service-type	enumeration	nudm-sdm nudm-uecm namf-comm npcf-policycontrol nchf-converged-charging nnrf-disc nnrf-nfm namf-evts		NF Service Type
nfset-peer-retry-response-code-template	leafref			template to configure nf reponse codes which are considered as okay to retry the next available nf of same type Leaf Reference Path: cna-smf:/smf/service-construct/nfset-nf-response-code-template/name

Attribute	Type	Range/Length/Pattern	Default	Description
oauth-required	boolean		false	OAuth Required
admin-state	enumeration	disabled enabled	enabled	admin-state for nf-service to disable or enable the NF communication
response-timeout	uint16	500 .. 30000	3000	Time period (in milliseconds) to wait for response after a request has been sent, after which it will be treated as no response from peer
expiry-time	uint32	300 .. 4294967295	604800	Time period (in seconds) to be send in event subscription message
deactivate-session-on-failure	enumeration	disabled enabled	enabled	Deactivate the session on failure
version	enumeration	16.12.0	16.12.0	Applicable spec is 3GPP TS 29.518; Access and Mobility Management Services.
version	enumeration	16.9.0	16.9.0	Applicable spec is 3GPP TS 29.518; Access and Mobility Management Services.
version	enumeration	15.7.0	15.7.0	Applicable spec is 3GPP TS 32.291; Telecommunication management; Charging management.
version	enumeration	16.5.0	16.5.0	Applicable spec is 3GPP TS 29.510; Network Function Repository Services.
version	enumeration	16.5.0	16.5.0	Applicable spec is 3GPP TS 29.510; Network Function Repository Services.
version	enumeration	16.10.0	16.10.0	Applicable spec is 3GPP TS 29.512; Session Management Policy Control Service.
version	enumeration	15.7.0	15.7.0	Applicable spec is 3GPP TS 29.503; Unified Data Management Services.
version	enumeration	15.4.0	15.4.0	Applicable spec is 3GPP TS 29.503; Unified Data Management Services.

Nf Service Discovery Parameters

Parameters to be used while discovering a given nf-service

XPath: *cna-smf:/smf/service-construct/nf-service-profile/nf-service-name/nf-service-discovery-parameters*

CLI Path: *smf service-construct nf-service-profile nf-service-name nf-service-discovery-parameters*

List Properties:

Keys: nfservice-type

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
nfservice-type	enumeration	nudm-sdm		NF Service Type
		nudm-uecm		
		namf-comm		
		npcf-policycontrol		
		nchf-converged-charging		
		nnrf-disc		
		nnrf-nfm		
		namf-evts		
preferred-locality	string			Name of the Preferred target NF location

Sepp Response Code Template

Configure error codes from sepp

XPath: *cna-smf:/smf/service-construct/sepp-response-code-template*

CLI Path: *smf service-construct sepp-response-code-template*

List Properties:

Keys: name

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
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Attribute	Type	Range/Length/Pattern	Default	Description
name	string			

Sepp Response Code

Configure sepp response code

XPath: [cna-smf:/smf/service-construct/sepp-response-code-template/sepp-response-code](#)

CLI Path: [smf service-construct sepp-response-code-template sepp-response-code](#)

List Properties:

Keys: sepp-message-response-code

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
sepp-message-response-code	uint16			Configure response code

Local Pfd App Id List

Contains the application-ids configured in UPF

XPath: [cna-smf:/smf/service-construct/local-pfd-app-id-list](#)

CLI Path: [smf service-construct local-pfd-app-id-list](#)

List Properties:

Keys: application-id

Maximum number of elements: 5120

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
application-id	string			Application id for which pfd fetch is not required.

Service

SMF Service Configuration

XPath: [cna-smf:/smf/service](#)

CLI Path: [smf service](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
admin-state	enumeration	disabled	enabled	
		enabled		
		maintenance		
instance-id	string			Unique Instance Id of the SMF Service. This will be registered with NRF.
smf-fqdn	string			Fully Qualified Domain Name of the SMF
smf-inter-plmn-fqdn	string			Inter Plmn Fully Qualified Domain Name of the SMF
supported-features	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/supported-features/name
bulk-deactivation-rate	uint32	0 .. 65535	1000	Maximum number of session deactivation requests initiated per second during bulk deactivation

Attribute	Type	Range/Length/Pattern	Default	Description
max-session-ids-in-one-db-fetch	uint32	100 .. 5000000	10000	Maximum number of session ids fetched from DB in one go for displaying on CLI or bulk deactivation.Note: This parameter should be modified to larger value only in SMF maintenance mode case or SMF not handling any traffic.It will lead to unexpected SMF behaviour otherwise.Warning: Please use extreme caution while changing this parameter
capacity	uint32	0 .. 65535		
priority	uint32	0 .. 65535		
home-plmn-list	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/plmn-list/name
allowed-visited-plmn-list	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/plmn-list/name
allowed-nssai	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/nssai/name
service-area-list	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/service-area-list/name
subscriber-policy-profile-analyzer-instance	string	(a-zA-Z{0,63})		Subscriber Policy Profile Analyzer Instance
subscriber-management-profile-analyzer-instance	string	(a-zA-Z{0,63})		Subscriber Management Profile Analyzer Instance
network-instance-profile-analyzer-instance	string	(a-zA-Z{0,63})		Network Instance Profile Analyzer Instance
non-sbi-based-legacy-dnn-list	leaf-list of leafref			dnn list for non-sbi based legacy subscribers Leaf Reference Path: cna-smf:/smf/service-construct/dnn-list/name
subscriber-charging-profile-analyzer-instance	string	(a-zA-Z{0,63})		Subscriber Charging Profile Analyzer Instance
guard-timer	uint32	1000 .. 30000	10000	Amount of time to wait in milliseconds if temporary reject is received, either to discard packets in network initiated service request procedure or retry network initiated request during handover procedure
network-initiated-service-request-n2-failure-guard-timer	uint32	100 .. 10000	500	Amount of time to wait in milliseconds if during network initiated service request procedure N2 PDU session resource setup unsuccessful transfer is received due to some handover in progress
guard-retry-count	uint32	1 .. 5	3	Number of retry attempts for a network initiated request when Guard timer is in use
neighbour-discovery-profile	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/neighbour-discovery-profile/name
pgw-support	enumeration	disabled enabled	disabled	Enable/Disable PGW functionality

Attribute	Type	Range/Length/Pattern	Default	Description
weighted-upf-load-balancing	enumeration	disabled enabled	disabled	Enable or disable UPF load balancing based on UPF capacity and load learned from NRF
overload-control-profile	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/overload-control-profile/name
overload-action-profile	leafref			Leaf Reference Path: cna-smf:/smf/service-construct/overload-action-profile/name
udm-sdm-subscription	enumeration	DISABLE_SDM ENABLE_GET_SM_DATA ENABLE_SDM_SUBSCRIPTION	ENABLE_GET_SM_DATA	Whether SMF should send sm-data or/and sdm-subscription query to UDM
default-edr-profile-name	leafref			Default EDR Profile name Leaf Reference Path: cna-smf:/smf/service-construct/edr-profile/name
peer-nf-service-profile	leafref			Profile to control SMF interactions with various NF Services Leaf Reference Path: cna-smf:/smf/service-construct/nf-service-profile/name
include-ebi-param-in-pcoie-in-cbr	boolean		false	Indicates if the EBI value of 0 need to be included in QosDlowDescs container of PCO IE in Bearer Context IE in Create Bearer Request message.
pfid-fetch-from-nef	enumeration	disabled enabled	disabled	Indicates whether pfid fetch from nef by smf is enabled or disbaled
session-id-quarantine-period	uint32	60 .. 86400	1800	Session Id Quarantine Period for reusing ids after session release (in seconds)
suppress-bulk-deactivation-on-gtpc-peer-restart	boolean		false	Indicates whether to suppress bulk-deactivation of sessions on detection of GTPC Peer Restart
dnn-matching-type	enumeration	exact-match ni-only ni-and-malformed-oi	exact-match	DNN Matching Type to be used

Pgw Profile

pgw service related configuration

XPath: [cna-smf:/smf/service/pgw-profile](#)

CLI Path: [smf service pgw-profile](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
pgwc-set-instance-info	enumeration	PGW_SET_FQDN PGW_ALT_FQDN PGW_ALT_IP	PGW_ALT_FQDN	PGW set identity field to be sent in PGW Change Info IE

Service Profile

smf service profile configuration

XPath: [cna-smf:/smf/service/service-profile](#)

CLI Path: [smf service service-profile](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
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Attribute	Type	Range/Length/Pattern	Default	Description
duplicate-pdu-session-handling	enumeration	disabled enabled	disabled	enable/disable duplicate session handling for 5G sessions
send-user-id-to-upf	enumeration	disabled enabled	disabled	enable/disable sending User ID IE to UPF
prioritized-tai-registration	enumeration	disabled enabled	disabled	Enable/Disable SMF's prioritized tai based registration with NRF
http-error-code-in-maintenance-and-disabled-admin-state	uint32	500 .. 501	503 .. 504	504
http-error-code-in-dnn-maintenance-state	uint32	500 .. 501	503 .. 504	504
sm-5g-cause-in-dnn-maintenance-state	uint32	26 .. 111	27	5G SM cause code to be used in session rejects when a dnn is in maintenance mode, applicable to all dnns
sm-4g-cause-in-dnn-maintenance-state	uint32	64 .. 129	78	4G SM cause code to be used in session rejects when a dnn is in maintenance mode, applicable to all dnns
http-error-code-for-ue-ip-pool-exhaust	uint32	400 .. 504	504	HTTP Error code to use for session rejects when ue ip pool exhausted for a dnn, applicable to all dnns
sm-5g-cause-for-ue-ip-pool-exhaust	uint32	26 .. 111	26	5G SM cause code to be used in session rejects when ue ip pool exhausted for a dnn, applicable to all dnns
sm-4g-cause-for-ue-ip-pool-exhaust	uint32	64 .. 129	73	4G cause code to be used in session rejects when ue ip pool exhausted for a dnn, applicable to all dnns
session-reject-for-ue-ip-pool-exhaust-detection	enumeration	disabled enabled	disabled	Enable/Disable session reject on UE IP Pool Exhaust Monitoring feature in SMF
locality	string			Location of the SMF to be registered with NRF

Peer Nf Profile

peer NF configuration

XPath: *cna-smf:/smf/service/service-profile/peer-nf-profile*

CLI Path: *smf service service-profile peer-nf-profile*

List Properties:

Keys: peer-nf-type

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
peer-nf-type	enumeration	udm amf pcf upf smf chf pcscf sepproxy		Peer NF Type
dscp-marking	string	([0-9] or [1-5][0-9] or [6][0-3]) or df or af[1-4][1-3] or cs[1-7] or ef	df	DSCP marking, Default: df; Valid values are: 0.63, df (DSCP-0), af11 (DSCP-10), af12 (DSCP-12), af13 (DSCP-14), af21 (DSCP-18), af22 (DSCP-20), af23 (DSCP-22), af31 (DSCP-26), af32 (DSCP-28), af33 (DSCP-30), af41 (DSCP-34), af42 (DSCP-36), af43 (DSCP-38), cs1 (DSCP-8), cs2 (DSCP-16), cs3 (DSCP-24), cs4 (DSCP-32), cs5 (DSCP-40), cs6 (DSCP-48), cs7 (DSCP-56), ef (DSCP-46)
subscription-required	boolean		false	NRF subscription Required
max-load-threshold	uint32	0 .. 100	100	NF's maximum load threshold value to be considered for selection
peer-nf-set-support	boolean		false	Support for Peer NF Set

Attribute	Type	Range/Length/Pattern	Default	Description
include-3gpp-peer-info	boolean		false	Support to send 3gpp peer info header
sepp-response-code-template	leafref			template to configure SEPP response codes Leaf Reference Path: cna-smf:/smf/service-construct/sepp-response-code-template/name

Service Timers

service level timers configuration

XPath: [cna-smf:/smf/service/service-profile/service-timers](#)

CLI Path: [smf service service-profile service-timers](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
threegpp-procedure-guard-timer	uint32	3 .. 120	60	The time duration (in seconds) from 3gpp procedure creation, after which 3gpp procedure is cleaned up
gtpv2-procedure-start-delay-duration-threshold	uint32	5 .. 120	30	The threshold for time duration (in seconds) from GtpV2 3gpp procedure creation to start of procedure. The procedure is aborted if the durations exceeds threshold
five-g-procedure-start-delay-duration-threshold	uint32	5 .. 120	30	The threshold for time duration (in seconds) from 5GS 3gpp procedure creation to start of procedure. The procedure is aborted if the durations exceeds threshold
pdu-session-cache-duration	uint32	10 .. 86400	600	The time duration (in seconds) after which pdu session is cleaned up from session-cache
restart-session-cache-timer-upon-activity	boolean		true	enable/disable refreshing session cache timer upon activity

N1 Profile

n1 protocol related configuration parameters

XPath: [cna-smf:/smf/service/service-profile/n1-profile](#)

CLI Path: [smf service service-profile n1-profile](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
n1-pdu-session-modify-command-timeout	uint16	500 .. 30000	10000	Time period (in milliseconds) to wait for n1 message in update-sm-context after n1 pdu-session-modify-command message is sent
n1-pdu-session-release-command-timeout	uint16	500 .. 30000	1500	Time period (in milliseconds) to wait for n1 message in update-sm-context after n1 pdu-session-release-command message is sent
n1-pdu-session-modify-command-timeout-during-eps-fallback	uint16	10 .. 30000	30	Time period (in milliseconds) to wait for n2 message in update-sm-context after n2 message(not specifically mentioned above) is sent

N2 Profile

n2 protocol related configuration parameters

XPath: [cna-smf:/smf/service/service-profile/n2-profile](#)

CLI Path: [smf service service-profile n2-profile](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
n2-pdu-session-resource-setup-request-timeout	uint16	500 .. 30000	5000	Time period (in milliseconds) to wait for n2 message in update-sm-context after n2 pdu-session-resource-setup-request message is sent

Attribute	Type	Range/Length/Pattern	Default	Description
n2-pdu-session-resource-modify-request-timeout	uint16	500 .. 30000	5000	Time period (in milliseconds) to wait for n2 message in update-sm-context after n2 pdu-session-resource-modify-request message is sent
n2-pdu-session-resource-release-command-timeout	uint16	500 .. 30000	1500	Time period (in milliseconds) to wait for n2 message in update-sm-context after n2 pdu-session-resource-release-command message is sent
n2-message-timeout-during-handovers	uint16	500 .. 30000	5000	Time period (in milliseconds) to wait for n2 message in update-sm-context after n2 handoff related messages are sent
n2-message-timeout-other	uint16	500 .. 30000	5000	Time period (in milliseconds) to wait for n2 message in update-sm-context after n2 message(not specifically mentioned in n2-profile messages) is sent

N16 Profile

n16 protocol related configuration parameters

XPath: *cna-smf:/smf/service/service-profile/n16-profile*

CLI Path: *smf service service-profile n16-profile*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
n16-vsmf-pdu-session-create-response-timeout	uint16	500 .. 30000	5000	Time period (in milliseconds) for V-SMF to wait for pdu session create response from H-SMF
n16-vsmf-pdu-session-update-response-timeout	uint16	500 .. 30000	3000	Time period (in milliseconds) for V-SMF to wait for a response from H-SMF to a pdu session update request that V-SMF sent
n16-vsmf-pdu-session-update-request-timeout	uint16	500 .. 30000	3000	Time period (in milliseconds) for V-SMF to wait for a subsequent pdu session update request that H-SMF should send
n16-vsmf-pdu-session-release-response-timeout	uint16	500 .. 30000	1000	Time period (in milliseconds) for V-SMF to wait for a response from H-SMF to a pdu session release request that V-SMF sent
n16-vsmf-pdu-session-status-notify-timeout	uint16	500 .. 30000	1000	Time period (in milliseconds) for V-SMF to wait for a pdu session statue notify from H-SMF during some session release procedures
n16-hsmf-pdu-session-update-response-modification-timeout	uint16	500 .. 30000	3000	Time period (in milliseconds) for H-SMF to wait for a pdu session update response from V-SMF during some session modification procedures
n16-hsmf-pdu-session-update-response-release-timeout	uint16	500 .. 30000	3000	Time period (in milliseconds) for H-SMF to wait for a pdu session update response from V-SMF during session release procedures
n16-hsmf-pdu-session-update-response-ebi-timeout	uint16	500 .. 30000	3000	Time period (in milliseconds) for H-SMF to wait for a pdu session update response from V-SMF during ebi assignment procedures
n16-hsmf-pdu-session-status-notify-response-timeout	uint16	500 .. 30000	1000	Time period (in milliseconds) for H-SMF to wait for a pdu session statue notify response from V-SMF
n16-hsmf-guard-retry-timer	uint16	100 .. 30000	700	Time period (in milliseconds) for H-SMF to wait before retrying an update request to V-SMF which was rejected due to handover in progress or other higher priority events on V-SMF
n16-hsmf-guard-retry-count	uint16	1 .. 5	3	Number of retry attempts by H-SMF to send an update request to V-SMF that was rejected due to handover in progress or other higher priority events on V-SMF

Nsmf Pdusession

NSMF Pdusession (Nsfm_Pdusession) Service

XPath: *cna-smf:/smf/service/service-profile/nsmf-pdusession*

CLI Path: *smf service service-profile nsmf-pdusession*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
version	enumeration	17.4.0	17.4.0	3GPP TS 29.502 V17.4.0; 5G System; Session Management Services.

Nsmf Event Exposure

NSMF Event Exposure (Nsfm_Exposure_Event) Service

XPath: [cna-smf:/smf/service/service-profile/nsmf-event-exposure](#)

CLI Path: [smf service service-profile nsmf-event-exposure](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
version	enumeration	16.4.0	16.4.0	3GPP TS 29.508 V16.4.0; 5G System; Session Management Event Exposure Service.

Vtap Profile

vTap configuration for streaming packets to vProbe NF

XPath: [cna-smf:/smf/service/vtap-profile](#)

CLI Path: [smf service vtap-profile](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
vprobe-fqdn	string			A valid Fully Qualified Domain Name of the vProbe server
vprobe-topic	string			A valid Topic to send kafka message

Peer

Peer NF type

The 'vtap-profile' fields vprobe-fqdn and vprobe-topic must be configured

XPath: [cna-smf:/smf/service/vtap-profile/peer](#)

CLI Path: [smf service vtap-profile peer](#)

List Properties:

Keys: peer-vtap-nf-type

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
peer-vtap-nf-type	enumeration	udm amf pcf nef sepp chf nrf		Peer NF Type
enable-vtap	boolean		false	enable/disable vtap

Rate Limit Profile

smf rate limit profile for access side initiated requests

XPath: [cna-smf:/smf/service/rate-limit-profile](#)

CLI Path: [smf service rate-limit-profile](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
max-create-session-request-rate	uint32	0 .. 100	0	Maximum allowed session setup requests per second for a pod
gtp-reject-cause	uint8	64..255	73	gtpc reject cause in create session response
min-exempt-gtp-message-priority	uint32	0 .. 15	15	Minimum GTP message priority value above which messages will be exempt from rate limiting

Trace

SMF Tracing Configuration

XPath: *cna-smf:/smf/trace*

CLI Path: *smf trace*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
admin-state	enumeration	disabled enabled	enabled	Enable or Disable Tracing Keys
verbose	enumeration	disabled enabled	disabled	SMF Verbose Tracing
probability	string		0	Probability of sampling the trace for SMF The 'probability' value must be less than 1.0 The 'probability' value must be greater than 0

Trace Keys

Name

Tracing Key List

Atleast one of SUPI, DNN or PEI should be configured for the trace keys

XPath: *cna-smf:/smf/trace/trace-keys*

CLI Path: *smf trace trace-keys*

List Properties:

Keys: name

Maximum number of elements: 128

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	(a-zA-Z{0,63})		
priority	uint32	1 .. 255	255	
probability	string		0	Probability of sampling the trace for the key The 'probability' value must be less than 1.0 The 'probability' value must be greater than 0
supi	string	((imsi-)([0-9]){1,15})		Subscription Permanent Identifier (SUPI) value to match traces
dnn	string	(a-zA-Z{0,63})		Data Network Name value to match traces
pei	string	((imei-)([0-9]){1,15})		Permanent Equipment Identifier (PEI) value to match traces
gpsi	string	((msisdn-)([0-9]){1,15})		Generic Public Subscription Identifier (GPSI) value to match traces

Smf Gauges

SMF Supported gauges

XPath: *cna-smf:/smf/smf-gauges*

CLI Path: *smf smf-gauges*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
smf_simultaneous_attached_users	enumeration	disabled enabled	disabled	Number of unique attached subscribers in SMF
smf_simultaneous_attached_active_user	enumeration	disabled enabled	disabled	Number of unique attached subscribers with Active state in SMF
smf_current_pdu_sessions	enumeration	disabled enabled	enabled	Total number of pdu sessions in SMF

Attribute	Type	Range/Length/Pattern	Default	Description
smf_pgw_current_bearers	enumeration	disabled enabled	disabled	Total number of bearers in SMF+PGW-C per fiveQi, arp PI
smf_current_qos_flows	enumeration	disabled enabled	disabled	Total number of qos flows per fiveQi and arp pl
smf_current_pdu_sessions_using_default_quota	enumeration	disabled enabled	disabled	Total number of current pdu sessions using default quota
smf_current_pdu_sessions_using_default_policy	enumeration	disabled enabled	disabled	Total number of current pdu sessions using default policy
smf_current_pdu_sessions_per_upf	enumeration	disabled enabled	disabled	Total number of pdu sessions per upf
smf_current_pdu_sessions_charging_disabled	enumeration	disabled enabled	disabled	Total number of current pdu sessions charging disabled or no charging
smf_current_pdu_session_using_udm_allocated_static_ue_ip_address	enumeration	disabled enabled	disabled	Total number of current pdu session using udm allocated static ue ip-address.
smf_current_pdu_session_using_radius_allocated_ue_ip_address	enumeration	disabled enabled	disabled	Total number of current pdu session using radius allocated ue ip-address.
smf_current_pdu_sessions_per_dnn	enumeration	disabled enabled	disabled	Current pdu sessions count for each dnn
smf_current_pdu_sessions_per_rat_type	enumeration	disabled enabled	disabled	Current pdu sessions count per rat type
smf_current_outbound_roaming_sessions	enumeration	disabled enabled	disabled	Total number of outbound roamers in SMF
smf_current_inbound_roaming_home_routed_sessions	enumeration	disabled enabled	disabled	Total number of inbound roaming home routed sessions in SMF
smf_current_pdu_sessions_per_mapped_dnn	enumeration	disabled enabled	disabled	Current pdu sessions count for each mapped dnn
smf_current_idle_pdu_sessions	enumeration	disabled enabled	disabled	Current number of idle pdu sessions in SMF
smf_current_pdu_sessions_using_gx_policy	enumeration	disabled enabled	disabled	Current number of pdu sessions using Gx policy interface in SMF
smf_sgw_simultaneous_attached_active_users	enumeration	disabled enabled	disabled	Number of unique sgw attached subscribers with Active state in SMF
smf_sgw_current_idle_pdu_sessions	enumeration	disabled enabled	disabled	Number of unique sgw idle pdu sessions in SMF
smf_sgw_idle_subscribers	enumeration	disabled enabled	disabled	Number of unique sgw idle subscribers in SMF

Subscriber Logging

SMF Key Based Logging

XPath: [cna-smf:/smf/subscriber-logging](#)

CLI Path: [smf subscriber-logging](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
is-enabled	boolean		false	Enable or Disable Subscriber Logging

Supi

Configure SUPI for subscriber logging

XPath: *cna-smf:/smf/subscriber-logging/supi*

CLI Path: *smf subscriber-logging supi*

List Properties:

Keys: value

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
value	string	((imsi-){[0-9]}){1,15})		Configure SUPI for subscriber logging
logging-enabled	boolean		true	Enable or disable subscriber logging for this supi

SUB_ANALYZER Configuration

The module for entities implementing cna-sub_analyzer.

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Analyzer

Analyzer

XPath: *cna-sub_analyzer:/analyzer*

CLI Path: *analyzer*

Serving Area Group

List of service areas

XPath: *cna-sub_analyzer:/analyzer/serving-area-group*

CLI Path: *analyzer serving-area-group*

List Properties:

Keys: name

Maximum number of elements: 100

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string	1..255		Unique identity of the service area
description	string			Description about the service-area

Serving Plmn

XPath: *cna-sub_analyzer:/analyzer/serving-area-group/serving-plmn*

CLI Path: *analyzer serving-area-group serving-plmn*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
mcc	string	(([0-9]{3}))		Mobile Country Code of the PLMN
mnc	string	(([0-9]{2,3}))		Mobile Network Code of the PLMN

Tac List 5G

List of Tracking Area Codes (TACs) associated with the PLMN

XPath: *cna-sub_analyzer:/analyzer/serving-area-group/serving-plmn/tac-list-5g*

CLI Path: *analyzer serving-area-group serving-plmn tac-list-5g*

List Properties:

Keys: tac-5g

Maximum number of elements: 100

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
tac-5g	string	(0x[A-Fa-f0-9]{1,6})		Tracking Area Code (TAC); Valid values are hexadecimal values of length 4 or 6 including a leading '0x'

Tac List 4G

List of Tracking Area Codes (TACs) associated with the PLMN

XPath: *cna-sub_analyzer:/analyzer/serving-area-group/serving-plmn/tac-list-4g*

CLI Path: *analyzer serving-area-group serving-plmn tac-list-4g*

List Properties:

Keys: tac-4g

Maximum number of elements: 100

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
tac-4g	string	(0x[A-Fa-f0-9]{1,4})		Tracking Area Code (TAC); Valid values are hexadecimal values of length 4 including a leading '0x'

Lac List

List of Location Area Codes (LACs) associated with the PLMN

XPath: *cna-sub_analyzer:/analyzer/serving-area-group/serving-plmn/lac-list*

CLI Path: *analyzer serving-area-group serving-plmn lac-list*

List Properties:

Keys: lac
Maximum number of elements: 100

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
lac	string	(0x[A-Fa-f0-9]{1,4})		Location Area Code (LAC); Valid values are hexadecimal values of length 4 including a leading '0x'

Rac List

List of Routing Area Codes (RACs) associated with LAC and PLMN

XPath: *cna-sub_analyzer:/analyzer/serving-area-group/serving-plmn/lac-list/rac-list*

CLI Path: *analyzer serving-area-group serving-plmn lac-list rac-list*

List Properties:

Keys: rac
Maximum number of elements: 100

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
rac	string	(0x[A-Fa-f0-9]{1,2})		Routing Area Code (RAC); Valid values are hexadecimal values of length 2 including a leading '0x'

Sac List

List of Serving Area Codes (SACs) associated with LAC the PLMN

XPath: *cna-sub_analyzer:/analyzer/serving-area-group/serving-plmn/lac-list/sac-list*

CLI Path: *analyzer serving-area-group serving-plmn lac-list sac-list*

List Properties:

Keys: sac
Maximum number of elements: 100

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
sac	string	(0x[A-Fa-f0-9]{1,4})		Service Area Code (SAC); Valid values are hexadecimal values of length 4 including a leading '0x'

Cell Id List

List of Cell IDs associated with the PLMN

XPath: *cna-sub_analyzer:/analyzer/serving-area-group/serving-plmn/lac-list/cell-id-list*

CLI Path: *analyzer serving-area-group serving-plmn lac-list cell-id-list*

List Properties:

Keys: cell-id
Maximum number of elements: 100

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
cell-id	string	(0x[A-Fa-f0-9]{1,4})		Cell Id; Valid values are hexadecimal values of length 4 including a leading '0x'

Eutran Cell Id List

List of E-UTRAN Cell IDs associated with the PLMN

XPath: *cna-sub_analyzer:/analyzer/serving-area-group/serving-plmn/eutran-cell-id-list*

CLI Path: *analyzer serving-area-group serving-plmn eutran-cell-id-list*

List Properties:

Keys: eutran-cell-id
Maximum number of elements: 100

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
eutran-cell-id	string	(0x[A-Fa-f0-9]{1,7})		E-UTRAN Cell Id; Valid values are hexadecimal values of length 7 including a leading '0x'

Nr Cell Id List

List of NR Cell IDs associated with the PLMN

XPath: *cna-sub_analyzer:/analyzer/serving-area-group/serving-plmn/nr-cell-id-list*

CLI Path: *analyzer serving-area-group serving-plmn nr-cell-id-list*

List Properties:

Keys: nr-cell-id
Maximum number of elements: 100

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
nr-cell-id	string	(0x[A-Fa-f0-9]{1,9})		NR Cell Id; Valid values are hexadecimal values of length 9 including a leading '0x'

Instance

List of analyzer instances in the analyzer service

XPath: *cna-sub_analyzer:/analyzer/instance*

CLI Path: *analyzer instance*

List Properties:

Keys: analyzer-name
Maximum number of elements: 10

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
analyzer-name	string	([0-9a-zA-Z_-]*)		Unique identity of the analyzer

Class

List of classes in the analyzer

XPath: *cna-sub_analyzer:/analyzer/instance/class*

CLI Path: *analyzer instance class*

List Properties:

Keys: class-name
Minimum number of elements: 1
Maximum number of elements: 10000

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
class-name	string	([0-9a-zA-Z_-]*)		Unique identity of the class
precedence	uint32	0..65535		Defines the precedence of this class. A smaller value is higher priority.

Filterset

List of filterset for the given class. A filterset can have multiple filters. All filters in filterset have to match for a class name to be chosen. If any one filterset matches, the class Nme associated with this filterset will be chosen

XPath: *cna-sub_analyzer:/analyzer/instance/class/filterset*

CLI Path: *analyzer instance class filterset*

List Properties:

Keys: filterset-name
Minimum number of elements: 1
Maximum number of elements: 1500

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
filterset-name	string	([0-9a-zA-Z_-])*		filter set name for the given class

Filter

Filters to be applied to match the filterset. All filters needs to be matched.

XPath: *cna-sub_analyzer:/analyzer/instance/class/filterset/filter*

CLI Path: *analyzer instance class filterset filter*

List Properties:

Keys: filter-name
Minimum number of elements: 1
Maximum number of elements: 20

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
filter-name	enumeration	imsi		Filter name - predefined set of supported filters
		msisdn		
		imei		
		dnn		
		dnn-type		
		s-nssai		
		classifier-name		
		nssai		
		requested-nssai		
		subscribed-nssai		
		current-tai-list		
		serving-location		
		event-type		
		rat-type		
		apn		
		requested-apn		
		pdn-type		
		charging-characteristics		
		roamer-type		
		custom-filters		

Imsi

XPath: *cna-sub_analyzer:/analyzer/instance/class/filterset/filter/imsi*

CLI Path: *analyzer instance class filterset filter imsi*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
match-criteria	enumeration	presence		match-criteria for IMSI filter
		in-range		
		out-of-range		
low	string	[0-9]{1,15}		IMSI start value
high	string	[0-9]{1,15}		IMSI end value

Iimei

XPath: *cna-sub_analyzer:/analyzer/instance/class/filterset/filter/imei*

CLI Path: *analyzer instance class filterset filter imei*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
match-criteria	enumeration	presence		match-criteria for IMEI filter
		in-range		
		out-of-range		
low	string	[0-9]{1,16}		IMEI start value
high	string	[0-9]{1,16}		IMEI end value

Msisdn

XPath: *cna-sub_analyzer:/analyzer/instance/class/filterset/filter/msisdn*

CLI Path: *analyzer instance class filterset filter msisdn*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
match-criteria	enumeration	presence		match-criteria for MSISDN filter
		in-range		
		out-of-range		
low	string	[0-9]{1,15}		MSISDN start value
high	string	[0-9]{1,15}		MSISDN end value

Dnn

XPath: *cna-sub_analyzer:/analyzer/instance/class/filterset/filter/dnn*

CLI Path: *analyzer instance class filterset filter dnn*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
match-criteria	enumeration	contains		match-criteria for dnn filter
		equals		
		does-not-equal		
value	string	([0-9a-zA-Z_-]*)*		Dnn filter value

Dnn Type

XPath: *cna-sub_analyzer:/analyzer/instance/class/filterset/filter/dnn-type*

CLI Path: *analyzer instance class filterset filter dnn-type*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
match-criteria	enumeration	presence		match-criteria for dnn type filter
		equals		
		does-not-equal		
value	enumeration	ipv4v6		Dnn type filter value
		ipv4		
		ipv6		
		unstr		
		ether		

S Nssai

Value FFFFFF is reserved and cannot be used to configure slice-differentiator

XPath: *cna-sub_analyzer:/analyzer/instance/class/filterset/filter/s-nssai*

CLI Path: *analyzer instance class filterset filter s-nssai*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
match-criteria	enumeration	presence equals does-not-equal		match-criteria for S-NSSAI filter
sst	union	['uint8,0 or 4 .. 255', 'Enumerations: eMBB URLLC MlIoT ']		SST (Slice Service Type) value for S-NSSAI Filter
sd	union	['string,([0-9a-fA-F]{6})', 'Enumerations: not-specified ']		SD (Slice Differentiator) value for S-NSSAI filter

Classifier Name

XPath: *cna-sub_analyzer:/analyzer/instance/class/filterset/filter/classifier-name*

CLI Path: *analyzer instance class filterset filter classifier-name*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
match-criteria	enumeration	contains equals does-not-equal		match-criteria for classifier-name filter
value	string	([0-9a-zA-Z_-]*)		Classifier name filter value

Nssai

At least one 's-nssai' entry must be configured when 'match-criteria' is configured as either equals or does-not-equals.

XPath: *cna-sub_analyzer:/analyzer/instance/class/filterset/filter/nssai*

CLI Path: *analyzer instance class filterset filter nssai*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
match-criteria	enumeration	contains equals does-not-equal		match-criteria for nssai filter

S Nssai

List of Supported slices per PLMN

Value FFFFFFFF is reserved and cannot be used to configure slice-differentiator

XPath: *cna-sub_analyzer:/analyzer/instance/class/filterset/filter/nssai/s-nssai*

CLI Path: *analyzer instance class filterset filter nssai s-nssai*

List Properties:

Keys: sst, sd

Maximum number of elements: 20

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
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Attribute	Type	Range/Length/Pattern	Default	Description
sst	union	['uint8,0 or 4 .. 255', 'Enumerations: eMBB URLLC MIoT']		SST (Slice Service Type) value for S-NSSAI list in NSSAI filter
sd	union	['string,([0-9a-fA-F]{6})', 'Enumerations: not-specified']		SD (Slice Differentiator) value for S-NSSAI list in NSSAI filter

Serving Location

XPath: [cna-sub_analyzer:/analyzer/instance/class/filterset/filter/serving-location](#)

CLI Path: [analyzer instance class filterset filter serving-location](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
match-criteria	enumeration	contains equals does-not-equal		match-criteria for serving location filter
zone	string	1..255		Zone name
serving-area-group	leafref			Serving Area Group name Leaf Reference Path: cna-sub_analyzer:/analyzer/serving-area-group/name

Serving Plmn

XPath: [cna-sub_analyzer:/analyzer/instance/class/filterset/filter/serving-location/serving-plmn](#)

CLI Path: [analyzer instance class filterset filter serving-location serving-plmn](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
mcc	string	(([0-9]{3}))		Mobile Country Code of the PLMN
mnc	string	(([0-9]{2,3}))		Mobile Network Code of the PLMN

Area (choice)

XPath: [cna-sub_analyzer:/analyzer/instance/class/filterset/filter/serving-location/serving-plmn](#)

CLI Path: [analyzer instance class filterset filter serving-location serving-plmn](#)

Case : tracking-area

Tacs (choice)

Choice: This tacs attribute is mutually exclusive with attributes [['lac', 'cell-id']]

XPath: [cna-sub_analyzer:/analyzer/instance/class/filterset/filter/serving-location/serving-plmn](#)

CLI Path: [analyzer instance class filterset filter serving-location serving-plmn](#)

Case : tac-5g

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
tac-5g	string	(0x[A-Fa-f0-9]{1,6})		Tracking Area Code (TAC); Valid values are hexadecimal values of length 4 or 6 prefix with '0x' Choice: This tac-5g attribute is mutually exclusive with attributes [['lac', 'cell-id'], ['tac-4g']]

Case : tac-4g

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
tac-4g	string	(0x[A-Fa-f0-9]){1,4})		Tracking Area Code (TAC); Valid values are hexadecimal values of length 4 prefix with '0x' Choice: This tac-4g attribute is mutually exclusive with attributes [[]], ['lac', 'cell-id'], ['tac-5g']]

Case : location-area

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
lac	string	(0x[A-Fa-f0-9]){1,4})		Location Area Code (LAC); Valid values are hexadecimal values of length 4 prefix with '0x'
cell-id	string	(0x[A-Fa-f0-9]){1,4})		Cell Id; Valid values are hexadecimal values of length 4 prefix with '0x'

Rac Sac (choice)

XPath: [cna-sub_analyzer:/analyzer/instance/class/filterset/filter/serving-location/serving-plmn](#)

***CLI Path: [analyzer instance class filterset filter serving-location serving-plmn](#) ***

Case : rac

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
rac	string	(0x[A-Fa-f0-9]){1,2})		Routing Area Code (RAC); Valid values are hexadecimal values of length 2 prefix with '0x' Choice: This rac attribute is mutually exclusive with attributes [['sac']]

Case : sac

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
sac	string	(0x[A-Fa-f0-9]){1,4})		Service Area Code (SAC); Valid values are hexadecimal values of length 4 prefix with '0x' Choice: This sac attribute is mutually exclusive with attributes [['rac']]

Cell Ids (choice)

XPath: [cna-sub_analyzer:/analyzer/instance/class/filterset/filter/serving-location/serving-plmn](#)

***CLI Path: [analyzer instance class filterset filter serving-location serving-plmn](#) ***

Case : eutran-cell-id

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
eutran-cell-id	string	(0x[A-Fa-f0-9]){1,7})		E-UTRAN Cell Id; Valid values are hexadecimal values of length 7 prefix with '0x' Choice: This eutran-cell-id attribute is mutually exclusive with attributes [['nr-cell-id']]

Case : nr-cell-id

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
nr-cell-id	string	(0x[A-Fa-f0-9]){1,9})		NR Cell Id; Valid values are hexadecimal values of length 9 prefix with '0x' Choice: This nr-cell-id attribute is mutually exclusive with attributes [['eutran-cell-id']]

Requested Nssai

At least one 's-nssai' entry must be configured when ' match-criteria' is configured as either equals or does-not-equals.

XPath: *cna-sub_analyzer:/analyzer/instance/class/filterset/filter/requested-nssai*

CLI Path: *analyzer instance class filterset filter requested-nssai*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
match-criteria	enumeration	contains contains-all equals		match-criteria for requested nssai filter

S Nssai

List of requested NSSAI
Value FFFFFFFF is reserved and cannot be used to configure slice-differentiator

XPath: *cna-sub_analyzer:/analyzer/instance/class/filterset/filter/requested-nssai/s-nssai*

CLI Path: *analyzer instance class filterset filter requested-nssai s-nssai*

List Properties:
Keys: sst, sd
Maximum number of elements: 8

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
sst	union	['uint8,0 or 4 .. 255', 'Enumerations: eMBB URLLC MlIoT ']		SST (Slice Service Type) value for S-NSSAI
sd	union	['string,([0-9a-fA-F]{6})', 'Enumerations: not-specified ']		SD (Slice Differentiator) value for S-NSSAI

Subscribed Nssai

At least one 's-nssai' entry must be configured when ' match-criteria' is configured as either equals or does-not-equals.

XPath: *cna-sub_analyzer:/analyzer/instance/class/filterset/filter/subscribed-nssai*

CLI Path: *analyzer instance class filterset filter subscribed-nssai*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
match-criteria	enumeration	contains contains-all equals		match-criteria for subscribed nssai filter

S Nssai

List of subscribed NSSAI
Value FFFFFFFF is reserved and cannot be used to configure slice-differentiator

XPath: *cna-sub_analyzer:/analyzer/instance/class/filterset/filter/subscribed-nssai/s-nssai*

CLI Path: *analyzer instance class filterset filter subscribed-nssai s-nssai*

List Properties:
Keys: sst, sd
Maximum number of elements: 20

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
sst	union	['uint8,0 or 4 .. 255', 'Enumerations: eMBB URLLC MIoT ']		SST (Slice Service Type) value for S-NSSAI
sd	union	['string,([0-9a-fA-F]{6})', 'Enumerations: not-specified ']		SD (Slice Differentiator) value for S-NSSAI

Current Tai List

XPath: *cna-sub_analyzer:/analyzer/instance/class/filterset/filter/current-tai-list*

CLI Path: *analyzer instance class filterset filter current-tai-list*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
match-criteria	enumeration	contains		match-criteria for current tai

Tai 5G

List of current TAIs

XPath: *cna-sub_analyzer:/analyzer/instance/class/filterset/filter/current-tai-list/tai-5g*

CLI Path: *analyzer instance class filterset filter current-tai-list tai-5g*

List Properties:

Keys: mcc, mnc, tac-5g

Maximum number of elements: 20

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
mcc	string	((([0-9]{3})))		Mobile Country Code of the PLMN
mnc	string	((([0-9]{2,3})))		Mobile Network Code of the PLMN
tac-5g	string	(0x[A-Fa-f0-9]{4,6})		Tracking Area Code (TAC); Valid values are hexadecimal values of length 4 or 6 prefix with '0x'

Tai 4G

List of current TAIs

XPath: *cna-sub_analyzer:/analyzer/instance/class/filterset/filter/current-tai-list/tai-4g*

CLI Path: *analyzer instance class filterset filter current-tai-list tai-4g*

List Properties:

Keys: mcc, mnc, tac-4g

Maximum number of elements: 20

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
mcc	string	((([0-9]{3})))		Mobile Country Code of the PLMN
mnc	string	((([0-9]{2,3})))		Mobile Network Code of the PLMN
tac-4g	string	(0x[A-Fa-f0-9]{4})		Tracking Area Code (TAC); Valid values are hexadecimal values of length 4 prefix with '0x'

Event Type

XPath: *cna-sub_analyzer:/analyzer/instance/class/filterset/filter/event-type*

CLI Path: *analyzer instance class filterset filter event-type*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
match-criteria	enumeration	equals		match-criteria for evnt type filter
value	enumeration	registration pdu-establishment		Event type filter value

Rat Type

XPath: [cna-sub_analyzer:/analyzer/instance/class/filterset/filter/rat-type](#)

CLI Path: [analyzer instance class filterset filter rat-type](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
match-criteria	enumeration	presence equals does-not-equal		match-criteria for rat type filter
value	enumeration	geran utran eutran-nb eutran-wb ng-ran wlan virtual		Radio Access Technology (RAT) filter value

Apn

XPath: [cna-sub_analyzer:/analyzer/instance/class/filterset/filter/apn](#)

CLI Path: [analyzer instance class filterset filter apn](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
match-criteria	enumeration	contains equals does-not-equal		match-criteria for apn filter
value	string	([0-9a-zA-Z_-]*)		Access Point Name filter value

Requested Apn

XPath: [cna-sub_analyzer:/analyzer/instance/class/filterset/filter/requested-apn](#)

CLI Path: [analyzer instance class filterset filter requested-apn](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
match-criteria	enumeration	contains equals does-not-equal		match-criteria for requested-apn filter
value	string	([0-9a-zA-Z_-]*)		Requested Access Point Name filter value

Pdn Type

XPath: [cna-sub_analyzer:/analyzer/instance/class/filterset/filter/pdn-type](#)

CLI Path: [analyzer instance class filterset filter pdn-type](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
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Attribute	Type	Range/Length/Pattern	Default	Description
match-criteria	enumeration	presence equals does-not-equal		match-criteria for pdn type filter
value	enumeration	ipv4 ipv6 ipv4v6 non-ip		Packet Data Network type filter value

Charging Characteristics

XPath: [cna-sub_analyzer:/analyzer/instance/class/filterset/filter/charging-characteristics](#)

CLI Path: [analyzer instance class filterset filter charging-characteristics](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
match-criteria	enumeration	is-any-set is-not-set equals does-not-equal		match-criteria for Charging Characteristics filter
value	string	(0x[A-Fa-f0-9]{4})		Charging Characteristics; Valid values are hexadecimal values of length 4 or 6 prefix with '0x'

Roamer Type

XPath: [cna-sub_analyzer:/analyzer/instance/class/filterset/filter/roamer-type](#)

CLI Path: [analyzer instance class filterset filter roamer-type](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
match-criteria	enumeration	presence equals does-not-equal		match-criteria for roamer type filter
value	enumeration	home-user-roaming visiting-user-local-breakout visiting-user-home-routed		Roamer filter value

Custom Filters

XPath: [cna-sub_analyzer:/analyzer/instance/class/filterset/filter/custom-filters](#)

CLI Path: [analyzer instance class filterset filter custom-filters](#)

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
match-criteria	enumeration	presence contains equals does-not-equal		Match criteria for custom-filters filter

Custom Filter

Custom filter list

XPath: [cna-sub_analyzer:/analyzer/instance/class/filterset/filter/custom-filters/custom-filter](#)

CLI Path: [analyzer instance class filterset filter custom-filters custom-filter](#)

List Properties:

Keys: name

Minimum number of elements: 1

Maximum number of elements: 10

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
name	string			Custom filter name
value	string			Value of the custom filter

UPFMONITOR Configuration

The module for entities implementing cna-upfmonitor.

Table of Contents

- upfmonitor
 - load-profile
 - critical-service-frequent-restart
 - deregistration-time
 - application-deregistration-time

Upfmonitor

UPF Monitor interface object

XPath: *cna-upfmonitor:/upfmonitor*

CLI Path: *upfmonitor*

Load Profile

Configure UPF Load Profile

XPath: *cna-upfmonitor:/upfmonitor/load-profile*

CLI Path: *upfmonitor load-profile*

List Properties:

Keys: load-profile

Maximum number of elements: 1

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
load-profile	string			UPF Load Profile Name
reporting-frequency-interval	uint64	60 .. 36000	120	Frequency interval(seconds) at which the UPF load should be reported to NRF
reporting-frequency-change-factor	uint64			Load change factor(+/- load difference) which should result in reporting the UPF load to the NRF

Critical Service Frequent Restart

Configure critical service frequent Restart monitoring feature
status-check-interval should be always greater than stable-interval
when monitoring is enabled, critical-flip-count should not be 0
when monitoring is enabled, threshold-flip-count should not be 0
critical-flip-count should be always greater than threshold-flip-count

XPath: *cna-upfmonitor:/upfmonitor/critical-service-frequent-restart*

CLI Path: *upfmonitor critical-service-frequent-restart*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
monitoring	string	enabled or disabled	disabled	Feature enabled or disabled
stable-interval	uint16	1..3600	120	The minimum runtime in seconds for a critical service to be considered as stable
status-check-interval	uint16	1..3600	180	The duration in seconds after which the critical service status should be rechecked.

Attribute	Type	Range/Length/Pattern	Default	Description
threshold-flip-count	uint16	1..3600	3	Number of times a critical service is found unstable after which alarm/alert is generated
critical-flip-count	uint16	1..3600	20	Number of times a critical service is found unstable after which vnf restart is triggered

Deregistration Time

Duration(in Secs), Monitor waits for a Critical Service to become available

XPath: *cna-upfmonitor:/upfmonitor/deregistration-time*

CLI Path: *upfmonitor deregistration-time*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
value	uint16	1..3600	300	Duration(in Secs), Monitor waits for a Critical Service to become available

Application Deregistration Time

Duration(in Secs), Monitor waits for the application to become available

XPath: *cna-upfmonitor:/upfmonitor/application-deregistration-time*

CLI Path: *upfmonitor application-deregistration-time*

Attributes

Attribute	Type	Range/Length/Pattern	Default	Description
value	uint16	1..3600	300	Duration(in Secs), Monitor waits for the application to become available